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(54) **CAS9 PROTEINS INCLUDING
LIGAND-DEPENDENT INTEINS**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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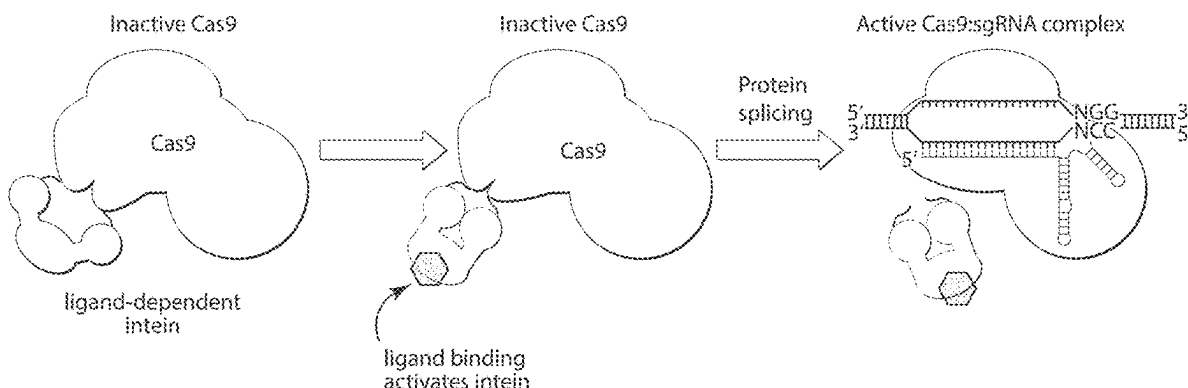
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(57) **ABSTRACT**

Some aspects of this disclosure provide compositions, meth-
ods, systems, and kits for controlling the activity of RNA-
programmable endonucleases, such as Cas9, or for control-
ling the activity of proteins comprising a Cas9 variant fused
to a functional effector domain, such as a nuclease, nickase,
recombinase, deaminase, transcriptional activator, transcrip-
tional repressor, or epigenetic modifying domain. For
example, the inventive proteins provided comprise a ligand-
dependent intein, the presence of which inhibits one or more
activities of the protein (e.g., gRNA binding, enzymatic
activity, target DNA binding). The binding of a ligand to the
intein results in self-excision of the intein, restoring the
activity of the protein.

33 Claims, 14 Drawing Sheets

Specification includes a Sequence Listing.



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continuation of application No. 16/132,276, filed on Sep. 14, 2018, now Pat. No. 10,704,062, which is a continuation of application No. 15/329,925, filed as application No. PCT/US2015/042770 on Jul. 30, 2015, now Pat. No. 10,077,453.

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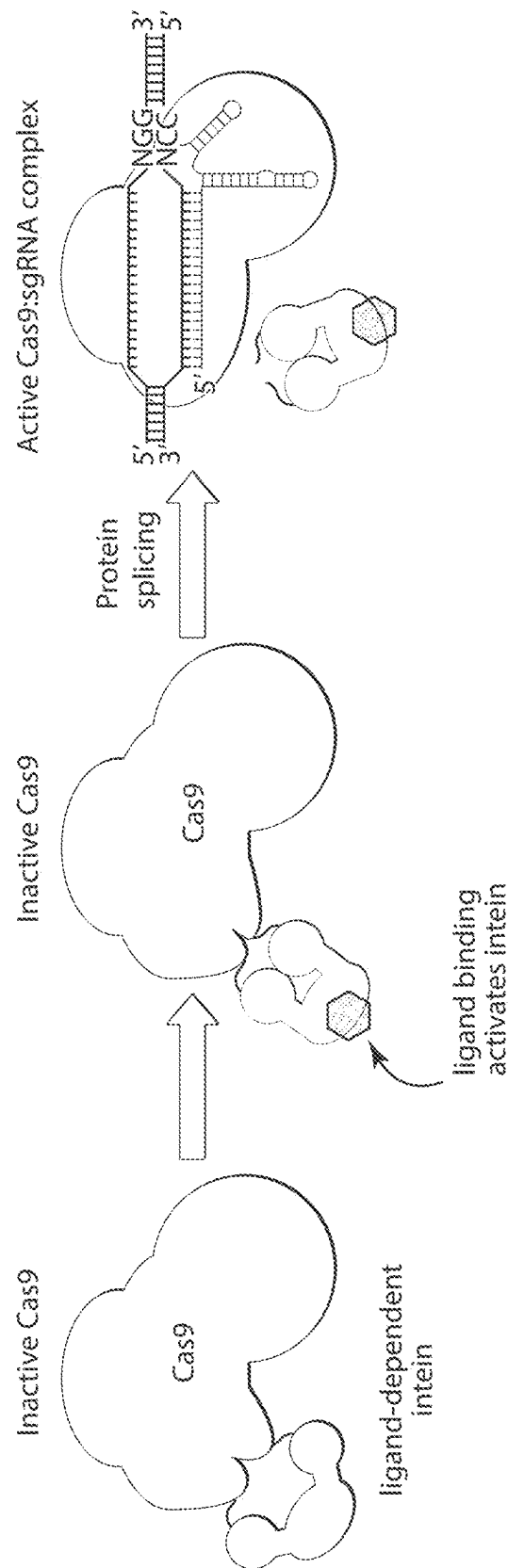


FIG. 1

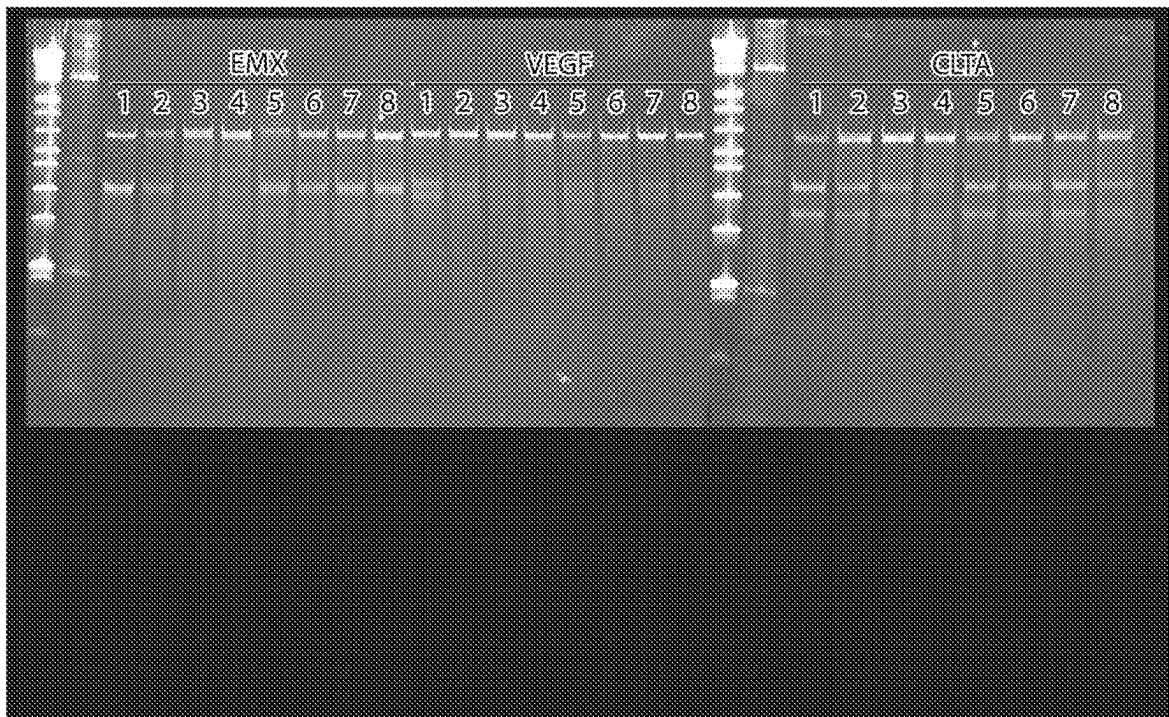


FIG. 2

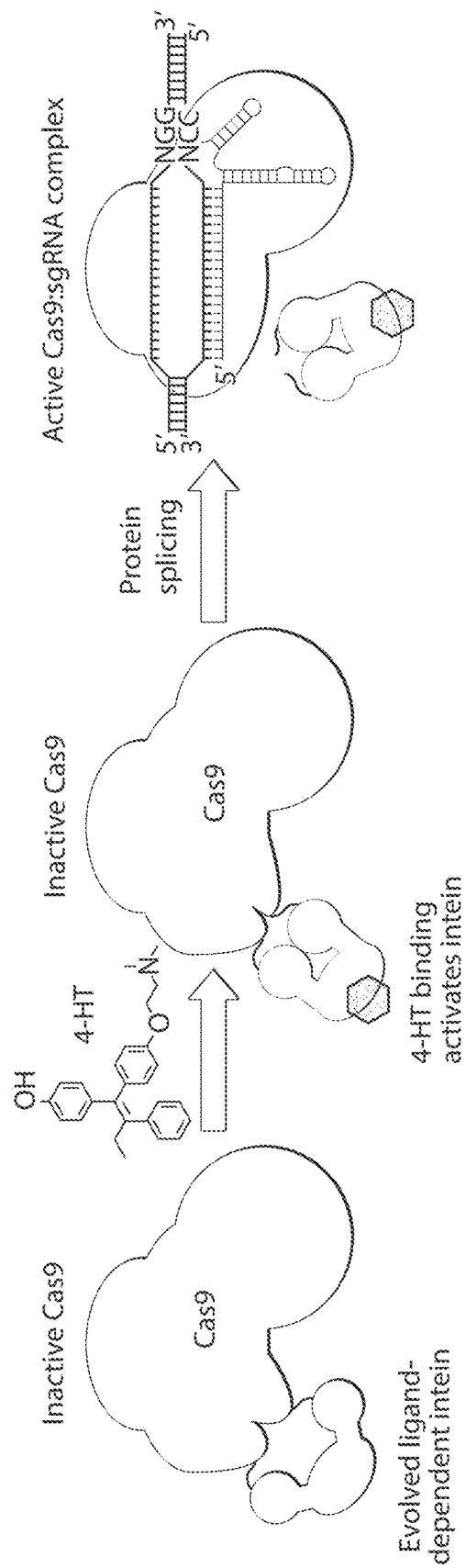


FIG. 3A



FIG. 3B

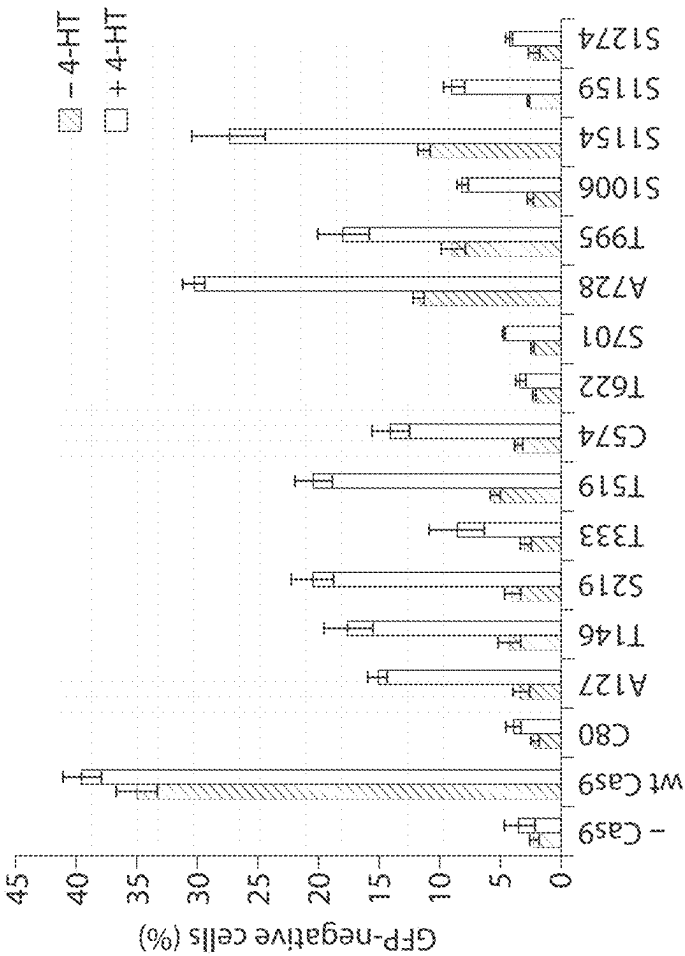


FIG. 3C

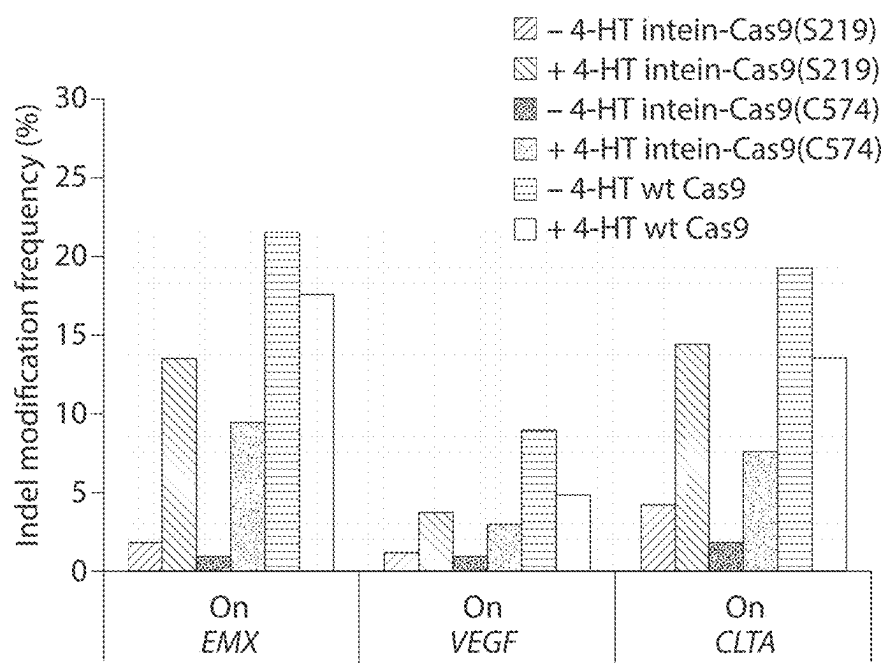


FIG. 4A

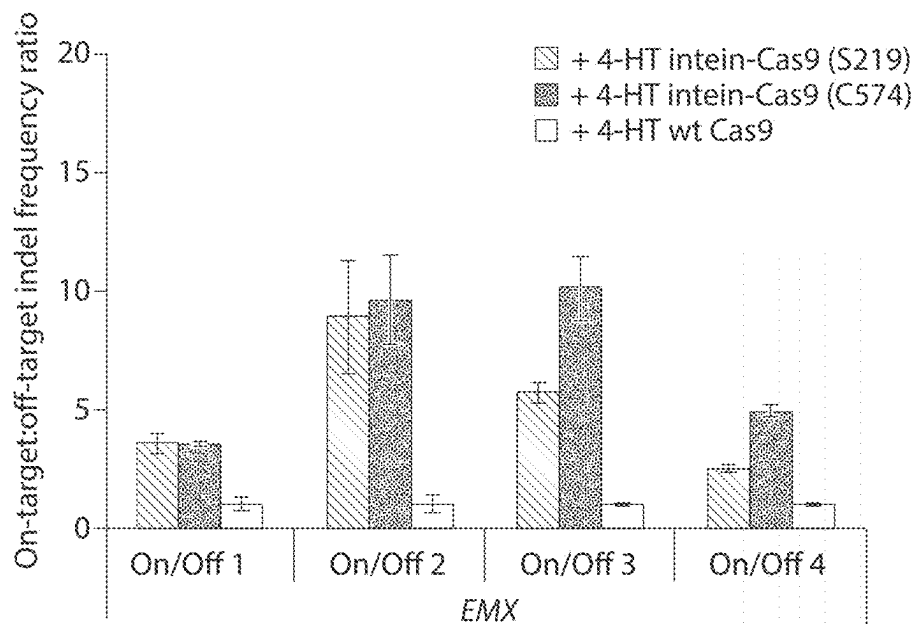


FIG. 4B

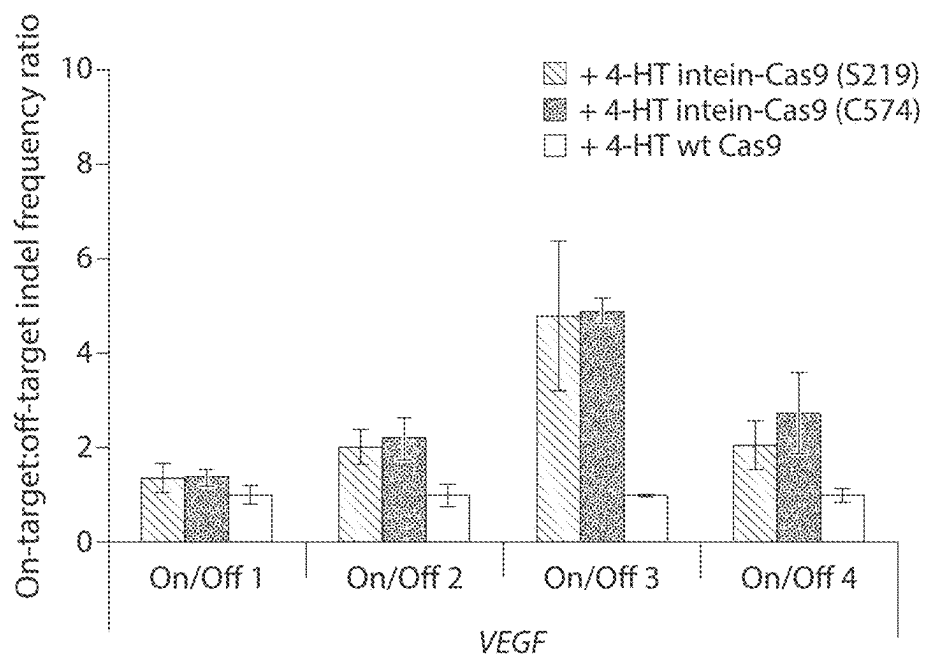


FIG. 4C

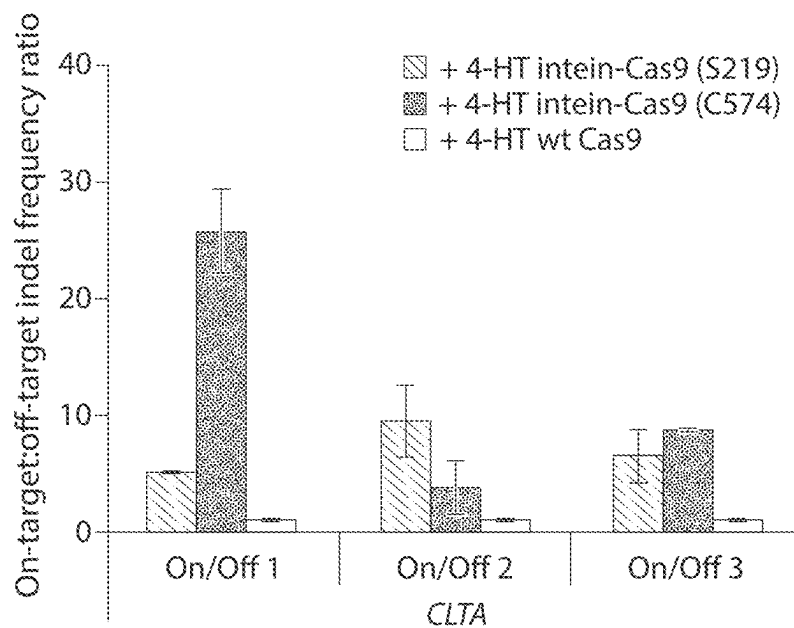


FIG. 4D

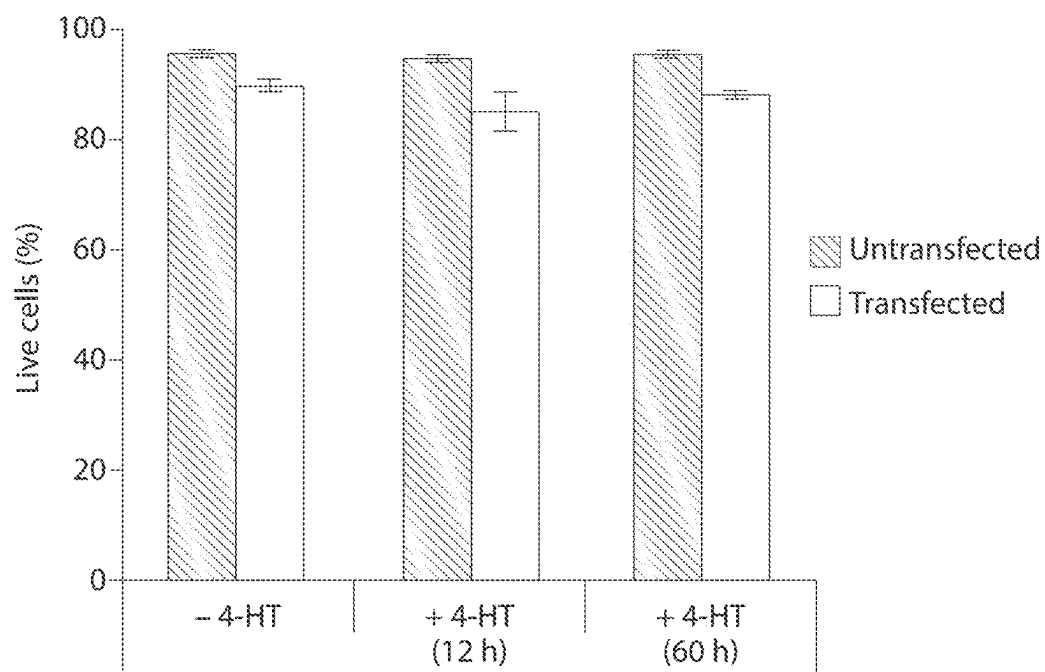


FIG. 5

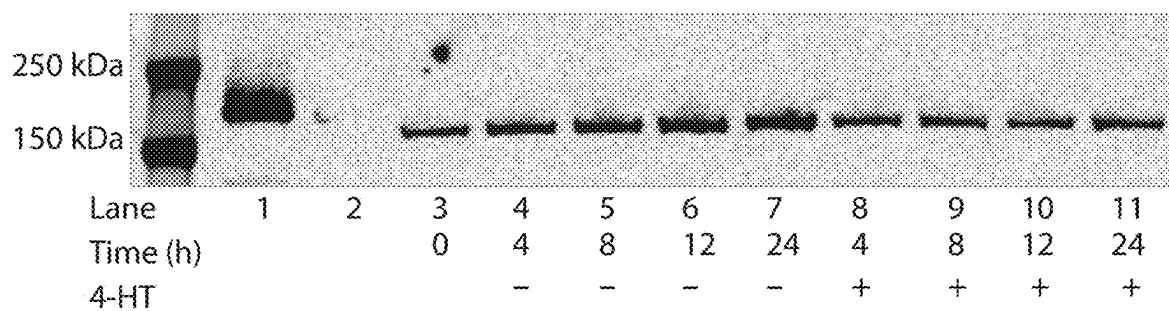


FIG. 6A

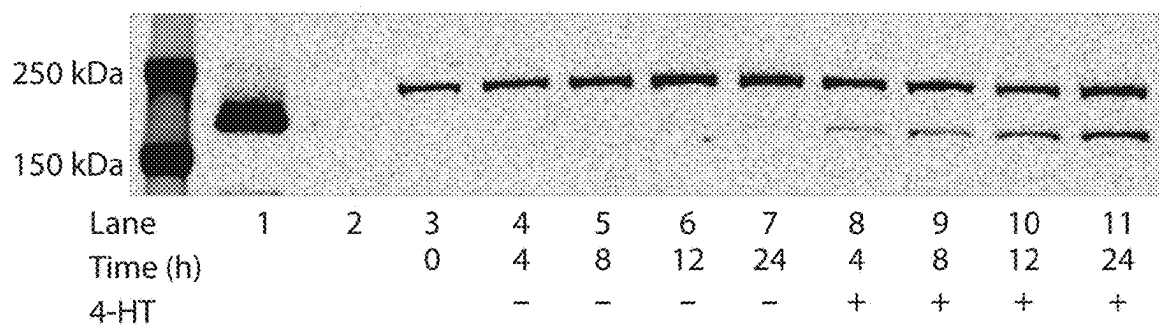


FIG. 6B

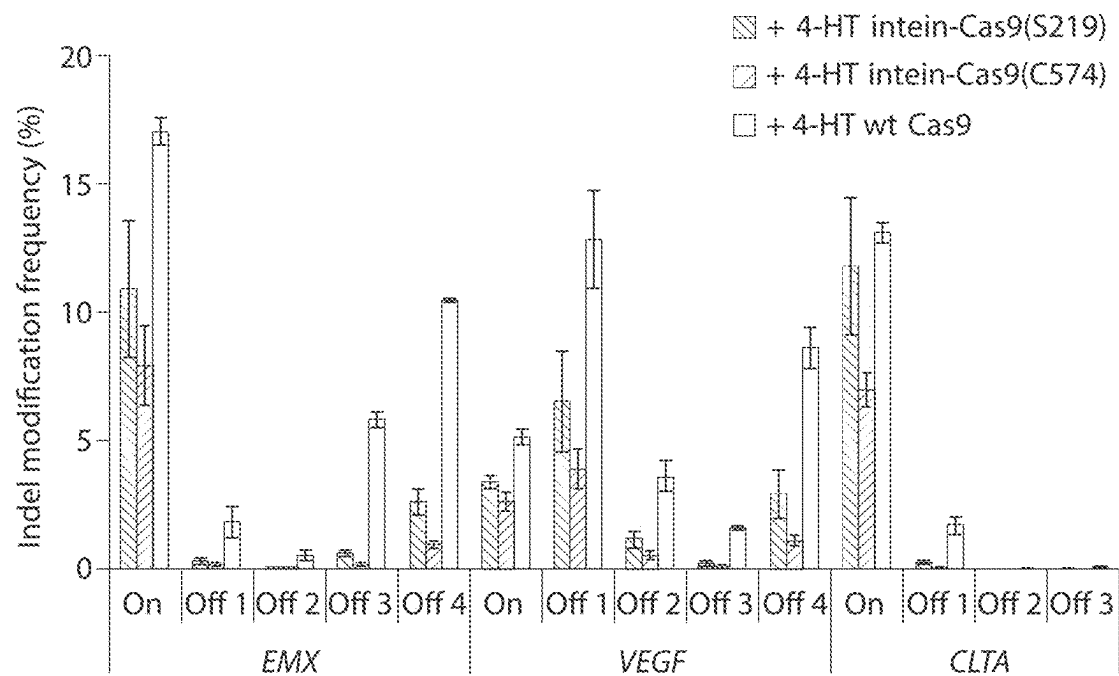


FIG. 7

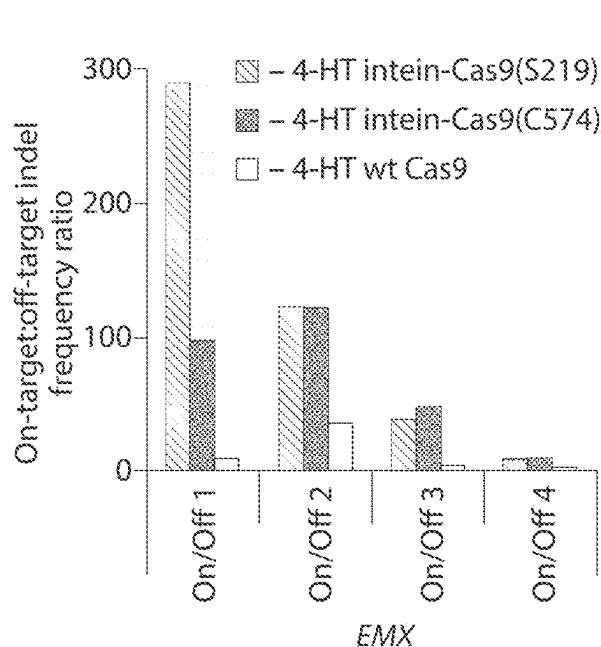


FIG. 8A

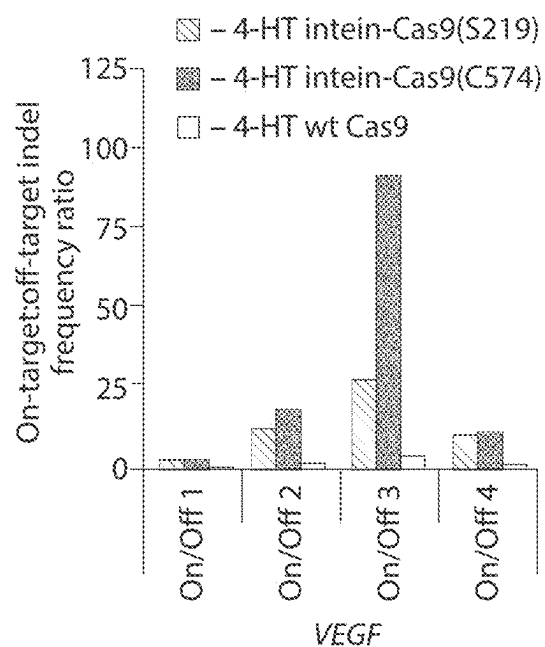


FIG. 8B

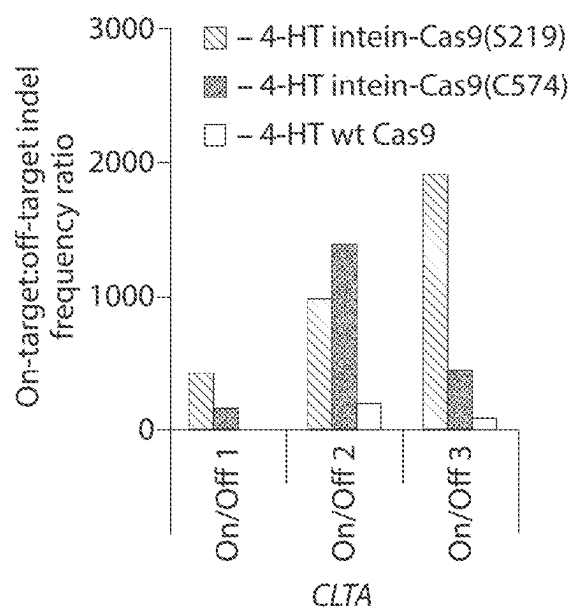


FIG. 8C

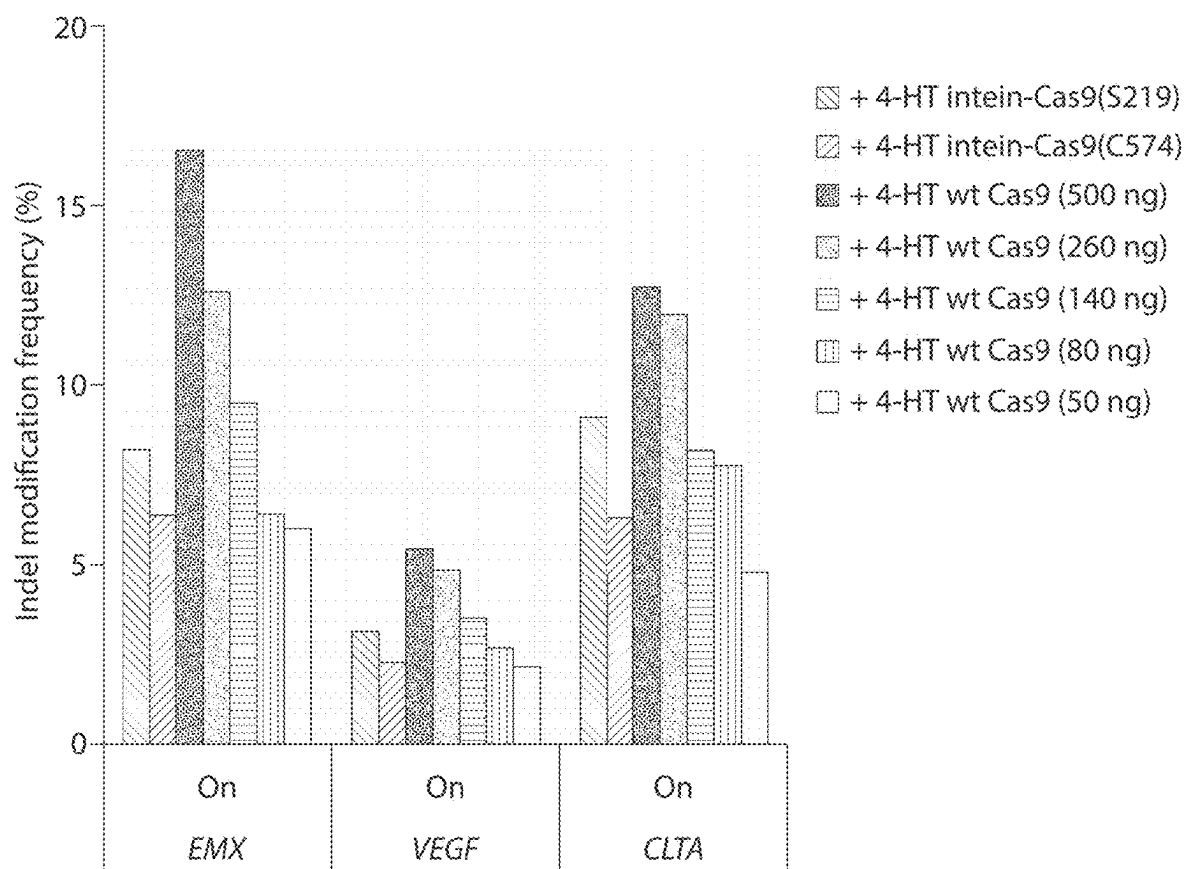


FIG. 9

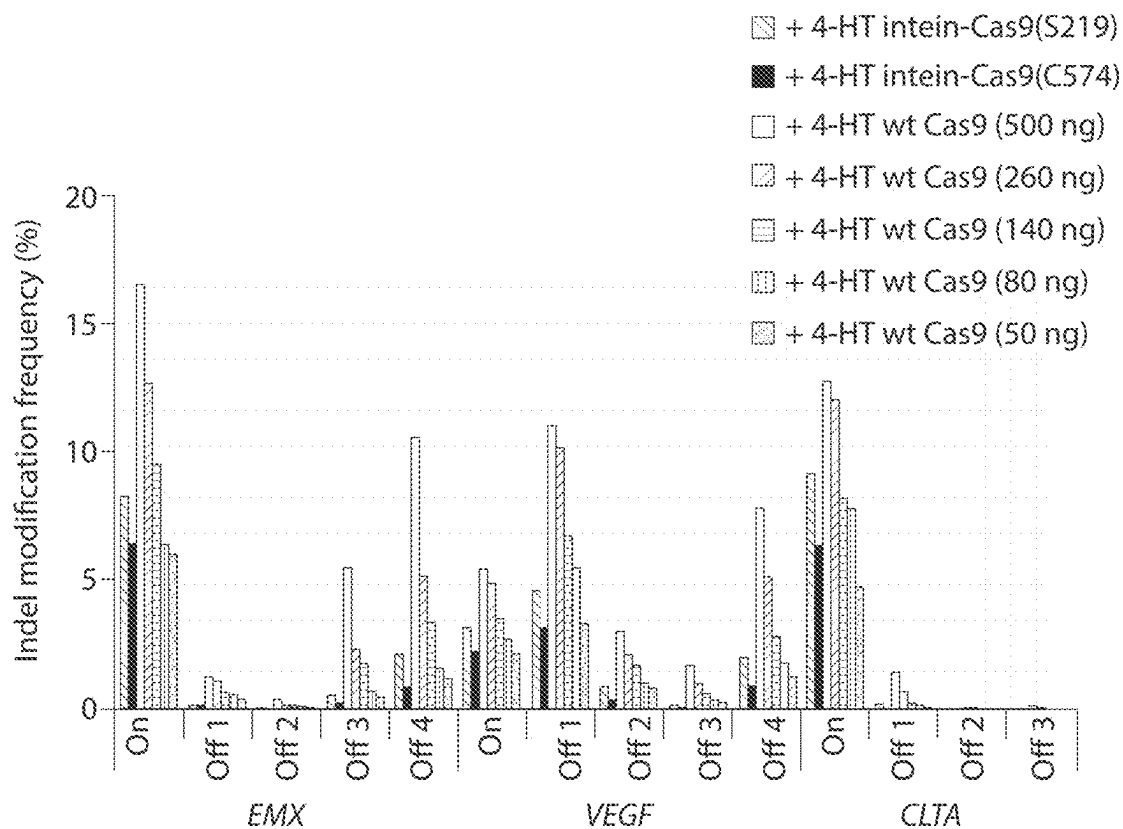


FIG. 10A

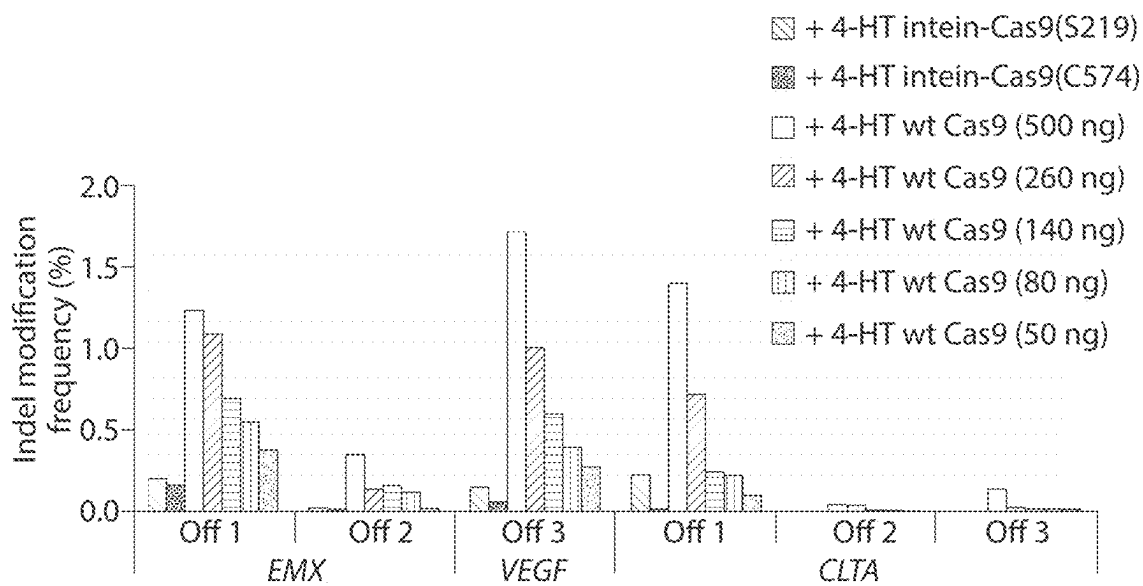


FIG. 10B

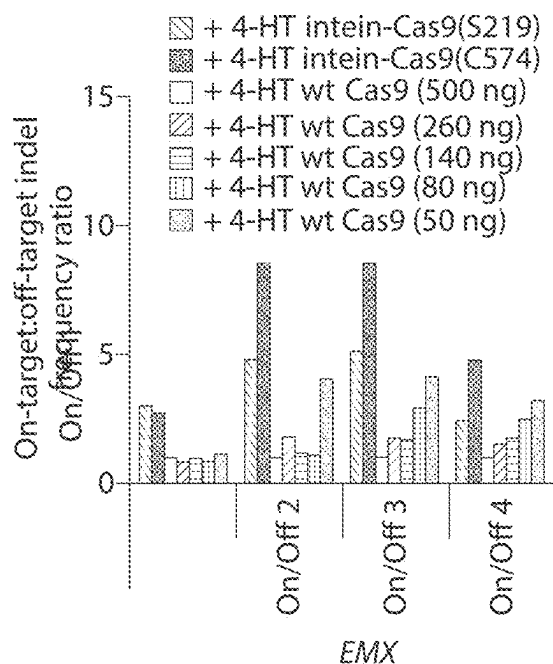


FIG. 11A

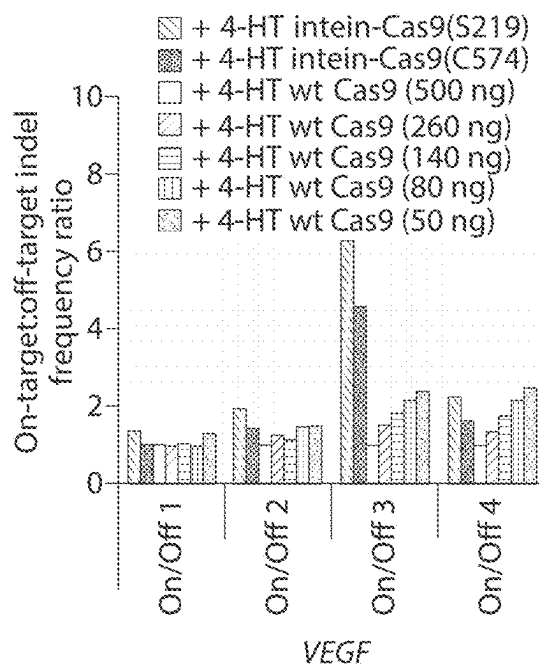


FIG. 11B

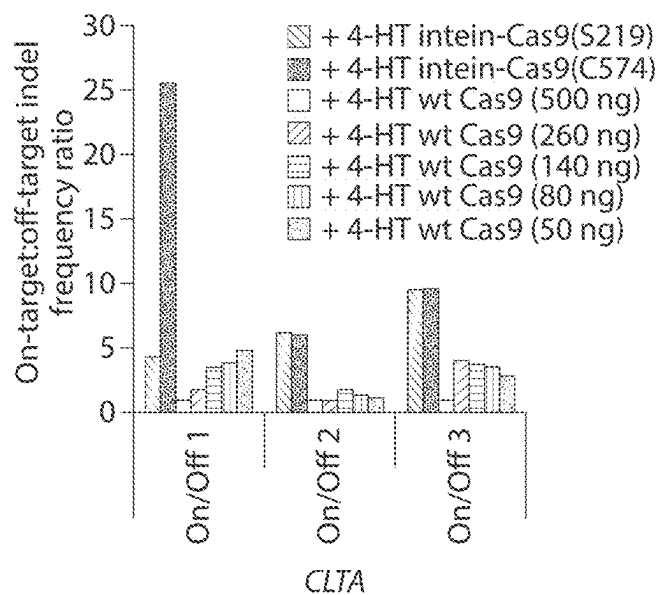


FIG. 11C

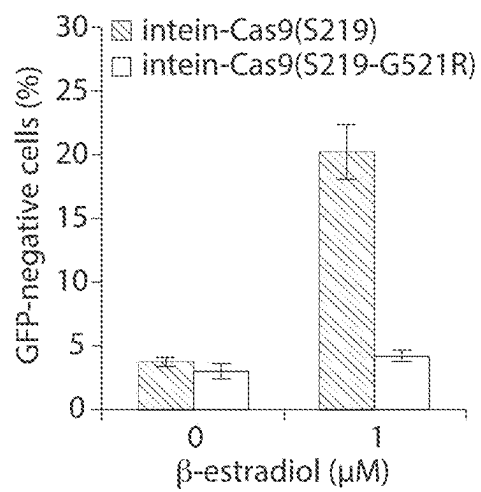


FIG. 12A

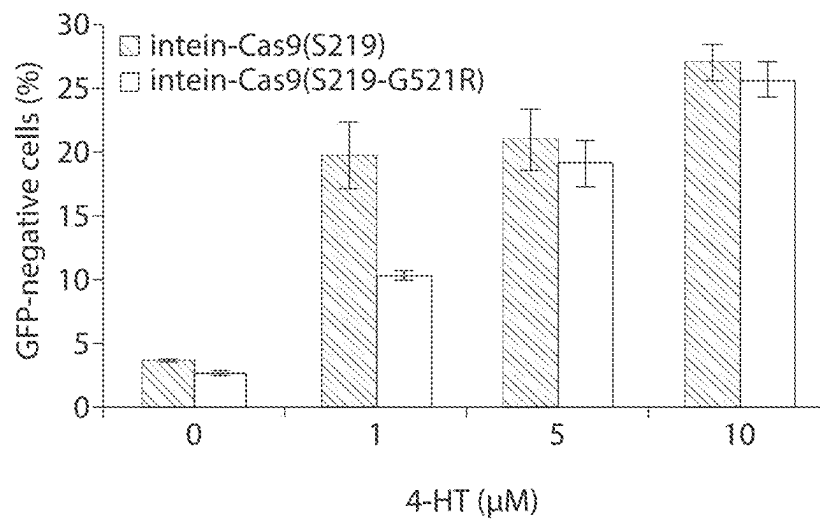


FIG. 12B

1

**CAS9 PROTEINS INCLUDING
LIGAND-DEPENDENT INTEINS****RELATED APPLICATIONS**

This application is a continuation of and claims priority under 35 U.S.C. § 120 to U.S. non-provisional patent application U.S. Ser. No. 16/888,646, filed May 29, 2020, which is a continuation of and claims priority under 35 U.S.C. § 120 to U.S. non-provisional patent application U.S. Ser. No. 16/132,276, filed Sep. 14, 2018, which is a continuation of U.S. non-provisional patent application U.S. Ser. No. 15/329,925, filed Jan. 27, 2017, which is a national stage filing under 35 U.S.C. § 371 of international PCT application, PCT/US2015/042770, filed Jul. 30, 2015, which claims priority under 35 U.S.C. § 119(e) to U.S. provisional patent application, U.S. Ser. No. 62/030,943, filed Jul. 30, 2014, and to U.S. provisional patent application, U.S. Ser. No. 62/135,629, filed Mar. 19, 2015, the entire contents of each of which are incorporated herein by reference.

GOVERNMENT SUPPORT

This invention was made with government support under HR0011-11-2-0003 awarded by the Department of Defense/ Defense Advanced Research Projects Agency; GM095501 and GM106601 awarded by the National Institutes of Health; and N66001-12-C-4207 awarded by the Space and Naval Warfare Systems Command. The government has certain rights in the invention.

**REFERENCE TO AN ELECTRONIC SEQUENCE
LISTING**

The contents of the electronic sequence listing (H082470183US05-SUBSEQ-JQM.xml; Size: 328,367 bytes; and Date of Creation: Jan. 1, 2023) is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Site-specific enzymes theoretically allow for the targeted manipulation of a single site within a genome and are useful in the context of gene targeting as well as for therapeutic applications. In a variety of organisms, including mammals, site-specific enzymes such as endonucleases have been used for genome engineering by stimulating either non-homologous end joining or homologous recombination. In addition to providing powerful research tools, site-specific nucleases also have potential as gene therapy agents, and two site-specific endonucleases have recently entered clinical trials: one, CCR5-2246, targeting a human CCR-5 allele as part of an anti-HIV therapeutic approach (clinical trials NCT00842634, NCT01044654, NCT01252641), and the other one, VF24684, targeting the human VEGF-A promoter as part of an anti-cancer therapeutic approach (clinical trial NCT01082926).

Specific manipulation of the intended target site without or with only minimal off-target activity is a prerequisite for clinical applications of site-specific enzymes, and also for high-efficiency genomic manipulations in basic research applications. For example, imperfect specificity of engineered site-specific binding domains of certain nucleases has been linked to cellular toxicity and undesired alterations of genomic loci other than the intended target. Most nucleases available today, however, exhibit significant off-target activity, and thus may not be suitable for clinical applications. An

2

emerging nuclease platform for use in clinical and research settings are the RNA-guided nucleases, such as Cas9. While these nucleases are able to bind guide RNAs (gRNAs) that direct cleavage of specific target sites, off-target activity is still observed for certain Cas9:gRNA complexes (Pat-
tanayak et al., “High-throughput profiling of off-target DNA cleavage reveals RNA-programmed Cas9 nuclease specificity.” *Nat Biotechnol.* 2013; doi: 10.1038/nbt.2673). Technology for engineering site-specific enzymes with reduced off-target effects is therefore needed.

SUMMARY OF THE INVENTION

The reported toxicity of some engineered site-specific enzymes such as endonucleases is thought to be based on off-target DNA cleavage. Further, the activity of existing RNA-guided nucleases generally cannot be controlled at the molecular level, for example, to switch a nuclease from an “off” to an “on” state. Controlling the activity of nucleases and other site-specific enzymes suitable for nucleic acid manipulations or modifications could decrease the likelihood of incurring off-target effects. Some aspects of this disclosure provide strategies, compositions, systems, and methods to control the binding and/or enzymatic activity of RNA-programmable enzymes, such as Cas9 endonuclease, nickases, deaminases, recombinases, transcriptional activators and repressors, epigenetic modifiers variants and fusions thereof.

Accordingly, one aspect of the present disclosure provides Cas9 proteins (including fusions of Cas9 proteins and functional domains) comprising inteins, for example, ligand-dependent inteins. The presence of the intein inhibits one or more activities of the Cas9 proteins, for example, nucleic acid binding activity (e.g., target nucleic acid binding activity and/or gRNA binding activity), a nuclease activity, or another enzymatic activity (e.g., nucleic acid modifying activity, transcriptional activation and repression, etc.) for which the Cas9 protein (e.g., Cas9 fusion protein) is engineered to undertake (e.g., nuclease activity, nickase activity, recombinase activity, deaminase activity, transcriptional activator/repressor activity, epigenetic modification, etc.). In some embodiments, the Cas9 protein is a Cas9 nickase. The Cas9 fusions are typically between a nuclease inactivated Cas9 (“dCas”) and one or more functional domains. The intein may be inserted into any location of a Cas9 protein, including one or more domains of a Cas9 protein or Cas9 fusion (including in a functional domain), such as the HNH nuclease domain or the RuvC nuclease domain. In some embodiments, the intein replaces amino acid residue Cys80, Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622, Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, or Ser1274 in the Cas9 polypeptide sequence set forth as SEQ ID NO:2, in the dCas9 polypeptide sequence set forth as SEQ ID NO:5, or in the Cas9 nickase polypeptide sequence set forth as SEQ ID NO:4. In some embodiments, the intein replaces or is inserted at an amino acid residue that is within 5, within 10, within 15, or within 20 amino acid residues of Cys80, Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622, Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, or Ser1274 in the Cas9 polypeptide sequence set forth as SEQ ID NO:2, in the dCas9 polypeptide sequence set forth as SEQ ID NO:5, or in the Cas9 nickase polypeptide sequence set forth as SEQ ID NO:4. The intein replaces amino acid residue Ala127, Thr146, Ser219, Thr519, or Cys574 in the Cas9 polypeptide sequence set forth as SEQ ID NO:2, in the dCas9 polypeptide sequence set forth as SEQ ID NO:5, or in the Cas9 nickase polypeptide sequence

set forth as SEQ ID NO:4. Typically the intein is a ligand-dependent intein which exhibits no or minimal protein splicing activity in the absence of ligand (e.g., small molecules such as 4-hydroxytamoxifen, peptides, proteins, polynucleotides, amino acids, and nucleotides). Ligand-dependent inteins are known, and include those described in U.S. patent application U.S. Ser. No. 14/004,280, published as U.S. 2014/0065711 A1, the entire contents of which are incorporated herein by reference. In some embodiments, the intein comprises an amino acid sequence selected from the group consisting of SEQ ID NO:7-14.

In one aspect, a Cas9 protein is provided that comprises: (i) a nuclease-inactivated Cas9 (e.g., dCas9 (SEQ ID NO:5)) domain; (ii) a ligand-dependent intein; and (iii) a recombinase catalytic domain. In some embodiments, the ligand-dependent intein domain is inserted into the dCas9 domain as described herein. Typically, the presence of the intein in the Cas9 protein inhibits one or more activities of the Cas9 protein, such as gRNA binding activity, target nucleic acid binding activity, and/or recombinase activity. Accordingly, upon self-excision of the intein (e.g., induced by ligand binding the intein) the one or more activities of the Cas9 protein is/are restored. In some embodiments, the recombinase catalytic domain is a monomer of the recombinase catalytic domain of Hin recombinase, Gin recombinase, or Tn3 recombinase.

According to another aspect, a Cas9 protein is provided that comprises: (i) a nuclease-inactivated Cas9 (e.g., dCas9 (SEQ ID NO:5)) domain; (ii) a ligand-dependent intein; and (iii) a deaminase catalytic domain. In some embodiments, the ligand-dependent intein domain is inserted into the dCas9 domain as described herein. Typically, the presence of the intein in the Cas9 protein inhibits one or more activities of the Cas9 protein, such as gRNA binding activity, target nucleic acid binding activity, and/or deaminase activity. Accordingly, upon self-excision of the intein (e.g., induced by ligand binding of the intein) the one or more activities of the Cas9 protein is/are restored. In some embodiments, the deaminase catalytic domain comprises a cytidine deaminase (e.g., of apolipoprotein B mRNA-editing complex (APOBEC) family deaminases such as APOBEC1 or activation-induced cytidine deaminase (AID)). In some embodiments, the deaminase catalytic domain comprises a ACF1/ASE deaminase or an adenosine deaminase, such as a ADAT family deaminase.

According to another aspect, a Cas9 protein is provided that comprises: (i) a nuclease-inactivated Cas9 (e.g., dCas9 (SEQ ID NO:5)) domain; (ii) a ligand-dependent intein; and (iii) a transcriptional activator domain. In some embodiments, the ligand-dependent intein domain is inserted into the dCas9 domain as described herein. Typically, the presence of the intein in the Cas9 protein inhibits one or more activities of the Cas9 protein, such as gRNA binding activity, target nucleic acid binding activity, and/or transcriptional activation. Accordingly, upon self-excision of the intein (e.g., induced by ligand binding the intein) the one or more activities of the Cas9 protein is/are restored. In some embodiments, the transcriptional activator domain is VP64, CP16, and p65.

According to yet another aspect, a Cas9 protein is provided that comprises: (i) a nuclease-inactivated Cas9 (e.g., dCas9 (SEQ ID NO:5)) domain; (ii) a ligand-dependent intein; and (iii) a transcriptional repressor domain. In some embodiments, the ligand-dependent intein domain is inserted into the dCas9 domain as described herein. Typically, the presence of the intein in the Cas9 protein inhibits one or more activities of the Cas9 protein, such as gRNA

binding activity, target nucleic acid binding activity, and/or transcriptional repression. Accordingly, upon self-excision of the intein (e.g., induced by ligand binding the intein) the one or more activities of the Cas9 protein is/are restored. In some embodiments, the transcriptional repressor domain is KRAB, SID, or SID4x. According to yet another aspect, a Cas9 protein is provided that comprises: (i) a nuclease-inactivated Cas9 (e.g., dCas9 (SEQ ID NO:5)) domain; (ii) a ligand-dependent intein; and (iii) an epigenetic modifier domain. In some embodiments, the ligand-dependent intein domain is inserted into the dCas9 domain as described herein. Typically, the presence of the intein in the Cas9 protein inhibits one or more activities of the Cas9 protein, such as gRNA binding activity, target nucleic acid binding activity, and/or epigenetic modification activity. Accordingly, upon self-excision of the intein (e.g., induced by ligand binding the intein) the one or more activities of the Cas9 protein is/are restored. In some embodiments, the epigenetic modifier domain is epigenetic modifier is selected from the group consisting of histone demethylase, histone methyltransferase, hydroxylase, histone deacetylase, and histone acetyltransferase. In some embodiments, the epigenetic modifier comprises the LSD1 histone demethylase or TET1 hydroxylase.

According to another aspect, methods of using Cas9 proteins are provided. In some embodiments involving site-specific DNA cleavage, the methods comprise (a) contacting a Cas9 protein (e.g., having nuclease activity) comprising a ligand-dependent intein with a ligand, wherein binding of the ligand to the intein induces self-excision of the intein; and (b) contacting a DNA with the Cas9 protein, wherein the Cas9 protein is associated with a gRNA; whereby self-excision of the intein from the Cas9 protein in step (a) allows the Cas9 protein to cleave the DNA, thereby producing cleaved DNA. In some embodiments, the Cas9 protein first binds a gRNA and optionally the target DNA prior to excision of the intein, but is unable to cleave the DNA until excision of the intein occurs. Any of the Cas9 proteins having nuclease activity and comprising a ligand-dependent intein, as described herein, can be used in the inventive methods.

According to another aspect, methods of using any of the ligand-dependent intein-containing Cas9 proteins comprising a recombinase catalytic domain are provided. In some embodiments, the method is useful for recombining two nucleic acids, such as two DNAs, and comprises (a) contacting a first DNA with a first ligand-dependent dCas9-recombinase fusion protein (e.g., any of those described herein), wherein the dCas9 domain of the first fusion protein binds a first gRNA that hybridizes to a region of the first DNA; (b) contacting the first DNA with a second ligand-dependent dCas9-recombinase fusion protein, wherein the dCas9 domain of the second fusion protein binds a second gRNA that hybridizes to a second region of the first DNA; (c) contacting a second DNA with a third ligand-dependent dCas9-recombinase fusion protein, wherein the dCas9 domain of the third fusion protein binds a third gRNA that hybridizes to a region of the second DNA; and (d) contacting the second DNA with a fourth ligand-dependent dCas9-recombinase fusion protein, wherein the dCas9 domain of the fourth fusion protein binds a fourth gRNA that hybridizes to a second region of the second DNA; whereby the binding of the fusion proteins in steps (a)-(d) results in the tetramerization of the recombinase catalytic domains of the fusion proteins, under conditions such that the DNAs are recombined. In some embodiments, the methods are useful for site-specific recombination between two regions of a

single DNA molecule, and comprise (a) contacting the DNA with a first ligand-dependent dCas9-recombinase fusion protein, wherein the dCas9 domain of the first fusion protein binds a first gRNA that hybridizes to a region of the DNA; (b) contacting the DNA with a second ligand-dependent dCas9-recombinase fusion protein, wherein the dCas9 domain of the second fusion protein binds a second gRNA that hybridizes to a second region of the DNA; (c) contacting the DNA with a third ligand-dependent dCas9-recombinase fusion protein, wherein the dCas9 domain of the third fusion protein binds a third gRNA that hybridizes to a third region of the DNA; (d) contacting the DNA with a fourth ligand-dependent dCas9-recombinase fusion protein, wherein the dCas9 domain of the fourth fusion protein binds a fourth gRNA that hybridizes to a fourth region of the DNA; whereby the binding of the fusion proteins in steps (a)-(d) results in the tetramerization of the recombinase catalytic domains of the fusion proteins, under conditions such that the DNA is recombined. In some embodiment, any of the methods first comprise contacting the fusion proteins with a ligand that induces self-excision of the intein. In some embodiments, the fusion proteins are contacted with the ligand after: (i) the fusion proteins bind a gRNA; (ii) the fusion proteins bind the DNA; or (iii) after the recombinase domains form a tetramer. In some embodiments, the gRNAs in any step (a)-(d) of the inventive methods hybridize to the same strand or to opposing strands in the DNA(s). In some embodiments, the gRNAs hybridize to regions of their respective DNAs that are no more than 10, no more than 15, no more than 20, no more than 25, no more than 30, no more than 40, no more than 50, no more than 60, no more than 70, no more than 80, no more than 90, or no more than 100 base pairs apart.

According to yet another aspect, methods of using any of the ligand-dependent intein Cas9 proteins comprising deaminase catalytic domains are provided. The methods comprise contacting a DNA molecule with (a) a ligand-dependent Cas9 protein comprising deaminase catalytic domain as provided herein; and (b) a gRNA targeting the Cas9 protein of step (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the Cas9 protein, and the gRNA in an amount effective and under conditions suitable for the deamination of a nucleotide base. In some embodiments, the methods comprise contacting the Cas9 protein with a ligand that induces self-excision of the intein either before or after the Cas9 protein binds the gRNA. In some embodiments, the target DNA sequence comprises a sequence associated with a disease or disorder, and wherein the deamination of the nucleotide base results in a sequence that is not associated with a disease or disorder. In some embodiments, the DNA sequence to be modified comprises a T→C or A→G point mutation associated with a disease or disorder, and the deamination of the mutant C or G base results in a sequence that is not associated with a disease or disorder (e.g., the deamination corrects the mutation that caused the disease or disorder). In some embodiments, the deamination corrects a point mutation in the sequence associated with the disease or disorder. In some embodiments, the sequence associated with the disease or disorder encodes a protein, and wherein the deamination introduces a stop codon into the sequence associated with the disease or disorder, resulting in a truncation of the encoded protein. In some embodiments, the deamination corrects a point mutation in the PI3KCA gene, thus correcting an H1047R and/or a A3140G mutation. In some embodiments, the contacting is performed in vivo in a subject susceptible to having or diagnosed with the disease

or disorder. In some embodiments, the disease or disorder is a disease associated with a point mutation, or a single-base mutation, in the genome. In some embodiments, the disease is a genetic disease, a cancer, a metabolic disease, or a lysosomal storage disease.

According to another aspect, methods for transcriptional activation of a gene are provided. In some embodiments, the methods comprise contacting a DNA molecule comprising a gene with (a) a ligand-dependent dCas9 fusion protein comprising a transcriptional activator (e.g., any of those provided herein) and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the transcriptional activation of the gene. In some embodiments, the method further comprises contacting the fusion protein with a ligand that induces self-excision of the intein. In some embodiments, the fusion protein is contacted with the ligand prior to forming a complex with a gRNA. In some embodiments, the fusion protein is contacted with the ligand after forming a complex with a gRNA. In some embodiments, the gRNA targets the promoter region of a gene.

According to another aspect, methods for transcriptional repression of a gene are provided. In some embodiments, the methods comprise contacting a DNA molecule comprising a gene with (a) a ligand-dependent dCas9 fusion protein comprising a transcriptional repressor (e.g., any of those provided herein) and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the transcriptional repression of the gene. In some embodiments, the method further comprises contacting the fusion protein with a ligand that induces self-excision of the intein. In some embodiments, the fusion protein is contacted with the ligand prior to forming a complex with a gRNA. In some embodiments, the fusion protein is contacted with the ligand after forming a complex with a gRNA. In some embodiments, the gRNA targets the promoter region of a gene.

According to another aspect, methods for epigenetic modification of DNA are provided. In some embodiments, the DNA is chromosomal DNA. In some embodiments, the methods comprise contacting a DNA molecule with (a) a ligand-dependent dCas9 fusion protein comprising an epigenetic modifier (e.g., any of those provided herein) and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the epigenetic modification of the DNA. In some embodiments, the method further comprises contacting the fusion protein with a ligand that induces self-excision of the intein. In some embodiments, the fusion protein is contacted with the ligand prior to forming a complex with a gRNA. In some embodiments, the fusion protein is contacted with the ligand after forming a complex with a gRNA. In some embodiments, the gRNA targets the promoter region of a gene in the DNA.

Any of the methods provided herein can be performed on DNA in a cell, for example, a cell in vitro or in vivo. In some embodiments, any of the methods provided herein are performed on DNA in a eukaryotic cell. In some embodiments, the eukaryotic cell is in an individual, for example, a human.

According to some embodiments, polynucleotides are provided, for example, that encode any of the proteins (e.g., proteins comprising ligand-dependent Cas9 proteins or vari-

Other advantages, features, and uses of the invention will be apparent from the Detailed Description of Certain Embodiments of the Invention; the Drawings, which are schematic and not intended to be drawn to scale; and the Claims.

FIGS. 4A-4D. Genomic DNA modification by intein-Cas9(S219), intein-Cas9(C574), and wild-type Cas9. (FIG. 4A) Indel frequency from high-throughput DNA sequencing of amplified genomic on-target sites in the absence or presence of 4-HT. Note that a significant number of indels were observed at the CLTA on-target site even in the absence of a targeting sgRNA (Table 9). (FIGS. 4B-4D) DNA modification specificity, defined as on-target:off-target indel frequency ratio⁴⁻⁶, normalized to wild-type Cas9. Cells were transfected with 500 ng of the Cas9 expression plasmid.

FIGS. 10A-10B. Indel frequency from high-throughput DNA sequencing of amplified genomic on-target sites ("On") and off-target sites ("Off 1-Off 4") by intein-Cas9 (S219), intein-Cas9(C574), and wild-type Cas9 in the presence of 4-HT. Five different amounts of wild-type Cas9 expression plasmid, specified in parenthesis, were transfected (FIG. 10A). Genomic sites with low modification

frequencies are enlarged in (FIG. 10B). P-values for comparisons between conditions (Table 8) were obtained using the Fisher exact test and adjusted for multiple comparisons using the Benjamini-Hochberg Method.

FIGS. 11A-11C. DNA modification specificity of intein-Cas9(S219), intein-Cas9(C574), and wild-type Cas9 in the presence of 4-HT. (FIGS. 11A-11C) On-target:off-target indel frequency ratio normalized to wild-type Cas9 (500 ng). Five different amounts of wild-type Cas9 expression plasmid, specified in parenthesis, were transfected.

FIGS. 12A-12B. Genomic EGFP disruption activity of intein-Cas9(S219) and intein-Cas9(S219-G521R) in the presence of (FIG. 12A) β -estradiol or (FIG. 12B) 4-HT. Error bars reflect the standard deviation of three technical replicates.

DEFINITIONS

As used herein and in the claims, the singular forms “a,” “an,” and “the” include the singular and the plural reference unless the context clearly indicates otherwise. Thus, for example, a reference to “an agent” includes a single agent and a plurality of such agents.

The term “Cas9” or “Cas9 nuclease” refers to an RNA-guided nuclease comprising a Cas9 protein, or a fragment thereof. A Cas9 nuclease is also referred to sometimes as a casn1 nuclease or a CRISPR (clustered regularly interspaced short palindromic repeat)—associated nuclease. CRISPR is a prokaryotic adaptive immune system that provides protection against mobile genetic elements (e.g., viruses, transposable elements, and conjugative plasmids). CRISPR clusters contain spacers, sequences complementary to antecedent mobile elements, and target invading nucleic acids. CRISPR clusters are transcribed and processed into CRISPR RNA (crRNA). In type II CRISPR systems correct processing of pre-crRNA requires a trans-encoded small RNA (tracrRNA), endogenous ribonuclease 3 (rnc), and a Cas9 protein. The tracrRNA serves as a guide for ribonuclease 3-aided processing of pre-crRNA. Subsequently, Cas9/crRNA/tracrRNA endonucleolytically cleaves linear or circular dsDNA target complementary to the spacer. The target strand not complementary to crRNA is first cut endonucleolytically, then trimmed 3'→5' exonucleolytically. In nature, DNA-binding and cleavage typically requires protein and both RNA species. However, single guide RNAs (“sgRNA”, or simply “gNRA”) can be engineered so as to incorporate aspects of both the crRNA and tracrRNA into a single RNA molecule. See, e.g., Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. *Science* 337:816-821(2012), the entire contents of which is hereby incorporated by reference. Cas9 recognizes a short motif in the CRISPR repeat sequences (the PAM or protospacer adjacent motif) to help distinguish self versus non-self. Cas9 nuclease sequences and structures are well known to those of skill in the art (see, e.g., “Complete genome sequence of an M1 strain of *Streptococcus pyogenes*.” Ferretti J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najjar F. Z., Ren Q., Zhu H., Song L. expand/collapse author list McLaughlin R. E., *Proc. Natl. Acad. Sci. U.S.A.* 98:4658-4663(2001); “CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III.” Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., *Nature* 471:602-607(2011); and “A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity.” Jinek M., Chylinski K., Fonfara I.,

Hauer M., Doudna J. A., Charpentier E. *Science* 337:816-821(2012), the entire contents of each of which are incorporated herein by reference). Cas9 orthologs have been described in various species, including, but not limited to, *S. pyogenes* and *S. thermophilus*. Additional suitable Cas9 nucleases and sequences will be apparent to those of skill in the art based on this disclosure, and such Cas9 nucleases and sequences include Cas9 sequences from the organisms and loci disclosed in Chylinski, Rhun, and Charpentier, “The tracrRNA and Cas9 families of type II CRISPR-Cas immunity systems” (2013) *RNA Biology* 10:5, 726-737; the entire contents of which are incorporated herein by reference. In some embodiments, proteins comprising Cas9 proteins or fragments thereof are referred to as “Cas9 variants.” A Cas9 variant shares homology to Cas9, or a fragment thereof. For example, a Cas9 variant may be at least about 70% identical, at least about 80% identical, at least about 90% identical, at least about 95% identical, at least about 98% identical, at least about 99% identical, at least about 99.5% identical, or at least about 99.9% to wild type Cas9. In some embodiments, the Cas9 variant comprises a fragment of Cas9 (e.g., a gRNA binding domain or a DNA-cleavage domain, an N-terminal domain or a C-terminal domain, etc.), such that the fragment is at least about 70% identical, at least about 80% identical, at least about 90% identical, at least about 95% identical, at least about 98% identical, at least about 99% identical, at least about 99.5% identical, or at least about 99.9% to the corresponding fragment of wild type Cas9. In some embodiments, wild type Cas9 corresponds to Cas9 from *Streptococcus pyogenes* (NCBI Reference Sequences: NC_017053.1 and NC_002737.1). In some embodiments, wild type Cas9 corresponds to SEQ ID NO:1 (nucleotide); SEQ ID NO:2 (amino acid). In some embodiments, Cas9 corresponds to a human codon optimized sequence of Cas9 (e.g., SEQ ID NO:3; See, e.g., Fu et al. High-frequency off-target mutagenesis induced by CRISPR-Cas nucleases in human cells. *Nat. Biotechnol.* 2013; 31, 822-826). In some embodiments, a Cas9 nuclease has an inactive (e.g., an inactivated) DNA cleavage domain. A nuclease-inactivated Cas9 protein may also be referred to as a “dCas9” protein (for nuclease “dead” Cas9). In some embodiments, dCas9 corresponds to, or comprises in part or in whole, the amino acid set forth as SEQ ID NO:5, below. In some embodiments, variants of dCas9 (e.g., variants of SEQ ID NO:5) are provided. For example, in some embodiments, variants having mutations other than D10A and H840A are provided, which e.g., result in a nuclease inactivated Cas9 (dCas9). Such mutations, by way of example, include other amino acid substitutions at D10 and H840, or other substitutions within the nuclease domains of Cas9 (e.g., substitutions in the HNH nuclease subdomain and/or the RuvC1 subdomain). In some embodiments, a Cas9 protein variant is a Cas9 nickase, which includes a mutation which abolishes the nuclease activity of one of the two nuclease domains of the protein. In some embodiments, a Cas9 nickase has one, but not both of a D10A and H840A substitution. In some embodiments, a Cas9 nickase corresponds to, or comprises in part or in whole, the amino acid set forth as SEQ ID NO:4, below. In some embodiments, variants or homologues of dCas9 or Cas9 nickase are provided which are at least about 70% identical, at least about 80% identical, at least about 90% identical, at least about 95% identical, at least about 98% identical, at least about 99% identical, at least about 99.5% identical, or at least about 99.9% to SEQ ID NO:5 or SEQ ID NO:4, respectively. In some embodiments, variants of dCas9 or

11

Cas9 nickase (e.g., variants of SEQ ID NO:5 and SEQ ID NO:4, respectively) are provided having amino acid sequences which are shorter, or longer than SEQ ID NO:5 or SEQ ID NO:4, by about 5 amino acids, by about 10 amino

12

acids, by about 15 amino acids, by about 20 amino acids, by about 25 amino acids, by about 30 amino acids, by about 40 amino acids, by about 50 amino acids, by about 75 amino acids, by about 100 amino acids or more.

Cas9; nucleotide (*Streptococcus pyogenes*)

(SEQ ID NO: 1)

ATGGATAAGAAATACTCAATAGGCTTAGATATCGGCACAAATAGCGTCGGATGGGCGGTGATCACTGATGATTAT
AAGGTTCCGTCTAAAAAGTTCAAGGTTCTGGGAAATACAGACCGCCACAGTATCAAAAAAATCTTATAGGGGCT
CTTTTATTTGGCAGTGGAGAGACAGCGGAAGCGACTCGTCTCAAACGGACAGCTCGTAGAAGGTATACACGTCGG
AAGAATCGTATTTGTTATCTACAGGAGATTTTTCAAATGAGATGGCGAAAGTAGATGATAGTTTCTTTCATCGA
CTTGAAGAGTCTTTTTTGGTGGAAAGACAAGAAGCATGAACGTCATCCTATTTTTGGAAATATAGTAGATGAA
GTTGCTTATCATGAGAAATATCCAACATCTATCATCTGCGAAAAAATGGCAGATTCTACTGATAAAGCGGAT
TTGCGCTTAATCTATTTGGCCTTAGCGCATATGATTAAGTTTCGTGGTCATTTTTTGATTGAGGGAGATTTAAAT
CCTGATAATAGTGATGTGGACAAACTATTTATCCAGTTGGTACAAATCTACAATCAATTATTTGAAGAAAACCTT
ATTAACGCAAGTAGAGTAGATGCTAAAGCGATTCTTCTGACGATTGAGTAAATCAAGACGATTAGAAAATCTC
ATTGCTCAGCTCCCCGGTGAGAAGAGAAATGGCTTGTTTGGGAATCTCATTGCTTTGTGATTGGGATTGACCCCT
AATTTTAAATCAAATTTTGATTGGCAGAAGATGCTAAATTACAGCTTTCAAAGATACCTACGATGATGATTTA
GATAATTTATGGCGCAAATGGAGATCAATATGCTGATTTGTTTTGGCAGCTAAGAATTTATCAGATGCTATT
TTACTTTGAGATATCCTAAGAGTAAATAGTGAAATAACTAAGGCTCCCTATCAGCTTCAATGATTAAGCGCTAC
GATGAACATCATCAAGACTTGACTCTTTTAAAAGCTTTAGTTGACAACAACCTCCAGAAAAGTATAAAGAAATC
TTTTTTGATCAATCAAAAACGGATATGCAGGTTATATTGATGGGGAGCTAGCCAAGAAGATTTTATAAATTT
ATCAAACCAATTTTAGAAAAATGGATGTTACTGAGGAATTATTGGTGAACTAAATCGTGAAGATTTGCTGCGC
AAGCAACGGACCTTGACAACGGCTCTATTCCCATCAAATCACTTGGTGAGCTGCATGCTATTTTGAGAAGA
CAAGAAGACTTTTATCATTTTTAAAAGACAATCGTGAGAAGATTGAAAAATCTTGACTTTTCGAATTCCTTAT
TATGTTGGTCCATTGGCGCGTGGAATAGTCGTTTTGCATGGATGACTCGGAAGTCTGAAGAAACAATACCCCA
TGGAATTTGAAGAAGTTGTCGATAAAGGTGCTTCAGCTCAATCATTTATTGAACGCATGACAACTTTGATAAA
AATCTTCAAATGAAAAAGTACTACCAAAACATAGTTTGCTTTATGAGTATTTACGGTTTATAACGAATTGACA
AAGGTCAAATATGTTACTGAGGGAATGCGAAAACAGCATTTCTTTCAGGTGAACAGAAGAACCCATTGTTGAT
TTACTCTTCAAAACAAATCGAAAAGTAACGTTAAGCAATTAAAAGAAGATTATTTCAAAAAATAGAATGTTTT
GATAGTGTGAAATTTGAGAGTTGAAGATAGATTTAATGCTTCATTAGGCGCTACCATGATTTGCTAAAAATT
ATTAAAGATAAAGATTTTTTGGATAATGAAGAAAATGAAGATATCTTAGAGGATATTGTTTTAACATTGACCTTA
TTGAAGATAGGGGATGATTGAGGAAAGACTTAAAACATATGCTCACCTCTTTGATGATAAGGTGATGAAACAG
CTTAAACGTCGCCGTTATACTGGTTGGGGACGTTTGCTCGAAAAATGATTAATGGTATTAGGGATAAGCAATCT
GGCAAAACAATATTAGATTTTTTGAAATCAGATGGTTTTGCCAATCGCAATTTTATGCAGCTGATCCATGATGAT
AGTTTGACATTTAAGAAGATATTCAAAAGCACAGGTGTCTGGACAAGGCCATAGTTTACATGAACAGATTGCT
AACTTAGCTGGCAGTCTGCTATTAAAAAGGTATTTACAGACTGTAAAAATTTGTTGATGAACGTTGTTGAAAGTA
ATGGGGCATAAGCCAGAAAATATCGTTATTGAAATGGCACGTGAAAAATCAGACAACTCAAAAGGGCCAGAAAAAT
TCGCGAGAGCGTATGAACGAATCGAAGAAGGTATCAAGAATTAGGAAGTCAGATTCTTAAAGAGCATCCTGTT
GAAAATACTCAATTGCAAAATGAAAAGCTCTATCTCTATTATCTACAAAATGGAAGAGACATGTATGTGGACCA
GAATTAGATATTAATCGTTTAAGTGATTATGATGTCGATCACAATTGTTCCACAAGTTTCATTAAAGACGATTCA
ATAGACAATAAGGTACTAACGCGTTCTGATAAAAAATCGTGGTAAATCGGATAACGTTCCAAGTGAAGAAGTAGTC
AAAAAGATGAAAACTATTGGAGACAACCTCTAAACGCCAAGTTAATCACTCAACGTAAGTTTGATAATTTAACG
AAAGCTGAACGTGGAGGTTTGAAGTGAACCTTGATAAAGCTGGTTTTATCAACGCCAATTTGGTTGAACTCGCCAA

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ATCACTAAGCATGTGGCACAAATTTTGGATAGTCGCATGAATACTAAATACGATGAAAATGATAAACTTATTTCGA
 GAGGTTAAAGTGATTACCTTAAATCTAAATTAGTTTCTGACTCCGAAAAGATTTCCAATTCTATAAAGTACGT
 GAGATTAAACAATTACCATCATGCCATGATGCGTATCTAAATGCCGTCGTTGGAAGTCTTTGATTAAAGAAATAT
 CCAAAACTTGAATCGGAGTTTGTCTATGGTGATTATAAGTTTATGATGTTCTGTAATGATTGCTAAGTCTGAG
 CAAGAAATAGGCAAAGCAACCGCAAAATATTTCTTTTACTCTAATATCATGAACTTCTTCAAAACAGAAATTACA
 CTGCAATGAGAGATTTCGCAACGCCCTCTAATCGAACTAATGGGAACTGGAGAAATTGTCTGGGATAAA
 GGGCGAGATTTTGCCACAGTGCAGCAAGTATTTGTCATGCCCAAGTCAATATTGTCAAGAAAACAGAAGTACAG
 ACAGGCGGATTCTCCAAGGAGTCAATTTTACCAAAAAGAAATTCGGACAAGCTTATTGCTCGTAAAAAGACTGG
 GATCCAAAAAATATGGTGGTTTGTAGTAGTCCACGGTAGCTTATTAGTCTTAGTGGTGTCTAAGGTGAAAAA
 GGGAAATCGAAGAAGTTAAATCCGTTAAAGAGTTACTAGGGATCACAATTATGGAAAGAAGTTCCTTTGAAAA
 AATCCGATTGACTTTTTAGAGCTAAAGGATATAAGGAAGTTAAAAAGACTTAATCATTAACTACCTAAATAT
 AGTCTTTTTGAGTTAGAAAACGGTCGTAACGGATGCTGGCTAGTGCCGAGAAATTACAAAAGGAAATGAGCTG
 GCTCTGCCAAGCAATATGTGAATTTTTATATTTAGCTAGTCATTATGAAAAGTTGAAGGGTAGTCCAGAAGAT
 AACGAACAAAAACAATTGTTTGTGGAGCAGCATAAGCATTATTTAGATGAGATTATTGAGCAATCAGTGAATTT
 TCTAAGCGTGTATTTTAGCAGATGCCAATTTAGATAAGTTCTTAGTGTCATATAACAAACATAGAGACAAACCA
 ATACGTGAACAAGCAGAAAATATTATTCATTTATTTACGTTGACGAATCTTGAGCTCCCGCTGCTTTTAAATAT
 TTTGATACAACAATTGATCGTAAACGATATACGCTCTACAAAAGAAGTTTATAGTGCCACTCTTATCCATCAATCC
 ATCACTGGTCTTTATGAAACACGCATTGATTTGAGTCAGCTAGGAGGTGACTGA

Cas9 (human codon optimized)

(SEQ ID NO: 3)

ATGGATAAAAAGTATTCTATTGGTTTAGACATCGGCACATAATCCGTTGGATGGGCTGTACATAACCGATGAATAC
 AAAGTACCTTCAAAGAAATTTAAGGTGTTGGGGAACACAGACCGTCATTGATTTAAAGAAATCTTATCGGTGCC
 CTCTATTTCGATAGTGGCGAAACGCGAGGCGACTCGCCTGAAACGAACCGCTCGGAGAAGGTATACACGTCGC
 AAGAACCGAATATGTTACTTACAAGAAATTTTAGCAATGAGATGGCCAAAGTTGACGATTCTTCTTTTCCCGT
 TTGGAAGAGTCTCTCTTGTGCAAGAGGACAAGAAACATGAACGGCACCCCATCTTTGGAACATAGTAGATGAG
 GTGGCATATCATGAAAAGTACCAACGATTATCACCTCAGAAAAAGCTAGTTGACTCAACTGATAAAGCGGAC
 CTGAGGTTAATCTACTTGGCTCTTGCCCATATGATAAAGTTCCGTGGGCACTTTCTCATTGAGGGTGATCTAAAT
 CCGGACAACCTCGGATGTGACAAACTGTTTCATCCAGTTAGTACAAACCTATAATCAGTTGTTTGAAGAGAACCCT
 ATAAATGCAAGTGGCGTGGATGCGAAGGCTATTCTTAGCGCCCGCTCTCTAAATCCGACGGCTAGAAAACCTG
 ATCGCACAATTACCCGGAGAGAAGAAAAATGGGTTGTTTCGGTAACCTTATAGCGCTCTCACTAGGCCTGACACCA
 AATTTTAAGTCGAACCTTCGACTTAGCTGAAGATGCCAATTCAGCTTAGTAAGGACACGTACGATGACGATCTC
 GACAATCTACTGGCACAATTTGGAGATCAGTATGCGGACTTATTTTGGCTGCCAAAACCTTAGCGATGCAATC
 CTCTATCTGACATACTGAGAGTTAATACTGAGATTACCAAGGCGCGTTATCCGCTTCAATGATCAAAAGGTAC
 GATGAACATCACCAAGACTTGACACTTCTCAAGGCCCTAGTCCGTGAGCAACTGCCTGAGAAATATAAGGAAATA
 TTCTTTGATCAGTCGAAAAACGGTACGAGGTTATATTGACGGCGGAGCGAGTCAAGAGGAATCTTACAAGTTT
 ATCAAAACCATATTAGAGAAGATGGATGGGACGGAAGAGTTGCTTGTAAGAACTCAATCGCGAAGATCTACTGCGA
 AAGCAGCGGACTTTTCGACAACGGTAGCATTCCACATCAAATCCACTTAGGCGAATTGCATGCTATACCTTAGAAGG
 CAGGAGGATTTTTATCCGTTCTCTCAAAGACAATCGTGAAAAGATTGAGAAAATCCTAACCTTTTCGCATACCTTAC
 TATGTGGGACCCCTGGCCGAGGGAACCTCGGTTTCGCATGGATGACAAGAAAGTCCGAAGAAACGATTACTCCA
 TGAATTTTGAGGAAGTTGTCGATAAAGGTGCGTCAGCTCAATCGTTCATCGAGAGGATGACCAACTTTGACAAG
 AATTTACCGAACGAAAAAGTATTGCCTAAGCACAGTTTACTTTACAGTATTTACAGTGTACAATGAACTCACG

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AAAGTTAAGTATGTCACTGAGGGCATGCGTAAACCCGCTTCTAAGCGGAGAACAGAAAGCAATAGTAGAT
 CTGTTATTCAAGACCAACCGCAAAGTGACAGTTAAGCAATTGAAAGAGGACTACTTTAAGAAAATTGAATGCTTC
 GATTCTGTGAGATCTCCGGGGTAGAAGATCGATTAAATGCGTCACTTGGTACGTATCATGACCTCCTAAAGATA
 ATTAAGATAAGGACTTCTCGGATAACGAAGAGAATGAAGATATCTTAGAAGATATAGTGTGACTCTTACCCTC
 TTTGAAGATCGGGAATGATTGAGGAAAGACTAAAACATACGCTCACCTGTTTCGACGATAAGGTTATGAAACAG
 TTAAGAGAGCGTCGTATACGGGCTGGGACGATTGTCGCGGAACTTATCAACGGGATAAGAGACAAGCAAAGT
 GGTAAGTATTCTCGATTTCTAAAGAGCGACGGCTTCGCCAATAGGAACCTTATGCAGCTGATCCATGATGAC
 TCTTTAACCTTCAAAGAGGATATACAAAAGGCACAGGTTCCGGACAAGGGGACTCATTGCACGAACATATTGCG
 AATCTTGCTGGTTTCGACGCCATCAAAAAGGGCATACTCCAGACAGTCAAAGTAGTGGATGAGCTAGTTAAGGTC
 ATGGGACGTCACAAACCGGAAACATTGTAATCGAGATGGCACGCGGAAATCAAACGACTCAGAAGGGGCAAAAA
 AACAGTCGAGAGCGGATGAAGAGAATAGAAGAGGTATTAAAGAACTGGGCAGCCAGATCTTAAAGGAGCATCCT
 GTGGAAAATACCCAATTGCAGAACGAGAACTTTACCTCTATTACCTACAAAATGGAAGGGACATGTATGTTGAT
 CAGGAACCTGGACATAAACCGTTTATCTGATTACGACGTCGATCACATTGTACCCCAATCCTTTTGAAGGACGAT
 TCAATCGACAATAAAGTGCTTACACGCTCGGATAAGAACCGAGGGAAGTGAATGTTCCAAGCGAGGAAGTC
 GTAAAGAAAATGAAGAACTATTGGCGGCAGCTCCTAAATGCGAACTGATAACGCAAAGAAAGTTCGATAACTTA
 ACTAAAGCTGAGAGGGGTGGCTTGTCTGAAGTTGACAAGGCCGGATTTATTAAACGTCAGCTCGTGGAAACCCGC
 CAAATCACAAAGCATGTTGCACAGATACTAGATTTCCGAATGAATACGAAATACGACGAGAACGATAAGCTGATT
 CGGGAAGTCAAAGTAATCACTTTAAAGTCAAATTTGGTGTGGGACTTCAGAAAGGATTTTCAATCTATAAGTT
 AGGGAGATAAATACTACCACCATGCGCACGACGCTTATCTTAATGCCGTCGTAGGGACCGCACTCATTAAGAAA
 TACCCGAAGCTAGAAGTGAGTTTGTGTATGGTGATTACAAAGTTTATGACGTCGTAAGATGATCGCGAAAAGC
 GAACAGGAGATAGGCAAGGCTACAGCCAAATACTTCTTTTATTCTAACATTATGAATTTCTTTAAGACGGAATC
 ACTCTGGCAAACGGAGAGATACGCAAACGACCTTTAATTGAAACCAATGGGGAGACAGGTGAAATCGTATGGGAT
 AAGGGCCGGGACTTCGCGACGGTGAGAAAAGTTTGTCCATGCCCAAGTCAACATAGTAAGAAAAGTGAAGTG
 CAGACCGGAGGGTTTTCAAAGGAATCGATTCTTCAAAGGAATAGTGATAAGCTCATCGCTCGTAAAAAGGAC
 TGGGACCCGAAAAGTACGGTGGCTTCGATAGCCCTACAGTTGCCATTCTGTCTAGTAGTGGCAAAGTTGAG
 AAGGGAAAATCCAAGAACTGAAGTCAGTCAAAGAATTATTGGGGATAACGATTATGGAGCGCTCGTCTTTTGAA
 AAGAACCCTCGACTTCTTGAGCGGAAAGTTACAAGGAAGTAAAAAGGATCTCATAATTAACTACCAAAG
 TATAGTCTGTTTGAAGTAGAAAATGGCCGAAACGGATGTTGGCTAGCGCCGAGAGCTTCAAAGGGGAACGAA
 CTCGCACTACCGCTTAAATACGTGAATTTCTGTATTTAGCGTCCATTACGAGAAGTTGAAAGGTTACCTGAA
 GATAACGAACAGAAGCAACTTTTGTGAGCAGCACAAACATTATCTCGACGAAATCATAGAGCAAAATTCGGAA
 TTCAGTAAGAGAGTCATCTAGCTGATGCCAATCTGGACAAAGTATTAAGCGCATACAACAGCACAGGGATAAA
 CCCATACGTGAGCAGCGGAAATATTATCCATTGTTTACTCTTACCAACCTCGCGCTCCAGCCGATTCAAG
 TATTTTGACACAAACGATAGATCGCAAACGATACACTTCTACCAAGGAGGTCTAGACGCGACACTGATTACCAA
 TCCATCACGGGATTATATGAACTCGGATAGATTGTGCAGCTTGGGGTGAC

Cas9; amino acid (*Streptococcus pyogenes*)

(SEQ ID NO: 2)

MDKKYISGLDIGTNSVGVAVITDEYKVPKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRR
 KNRICYLQEIFSNEMAKVDDSPFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKAD
 LRLLIYLAHMIKFRGHFLIEGDLNPDNSDVKLFILVQTYNQLFEENPINASGVDAKAILSARLSKSRRLLENL
 IAQLPGEKKNLGNLIALSLGLTPNFKSNFDLAEDAKQLSKDYYDDLDNLLAQIGDQYADLFLAAKNLSDAI
 LLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKF
 IKPILEKMDGTEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPY

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YVGPLARGNSRFAMWTRKSEETITPWNFEVVDKGASQSFIERMTNFDKNLPNEKVLPHKSLLEYEFTVYNELT
 VKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKKI
 IKDKDFLDNEENEDILEDIVLTLTLFEDREMIERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQS
 GKTILDFLKSDGFANRNFQMQLIHDDSLTFKEDIQKAQVSGQGSSLHEHIANLAGSPAIKKGILQTVKVVDELVKV
MGRHKPENIVIAMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVD
QELDINRLSDYDVDHIVPQSFLKDDSIDNKVLRSDKNRGKSDNVPSEEVVKMKMNYWRQLLNAKLITQRKFDNL
TKAERGGLSELDKAGFIKRLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKV
REINNYHHAHDAYLNAVVGTALIKKYPKLESEFVYGDYKVYDVRKMIAKSEQEIGKATAKYFFYSNIMNFFKTEI
TLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLSMPQVNIKKTEVQTTGGFSKESILPKRNSDKLIARKKD
 WDPKKYGGFDSPTVAYSVLVAVKEKGSKKLKSVKELLGITIMERSSEFKNPIDFLEAKGYKEVKKDLIIKLPK
 YSLFELENGRKRMLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDEQKQLFVEQHKHYLDEIEQISE
 FSKRVILADANLDKVL SAYNKHDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQ
 SITGLYETRIDLSQLGGD
 (single underline: HNH domain; double underline: RuvC domain)

dCas9 (D10A and H840A)

(SEQ ID NO: 5)

MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRR
 KNRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLV DSTKAD
 LRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFQILVQTYNQLFEENPINASGVDAKAILSARLSKSRRLLENL
 IAQLPGEKKNGLFGNLIALLSLGLTPNFKSNFDLAEDAKLQLSKDYYDDDLNLLAQIGDQYADLFLAAKNLSDAI
 LLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKF
 IKPILEKMDGTEELVVKLNREDLRLKQRTFDNGSIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPY
 YVGPLARGNSRFAMWTRKSEETITPWNFEVVDKGASQSFIERMTNFDKNLPNEKVLPHKSLLEYEFTVYNELT
 VKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKKI
 IKDKDFLDNEENEDILEDIVLTLTLFEDREMIERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQS
 GKTILDFLKSDGFANRNFQMQLIHDDSLTFKEDIQKAQVSGQGSSLHEHIANLAGSPAIKKGILQTVKVVDELVKV
MGRHKPENIVIAMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVD
QELDINRLSDYDVDAIVPQSFLKDDSIDNKVLRSDKNRGKSDNVPSEEVVKMKMNYWRQLLNAKLITQRKFDNL
TKAERGGLSELDKAGFIKRLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKV
REINNYHHAHDAYLNAVVGTALIKKYPKLESEFVYGDYKVYDVRKMIAKSEQEIGKATAKYFFYSNIMNFFKTEI
TLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLSMPQVNIKKTEVQTTGGFSKESILPKRNSDKLIARKKD
 WDPKKYGGFDSPTVAYSVLVAVKEKGSKKLKSVKELLGITIMERSSEFKNPIDFLEAKGYKEVKKDLIIKLPK
 YSLFELENGRKRMLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDEQKQLFVEQHKHYLDEIEQISE
 FSKRVILADANLDKVL SAYNKHDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQ
 SITGLYETRIDLSQLGGD
 (single underline: HNH domain; double underline: RuvC domain)

Cas9 nickase (D10A) (amino acid sequence)

(SEQ ID NO: 4)

MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRR
 KNRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLV DSTKAD
 LRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFQILVQTYNQLFEENPINASGVDAKAILSARLSKSRRLLENL
 IAQLPGEKKNGLFGNLIALLSLGLTPNFKSNFDLAEDAKLQLSKDYYDDDLNLLAQIGDQYADLFLAAKNLSDAI
 LLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKF

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IKPILEKMDGTEELLVKNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPFLKDNREKIEKILTFRIPIY
 YVGPLARGNSRFAMWTRKSEETITPWNFEFVVDKGASQSFIERMTNFDKINLPNEKVLPHKSLLEYEFTVYNELT
 KVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKKI
 IKDKDFLDNEENEDILEDIVLTLTLFEDREMIERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQS
 GKTILDFLKSDFANRNFMLIHDDSLTFKEDIQKAQVSGQDSLHEHIANLAGSPAIIKKGILQTVKVVDELVKV
 MGRHKPENIVIAMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVD
 QELDINRLSDYDVIDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVKKMKNYWRQLLNAKLITQRKFDNL
 TKAERGGLSELDKAGFIKRLVETRQITKHVAQIILSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKV
 REINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVYDVRKMIKSEQEI GKATAYFFYSNIMNFFKTEI
 TLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLSPQVNIKKTEVQTGGFSKESILPKRNSDKLIARKKD
 WDPKKYGGFDSPTVAYSVLVAKVEKGSKKLKSVKELLGITIMERSSEKPNIDFLEAKGYKEVKKDLIIKLPK
 YSLFELENGRKRMLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEQKQFLVEQHKHYLDEIEQISE
 FSKRVILADANLDKVL SAYNKHDKPIREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQ
 SITGLYETRIDLSQLGGD

Cas9 variants are provided comprising an intein (e.g., a
 ligand-dependent intein) inserted within the Cas9 sequence
 and may be referred to as small-molecule-controlled Cas9 or
 ligand-dependent Cas9. In some embodiments, the intein is
 inserted into any location (e.g., at any amino acid position)
 in Cas9. In some embodiments, the inserted intein sequence
 replaces one or more amino acids in Cas9. For example, in
 some embodiments the inserted intein sequence replaces any
 cysteine, any alanine, any threonine, or any serine in Cas9
 or a Cas9 variant such as dCas9 or Cas9 nickase. In some
 embodiments the inserted intein sequence replaces Cys80,
 Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622,
 Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, or
 Ser1274 in Cas9 (SEQ ID NO:2), dCas9 (SEQ ID NO:5), or
 Cas9 nickase (SEQ ID NO:4).

The terms “conjugating,” “conjugated,” and “conjugation”
 refer to an association of two entities, for example, of
 two molecules such as two proteins, two domains (e.g., a
 binding domain and a cleavage domain), or a protein and an
 agent, e.g., a ligand binding domain and a small molecule.
 In some aspects, the association is between a protein (e.g.,
 RNA-programmable nuclease) and a nucleic acid (e.g., a
 guide RNA). The association can be, for example, via a
 direct or indirect (e.g., via a linker) covalent linkage. In
 some aspects, the association is between two or more
 proteins, for example, an RNA-programmable nuclease
 (e.g., Cas9) and an intein protein. In some embodiments, the
 association is covalent. In some embodiments, two mol-
 ecules are conjugated via a linker connecting both mol-
 ecules.

The term “consensus sequence,” as used herein in the
 context of nucleic acid sequences, refers to a calculated
 sequence representing the most frequent nucleotide residues
 found at each position in a plurality of similar sequences.
 Typically, it represents the results of a multiple sequence
 alignments in which related sequences are compared to each
 other and similar sequence motifs are calculated. Methods
 and software for determining a consensus sequence are
 known in the art (See, e.g., JalCiew (jalview.org); and
 UGENE; Okonechnikov, K.; Golosova, O.; Fursov, M.; the
 UGENE team. “Unipro UGENE: a unified bioinformatics
 toolkit”. *Bioinformatics*. 2012; doi:10.1093/bioinformatics/
 bts091).

The term “deaminase” refers to an enzyme that catalyzes
 a deamination reaction. In some embodiments, the deami-
 nase is a cytidine deaminase, catalyzing the hydrolytic
 deamination of cytidine or deoxycytidine to uracil or
 deoxyuracil, respectively.

The term “effective amount,” as used herein, refers to an
 amount of a biologically active agent (e.g., a ligand-depen-
 dent Cas9) that is sufficient to elicit a desired biological
 response. For example, in some embodiments, an effective
 amount of a nuclease may refer to the amount of the
 nuclease that is sufficient to induce cleavage of a desired
 target site-specifically bound and cleaved by the nuclease,
 preferably with minimal or no off-target cleavage. In some
 embodiments, an effective amount of another ligand-depen-
 dent Cas9 protein having other nucleic acid modifying
 activities may refer to the amount of the protein that is
 sufficient to induce the nucleic acid modification. As will be
 appreciated by the skilled artisan, the effective amount of an
 agent, e.g., a ligand-dependent nuclease, deaminase, recom-
 binase, nickase, or a hybrid protein, a fusion protein, a
 protein dimer, a complex of a protein (or protein dimer) and
 a polynucleotide, or a polynucleotide, may vary depending
 on various factors as, for example, on the desired biological
 response, the specific allele, genome, target site, cell, or
 tissue being targeted, and the agent being used.

The term “engineered,” as used herein, refers to a nucleic
 acid molecule, a protein molecule, complex, substance, or
 entity that has been designed, produced, prepared, synthe-
 sized, and/or manufactured by a human. Accordingly, an
 engineered product is a product that does not occur in nature.

The term “epigenetic modifier,” as used herein, refers to
 a protein or catalytic domain thereof having enzymatic
 activity that results in the epigenetic modification of DNA,
 for example, chromosomal DNA. Epigenetic modifications
 include, but are not limited to, DNA methylation and dem-
 ethylation; histone modifications including methylation and
 demethylation (e.g., mono-, di- and tri-methylation), histone
 acetylation and deacetylation, as well as histone ubiquity-
 lation, phosphorylation, and sumoylation.

The term “extein,” as used herein, refers to a polypeptide
 sequence that is flanked by an intein and is ligated to another
 extein during the process of protein splicing to form a

mature, spliced protein. Typically, an intein is flanked by two extein sequences that are ligated together when the intein catalyzes its own excision. Exteins, accordingly, are the protein analog to exons found in mRNA. For example, a polypeptide comprising an intein may be of the structure extein(N)—intein—extein(C). After excision of the intein and splicing of the two exteins, the resulting structures are extein(N)—extein(C) and a free intein.

The term “hybrid protein,” as used herein, refers to a protein that comprises the amino acid sequence of a target protein (e.g., a Cas9 protein) and, embedded in that amino acid sequence, a ligand-dependent intein as described herein. Accordingly, a hybrid protein generally comprises the structure: target protein(N)—intein—target protein(C). Typically, a hybrid protein comprises a Cas9 protein (e.g., Cas9, Cas9 variants such as dCas9, fragments of Cas9 or Cas9 variants, etc.) and a ligand-dependent intein. In some embodiments, a hybrid protein is encoded by a recombinant nucleic acid, in which a nucleic acid sequence encoding an intein is inserted in frame into a nucleic acid sequence encoding a target protein. In certain embodiments, the target protein exhibits a desired activity or property that is absent or reduced in the hybrid protein. In some embodiments, excision of the intein from the hybrid protein results in a restoration of the desired activity or property in the mature, spliced target protein. Non-limiting examples of desired activities or properties of target proteins are binding activities, enzymatic activities (e.g., nuclease activities, gene editing activities, deaminase activities, recombinase activities), reporter activities (e.g., fluorescent activity), therapeutic activities, size, charge, hydrophobicity, hydrophilicity, or 3D-structure. In some embodiments, excision of the intein from a hybrid protein results in a mature, spliced target protein that exhibits the same or similar levels of a desired activity as the native target protein. A hybrid protein may be created from any target protein by embedding an intein sequence into the amino acid sequence of the target protein, for example, by generating a recombinant, hybrid protein-encoding nucleic acid molecule and subsequent transcription and translation, or by protein synthesis methods known to those of skill in the art.

The term “intein,” as used herein, refers to an amino acid sequence that is able to excise itself from a protein and to rejoin the remaining protein segments (the exteins) via a peptide bond in a process termed protein splicing. Inteins are analogous to the introns found in mRNA. Many naturally occurring and engineered inteins and hybrid proteins comprising such inteins are known to those of skill in the art, and the mechanism of protein splicing has been the subject of extensive research. As a result, methods for the generation of hybrid proteins from naturally occurring and engineered inteins are well known to the skilled artisan. For an overview, see pages 1-10, 193-207, 211-229, 233-252, and 325-341 of Gross, Belfort, Derbyshire, Stoddard, and Wood (Eds.) *Homing Endonucleases and Inteins* Springer Verlag Heidelberg, 2005; ISBN 9783540251064; the contents of which are incorporated herein by reference for disclosure of inteins and methods of generating hybrid proteins comprising natural or engineered inteins. As will be apparent to those of skill in the art, an intein may catalyze protein splicing in a variety of extein contexts. Accordingly, an intein can be introduced into virtually any target protein sequence to create a desired hybrid protein, and the invention is not limited in the choice of target proteins.

The term “intein domain,” as used herein, refers to the amino acid sequence of an intein that is essential for self-excision and extein ligation. For example, in some

inteins, the entire intein amino acid sequence, or part(s) thereof, may constitute the intein domain, while in ligand-dependent inteins, the ligand-binding domain is typically embedded into the intein domain, resulting in the structure: intein domain (N)—ligand-binding domain—intein domain (C).

The term “ligand binding domain,” as used herein, refers to a peptide or protein domain that binds a ligand. A ligand binding domain may be a naturally occurring domain or an engineered domain. Examples of ligand-binding domains referred to herein are the ligand binding domain of a native estrogen receptor, e.g., the ligand-binding domain of the native human estrogen receptor, and engineered, evolved, or mutated derivatives thereof. Other suitable ligand binding domains include the human thyroid hormone receptor (see, e.g., Skretas et al., “Regulation of protein activity with small-molecule-controlled inteins.” *Protein Sci.* 2005; 14, 523-532) and members of the ribose-binding protein family (see, e.g., Björkman et al., “Multiple open forms of ribose-binding protein trace the path of its conformational change.” *J Mol Biol.* 1998 12; 279(3):651-64). Typically, a ligand-binding domain useful in the context of ligand-dependent inteins, as provided herein, exhibits a specific three-dimensional structure in the absence of the ligand, which inhibits intein self-excision, and undergoes a conformational change upon binding of the ligand, which promotes intein self-excision. Some of the ligand-dependent inteins provided herein comprise a ligand-binding domain derived from the estrogen receptor that can bind 4-HT and other estrogen-receptor ligands, e.g., ligands described in more detail elsewhere herein, and undergo a conformational change upon binding of the ligand. An appropriate ligand may be any chemical compound that binds the ligand-binding domain and induces a desired conformational change. In some embodiments, an appropriate ligand is a molecule that is bound by the ligand-binding domain with high specificity and affinity. In some embodiments, the ligand is a small molecule. In some embodiments, the ligand is a molecule that does not naturally occur in the context (e.g., in a cell or tissue) that a ligand-dependent intein is used in. For example, in some embodiments, the ligand-binding domain is a ligand-binding domain derived from an estrogen receptor, and the ligand is tamoxifen, or a derivative or analog thereof (e.g., 4-hydroxytamoxifen, 4-HT).

The term “ligand-dependent intein,” as used herein refers to an intein that comprises a ligand-binding domain. Typically, the ligand-binding domain is inserted into the amino acid sequence of the intein, resulting in a structure intein (N)—ligand-binding domain—intein (C). Typically, ligand-dependent inteins exhibit no or only minimal protein splicing activity in the absence of an appropriate ligand, and a marked increase of protein splicing activity in the presence of the ligand. In some embodiments, the ligand-dependent intein does not exhibit observable splicing activity in the absence of ligand but does exhibit splicing activity in the presence of the ligand. In some embodiments, the ligand-dependent intein exhibits an observable protein splicing activity in the absence of the ligand, and a protein splicing activity in the presence of an appropriate ligand that is at least 5 times, at least 10 times, at least 50 times, at least 100 times, at least 150 times, at least 200 times, at least 250 times, at least 500 times, at least 1000 times, at least 1500 times, at least 2000 times, at least 2500 times, at least 5000 times, at least 10000 times, at least 20000 times, at least 25000 times, at least 50000 times, at least 100000 times, at least 500000 times, or at least 1000000 times greater than the activity observed in the absence of the ligand. In some

23

embodiments, the increase in activity is dose dependent over at least 1 order of magnitude, at least 2 orders of magnitude, at least 3 orders of magnitude, at least 4 orders of magnitude, or at least 5 orders of magnitude, allowing for fine-tuning of intein activity by adjusting the concentration of the ligand. Suitable ligand-dependent inteins are known in the art, and in include those provided below and those described in published U.S. Patent Application U.S. 2014/0065711 A1; Mootz et al., "Protein splicing triggered by a small molecule." *J. Am. Chem. Soc.* 2002; 124, 9044-9045; Mootz et al., "Conditional protein splicing: a new tool to control

24

protein structure and function in vitro and in vivo." *J. Am. Chem. Soc.* 2003; 125, 10561-10569; Buskirk et al., *Proc. Natl. Acad. Sci. USA.* 2004; 101, 10505-10510; Skretas & Wood, "Regulation of protein activity with small-molecule-controlled inteins." *Protein Sci.* 2005; 14, 523-532; Schwartz, et al., "Post-translational enzyme activation in an animal via optimized conditional protein splicing." *Nat. Chem. Biol.* 2007; 3, 50-54; Peck et al., *Chem. Biol.* 2011; 18 (5), 619-630; the entire contents of each are hereby incorporated by reference.

2-4 intein:

(SEQ ID NO: 7)

CLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSWFDQGTDRDVGRLIAGGAIWVATPDHKV
 LTEYGWRAAGELRKGRVAGPGGSGNSLALSLTADQMVSALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRE
 LVHMINWAKRVPGFVDLTLDQAHLLECAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCKVEGMVEIFDML
 LATSSRFMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAGLTLQQQHQR
 LAQLLLLILSHIRHMSNKGMEHLYSMKYKNVVPLYDLLLLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMLAEE
 LRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNC

3-2 intein:

(SEQ ID NO: 8)

CLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSWFDQGTDRDVGRLIAGGAIWVATPDHKV
 LTEYGWRAAGELRKGRVAGPGGSGNSLALSLTADQMVSALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRE
 LVHMINWAKRVPGFVDLTLDQAHLLECAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCKVEGMVEIFDML
 LATSSRFMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAGLTLQQQHQR
 LAQLLLLILSHIRHMSNKGMEHLYSMKYKNVVPLYDLLLLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMLAEE
 LRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNC

3OR3-1 intein:

(SEQ ID NO: 9)

CLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSWFDQGTDRDVGRLIAGGATVWATPDHKV
 LTEYGWRAAGELRKGRVAGPGGSGNSLALSLTADQMVSALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRE
 LVHMINWAKRVPGFVDLTLDQAHLLECAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCKVEGMVEIFDML
 LATSSRFMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAGLTLQQQHQR
 LAQLLLLILSHIRHMSNKGMEHLYSMKYKNVVPLYDLLLLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMLAEG
 LRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNC

3OR3-2 intein:

(SEQ ID NO: 10)

CLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSWFDQGTDRDVGRLIAGGATVWATPDHKV
 LTEYGWRAAGELRKGRVAGPGGSGNSLALSLTADQMVSALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRE
 LVHMINWAKRVPGFVDLTLDQAHLLECAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCKVEGMVEIFDML
 LATSSRFMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAGLTLQQQHQR
 LAQLLLLILSHIRHMSNKGMEHLYSMKYKNVVPLYDLLLLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMLAEE
 LRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNC

3OR3-3 intein:

(SEQ ID NO: 11)

CLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSWFDQGTDRDVGRLIAGGATVWATPDHKV
 LTEYGWRAAGELRKGRVAGPGGSGNSLALSLTADQMVSALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRE
 LVHMINWAKRVPGFVDLTLDQAHLLECAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCKVEGMVEIFDML
 LATSSRFMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAGLTLQQQHQR

-continued

LAQLLILSHIRHMSNKGMEHLYSMKYKNVPLYDLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMMLAEE

LRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNC

37R3-1 intein:

(SEQ ID NO: 12)

CLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSFWDQGTDRDVLRIAGGATVWATPDHKV

LTEYGWRAAGELRKGDRVAGPGGSGNSLALSLTADQMVSAALLDAEPPILYSEYNPTSPFSEASMMGLLTNLADRE

LVHMINWAKRVPGFVDLTLDHQAHLLEAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCEGMVEIFDML

LATSSRFRMMNLQGEFVCLKSIILLNSGVYTFLLSSTLKSLEEKDHIHRALDKITDTLIHLMAGLTLQQQHQR

LAQLLILSHIRHMSNKGMEHLYSMKYKNVPLYDLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMMLAEG

LRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNC

37R3-2 intein:

(SEQ ID NO: 13)

CLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSFWDQGTDRDVLRIAGGATVWATPDHKV

LTEYGWRAAGELRKGDRVAGPGGSGNSLALSLTADQMVSAALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRE

LVHMINWAKRVPGFVDLTLDHQAHLLEAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCEGMVEIFDML

LATSSRFRMMNLQGEFVCLKSIILLNSGVYTFLLSSTLKSLEEKDHIHRALDKITDTLIHLMAGLTLQQQHQR

LAQLLILSHIRHMSNKGMEHLYSMKYKNVPLYDLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMMLAEG

LRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNC

37R3-3 intein:

(SEQ ID NO: 14)

CLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSFWDQGTDRDVLRIAGGATVWATPDHKV

LTEYGWRAAGELRKGDRVAGPGGSGNSLALSLTADQMVSAALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRE

LVHMINWAKRVPGFVDLTLDHQAHLLEAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCEGMVEIFDML

LATSSRFRMMNLQGEFVCLKSIILLNSGVYTFLLSSTLKSLEEKDHIHRALDKITDTLIHLMAGLTLQQQHQR

LAQLLILSHIRHMSNKGMEHLYSMKYKNVPLYDLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMMLAEE

LRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNC

The term “linker,” as used herein, refers to a chemical group or a molecule linking two adjacent molecules or moieties, e.g., two polypeptides. Typically, the linker is positioned between, or flanked by, two groups, molecules, or other moieties and connected to each one via a covalent bond, thus connecting the two. In some embodiments, the linker is an amino acid linker. In some embodiments, the amino acid linker comprises at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 15, at least 20, at least 25, or at least 30 amino acids. In some embodiments, the linker is a divalent organic molecule, group, polymer, or chemical moiety. In some embodiments, the peptide linker comprises repeats of the tri-peptide Gly-Gly-Ser, e.g., comprising the sequence (GGG)_n (SEQ ID NO: 203), wherein n represents at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more repeats. In some embodiments, the linker comprises the sequence (GGG)₆ (SEQ ID NO:15). In some embodiments, the peptide linker is the 16 residue “XTEN” linker, or a variant thereof (See, e.g., Schellenberger et al. A recombinant polypeptide extends the in vivo half-life of peptides and proteins in a tunable manner. *Nat. Biotechnol.* 27, 1186-1190 (2009)). In some embodiments, the XTEN linker comprises the sequence SGSETPGTSESATPES (SEQ ID NO:16), SGSETPGTSESA (SEQ ID NO:17), or SGSETPGTSESAT-PEGGSGGS (SEQ ID NO:18). In some embodiments, the peptide linker is one or more selected from VPFLLEPDN-

40 INGKTC (SEQ ID NO:19), GSAGSAAGSGEF (SEQ ID NO:20), SIVAQLSRPDPA (SEQ ID NO:21), MKIEQLPSA (SEQ ID NO:22), VRHKLKRVGS (SEQ ID NO:23), GHGTGSTGSGSS (SEQ ID NO:24), MSRPDPA (SEQ ID NO:25); or GGSM (SEQ ID NO:26).

45 The term “mutation,” as used herein, refers to a substitution of a residue within a sequence, e.g., a nucleic acid or amino acid sequence, with another residue, or a deletion or insertion of one or more residues within a sequence. Mutations are typically described herein by identifying the original residue followed by the position of the residue within the sequence and by the identity of the newly substituted residue. Methods for making the amino acid substitutions (mutations) provided herein are known in the art and are provided by, for example, Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)).

60 The term “nuclease,” as used herein, refers to an agent, for example, a protein, capable of cleaving a phosphodiester bond connecting two nucleotide residues in a nucleic acid molecule. In some embodiments, “nuclease” refers to a protein having an inactive DNA cleavage domain, such that the nuclease is incapable of cleaving a phosphodiester bond. In some embodiments, the nuclease is a protein, e.g., an enzyme that can bind a nucleic acid molecule and cleave a phosphodiester bond connecting nucleotide residues within the nucleic acid molecule. A nuclease may be an endonu-

cleave, cleaving a phosphodiester bonds within a polynucleotide chain, or an exonuclease, cleaving a phosphodiester bond at the end of the polynucleotide chain. In some embodiments, a nuclease is a site-specific nuclease, binding and/or cleaving a specific phosphodiester bond within a specific nucleotide sequence, which is also referred to herein as the "recognition sequence," the "nuclease target site," or the "target site." In some embodiments, a nuclease is an RNA-guided (i.e., RNA-programmable) nuclease, which is associated with (e.g., binds to) an RNA (e.g., a guide RNA, "gRNA") having a sequence that complements a target site, thereby providing the sequence specificity of the nuclease. In some embodiments, a nuclease recognizes a single stranded target site. In some embodiments, a nuclease recognizes a double-stranded target site, for example, a double-stranded DNA target site. The target sites of many naturally occurring nucleases, for example, many naturally occurring DNA restriction nucleases, are well known to those of skill in the art. In many cases, a DNA nuclease, such as EcoRI, HindIII, or BamHI, recognize a palindromic, double-stranded DNA target site of 4 to 10 base pairs in length, and cut each of the two DNA strands at a specific position within the target site. Some endonucleases cut a double-stranded nucleic acid target site symmetrically, i.e., cutting both strands at the same position so that the ends comprise base-paired nucleotides, also referred to herein as blunt ends. Other endonucleases cut a double-stranded nucleic acid target site asymmetrically, i.e., cutting each strand at a different position so that the ends include unpaired nucleotides. Unpaired nucleotides at the end of a double-stranded DNA molecule are also referred to as "overhangs," e.g., as "5'-overhang" or as "3'-overhang," depending on whether the unpaired nucleotide(s) form(s) the 5' or the 3' end of the respective DNA strand. Double-stranded DNA molecule ends ending with unpaired nucleotide(s) are also referred to as sticky ends, as they can "stick to" other double-stranded DNA molecule ends comprising complementary unpaired nucleotide(s). A nuclease protein typically comprises a "binding domain" that mediates the interaction of the protein with the nucleic acid substrate, and also, in some cases, specifically binds to a target site, and a "cleavage domain" that catalyzes the cleavage of the phosphodiester bond within the nucleic acid backbone. In some embodiments, a nuclease protein can bind and cleave a nucleic acid molecule in a monomeric form. In some embodiments, a nuclease protein has to dimerize or multimerize in order to cleave a target nucleic acid. Binding domains and cleavage domains of naturally occurring nucleases, as well as modular binding domains and cleavage domains that can be fused to create nucleases binding specific target sites, are well known to those of skill in the art. For example, the binding domain of RNA-programmable nucleases (e.g., Cas9), or a Cas9 protein having an inactive DNA cleavage domain (e.g., dCas9), can be used as a binding domain (e.g., that binds a gRNA to direct binding to a target site) to specifically bind a desired target site, and fused or conjugated to a cleavage domain, for example, the cleavage domain of FokI, to create an engineered nuclease cleaving the target site. In some embodiments, Cas9 fusion proteins provided herein comprise the cleavage domain of FokI, and are therefore referred to as "fCas9" proteins. In some embodiments, the cleavage domain of FokI, e.g., in a fCas9 protein corresponds to, or comprises in part or whole, the amino acid sequence (or variants thereof) set forth as SEQ ID NO:6, below. In some embodiments, variants or homologues of the FokI cleavage domain include any variant or homologue capable of dimerizing (e.g., as part of fCas9 fusion protein) with

another FokI cleavage domain at a target site in a target nucleic acid, thereby resulting in cleavage of the target nucleic acid. In some embodiments, variants of the FokI cleavage domain (e.g., variants of SEQ ID NO:6) are provided which are at least about 70% identical, at least about 80% identical, at least about 90% identical, at least about 95% identical, at least about 98% identical, at least about 99% identical, at least about 99.5% identical, or at least about 99.9% to SEQ ID NO:6. In some embodiments, variants of the FokI cleavage domain (e.g., variants of SEQ ID NO:6) are provided having an amino acid sequence which is shorter, or longer than SEQ ID NO:6, by about 5 amino acids, by about 10 amino acids, by about 15 amino acids, by about 20 amino acids, by about 25 amino acids, by about 30 amino acids, by about 40 amino acids, by about 50 amino acids, by about 75 amino acids, by about 100 amino acids, or more.

20 Cleavage domain of FokI: (SEQ ID NO: 6)
 GSQVLKSELEEKSELRLHKLKYPHEYLIEIARNSTQDRILEMKVMEF
 FMKVYGYRGKHLGGSRKPDGAIYTVGSPIDYGVIVDTKAYSGGYNLPIGQ
 25 ADEMQRVYEENQTRNKHINPNEWKVPSSVTEFKFLFVSGHFGKGYKQAQ
 LTRLNHTNCNGAVLSVEELLIGGEMIKAGTLTLEEVRKFNNGEINF

The terms "nucleic acid" and "nucleic acid molecule," as used herein, refer to a compound comprising a nucleobase and an acidic moiety, e.g., a nucleoside, a nucleotide, or a polymer of nucleotides. Typically, polymeric nucleic acids, e.g., nucleic acid molecules comprising three or more nucleotides are linear molecules, in which adjacent nucleotides are linked to each other via a phosphodiester linkage. In some embodiments, the term "nucleic acid" refers to individual nucleic acid residues (e.g. nucleotides and/or nucleosides). In some embodiments, "nucleic acid" refers to an oligonucleotide chain comprising three or more individual nucleotide residues. As used herein, the terms "oligonucleotide" and "polynucleotide" can be used interchangeably to refer to a polymer of nucleotides (e.g., a string of at least three nucleotides). In some embodiments, "nucleic acid" encompasses RNA as well as single and/or double-stranded DNA. Nucleic acids may be naturally occurring, for example, in the context of a genome, a transcript, an mRNA, tRNA, rRNA, siRNA, snRNA, a plasmid, cosmid, chromosome, chromatid, or other naturally occurring nucleic acid molecule. On the other hand, a nucleic acid molecule may be a non-naturally occurring molecule, e.g., a recombinant DNA or RNA, an artificial chromosome, an engineered genome, or fragment thereof, or a synthetic DNA, RNA, DNA/RNA hybrid, or including non-naturally occurring nucleotides or nucleosides. Furthermore, the terms "nucleic acid," "DNA," "RNA," and/or similar terms include nucleic acid analogs, i.e. analogs having other than a phosphodiester backbone. Nucleic acids can be purified from natural sources, produced using recombinant expression systems and optionally purified, chemically synthesized, etc. Where appropriate, e.g., in the case of chemically synthesized molecules, nucleic acids can comprise nucleoside analogs such as analogs having chemically modified bases or sugars, and backbone modifications. A nucleic acid sequence is presented in the 5' to 3' direction unless otherwise indicated. In some embodiments, a nucleic acid is or comprises natural nucleosides (e.g. adenosine, thymidine, guanosine, cytidine, uridine, deoxyadenosine, deoxythymidine, deoxyguanosine, and deoxycytidine);

nucleoside analogs (e.g., 2-aminoadenosine, 2-thiothymidine, inosine, pyrrolo-pyrimidine, 3-methyl adenosine, 5-methylcytidine, 2-aminoadenosine, C5-bromouridine, C5-fluorouridine, C5-iodouridine, C5-propynyl-uridine, C5-propynyl-cytidine, C5-methylcytidine, 2-aminoadenosine, 7-deazaadenosine, 7-deazaguanosine, 8-oxoadenosine, 8-oxoguanosine, 0(6)-methylguanine, and 2-thiocytidine); chemically modified bases; biologically modified bases (e.g., methylated bases); intercalated bases; modified sugars (e.g., 2'-fluororibose, ribose, 2'-deoxyribose, arabinose, and hexose); and/or modified phosphate groups (e.g., phosphorothioates and 5'-N-phosphoramidite linkages).

The term "pharmaceutical composition," as used herein, refers to a composition that can be administered to a subject in the context of treatment of a disease or disorder. In some embodiments, a pharmaceutical composition comprises an active ingredient, e.g., a nuclease or a nucleic acid encoding a nuclease, and a pharmaceutically acceptable excipient.

The term "proliferative disease," as used herein, refers to any disease in which cell or tissue homeostasis is disturbed in that a cell or cell population exhibits an abnormally elevated proliferation rate. Proliferative diseases include hyperproliferative diseases, such as pre-neoplastic hyperplastic conditions and neoplastic diseases. Neoplastic diseases are characterized by an abnormal proliferation of cells and include both benign and malignant neoplasias. Malignant neoplasia is also referred to as cancer.

The terms "protein," "peptide," and "polypeptide" are used interchangeably herein and refer to a polymer of amino acid residues linked together by peptide (amide) bonds. The terms refer to a protein, peptide, or polypeptide of any size, structure, or function. Typically, a protein, peptide, or polypeptide will be at least three amino acids long. A protein, peptide, or polypeptide may refer to an individual protein or a collection of proteins. One or more of the amino acids in a protein, peptide, or polypeptide may be modified, for example, by the addition of a chemical entity such as a carbohydrate group, a hydroxyl group, a phosphate group, a farnesyl group, an isofarnesyl group, a fatty acid group, a linker for conjugation, functionalization, or other modification, etc. A protein, peptide, or polypeptide may also be a single molecule or may be a multi-molecular complex. A protein, peptide, or polypeptide may be just a fragment of a naturally occurring protein or peptide. A protein, peptide, or polypeptide may be naturally occurring, recombinant, or synthetic, or any combination thereof.

The term "fusion protein" as used herein refers to a hybrid polypeptide which comprises protein domains from at least two different proteins. One protein may be located at the amino-terminal (N-terminal) portion of the fusion protein or at the carboxy-terminal (C-terminal) protein thus forming an "amino-terminal fusion protein" or a "carboxy-terminal fusion protein," respectively. A protein may comprise different domains, for example, a nucleic acid binding domain (e.g., the gRNA binding domain of Cas9 that directs the binding of the protein to a target site or a dCas9 protein) and a nucleic acid cleavage domain(s). In some embodiments, a protein is in a complex with, or is in association with, a nucleic acid, e.g., DNA or RNA. Any of the proteins provided herein may be produced by any method known in the art. For example, the proteins provided herein may be produced via recombinant protein expression and purification, which is especially suited for fusion proteins comprising a peptide linker. Methods for recombinant protein expression and purification are well known and include those described by Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor

Laboratory Press, Cold Spring Harbor, N.Y. (2012)), the entire contents of which are incorporated herein by reference.

The term "protein splicing," as used herein, refers to a process in which a sequence, an intein, is excised from within an amino acid sequence, and the remaining fragments of the amino acid sequence, the exteins, are ligated via an amide bond to form a continuous amino acid sequence.

The term "RNA-programmable nuclease" and "RNA-guided nuclease" are used interchangeably herein and refer to a nuclease that forms a complex with (e.g., binds or associates with) one or more RNAs that is not a target for cleavage. In some embodiments, an RNA-programmable nuclease, when in a complex with an RNA, may be referred to as a nuclease:RNA complex. Typically, the bound RNA(s) is referred to as a guide RNA (gRNA). gRNAs can exist as an association of two or more RNAs, or as a single RNA molecule. gRNAs that exist as a single RNA molecule may be referred to as single-guide RNAs (sgRNAs), though "gRNA" is used interchangeably to refer to guide RNAs that exist as either a single molecule or as a complex of two or more molecules. Typically, gRNAs that exist as single RNA species comprise at least two domains: (1) a domain that shares homology to a target nucleic acid and may direct binding of a Cas9 complex to the target; and (2) a domain that binds a Cas9 protein. In some embodiments, domain (2) corresponds to a sequence known as a tracrRNA and comprises a stem-loop structure. For example, in some embodiments, domain (2) is homologous to a tracrRNA as depicted in FIG. 1E of Jinek et al., *Science* 337:816-821(2012), the entire contents of which is incorporated herein by reference. In some embodiments, domain 2 is at least 90%, at least 95%, at least 98%, or at least 99% identical to the tracrRNA as described by Jinek et al., *Science* 337:816-821(2012). The gRNA comprises a nucleotide sequence that complements a target site, which mediates binding of the nuclease/RNA complex to said target site and the sequence specificity of the nuclease:RNA complex. The sequence of a gRNA that binds a target nucleic acid can comprise any sequence that complements a region of the target and is suitable for a nuclease:RNA complex to bind. In some embodiments, the RNA-programmable nuclease is the (CRISPR-associated system) Cas9 endonuclease, for example, Cas9 (Csn1) from *Streptococcus pyogenes* (see, e.g., "Complete genome sequence of an M1 strain of *Streptococcus pyogenes*." Ferretti J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najjar F. Z., Ren Q., Zhu H., Song L. expand/collapse author list McLaughlin R. E., *Proc. Natl. Acad. Sci. U.S.A.* 98:4658-4663(2001); "CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III." Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., *Nature* 471:602-607(2011); and "A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity." Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. *Science* 337:816-821(2012), the entire contents of each of which are incorporated herein by reference.

Because RNA-programmable nucleases (e.g., Cas9) use RNA:DNA hybridization to determine target DNA cleavage sites, these proteins are able to cleave, in principle, any sequence specified by the guide RNA. Methods of using RNA-programmable nucleases, such as Cas9, for site-specific cleavage (e.g., to modify a genome) are known in the art (see e.g., Cong et al. Multiplex genome engineering using CRISPR/Cas systems. *Science* 339, 819-823 (2013);

Mali et al. RNA-guided human genome engineering via Cas9. *Science* 339, 823-826 (2013); Hwang et al. Efficient genome editing in zebrafish using a CRISPR-Cas system. *Nature biotechnology* 31, 227-229 (2013); Jinek et al. RNA-programmed genome editing in human cells. *eLife* 2, e00471 (2013); Dicarolo et al. Genome engineering in *Saccharomyces cerevisiae* using CRISPR-Cas systems. *Nucleic acids research* (2013); Jiang et al. RNA-guided editing of bacterial genomes using CRISPR-Cas systems. *Nature biotechnology* 31, 233-239 (2013); the entire contents of each of which are incorporated herein by reference).

The term "recombinase," as used herein, refers to a site-specific enzyme that mediates the recombination of DNA between recombinase recognition sequences, which results in the excision, integration, inversion, or exchange (e.g., translocation) of DNA fragments between the recombinase recognition sequences. Recombinases can be classified into two distinct families: serine recombinases (e.g., resolvases and invertases) and tyrosine recombinases (e.g., integrases). Examples of serine recombinases include, without limitation, Hin, Gin, Tn3, β -six, CinH, ParA, $\gamma\delta$, Bxb1, Φ C31, TP901, TG1, ϕ BT1, R4, ϕ RV1, ϕ FC1, MR11, A118, U153, and gp29. Examples of tyrosine recombinases include, without limitation, Cre, FLP, R, Lambda, HK101, HK022, and pSAM2. The serine and tyrosine recombinase names stem from the conserved nucleophilic amino acid residue that the recombinase uses to attack the DNA and which becomes covalently linked to the DNA during strand exchange. Recombinases have numerous applications, including the creation of gene knockouts/knock-ins and gene therapy applications. See, e.g., Brown et al., "Serine recombinases as tools for genome engineering." *Methods*. 2011; 53(4):372-9; Hirano et al., "Site-specific recombinases as tools for heterologous gene integration." *Appl. Microbiol. Biotechnol.* 2011; 92(2):227-39; Chavez and Calos, "Therapeutic applications of the Φ C31 integrase system." *Curr. Gene Ther.* 2011; 11(5):375-81; Turan and Bode, "Site-specific recombinases: from tag-and-target- to tag-and-exchange-based genomic modifications." *FASEB J.* 2011; 25(12):4088-107; Venken and Bellen, "Genome-wide manipulations of *Drosophila melanogaster* with transposons, FLP recombinase, and Φ C31 integrase." *Methods Mol. Biol.* 2012; 859:203-28; Murphy, "Phage recombinases and their applications." *Adv. Virus Res.* 2012; 83:367-414; Zhang et al., "Conditional gene manipulation: Cre-ating a new biological era." *J. Zhejiang Univ. Sci. B.* 2012; 13(7):511-24; Karpenshif and Bernstein, "From yeast to mammals: recent advances in genetic control of homologous recombination." *DNA Repair (Amst).* 2012; 1; 11(10):781-8; the entire contents of each are hereby incorporated by reference in their entirety. The recombinases provided herein are not meant to be exclusive examples of recombinases that can be used in embodiments of the invention. The methods and compositions of the invention can be expanded by mining databases for new orthogonal recombinases or designing synthetic recombinases with defined DNA specificities (See, e.g., Groth et al., "Phage integrases: biology and applications." *J. Mol. Biol.* 2004; 335, 667-678; Gordley et al., "Synthesis of programmable integrases." *Proc. Natl. Acad. Sci. USA.* 2009; 106, 5053-5058; the entire contents of each are hereby incorporated by reference in their entirety). Other examples of recombinases that are useful in the methods and compositions described herein are known to those of skill in the art, and any new recombinase that is discovered or generated is expected to be able to be used in the different embodiments of the invention. In some embodiments, the catalytic domains of a recombinase are fused to a nuclease-

inactivated RNA-programmable nuclease (e.g., dCas9, or a fragment thereof), such that the recombinase domain does not comprise a nucleic acid binding domain or is unable to bind to a target nucleic acid (e.g., the recombinase domain is engineered such that it does not have specific DNA binding activity). Recombinases lacking DNA binding activity and methods for engineering such are known, and include those described by Klippel et al., "Isolation and characterization of unusual gin mutants." *EMBO J.* 1988; 7: 3983-3989; Burke et al., "Activating mutations of Tn3 resolvase marking interfaces important in recombination catalysis and its regulation." *Mol. Microbiol.* 2004; 51: 937-948; Olorunniji et al., "Synapsis and catalysis by activated Tn3 resolvase mutants." *Nucleic Acids Res.* 2008; 36: 7181-7191; Rowland et al., "Regulatory mutations in Sin recombinase support a structure-based model of the synaptosome." *Mol. Microbiol.* 2009; 74: 282-298; Akopian et al., "Chimeric recombinases with designed DNA sequence recognition." *Proc Natl Acad Sci USA.* 2003; 100: 8688-8691; Gordley et al., "Evolution of programmable zinc finger-recombinases with activity in human cells." *J. Mol. Biol.* 2007; 367: 802-813; Gordley et al., "Synthesis of programmable integrases." *Proc Natl Acad Sci USA.* 2009; 106: 5053-5058; Arnold et al., "Mutants of Tn3 resolvase which do not require accessory binding sites for recombination activity." *EMBO J.* 1999; 18: 1407-1414; Gaj et al., "Structure-guided reprogramming of serine recombinase DNA sequence specificity." *Proc Natl Acad Sci USA.* 2011; 108(2):498-503; and Proudfoot et al., "Zinc finger recombinases with adaptable DNA sequence specificity." *PLoS One.* 2011; 6(4):e19537; the entire contents of each are hereby incorporated by reference. For example, serine recombinases of the resolvase-invertase group, e.g., Tn3 and $\gamma\delta$ resolvases and the Hin and Gin invertases, have modular structures with autonomous catalytic and DNA-binding domains (See, e.g., Grindley et al., "Mechanism of site-specific recombination." *Ann Rev Biochem.* 2006; 75: 567-605, the entire contents of which are incorporated by reference). The catalytic domains of these recombinases are thus amenable to being recombined with nuclease-inactivated RNA-programmable nucleases (e.g., dCas9, or a fragment thereof) as described herein, e.g., following the isolation of 'activated' recombinase mutants which do not require any accessory factors (e.g., DNA binding activities) (See, e.g., Klippel et al., "Isolation and characterization of unusual gin mutants." *EMBO J.* 1988; 7: 3983-3989; Burke et al., "Activating mutations of Tn3 resolvase marking interfaces important in recombination catalysis and its regulation." *Mol. Microbiol.* 2004; 51: 937-948; Olorunniji et al., "Synapsis and catalysis by activated Tn3 resolvase mutants." *Nucleic Acids Res.* 2008; 36: 7181-7191; Rowland et al., "Regulatory mutations in Sin recombinase support a structure-based model of the synaptosome." *Mol. Microbiol.* 2009; 74: 282-298; Akopian et al., "Chimeric recombinases with designed DNA sequence recognition." *Proc Natl Acad Sci USA.* 2003; 100: 8688-8691). Additionally, many other natural serine recombinases having an N-terminal catalytic domain and a C-terminal DNA binding domain are known (e.g., ϕ HC31 integrase, TnpX transposase, IS607 transposase), and their catalytic domains can be co-opted to engineer programmable site-specific recombinases as described herein (See, e.g., Smith et al., "Diversity in the serine recombinases." *Mol. Microbiol.* 2002; 44: 299-307, the entire contents of which are incorporated by reference). Similarly, the core catalytic domains of tyrosine recombinases (e.g., Cre, λ integrase) are known, and can be similarly co-opted to engineer programmable site-specific recombinases as described herein (See, e.g., Guo et al., "Structure of Cre

recombinase complexed with DNA in a site-specific recombination synapse." *Nature*. 1997; 389:40-46; Hartung et al., "Cre mutants with altered DNA binding properties." *J Biol Chem* 1998; 273:22884-22891; Shaikh et al., "Chimeras of the F1p and Cre recombinases: Tests of the mode of cleavage by F1p and Cre." *J Mol Biol*. 2000; 302:27-48; Rongrong et al., "Effect of deletion mutation on the recombination activity of Cre recombinase." *Acta Biochim Pol*. 2005; 52:541-544; Kilbride et al., "Determinants of product topology in a hybrid Cre-Tn3 resolvase site-specific recombination system." *J Mol Biol*. 2006; 355:185-195; Warren et al., "A chimeric cre recombinase with regulated directionality." *Proc Natl Acad Sci USA*. 2008 105:18278-18283; Van Duyne, "Teaching Cre to follow directions." *Proc Natl Acad Sci USA*. 2009 Jan. 6; 106(1):4-5; Numrych et al., "A comparison of the effects of single-base and triple-base changes in the integrase arm-type binding sites on the site-specific recombination of bacteriophage λ ." *Nucleic Acids Res*. 1990; 18:3953-3959; Tirumalai et al., "The recognition of core-type DNA sites by λ integrase." *J Mol Biol*. 1998; 279:513-527; Aihara et al., "A conformational switch controls the DNA cleavage activity of λ integrase." *Mol Cell*. 2003; 12:187-198; Biswas et al., "A structural basis for allosteric control of DNA recombination by λ integrase." *Nature*. 2005; 435:1059-1066; and Warren et al., "Mutations in the amino-terminal domain of λ -integrase have differential effects on integrative and excisive recombination." *Mol Microbiol*. 2005; 55:1104-1112; the entire contents of each are incorporated by reference).

The term "recombine," or "recombination," in the context of a nucleic acid modification (e.g., a genomic modification), is used to refer to the process by which two or more nucleic acid molecules, or two or more regions of a single nucleic acid molecule, are modified by the action of a recombinase protein (e.g., an inventive recombinase fusion protein provided herein). Recombination can result in, inter alia, the insertion, inversion, excision, or translocation of nucleic acids, e.g., in or between one or more nucleic acid molecules.

The term "site-specific enzyme," as used herein, refers to any enzyme capable of binding a nucleic acid at a target site to mediate a modification of the nucleic acid. Typically, the site-specific enzymes provided herein comprise an intein (e.g., a ligand-dependent intein). In some embodiments, the site-specific enzyme is unable to bind a target site prior to excision of the intein. In some embodiments, the site-specific enzyme is able to bind a target site prior to excision of the intein but remains enzymatically inactive (e.g., cannot cleave, recombine, edit, or otherwise modify a nucleic acid) until excision of the intein.

The term "small molecule," as used herein, refers to a non-peptidic, non-oligomeric organic compound either prepared in the laboratory or found in nature. Small molecules, as used herein, can refer to compounds that are "natural product-like", however, the term "small molecule" is not limited to "natural product-like" compounds. Rather, a small molecule is typically a non-polymeric, non-oligomeric molecule that is characterized in that it contains several carbon-carbon bonds, and has a molecular weight of less than 2000 g/mol, preferably less than 1500 g/mol, although this characterization is not intended to be limiting for the purposes of the present invention. In certain embodiments, the ligand of a ligand-dependent inteins used in the present invention is a small molecule.

The term "subject," as used herein, refers to an individual organism. In some embodiments, the subject is a human. In some embodiments, the subject is a non-human mammal. In

some embodiments, the subject is a non-human primate. In some embodiments, the subject is a rodent. In some embodiments, the subject is a sheep, a goat, a cattle, a cat, or a dog. In some embodiments, the subject is a vertebrate, an amphibian, a reptile, a fish, an insect, a fly, or a nematode. In some embodiments, the subject is a research animal. In some embodiments, the subject is genetically engineered, e.g., a genetically engineered non-human subject. The subject may be of either sex and at any stage of development. The terms "target nucleic acid," and "target genome," as used herein in the context of nucleases, refer to a nucleic acid molecule or a genome, respectively, that comprises at least one target site of a given nuclease.

The term "target site," refers to a sequence within a nucleic acid molecule that is bound and (1) cleaved; (2) recombined; (3) edited; or (4) otherwise modified by a site-specific enzyme. In some embodiments, a target site refers to a "nuclease target site," which is a sequence within a nucleic acid molecule that is bound and cleaved by a nuclease. A target site may be single-stranded or double-stranded. In the context of RNA-guided (i.e., RNA-programmable) nucleases (e.g., a Cas9 protein, a Cas9 variant, fragments of Cas9 or fragments of Cas9 variants, etc.), a target site typically comprises a nucleotide sequence that is complementary to a gRNA of the RNA-guided nuclease, and a protospacer adjacent motif (PAM) at the 3' end adjacent to the gRNA-complementary sequence. For the RNA-guided nuclease Cas9, the target site may be, in some embodiments, 20 base pairs plus a 3 base pair PAM (e.g., NNN, wherein N represents any nucleotide). Typically, the first nucleotide of a PAM can be any nucleotide, while the two downstream nucleotides are specified depending on the specific RNA-guided nuclease. Exemplary target sites for RNA-guided nucleases, such as Cas9, are known to those of skill in the art and include, without limitation, NNG, NGN, NAG, and NGG, wherein N represents any nucleotide. In addition, Cas9 nucleases from different species (e.g., *S. thermophilus* instead of *S. pyogenes*) recognize a PAM that comprises the sequence: NGGNG. Additional PAM sequences are known, including, but not limited to, NNAGAAW and NAAR (see, e.g., Esvelt and Wang, *Molecular Systems Biology*, 9:641 (2013), the entire contents of which are incorporated herein by reference). For example, the target site of an RNA-guided nuclease, such as, e.g., Cas9, may comprise the structure $[N_z]$ -[PAM], where each N is, independently, any nucleotide, and z is an integer between 1 and 50. In some embodiments, z is at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 25, at least 30, at least 35, at least 40, at least 45, or at least 50. In some embodiments, z is 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50. In some embodiments, Z is 20. In some embodiments, "target site" may also refer to a sequence within a nucleic acid molecule that is bound but not cleaved by a nuclease.

The terms "transcriptional activator" and "transcriptional repressor," refer to agents which activate and repress the transcription of a gene, respectively. Typically, such activators and repressors are proteins, e.g., as provided herein.

The terms "treatment," "treat," and "treating," refer to a clinical intervention aimed to reverse, alleviate, delay the onset of, or inhibit the progress of a disease or disorder, or one or more symptoms thereof, as described herein. As used herein, the terms "treatment," "treat," and "treating" refer to

a clinical intervention aimed to reverse, alleviate, delay the onset of, or inhibit the progress of a disease or disorder, or one or more symptoms thereof, as described herein. In some embodiments, treatment may be administered after one or more symptoms have developed and/or after a disease has been diagnosed. In some embodiments, treatment may be administered in the absence of symptoms, e.g., to prevent or delay onset of a symptom or inhibit onset or progression of a disease. For example, treatment may be administered to a susceptible individual prior to the onset of symptoms (e.g., in light of a history of symptoms and/or in light of genetic or other susceptibility factors). Treatment may also be continued after symptoms have resolved, for example, to prevent or delay their recurrence.

The term “vector” refers to a polynucleotide comprising one or more recombinant polynucleotides of the present invention, e.g., those encoding or a Cas9 protein (e.g., a Cas9 protein comprising an intein) and/or a gRNA provided herein. Vectors include, but are not limited to, plasmids, viral vectors, cosmids, artificial chromosomes, and phagemids. The vector is one which is able to replicate in a host cell, and which is further characterized by one or more endonuclease restriction sites at which the vector may be cut and into which a desired nucleic acid sequence may be inserted. Vectors may contain one or more marker sequences suitable for use in the identification and/or selection of cells which have or have not been transformed or genomically modified with the vector. Markers include, for example, genes encoding proteins which increase or decrease either resistance or sensitivity to antibiotics (e.g., kanamycin, ampicillin) or other compounds, genes which encode enzymes whose activities are detectable by standard assays known in the art (e.g., β -galactosidase, alkaline phosphatase or luciferase), and genes which visibly affect the phenotype of transformed or transfected cells, hosts, colonies, or plaques. Any vector suitable for the transformation of a host cell, (e.g., *E. coli*, mammalian cells such as CHO cell, insect cells, etc.) as embraced by the present invention, for example vectors belonging to the pUC series, pGEM series, pET series, pBAD series, pTET series, or pGEX series. In some embodiments, the vector is suitable for transforming a host cell for recombinant protein production. Methods for selecting and engineering vectors and host cells for expressing gRNAs and/or proteins (e.g., those provided herein), transforming cells, and expressing/purifying recombinant proteins are well known in the art, and are provided by, for example, Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)).

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS OF THE INVENTION

Site-specific enzymes which catalyze nucleic acid modifications are powerful tools for targeted genome modification in vitro and in vivo. Some site-specific enzymes can theoretically achieve a level of specificity for a target site that would allow one to target a single unique site in a genome for modification without affecting any other genomic site. In the case of site-specific nucleases, it has been reported that nuclease cleavage in living cells triggers a DNA repair mechanism that frequently results in a modification of the cleaved and repaired genomic sequence, for example, via homologous recombination or non-homologous end-joining. Accordingly, the targeted cleavage of a specific unique sequence within a genome opens up new avenues for gene targeting and gene modification in living

cells, including cells that are hard to manipulate with conventional gene targeting methods, such as many human somatic cells or embryonic stem cells. Nuclease-mediated modification of disease-related sequences, e.g., the CCR-5 allele in HIV/AIDS patients, or of genes necessary for tumor neovascularization, can be used in the clinical context, and two site-specific nucleases are currently in clinical trials (Perez, E. E. et al., “Establishment of HIV-1 resistance in CD4+ T cells by genome editing using zinc-finger nucleases.” *Nature Biotechnology*. 26, 808-816 (2008); Clinical-Trials.gov identifiers: NCT00842634, NCT01044654, NCT01252641, NCT01082926). Other diseases that can be treated using site-specific nucleases or other site-specific DNA modifying enzymes include, for example, diseases associated with triplet expansion (e.g., Huntington’s disease, myotonic dystrophy, spinocerebellar ataxias, etc.), cystic fibrosis (by targeting the CFTR gene), cancer, autoimmune diseases, and viral infections.

One important problem with site-specific modification is off-target effects, e.g., the modification of genomic sequences that differ from the intended target sequence by one or more nucleotides. Undesired side effects of off-target modification range from insertion into unwanted loci during a gene targeting event to severe complications in a clinical scenario. For example, off-target modification of sequences encoding essential gene functions or tumor suppressor genes may result in disease or even the death of a subject. Accordingly, it is desirable to employ new strategies in designing site-specific enzymes having the greatest chance of minimizing off-target effects.

The systems, methods, and compositions of the present disclosure represent, in some aspects, an improvement over previous methods and compositions by providing means to control the spatiotemporal activity of site-specific enzymes, for example, RNA-guided nucleases and engineered RNA-guided nucleic acid modifying enzymes. For example, RNA-guided nucleases known in the art, both naturally occurring and those engineered, typically bind to and cleave DNA upon forming a complex with an RNA (e.g., a gRNA) that complements the target. Aspects of the present invention relate to the recognition that having spatiotemporal control of the enzymatic or nucleic acid binding properties of an RNA-guided nuclease and RNA-guided nucleic acid modifying enzymes by engineering variants to include an intein will decrease the likelihood of off-target effects by minimizing or controlling the time a RNA-guided nuclease or engineered RNA-guided nucleic acid modifying enzymes is active. Accordingly, the strategies, methods, compositions, kits, and systems provided herein can be used to control the activity of any site-specific enzyme (both naturally occurring and those engineered) such as RNA-guided nucleases (e.g., Cas9, Cas9 variants, fragments of Cas9 or Cas9 variants, etc.) or engineered nucleic acid modifying enzymes comprising a variant of an RNA-guided nuclease (e.g., dCas9).

Inteins are protein splicing elements that are able to catalyze their excision out of a single polypeptide and leave behind the flanking sequences, or exteins, precisely ligated together through a native peptide bond. Inteins are attractive tools for modulating protein structure and function because they do not require any other cellular components, are able to splice out of a wide variety of extein contexts, and can undergo splicing in minutes. Although natural inteins splice spontaneously, inteins that undergo splicing in a small molecule-dependent or ligand-dependent manner have been developed by fusing intein halves with proteins that dimerize in the presence of a small molecule, or by directed

evolution in which a library of intact inteins fused to a ligand-binding domain was screened to splice in the presence, but not the absence, of a small molecule or ligand. These ligand-dependent inteins have enabled protein function in cells to be controlled post-translationally by the addition of an exogenous, cell-permeable molecule (See e.g., published U.S. Patent Application US 2014/0065711 A1, the entire contents of which are hereby incorporated by reference). The inventors have found that the targeted insertion of ligand-dependent inteins into site-specific enzymes renders the enzymes, in some instances, inactive prior to the controlled excision of the intein through binding of a ligand specific for the intein. For example, the targeted insertion of a ligand-dependent intein into Cas9 at fifteen different positions resulted in a subset of Cas9 variants that were inactive in the absence of ligand, but upon addition of the ligand the intein self-excised resulting in an active Cas9 protein capable of site-specific cleavage of a target gene.

Some aspects of this disclosure are based on the surprising discovery that Cas9 proteins comprising an intein, for example, a ligand-dependent intein as described herein, exhibit an increased specificity as compared to constitutively active Cas9 proteins. For example, it was found that the conditionally active Cas9 proteins comprising an intein exhibit an activity in the “on” state that is comparable to wild-type Cas9 activity or only slightly decreased as compared to wild-type Cas9 activity, while exhibiting decreased off-target activity.

In addition, some aspects of this disclosure relate to the recognition that Cas9 off-target activity is at least in part related to the concentration of active Cas9 proteins, and that the off-target activity of the provided conditionally active Cas9 proteins, e.g., the provided ligand-dependent Cas9 proteins, can be modulated, e.g., further decreased, by contacting the Cas9 proteins with a minimal amount of ligand effecting the desired result, e.g., the minimal amount effecting intein excision from a Cas9 protein, or the minimal amount resulting in a desired level of Cas9 protein activity.

While of particular relevance to DNA and DNA-cleaving nucleases such as Cas9 and variants thereof, the inventive concepts, methods, compositions, strategies, kits, and systems provided herein are not limited in this respect, but can be applied to any nuclease or nucleic acid:enzyme system utilizing nucleic acid templates such as RNA to direct binding to a target nucleic acid. For example, the inventive concepts provided herein can be applied to RNA-guided nucleic acid-targeting protein, e.g., to RNA-guided nucleases, and to fusion proteins comprising nucleic acid-targeting domains of such nucleases, e.g., to fusion proteins comprising a Cas9 targeting domain (e.g., dCas9 domain), and a functional (effector) domain, such as, for example, a heterologous nuclease domain, recombinase domain, or other nucleic acid-editing domain.

Small Molecule Controlled Site-Specific Enzymes

Some aspects of this disclosure provide site-specific enzymes engineered to have both an “on” and “off” state which depends on the presence of a ligand such as a small molecule. The ligand binds and activates the enzyme through binding a ligand-dependent intein in the enzyme, whereby ligand binding induces self-excision of the intein thereby activating the enzyme (e.g., the presence of the intein in the enzyme disrupted one or more activities of the enzyme). In some aspects then, the enzymes may collectively be referred to as “small molecule controlled” or “ligand-dependent” site-specific enzymes. In some embodi-

ments, the site-specific enzyme that has been modified to include a ligand-dependent intein comprises Cas9, or a variant of Cas9.

Accordingly, in the absence of a ligand that binds the intein, the intein is not excised, and the protein comprising Cas9 or variant of Cas9 remains inactive. By “inactive” it is meant that the protein has no or minimal activity with respect to one or more activities described herein. In some embodiments, prior to intein excision, the protein has (i) no or minimal enzymatic activity; (ii) no or minimal gRNA binding activity; (iii) no or minimal target nucleic acid binding activity; or any combination of (i)-(iii), e.g., the protein has (i) and (ii); (i) and (iii); (ii) and (iii); or (i), (ii) and (iii). Enzymatic activities for (i), include, for example, nuclease activity, nickase activity, recombinase activity, nucleic acid editing (e.g., deaminase) activity, transcriptional activation, transcriptional repression, and epigenetic modification activity.

In some embodiments, by “minimal” activity, it is meant that the protein, prior to excision of the intein, exhibits less than 50%, less than 45%, less than 40%, less than 35%, less than 30%, less than 25%, less than 24%, less than 23%, less than 22%, less than 21%, less than 20, less than 19%, less than 18%, less than 17%, less than 16%, less than 15%, less than 14%, less than 13%, less than 12%, less than 11%, less than 10%, less than 9%, less than 8%, less than 7%, less than 6%, less than 5%, less than 4%, less than 3%, less than 2%, or less than 1% of a particular activity (e.g., nuclease activity, nickase activity, recombinase activity, deaminase activity, transcriptional activation, transcriptional repression, epigenetic modification activity, gRNA binding activity, and/or target nucleic acid binding activity) as compared to either the wild type counterpart of the protein or the intein-excised form of the protein. In some embodiments, following excision of the intein, the protein exhibits at least a 1.25-fold increase, at least a 1.5-fold increase, at least a 1.75-fold increase, at least a 2.0-fold increase, at least a 2.25-fold increase, at least a 2.5-fold increase, at least a 2.75-fold increase, at least a 3.0-fold increase, at least a 3.25-fold increase, at least a 3.5-fold increase, at least a 3.75-fold increase, at least a 4.0-fold increase, at least a 4.5-fold increase, at least a 5.0-fold increase, at least a 5.5-fold increase, at least a 6.0-fold increase, at least a 6.5-fold increase, at least a 7.0-fold increase, at least a 7.5-fold increase, at least a 8.0-fold increase, at least a 8.5-fold increase, at least a 9.0-fold increase, at least a 9.5-fold increase, or at least a 10.0-fold or more increase in activity (e.g., nuclease activity, nickase activity, recombinase activity, or deaminase activity) as compared to the intein-intact form of the protein. Methods for assessing the activity of any ligand-dependent site-specific Cas9-containing enzyme provided herein are well known to those of ordinary skill in the art, and in the context of nuclease activity include those described in the Examples.

In some embodiments, upon excision, the intein leaves a cysteine residue. Thus, if the intein is inserted such that it replaces a cysteine, the Cas9 protein, upon intein excision, will be unmodified as compared to the original protein. If the intein replaces any other amino acid, the Cas9 protein, upon intein excision, will contain a cysteine in place of the amino acid that was replaced. In some embodiments, the intein does not replace an amino acid residue in a Cas9 protein, but is inserted into the Cas9 protein (e.g., in addition to the amino acid residues of the Cas9 protein). In this aspect, upon excision, the protein will comprise an additional cysteine residue. While the presence of an additional cysteine residue (or the substitution of a residue for a cysteine upon excision)

is unlikely to affect the function of the Cas9 protein, in some embodiments where the intein does not replace a cysteine, the intein replaces an alanine, serine, or threonine amino acid, as these residues are similar in size and/or polarity to cysteine.

Accordingly, in some embodiments, the intein is inserted into one or both of the nuclease domains of Cas9 or a Cas9 variant (e.g., dCas9, Cas9 nickase), such as the HNH domain and/or the RuvC domain. In some embodiments, the intein is inserted into one or more other domains of Cas9 or a Cas9 variant (e.g., dCas9, Cas9 nickase), such as, REC1, REC2, PAM-interacting (PI), and/or bridge helix (BH) domain. The sequences and structure corresponding to these domains are known, and in some aspects are represented by the underlined segments of SEQ ID NO:2 (Cas9) and SEQ ID NO:5 (dCas9) above (See also, Nishimasu et al., "Crystal structure of Cas9 in complex with guide RNA and target DNA." Cell. 2014; 156(5), 935-949). In some embodiments, the intein is inserted into any location of Cas9, e.g., any location that disrupts one or more activities of Cas9 (e.g., enzymatic activity, gRNA binding activity, and/or DNA binding activity). In some embodiments, the intein is inserted into a sequence of Cas9 or a Cas9 variant such that the intein sequence replaces one or more amino acids in the protein. In some embodiments, the intein replaces any cysteine, any alanine, any threonine, or any serine residue in Cas9 or a Cas9 variant including Cas9 nickase and dCas9 (and fusions

thereof). In some embodiments the inserted intein sequence replaces Cys80, Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622, Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, or Ser1274 in Cas9 (SEQ ID NO:2), dCas9 (SEQ ID NO:5), or Cas9 nickase (SEQ ID NO:4). In some embodiments, the intein is inserted within 5, within 10, within 15, or within 20 amino acids of Cys80, Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622, Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, or Ser1274 in Cas9 (SEQ ID NO:2), dCas9 (SEQ ID NO:5), or Cas9 nickase (SEQ ID NO:4). In some embodiments, the inserted intein sequence replaces Ala127, Thr146, Ser219, Thr519, or Cys574 in Cas9 (SEQ ID NO:2), dCas9 (SEQ ID NO:5), or Cas9 nickase (SEQ ID NO:4). In some embodiments, a Cas9 protein comprising an intein comprises an amino acid sequence selected from the group consisting of SEQ ID NOs:27-41, or comprises an amino acid sequence that has at least 80%, at least 85%, at least 90%, at least 95%, or at least 99% sequence identity to any one of SEQ ID NOs:27-41. In some embodiments, the intein is inserted into the protein such that it does not replace any amino acid, but is added in addition to the amino acids of the protein. The intein that is inserted into the protein can be any ligand-dependent intein, e.g., those described herein. For example, in some embodiments, the intein that is inserted into the protein comprises, in part or in whole, a sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 99%, or 100% identical to any one of SEQ ID NO:27-41.

Cas9:Intein (37R3-2; in double underline) replacing Cys80
(SEQ ID NO: 27)
MDKKYISGLDIGTNSVGNVAVITDEYKVPSEKPKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
RRRYTRRNRICLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLARPVVSWFDQGTDRDVI
GLRIAGGAIWVATPDHKVLTEYGNRAAGELRKGDRVAGPGSGNSLALSITADQMVSALLDAEPPILY
SEYDPTSPFSEASMMGLLTNLADRELVHMNINWAKRVPGFVDLTLDHQAHLLEAWLEILMIGLVWRSM
EHPGKLLFAPNLLDRNQKGKVEGMVEIFDMLLATSSFRMMNLQGEFVCLKSIILLNSGVYTFLLS
TLKSLEEKDHIHRALDKITDTHLMAKAGLTLQQQHQLAQLLLILSHIRHMSNKGMEHLYSMKYKN
VVPLYDLLLEMLDAHRLHAGSGASRVQAFADALDDKFLHDLAEGRLYSVIREVLPTRRARTFDLEV
EELHTLVAEGVVVHNCYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHERHPFIGNIVDEVAYHEKY
PTIYHLRKKLVSDTKADLRILIYALAHMIKFRGHFLIEGLNPDNSDVKLFIQLVQTYNQLFEEPN
INASGVDAKAILSARLSKSRLENLIAQLPGEKKNLFGNLIASLGLTPNFKSNFDLAEDAKLQLSK
DTYDDDLNLLAQIGDQYADLFLLAAKNLSDAILLSDILRVNTEITKAPLSAMIKRYDEHHQDLTLK
ALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKNREDLLRKQRT
FDNGSIHQIHLGELHAILRQEDFYFPLKDNREKIEKILTFRIPIYYVGPLARGNSRPAWMTRKSEET
ITPWNFEVVDKGASQSFIERMTNFDKNLPNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPAFL
SGEQKKAIVDLLFKTNKRVTKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKLIKDKDFLD
NEENEDILEDIVLTLTLFEDREMIERLKYTAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQKSG
KTIIDFLKSDGFANRNFQILHDDSLTFKEDIQKAQVSGQDLSLHEHIANLAGSPAIIKGIQLQTVKVV
DELVKVMGRHKPENIVIMARENTTQKGQKNSRERMKRIEELGKELGSKILKEHPVENTQLQNEKLY
LYYLQNGRDMYVDQELDINRLSDYDVDHIVPQSFLKDDSIDNKVLTSDKNRGKSDNVPSEEVVKMK
NYWRQLLNAKLITQRKFDNLTKAERGGSELKAGFIKRLVETRQITKHVAQILDSRMNTKYDENK
LIREVKVITLKSCLVSDFRKDFQFYKREINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVY
DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKPLIETNGETGEIVWDKGRDFATVR

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KVLSMPQVNIKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKEKG
 KSKKLKSVKELLGITIMERSSEFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFYLYLASHYEKLGSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVLADANLD
 KVLSAYNKHDKPIREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQSIITGLY
 ETRIDLSQLGGD

Cas9:Intein (37R3-2; in double underline) replacing Ala127
 (SEQ ID NO: 28)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKKFVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRNKRICYLQEIFSNEMAKVDDSPFHRLEESFLVEEDKKHERHPIFGNIVDEVCLAEGRIPD
PVTGTTTHRIEDVVDGRKPIHVVAADGTLLARPVVSFWDQGTDRDVIGLRIAGGAIVWATPDHKVLTE
YGWRAAGELRKGDVRVAGPGGSGNSLALSLTADQMVSAALLDAEPPILYSEYDPTSPFSEASMMGLLTNL
ADRELVHMINWAKRVPGFVDLTLDQAHLLERAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKGC
VEGMVEIFDMLLATSSFRMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDT
LIHLMAGAGTLTQQQHQLAQLLLILSHIRHMSNKGMEHLYSMKYKNVPLYDLLLLMLDAHRLHAGG
SGASRVQAFADALDDKFLHDMLAEGLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNCYHEKY
 PTIYHLRKKLVSDTKADRLRIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEEENP
 INASGVDAKAILSARLSKSRLENLIAQLPGEKKNLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSK
 DTYDDDLNLLAQIGDQYADLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLK
 ALVRQQLPEKYKEIFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRT
 PDNGSIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEET
 ITPWNFEVVDKGASQSFIERMTNFDKNLPNEKVLPHKSLLYEYFTVYNELTKVKYVTEGMRKPAFL
 SGEQKKAIVDLLFKTRNKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKIIKDKDFLD
 NEENEDILEDIVLTLTLEFDEMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSG
 KTILDFLKSDFANRNFMLIHDDSLTFKEDIQKAQVSGQGSLSHEHIANLAGSPAICKGILQTVKVV
 DELVKVMGRHKPENIVIEARENQTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLY
 LYYLQNGRDMYVDQELDINRLSDYDVHDIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMK
 NYWRQLNKLITQRKFDNLTKAERGGLSELDKAGFIKRLVETRQITKHVAQILDSRMNTKYDENDK
 LIREVKVITLKSCLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVY
 DVRKMIKSEQEIQKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR
 KVLSMPQVNIKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKEKG
 KSKKLKSVKELLGITIMERSSEFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFYLYLASHYEKLGSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVLADANLD
 KVLSAYNKHDKPIREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQSIITGLY
 ETRIDLSQLGGD

Cas9:Intein (37R3-2; in double underline) replacing Thr146
 (SEQ ID NO: 29)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKKFVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRNKRICYLQEIFSNEMAKVDDSPFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY
 HLRKKLVDSCLAEGRIPDPVTGTTTHRIEDVVDGRKPIHVVAADGTLLARPVVSFWDQGTDRDVIGL
RIAGGAIVWATPDHKVLTEYGWRAAGELRKGDVRVAGPGGSGNSLALSLTADQMVSAALLDAEPPILYSE
YDPTSPFSEASMMGLLTNLADRELVHMINWAKRVPGFVDLTLDQAHLLERAWLEILMIGLVWRSMEH
PGKLLFAPNLLDRNQKGCVEGMVEIFDMLLATSSFRMMNLQGEFVCLKSIILLNSGVYTFLSSTL

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KSLEEKDHIHRALDKITDTLIHLMAKAGLTLOQQHQRLAQLLLILSHIRHMSNKGMEHLYSMKYKNV
PLYDLLLLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMLAEGRLYSVIREVLPTRRARTFDLEVEE
LHTLVAEGVVVHNCADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENP
 INASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSK
 DTYDDDLNLLAQIGDQYADLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLK
 ALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRT
 FDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKILTFRIPYYVGPLARGNSRFAWMTRKSEET
 ITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPHKSLLYEYFTVYNELTKVKYVTEGMRKPAFL
 SGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKLIKDKDFLD
 NEENEDILEDIVLTTLTFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSG
 KTILDPLKSDGFANRNFQLIHDDSLTFKEDIQKAQVSGQGSLEHEIANLAGSPAICKGILQTVKVV
 DELVKVMGRHKPENIVIMARENQTTQKGQKNSRERMKRIEEDIKELGSQILKEHPVENTQLQNEKLY
 LYYLQNGRDMYVDQELDINRLSDYVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMK
 NYWRQLLNAKLITQRKFDNLTAKERGGSELKAGFIKRLVETRQITKHVAQILDSRMNTKYDENDK
 LIREVKVITLKSFLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVV
 DVRKMIKSEQEIGKATAKYFFYSNIMNPFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR
 KVLMPQVNIKKTEVQTGGFSKESILPKRNSDKLIARKDWDPKKYGGFDSPTVAYSVLVAKVEKG
 KSKKLKSVKELLGITIMERSSFEKNPIDFLEAKGYEVKKDLIILPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFLYLASHYEKLGSPEDNEQQLFVEQHKHYLDEIEQISEFSKRVILADANLD
 KVL SAYNKHDKPIREQAENIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLHQSITGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Ser219
 (SEQ ID NO: 30)

MDKKYSIGLDIGTNSVGWAVITDEYKVPKSKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY
 HLRKKLVSDTKADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS
 GVDKAILSARLSKCLAEGTRIFDPVTGTHRIEDVVDGRKPIHVAAAKDGTLLARPVVSWFDQGTR
DVIGLRIAGGAIVWATPDHKVLTEYGWRAAGELRKGDRVAGPGGSGNSLALSLTADQMVSALLDAEPP
ILYSEYDPTSPFSEASMMGLLTNLADRELVHMINWAKRVPGFVDLTLHDQAHLLERAWLEILMIGLVW
RSMHPGKLLFAPNLLDRNQKCEGMVEIFDMLLATSSRFRMMNLQGEFVCLKSILLNSGVYTF
LSSTLKSLEEKDHIHRALDKITDTLIHLMAKAGLTLOQQHQRLAQLLLILSHIRHMSNKGMEHLYSMK
YKNVVPLYDLLLLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMLAEGRLYSVIREVLPTRRARTFD
LEVEELHTLVAEGVVVHNCRRLENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSK
 DTYDDDLNLLAQIGDQYADLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLK
 ALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRT
 FDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKILTFRIPYYVGPLARGNSRFAWMTRKSEET
 ITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPHKSLLYEYFTVYNELTKVKYVTEGMRKPAFL
 SGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKLIKDKDFLD
 NEENEDILEDIVLTTLTFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSG
 KTILDPLKSDGFANRNFQLIHDDSLTFKEDIQKAQVSGQGSLEHEIANLAGSPAICKGILQTVKVV
 DELVKVMGRHKPENIVIMARENQTTQKGQKNSRERMKRIEEDIKELGSQILKEHPVENTQLQNEKLY
 LYYLQNGRDMYVDQELDINRLSDYVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMK

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NYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRLVETRQITKHVAQILDSRMNTKYDENDK
 LIREVKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVY
 DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR
 KVLSPQVNIIVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKEKG
 KSKKLKSVKELLGITIMERSSEFEKNPIDFLEAGYKEVKKDLI IKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFYLYLASHYEKLGKSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVLADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQ SITGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Thr333

(SEQ ID NO: 31)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRNKRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS
 GVDAKAILSARLSKSRRLLENLIAQLPGEKKNGLFGNLI ALSGLTPNFKSNFDLAEDAKLQLSKDTYD
 DDLNLLAQIGDQYADFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDL CLAEGTRI
FDPVTGTTTHRIEDVDGRKPIHVVAADGTLARPVVSWFDQGT RDVIGLRIAGGAIVWATPDHKVL
TEYGWRAAGELRKGDRVAGPGSGNSLALS LTADQMVSALLDAEPPILYSEYDPTSPFSEASMMGLLT
NLADRELVHMINWAKRVPGFVDLT LHDAQHLLERAWLEILMIGLVWRSMHEPGKLLFAPNLLDNRQG
KCVGEMVEIFDMLLATSSFRPMNLQGEFVCLKSIILNSGVYTFLSSTLKSLEEKDHIHRALDKIT
DTLIHLMAKAGLTQQQHORLAQOLLILSHIRHMSNKGMEHLYSMKYKNVPLYDLLEMLDAHRLHA
GGSGASRVQAFADALDDKFLHDM LAEGLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNCCLK
 ALVRQQLPKEYKEIFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRT
 FDN GSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKILTFRIPIYYVGPLARGNSRFAMTRKSEET
 ITPWNFEVV DKGASQSFIERMTNFDKNLPNEKVL PKHSLLYEYFTVYNELTKVKYVTEGMRKPAFL
 SGEQKKAIVDLLFKTNKRVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDL LKIKDKDFLD
 NEENEDILEDIVLTLTLFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSG
 KTILDFLKS DGFANRNFMLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPA IKKGILQTVKVV
 DELVKVMGRHKPENIVIEARENQTQKGQKNSRERMKRIE EGIKELGSQILKEHPVENTQLQNEKLY
 LYYLQNGRDMYVDQELDINRLSDYDVHDIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMK
 NYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRLVETRQITKHVAQILDSRMNTKYDENDK
 LIREVKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVY
 DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR
 KVLSPQVNIIVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKEKG
 KSKKLKSVKELLGITIMERSSEFEKNPIDFLEAGYKEVKKDLI IKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFYLYLASHYEKLGKSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVLADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQ SITGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Thr519

(SEQ ID NO: 32)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRNKRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS

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GVDAKAILSARLSKSRLENLIAQLPGEKKNLFGNLIALLSLGLTPNFKSNFDLAEDAKLQLSKDITYD
 DDLNLLAQIGDQYADFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVR
 QQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNG
 SIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEETITPW
 NFEEVVDKGASQSFIERMTNFDKNLPNEKVLPHKSLLYEYFCLAEGTRIFDPVTGTTHRIEDVVDGR
KPIHVVAADGTLLARPVVSWFDQGTDRDVLGRIAGGAI VWATPDHKVLT EYGWRAAGELRKGRVA
GPGGSGNSLALS LTADQMV SAL LDAEPPILYSEYDPTS PFSEASMMGLL TNLADREL VHM INWAKRVP
GFVDLT LHDQAHLLERAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCV EGMVEIFDMLLATSS
RFRMMNLQGE E FVCLKS I ILLNSGVYTFLSSTLKSLEEKDHIHRALDKI TDTLIHMAKAGLT LQQQH
QRLAQLLLILSHIRHMSNKGMEHLYSMKYKNVPLYD LLEMLDAHRLHAGGSGASRVQAFADALDDK
FLHDM LAEGLRYSVIREVLPTRRARTF DLEVEELHTLV AEGVVVHNCVYNELTKVKYVTEGMRKPAFL
 SGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGT YHDLKIIKDKDFLD
 NEENEDI EDIVLT LTLFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSG
 KTILDFLKS DGFANRNFQQLIHDDSLTFKEDIQKAQVSGQGD SLHEHIANLAGSPA IKKGILQTVKVV
 DELVKVMGRHKPENIVIEMARENQTQKQKNSRERMKRIE EGIKELGSQILKEHPVENTQLQNEKLY
 LYYLQNGRDMYVDQELDINRLSDYVDVHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVKKMK
 NYWRQLLNAKLITQRKFDNLTKAERGG LSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDK
 LIREVKVITLKS KLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVG TALIKKYPKLESEFVYGDYKVY
 DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEI VWDKGRDFATVR
 KVLMPQVNI VVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKEVG
 KSKKLKSVKELGITIMERSSFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFLYLASHYEKLGSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQ SITGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Cys574

(SEQ ID NO: 33)

MDKKYSIGLDIGTNSVGWAVITDEYKVP SKFKVLGNTDRHSI KKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRKNRICYLQEIFSNEMAKVDDSFHRL EESFLVEEDKKHERHP IFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS
 GVDAKAILSARLSKSRLENLIAQLPGEKKNLFGNLIALLSLGLTPNFKSNFDLAEDAKLQLSKDITYD
 DDLNLLAQIGDQYADFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVR
 QQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNG
 SIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEETITPW
 NFEEVVDKGASQSFIERMTNFDKNLPNEKVLPHKSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQ
 KKAIVDLLFKTNRKVTVKQLKEDYFKKIECLAEGTRIFDPVTGTTHRIEDVVDGRKPIHVVAADGT
LLARPVVSWFDQGTDRDVLGRIAGGAI VWATPDHKVLT EYGWRAAGELRKGRVAGPGGSGNSLALS
TADQMV SAL LDAEPPILYSEYDPTS PFSEASMMGLL TNLADREL VHM INWAKRVP GFVDLT LHDQAH
LERAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCV EGMVEIFDMLLATSS RFRMMNLQGE E FV
CLKS I ILLNSGVYTFLSSTLKSLEEKDHIHRALDKI TDTLIHMAKAGLT LQQQH QRLAQLLLILSHI
RHMSNKGMEHLYSMKYKNVPLYD LLEMLDAHRLHAGGSGASRVQAFADALDDK FLHDM LAEGLRYS
VIREVLPTRRARTF DLEVEELHTLV AEGVVVHNCFD SVEISGVEDRFNASLGT YHDLKIIKDKDFLD
 NEENEDI EDIVLT LTLFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSG

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KTILDPLKSDGFANRNFQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKGILQTVKVV
 DELVKVMGRHKPENIVIEARENQTTQKGQKNSRERMKRIIEGKELGSQILKEHPVENTQLQNEKLY
 LYYLQNGRDMYVDQELDINRLSDYVDHIVPQSFLKDDSIDNKVLRSDKNRGKSDNVPSEEVVKMK
 NYWRQLLNAKLI TQRKFDNLTKAERGGSELKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDK
 LIREVKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVV
 DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR
 KVLSPQVNIIVKTEVQTGGFSKESILPKRNSDKLIARKKWDPKKYGGFDSPTVAYSVLVAKVEKG
 KSKKLKSVKELGITIMERSSEFKNPIDFLEAGYKEVKKDLIIKLPKYSLEFENGRKRLASAGEL
 QKGNELALPSKYVNFYLAHSHYEKLGKSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLD
 KVL SAYNKHRRDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQSI TGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Thr622

(SEQ ID NO: 34)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRNRIICYLQEIFSNEMAKVDDSPFHRLEESPLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS
 GVDKAILSARLSKSRLENLIAQLPGEKKNLFGNLI ALSLGLTPNPKSNFDLAEDAKLQLSKD TYD
 DDLNLLAQIGDQYADFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVR
 QQLPEKYKEIFPDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNG
 SIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEETITPW
 NFEEVVDKGSASQSFIERMTNFDKNLPNEKVLPHKSLLEYEFTVYNELTKVKYVTEGMRKPAFLSGEQ
 KKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKI IKDKDFLDNEEN
 EDILEDIVLCLAEAGTRIFDPVTGTTHRIEDVVDGRKPIHVVAADGDTLLARPVVS WFDQGT RDVIGL
RIAGGAIVWATPDHKVLT EYGWRAAGELRKGRVAGPGGSGNSLALS LTADQMVSALLDAEPPILYSE
YDPTSPFSEASMMGLLTNLADRELVHM INWAKRVPGFVDLT LHDQAHLLERAWLEILMIGLVWRSMEH
PGKLLFAPNLLDRNQKCGVEGMVEIFDMLLATSSRFRMMNLQGEFVCLKSII LLNSGVYTFLSSTL
KSLEEKDHIHRALDKITDTLIHLMAKAGLT LQQQHQLAQLLLI LSHIRHMSNKGMEHLYSMKYQNVV
PLYDLLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDMLAEGRLYSVIREVLPTRRARTFDLEVEE
LHTLVAEGVVVHNCLTLFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSG
 KTILDPLKSDGFANRNFQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKGILQTVKVV
 DELVKVMGRHKPENIVIEARENQTTQKGQKNSRERMKRIIEGKELGSQILKEHPVENTQLQNEKLY
 LYYLQNGRDMYVDQELDINRLSDYVDHIVPQSFLKDDSIDNKVLRSDKNRGKSDNVPSEEVVKMK
 NYWRQLLNAKLI TQRKFDNLTKAERGGSELKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDK
 LIREVKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVV
 DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR
 KVLSPQVNIIVKTEVQTGGFSKESILPKRNSDKLIARKKWDPKKYGGFDSPTVAYSVLVAKVEKG
 KSKKLKSVKELGITIMERSSEFKNPIDFLEAGYKEVKKDLIIKLPKYSLEFENGRKRLASAGEL
 QKGNELALPSKYVNFYLAHSHYEKLGKSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLD
 KVL SAYNKHRRDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQSI TGLY
 ETRIDLSQLGGD

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Cas9:Intein (37R3-2; in double underline) replacing Ser701
(SEQ ID NO: 35)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA

RRRYTRRKNRICYLQEIFSNEMAKVDDSFPHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY

HLRKKLVSDTKADRLRIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS

GVDAKAILSARLSKSRRLENLIAQLPGEKKNLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKDTYD

DDLNLQAIGIQYADLFLAAKNLSDAILSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVR

QQLPEKYKEIFPDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNG

SIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEETITPW

NFEEVVDKGASQSFIERMTNFDKNLPNEKVLPKHSLLEYEFTVYNELTKVKYVTEGMRKPAFLSGEQ

KKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKI IKDKDFLDNEEN

EDILEDIVLTLTLFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKITIL

DPLKSDGFANRNFMLIHDDCLAEGRIFDPVTGTTHRIEDVVDGRKPIHVVAAGDGTLLARPVVS

FDQGRDVI GLRIAGGAIVWATPDHKVLT EYGWRAAGELRK GDRVAGPGSGNSLALS LTADQMVSAL

LDAEPPILYSEYDPTSPFSEASMMGLLTNLADRELVHMINWAKRVPGFVDLTLHDQAHLLERAWLEIL

MIGLVWRSMEHPGKLLFAPNLLDRNQKCVEGMVEIFDMLLATSSRFRMMNLQGEFVCLKSIILLN

SGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAKAGLTQQQHQRLAQLLLLILSHIRHMSNKGME

HLYSMKYKNVPLYDLLLLLEMLDAHRLHAGSGSASRVQAFADALDDKFLHDM LAEGLRYSVIREVLPTR

RARTFDLEVEELHTLVAEGVVVHNCLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIKKGILQTVKVV

DELVKVMGRHKPENIVIMARENQTQKGQKNSRERMKRIE EGIKELGSQILKEHPVENTQLQNEKLY

LYYLQNGRDMYVDQELDINRLSDYVDVHIVPQSFLKDDSIDNKVLTSDKNRGKSDNVPSEEVKKMK

NYWRQLLNAKLI TQRKFDNLTKAERGGELSEDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDK

LIREVKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVG TALIKKYPKLESEFVYGDYKVY

DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR

KVLSMPQVNI VVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKEKG

KSKKLKSVKELGITIMERSSFEKNPIDFLEAKGYKEVKDLI IKLPKYSLFELENGRKRMLASAGEL

QKGNELALPSKYVNFLYLASHYEKLGSPEDNEQKQLFVEQHKHYLDEI IEQISEFSKRVILADANLD

KVLSAYNKHDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQSI TGLY

ETRIDLSQLGGD

Cas9:Intein (37R3-2; in double underline) replacing Ala728
(SEQ ID NO: 36)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA

RRRYTRRKNRICYLQEIFSNEMAKVDDSFPHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY

HLRKKLVSDTKADRLRIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS

GVDAKAILSARLSKSRRLENLIAQLPGEKKNLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKDTYD

DDLNLQAIGIQYADLFLAAKNLSDAILSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVR

QQLPEKYKEIFPDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNG

SIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEETITPW

NFEEVVDKGASQSFIERMTNFDKNLPNEKVLPKHSLLEYEFTVYNELTKVKYVTEGMRKPAFLSGEQ

KKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKI IKDKDFLDNEEN

EDILEDIVLTLTLFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKITIL

DPLKSDGFANRNFMLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLCLAEGRIFDPVTGTTHRIED

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VVDGRKPIHVVAADGTLARPVVSWFDQGTRDVI GLRIAGGAI VWATPDHKVLT EYGWRAAGELRK
GDRVAGPGGSGNSLALSLTADQMVSA LLD AEPPI LYSEYDPTSPFSEASMMGLLTNLADREL VHMINW
AKRVPGFVDLT LHDQAHLLERAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQ GKCV EGMVEIFDML
LATSSRFRMMNLQGEFVCLKSI ILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAGL T
LQQQHQLAQ LLLILSHIRHMSNKGMEHLYSMKYKNVPLYD LLEMLDAHRLHAGSGGASRVQAFAD
ALDDKFLHDM LAEGLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVHNCGSPA IKGILQTVKV V
 DELVKVMGRHKPENIVIEARENQTTQKGQKNSRERMKRIE EGIKELGSQILKEHPVENTQLQNEKLY
 LYYLQNGRDMYVDQELDINRLSDYVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMK
 NYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDK
 LIREVKVITLKSCLVDFRKFQFYKVRINNYHHAHDAYLNAVVG TALIKKYPKLESEFVYGDYKVY
 DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR
 KVLSPQVNI VVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKYGGFDSPTVAYSVLVAKVEKG
 KSKKLKSVKELGITIMERSSEFEKNPIDFLEAGYKEVKKDLI IKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFYLA SHYEKLGSPEDNEQQLFVEQHKHYLDEI IEQISEFSKRVLADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQSI TGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Thr995

(SEQ ID NO: 37)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKKFVLGNTDRHSIKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRNKRICYLQEIFSNEMAKVDDSPFHRLEESPLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS
 GVDAKAILSARLSKSRRL ENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYD
 DDLNLLAQIGDQYADFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVR
 QQLPEKYKEIFPDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKNREDLLRKQRTFDNG
 SIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPYYVGPLARGNSRFAWMTRKSEETITPW
 NFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLEYFTVYNELTKVKYVTEGMRKPAFLSGEQ
 KKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKI IKDKDFLDNEEN
 EDILEDIVLTTLTFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKITIL
 DFLKSDGFANRNFQMQLIHDDSLTFKEDIQKAVSGQGDSLHEHIANLAGSPA IKGILQTVKVVDLV
 KVMGRHKPENIVIEARENQTTQKGQKNSRERMKRIE EGIKELGSQILKEHPVENTQLQNEKLYLYYL
 QNGRDMYVDQELDINRLSDYVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKNYWR
 QLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIRE
 VKVITLKSCLVDFRKFQFYKVRINNYHHAHDAYLNAVVG CLAEGTRIFDPVTGTTHRIEDVVDGR
KPIHVVAADGTLARPVVSWFDQGTRDVI GLRIAGGAI VWATPDHKVLT EYGWRAAGELRK GDRVA
GPGGSGNSLALSLTADQMVSA LLD AEPPI LYSEYDPTSPFSEASMMGLLTNLADREL VHMINWAKRVP
GFVDLT LHDQAHLLERAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQ GKCV EGMVEIFDMLLATSS
RFRMMNLQGEFVCLKSI ILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAGL TLQQQH
QLAQ LLLILSHIRHMSNKGMEHLYSMKYKNVPLYD LLEMLDAHRLHAGSGGASRVQAFADALDDK
FLHDM LAEGLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVHNCALIKKYPKLESEFVYGDYKVY
 DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVR
 KVLSPQVNI VVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKYGGFDSPTVAYSVLVAKVEKG
 KSKKLKSVKELGITIMERSSEFEKNPIDFLEAGYKEVKKDLI IKLPKYSLFELENGRKRMLASAGEL

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QKGNELALPSKYVNFYLYLASHYEKLGKSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQ SITGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Ser1006
 (SEQ ID NO: 38)
 MDKKYSIGLDIGTNSVGWAVITDEYKVP SKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA

RRRYTRRKNRICYLQEIFSNEMAKVDDSFHRL EESFLVEEDKKHERHP IFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS
 GVDKAILSARLSKSRLENLIAQLPGEKKNGLFGNLI ALSLGLTPNFKSNFDLAEDAKLQLSKD TYD
 DDLNLLAQIGQYADLFLAAKNLSDAILSDILRVNTEITKAPLSAMIKRYDEHHQDLTLLKALVR
 QQLPEKYKEIFPDQSKNGYAGYIDGGASQE EYKFIKPILEKMDGTEELLVKNREDLLRKQRTFDNG
 SIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEETITPW
 NFEEVVDKGASAQSFIERMTNFDKNLPNEKVL PKHSLLEYEFTVYNELTKVKYVTEGMRKPAFLSGEQ
 KKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDL LKI IKDKDFLDNEEN
 EDILEDIVLTLTLPEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTIL
 DPLKSDGFANRNFMLIHDLSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPA I KKGILQTVKVVDDELV
 KVMGRHKPENIVIMARENQTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLYLYYL
 QNGRDMYVDQELDINRLSDYDVHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKNYWR
 QLLNAKLI TQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIRE
 VKVITL KSKLVSDFRKDFQFYKVR EINNYYHHAHDAYLNAVVG TALIKKYPKLECLAEGRIFDPVTGT
THRIEDVVDGRKPIHVVA AAKDGTLLARPVVS WFDQGTRDVIGLR IAGGAI VWATPDHKVLTEYGWRA
AGELRKGRVAGPGGSGNSLALS LTADQMVSALLDAEPPILYSEYDPTS PFSEASMMGLLTNLADREL
VHMINWAKRVP GFGVDLTLDHQAHL LERAWLEILMIGLVRSMEHPGKLLFAPNLLDRNQ GKCVEGMV
EIFDMLLATSSRFRMMNLQGE EYVCLKSI ILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLM
AKAGLT LQQQHQLAQLLLILSHIRHMSNKGMEHLYSMKYKNVVP LYD LLEMLDAHRLHAGGSGASR
VQAFADALDDKFLHDM LAEGLRYSVIREVLPTRRARTFDLEVEELHTLV AEGVVVHNCFEYVGYKVY
 DVRKMI AKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEI VDWKGRDFATVR
 KVLSPQVNI VVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKEKG
 KSKKLKSVKELGITIMERSSFEKNPIDFLEAKGYKEVKDLIIKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFYLYLASHYEKLGKSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQ SITGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Ser1154
 (SEQ ID NO: 39)
 MDKKYSIGLDIGTNSVGWAVITDEYKVP SKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA

RRRYTRRKNRICYLQEIFSNEMAKVDDSFHRL EESFLVEEDKKHERHP IFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS
 GVDKAILSARLSKSRLENLIAQLPGEKKNGLFGNLI ALSLGLTPNFKSNFDLAEDAKLQLSKD TYD
 DDLNLLAQIGQYADLFLAAKNLSDAILSDILRVNTEITKAPLSAMIKRYDEHHQDLTLLKALVR
 QQLPEKYKEIFPDQSKNGYAGYIDGGASQE EYKFIKPILEKMDGTEELLVKNREDLLRKQRTFDNG
 SIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEETITPW
 NFEEVVDKGASAQSFIERMTNFDKNLPNEKVL PKHSLLEYEFTVYNELTKVKYVTEGMRKPAFLSGEQ

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KKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKI IKDKDFLDNEEN
 EDILEDIVLTTLTFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQSGKITIL
 DFLKSDGFANRNFQMLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKKGILQTVKVVDELV
 KVMGRHKPENIVIMARENQTTQKGQKNSRERMKRIEEGIELGSQILKEHPVENTQLQNEKLYLYYL
 QNGRDMYVDQELDINRLSDYDVHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKNYWR
 QLLNAKLI TQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIRE
 VKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGTAIIKKYPKLESEFVYGDYKVYDVRK
 MIAKSEQEIGKATAKYFFYSNIMNPFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLS
 MPQVNIKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVAKVEKGKCLA
EGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLLARPVVSWFDOGTRDVI GLRIAGGAIWATP
DHKVLTEYGWRAAGELRKGRVAGPGSGNSLALS LTADQMVSA LLDAEPPILYSEYDPTSPFSEASM
MGLLTNLADREL VHMINWAKRVPGFVDLT LHDOAHL LERAWLEILMIGLVWRSM EHPGKLLFAPNLLL
DRNQKCKVEGMVEIFDMLLATSSRFMMNLQGEFVCLKSI ILLNSGVYTFLSSTLKSLEEKDHIHRA
LDKITD TLIHMAKAGLTQQQHQLAQLLLILSHIRHMSNKGMEHLYSMKYKNVPLYDLLEMLDA
HRLHAGGS GASRVQAFADALDDKFLHDM LAEGLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVH
NCKKLKSVKELGITIMERS SFEKNPIDFLEAKGYKEVKKD LIKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFYLA SHYEKLGKSPEDNEQQLFVEQHKHYLDEIIEQISEFSKRVLADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQSI TGLY
 ETRIDLSQLGGD

Cas9: Intein (37R3-2; in double underline) replacing Ser1159

(SEQ ID NO: 40)

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRNKRICYLQEIFSNEMAKVDDSPFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFENPINAS
 GVDAIL SARLSKSRRL ENLIAQLPGEKKNGLFGNLI ALSGLTPNFKSNFDLAEDAKLQLSKDTYD
 DDLNLLAQIGDQYADFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVR
 QQLPEKYKEIFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKNREDLLRKQRTFDNG
 SIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFPIPYVGPLARGNSRFAWMTRKSEETITPW
 NFEEVVDKGASAQSFIERMTNFDKNLPNEKVL PKHSLLEYFTVYNELTKVKYVTEGMRKPAFLSGEQ
 KKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKI IKDKDFLDNEEN
 EDILEDIVLTTLTFEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQSGKITIL
 DFLKSDGFANRNFQMLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKKGILQTVKVVDELV
 KVMGRHKPENIVIMARENQTTQKGQKNSRERMKRIEEGIELGSQILKEHPVENTQLQNEKLYLYYL
 QNGRDMYVDQELDINRLSDYDVHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKNYWR
 QLLNAKLI TQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIRE
 VKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGTAIIKKYPKLESEFVYGDYKVYDVRK
 MIAKSEQEIGKATAKYFFYSNIMNPFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLS
 MPQVNIKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVAKVEKGKSKK
LKCLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLLARPVVSWFDOGTRDVI GLRIAGGAI
WATPDHKVLTEYGWRAAGELRKGRVAGPGSGNSLALS LTADQMVSA LLDAEPPILYSEYDPTSPF
SEASMMGLLTNLADREL VHMINWAKRVPGFVDLT LHDOAHL LERAWLEILMIGLVWRSM EHPGKLLFA
PNLLDRNQKCKVEGMVEIFDMLLATSSRFMMNLQGEFVCLKSI ILLNSGVYTFLSSTLKSLEEKD

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HIHRALDKITDTLIHLMKAGLTLOQQHQRLAQQLLILSHIRHMSNKGMEHLYSMKYKNVPLYDL
LL
EMLDAHRLHAGGSGASRVQAFADALDDKFLHDMLAEGRLYSVIREVLPTRRARTFDLEVEELHTLV
AE
GVVVHNCVKELGITIMERSSEFEKNPIDFLEAKGYKEVKDLIIKLPKYSLFELENGRKRMLASAGEL
 QKGNELALPSKYVNFYLYLASHYEKLKGSPEDEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQ SITGLY
 ETRIDLSQLGGD
 Cas9: Intein (37R3-2; in double underline) replacing Ser1274
 (SEQ ID NO: 41)
 MDKKYSIGLDIGTNSVGWAVITDEYKVP SKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTA
 RRRYTRRNKRICYLQEIFSNEMAKVDDSFHRL EESFLVEEDKKHERHP IFGNIVDEVAYHEKYPTIY
 HLRKKLV DSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINAS
 GVDAKAILSARLSKSRLENLIAQLPGEKKNGLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKDTYD
 DDLNLLAQIGDQYADLFLAAKNLSDAILSDILRVNTEITKAPLSAMIKRYDEHHQDLTLKALVR
 QQLPEKYKEIFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNG
 SIPHQIHLGELHAILRRQEDFYPLKDNREKIEKILTFRIPIYVGPLARGNSRFAWMTRKSEETITPW
 NFEEVVDKGASQSFIERMTNFDKNLPNEKVL PKHSLLEYFTVYNELTKVKYVTEGMRKPAFLSSEQ
 KKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKI IKDKDFLDNEEN
 EDILEDIVLTLTLPEDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTIL
 DFLKSDGFANRNFMLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKKGILQTVKVVDELV
 KVMGRHKPENIVIMARENQTQKGQKNSRERMKRIE EGIKELGSQILKEHPVENTQLQNEKLYLYYL
 QNGRDMYVDQELDINRLSDYDVHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWR
 QLLNAKLI TQRKFDNLTKAERGGSEL DKA GFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIRE
 VKVITLKS KLVSDFRDKDFQFYKREINNYHHA DAYLNAVVG TALIKKYPKLESEFVYGDYKVYDVRK
 MIAKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDPATVRKVLS
 MPQVNI VKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVAKVEKGSKK
 LKSVKELLGITIMERSSEFEKNPIDFLEAKGYKEVKDLIIKLPKYSLFELENGRKRMLASAGELQKGN
 ELALPSKYVNFYLYLASHYEKLKGSPEDEQKQLFVEQHKHYLDEIEQICLAEGTRIFDPVTGTTHRI
EDVVDGRKPIHVVAAGKDTLLARPVVSWFQDQGRDVI GLRIAGGAI VWATPDHKVLTEYGWRAAGEL
RKGDRVAGPGSGNSLALSLTADQMVSA LLD AEPPILYSEYDPTSPFSEASMMGLLTNLADREL VMI
NWAKRVP GFVDLT LHDQAHLLERAWLEILMIGLVWRSMEHPGKLLFAPNLLDRNQKCV EGMVEIFD
MLLATSSFRMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMKAG
LTLOQQHQRLAQQLLILSHIRHMSNKGMEHLYSMKYKNVPLYDLLEMLDAHRLHAGGSGASRVQAF
ADALDDKFLHDMLAEGRLYSVIREVLPTRRARTFDLEVEELHTLV AEGVVVHNCEF SKRVILADANLD
 KVL SAYNKH RDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQ SITGLY
 ETRIDLSQLGGD

In some embodiments, the intein inserted into the Cas9
 protein is ligand-dependent. In some embodiments, the
 ligand-dependent inteins comprise a modified ligand-binding
 domain of the estrogen receptor protein, embedded into
 a modified RecA intein from *M. tuberculosis*. In some
 embodiments, the ligand-binding domain is derived from an
 estrogen receptor protein, for example, from the human
 estrogen receptor. The sequence of the human estrogen
 receptor and the location of the ligand-binding domain
 within the human estrogen receptor are known to those of

skill in the art. Non-limiting, exemplary sequences of the
 human estrogen receptor can be retrieved from RefSeq
 database entries NP_000116 (isoform 1); NP_001116212
 (isoform 2); NP_001116213 (isoform 3); and
 NP_001116214 (isoform 4) from the National Center for
 Biotechnology Information (NCBI, ncbi.nlm.nih.gov). In
 some embodiments, the ligand-binding domain of a ligand-
 dependent intein provided herein comprises or is derived
 from a sequence comprising amino acid residues 304-551 of
 the human estrogen receptor.

It will be appreciated by those of skill in the art that other ligand-dependent inteins are also suitable and useful in connection with the Cas9 proteins and methods provided herein. For example, some aspects of this invention provide Cas9 proteins comprising ligand-dependent inteins that comprise a ligand-binding domain of a hormone-binding protein, e.g., of an androgen receptor, an estrogen receptor, an ecdysone receptor, a glucocorticoid receptor, a mineralocorticoid receptor, a progesterone receptor, a retinoic acid receptor, or a thyroid hormone receptor protein. Ligand-binding domains of hormone-binding receptors, inducible fusion proteins comprising such ligand-binding domains, and methods for the generation of such fusion proteins are known to those of skill in the art (see, e.g., Becker, D., Hollenberg, S., and Ricciardi, R. (1989). Fusion of adenovirus E1A to the glucocorticoid receptor by high-resolution deletion cloning creates a hormonally inducible viral transactivator. *Mol. Cell. Biol.* 9, 3878-3887; Boehmelt, G., Walker, A., Kabrun, N., Mellitzer, G., Beug, H., Zenke, M., and Enrietto, P. J. (1992). Hormone-regulated v-rel estrogen receptor fusion protein: reversible induction of cell transformation and cellular gene expression. *EMBO J* 11, 4641-4652; Braselmann, S., Graninger, P., and Busslinger, M. (1993). A selective transcriptional induction system for mammalian cells based on Gal4-estrogen receptor fusion proteins. *Proc Natl Acad Sci USA* 90, 1657-1661; Furga G, Busslinger M (1992). Identification of Fos target genes by the use of selective induction systems. *J. Cell Sci. Suppl* 16, 97-109; Christopherson, K. S., Mark, M. R., Bajaj, V., and Godowski, P. J. (1992). Ecdysteroid-dependent regulation of genes in mammalian cells by a *Drosophila* ecdysone receptor and chimeric transactivators. *Proc Natl Acad Sci USA* 89, 6314-8; Eilers, M., Picard, D., Yamamoto, K., and Bishop, J. (1989). Chimeras of Myc oncoprotein and steroid receptors cause hormone-dependent transformation of cells. *Nature* 340, 66-68; Fankhauser, C. P., Briand, P. A., and Picard, D. (1994). The hormone binding domain of the mineralocorticoid receptor can regulate heterologous activities in cis. *Biochem Biophys Res Commun* 200, 195-201; Godowski, P. J., Picard, D., and Yamamoto, K. R. (1988). Signal transduction and transcriptional regulation by glucocorticoid receptor-LexA fusion proteins. *Science* 241, 812-816; Kellendonk, C., Tronche, F., Monaghan, A., Angrand, P., Stewart, F., and Schütz, G. (1996). Regulation of Cre recombinase activity by the synthetic steroid RU486. *Nuc. Acids Res.* 24, 1404-1411; Lee, J. W., Moore, D. D., and Heyman, R. A. (1994). A chimeric thyroid hormone receptor constitutively bound to DNA requires retinoid X receptor for hormone-dependent transcriptional activation in yeast. *Mol Endocrinol* 8, 1245-1252; No, D., Yao, T. P., and Evans, R. M. (1996). Ecdysone-inducible gene expression in mammalian cells and transgenic mice. *Proc Natl Acad Sci USA* 93, 3346-3351; and Smith, D., Mason, C., Jones, E., and Old, R. (1994). Expression of a dominant negative retinoic acid receptor g in *Xenopus* embryos leads to partial resistance to retinoic acid. *Roux's Arch. Dev. Biol.* 203, 254-265; all of which are incorporated herein by reference in their entirety). Additional ligand-binding domains useful for the generation of ligand-dependent inteins as provided herein will be apparent to those of skill in the art, and the invention is not limited in this respect.

Additional exemplary inteins, ligand-binding domains, and ligands suitable for use in the Cas9 proteins provided herein are described in International Patent Application, PCT/US2012/028435, entitled "Small Molecule-Dependent

which are incorporated herein by reference. Additional suitable inteins, ligand-binding domains, and ligands will be apparent to the skilled artisan based on this disclosure.

The ligand-dependent inteins provided herein are inactive (or only minimally active) in the absence of the appropriate ligand, but can be induced to be active, and, thus, to self-excise, by contacting them with a ligand that binds the ligand-binding domain of the human estrogen receptor. Small molecule ligands binding the ligand-binding domain of the estrogen receptor (e.g., the human estrogen receptor), and thus useful to induce the activity of the ligand-dependent inteins described herein, are known to those of skill in the art. In some embodiments, the ligand used to induce the activity of the ligand-dependent inteins described herein specifically binds to the ligand-binding domain of the estrogen receptor. In some embodiments, the ligand binds the ligand-binding domain of a ligand-dependent intein provided herein with high affinity, for example, with an affinity of at least about 10^{-10} M, at least about 10^{-9} M, at least about 10^{-8} M, at least about 10^{-7} M, at least about 10^{-6} M, or at least about 10^{-5} M. Examples of appropriate estrogen receptor-binding ligands that are useful to induce the activity of the ligand-dependent inteins provided herein, for example, the ligand-dependent inteins provided in SEQ ID NOs 3-8, include, but are not limited to, 17 β -estradiol, 17 α -ethynyl estradiol, tamoxifen and tamoxifen analogs (e.g., 4-hydroxytamoxifen (4-HT, 4-OHT), 3-hydroxytamoxifen (droloxifene)), tamoxifen metabolites (e.g., hydroxytamoxifen, endoxifen), raloxifene, toremifene, ICI-182, and ICI-780. Other useful ligands will be apparent to those of skill in the art, and the invention is not limited in this respect.

In some embodiments, any of the Cas9 proteins comprising inteins (e.g., SEQ ID NOs:27-41) can be modified so as to generate a Cas9 nickase comprising an intein (e.g., by making one of a D10A or H840A mutation relative to the Cas9 sequence lacking an intein), or to generate a dCas9 protein comprising an intein (e.g., by making both D10A and H840A mutations relative to the Cas9 sequence lacking an intein). In some embodiments, any of the Cas9 proteins comprising inteins (e.g., SEQ ID NOs:27-41) have additional features, for example, one or more linker sequences, localization sequences, such as nuclear localization sequences (NLS; e.g., MAPKKKRKVGIIHRGVP (SEQ ID NO:42)); cytoplasmic localization sequences; export sequences, such as nuclear export sequences; or other localization sequences, as well as sequence tags that are useful for solubilization, purification, or detection of the fusion proteins. Suitable localization signal sequences and sequences of protein tags are provided herein and are known in the art, and include, but are not limited to, biotin carboxylase carrier protein (BCCP) tags, myc-tags, calmodulin-tags, FLAG-tags (e.g., 3 $\times$ FLAG TAG: MDYKDHGDYKDHIDYKDDDDK (SEQ ID NO:43)), hemagglutinin (HA) tags, polyhistidine tags, also referred to as histidine tags or His-tags, maltose binding protein (MBP)-tags, nus-tags, glutathione-S-transferase (GST) tags, green fluorescent protein (GFP) tags, thioredoxin-tags, S-tags, Softags (e.g., Softag 1, Softag 3), strep-tags, biotin ligase tags, FAsH tags, V5 tags, and SBP-tags. Additional suitable sequences will be apparent to those of skill in the art.

In some embodiments, ligand-dependent site-specific enzymes (e.g., fusion proteins) are provided which comprise a Cas9 variant (e.g., dCas9), a ligand-dependent intein, and one or more other polypeptide domains having a particular enzymatic activity. In some embodiments, the fusion protein comprises a nuclease inactivated Cas9 domain (e.g., dCas9),

wherein the dCas9 domain comprises an intein sequence inserted in place of or in addition to any amino acid in dCas9. In some embodiments the inserted intein sequence replaces Cys80, Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622, Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, or Ser1274 of dCas9 (SEQ ID NO:5). In some embodiments, the inserted intein sequence replaces Ala127, Thr146, Ser219, Thr519, or Cys574 of dCas9 (SEQ ID NO:5). In some embodiments, the intein is inserted into another domain of the fusion protein (i.e., not in the Cas9 domain, e.g., not in the dCas9 domain), such as the domain having a particular enzymatic activity. In some embodiments, the domain having a particular enzymatic activity is a nuclease domain (e.g., FokI), a recombinase catalytic domain (e.g., Hin, Gin, or Tn3 recombinase domains), a nucleic acid-editing domain (e.g., a deaminase domain), a transcriptional activator domain (e.g., VP64, p65), a transcriptional repressor domain (e.g., KRAB, SID), or an epigenetic modifier (e.g., LSD1 histone demethylase, TET1 hydroxylase). The intein that is inserted into the fusion protein can be any ligand-dependent intein, e.g., those described herein. For example, in some embodiments, the intein that is inserted into a Cas9 protein comprises in part or in whole, a sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 99%, or 100% percent identical to any one of SEQ ID NO:7-14.

In some embodiments, the general architecture of exemplary fusion proteins provided herein comprises the structure:

[NH₂]-[enzymatic domain]-[dCas9]-[COOH] or
[NH₂]-[dCas9]-[enzymatic domain]-[COOH];

wherein NH₂ is the N-terminus of the fusion protein, COOH is the C-terminus of the fusion protein, dCas9 comprises an intein as provided herein, and the enzymatic domain comprises a nuclease domain (e.g., FokI), a recombinase catalytic domain (e.g., Hin, Gin, or Tn3 recombinase domains), a nucleic acid-editing domain (e.g., a deaminase domain), a transcriptional activator domain (e.g., VP64, p65), a transcriptional repressor domain (e.g., KRAB, SID), or an epigenetic modifier (e.g., LSD1 histone demethylase, TET1 hydroxylase). In some embodiments, the intein is comprised in a domain other than dCas9 (e.g., in an enzymatic domain), or is located between two domains.

Additional features may be present, for example, one or more linker sequences between certain domains. Other exemplary features that may be present are localization sequences, such as nuclear localization sequences (NLS; e.g., MAPKKKRVKGIHRGVP (SEQ ID NO:42)); cytoplasmic localization sequences; export sequences, such as nuclear export sequences; or other localization sequences, as well as sequence tags that are useful for solubilization, purification, or detection of the fusion proteins. Suitable localization signal sequences and sequences of protein tags are provided herein and are known in the art, and include, but are not limited to, biotin carboxylase carrier protein (BCCP) tags, myc-tags, calmodulin-tags, FLAG-tags (e.g., 3xFLAG TAG: MDYKDHGDYKDHDIDYKDDDDK (SEQ ID NO:43)), hemagglutinin (HA) tags, polyhistidine tags, also referred to as histidine tags or His-tags, maltose binding protein (MBP)-tags, nus-tags, glutathione-S-transferase (GST) tags, green fluorescent protein (GFP) tags, thioredoxin-tags, S-tags, Softags (e.g., Softag 1, Softag 3), strep-tags, biotin ligase tags, FIAsh tags, V5 tags, and SBP-tags. Additional suitable sequences will be apparent to those of skill in the art.

In some embodiments, the enzymatic domain comprises a nuclease or a catalytic domain thereof. For example, in some

embodiments, the general architecture of exemplary ligand-dependent dCas9 fusion proteins with a nuclease domain comprises the structure:

[NH₂]-[NLS]-[dCas9]-[nuclease]-[COOH],
[NH₂]-[NLS]-[nuclease]-[dCas9]-[COOH],
[NH₂]-[dCas9]-[nuclease]-[COOH], or
[NH₂]-[nuclease]-[dCas9]-[COOH];

wherein NLS is a nuclear localization signal, dCas9 comprises an intein as provided herein, NH₂ is the N-terminus of the fusion protein, and COOH is the C-terminus of the fusion protein. In some embodiments, a linker is inserted between the dCas9 and the nuclease domain. In some embodiments, a linker is inserted between the NLS and the nuclease and/or dCas9 domain. In some embodiments, the NLS is located C-terminal of the nuclease and/or the dCas9 domain. In some embodiments, the NLS is located between the nuclease and the dCas9 domain. Additional features, such as sequence tags, may also be present. In some aspects, the nuclease domain is a nuclease requiring dimerization (e.g., the coming together of two monomers of the nuclease) in order to cleave a target nucleic acid (e.g., DNA). In some embodiments, the nuclease domain is a monomer of the FokI DNA cleavage domain. The FokI DNA cleavage domain is known, and in some aspects corresponds to amino acids 388-583 of FokI (NCBI accession number J04623). In some embodiments, the FokI DNA cleavage domain corresponds to amino acids 300-583, 320-583, 340-583, or 360-583 of FokI. See also Wah et al., "Structure of FokI has implications for DNA cleavage" *Proc. Natl. Acad. Sci. USA*. 1998; 1; 95(18):10564-9; Li et al., "TAL nucleases (TALNs): hybrid proteins composed of TAL effectors and FokI DNA-cleavage domain" *Nucleic Acids Res.* 2011; 39(1):359-72; Kim et al., "Hybrid restriction enzymes: zinc finger fusions to FokI cleavage domain" *Proc. Natl. Acad. Sci. USA*. 1996; 93:1156-1160; the entire contents of each are herein incorporated by reference). In some embodiments, the FokI DNA cleavage domain corresponds to, or comprises in part or whole, the amino acid sequence set forth as SEQ ID NO:6. In some embodiments, the FokI DNA cleavage domain is a variant of FokI (e.g., a variant of SEQ ID NO:6), as described herein. Other exemplary compositions and methods of using dCas9-nuclease fusion proteins can be found in U.S. patent application U.S. Ser. No. 14/320,498; titled "Cas9-FokI fusion Proteins and Uses Thereof," filed Jun. 30, 2014; the entire contents of which are incorporated herein by reference.

In some embodiments, the enzymatic domain comprises a recombinase or other catalytic domain thereof. For example, in some embodiments, the general architecture of exemplary ligand-dependent dCas9 fusion proteins with a recombinase domain comprises the structure:

[NH₂]-[NLS]-[dCas9]-[recombinase]-[COOH],
[NH₂]-[NLS]-[recombinase]-[dCas9]-[COOH],
[NH₂]-[dCas9]-[recombinase]-[COOH], or
[NH₂]-[recombinase]-[dCas9]-[COOH];

wherein NLS is a nuclear localization signal, dCas9 comprises an intein as provided herein, NH₂ is the N-terminus of the fusion protein, and COOH is the C-terminus of the fusion protein. In some embodiments, a linker is inserted between the dCas9 and the recombinase domain. In some embodiments, a linker is inserted between the NLS and the recombinase and/or dCas9 domain. In some embodiments, the NLS is located C-terminal of the recombinase domain and/or the dCas9 domain. In some embodiments, the NLS is located between the recombinase domain and the dCas9 domain. Additional features, such as sequence tags, may also be present. By "catalytic domain of a recombinase," it

65

is meant that a fusion protein includes a domain comprising an amino acid sequence of (e.g., derived from) a recombinase, such that the domain is sufficient to induce recombination when contacted with a target nucleic acid (either alone or with additional factors including other recombinase catalytic domains which may or may not form part of the fusion protein). In some embodiments, a catalytic domain of a recombinase does not include the DNA binding domain of the recombinase. In some embodiments, the catalytic domain of a recombinase includes part or all of a recombinase, e.g., the catalytic domain may include a recombinase domain and a DNA binding domain, or parts thereof, or the catalytic domain may include a recombinase domain and a DNA binding domain that is mutated or truncated to abolish DNA binding activity. Recombinases and catalytic domains of recombinases are known to those of skill in the art, and include, for example, those described herein. In some embodiments, the catalytic domain is derived from any recombinase. In some embodiments, the recombinase catalytic domain is a catalytic domain of a Tn3 resolvase, a Hin recombinase, or a Gin recombinase. In some embodiments, the catalytic domain comprises a Tn3 resolvase (e.g., Stark Tn3 recombinase) that is encoded by a nucleotide sequence comprising, in part or in whole, SEQ ID NO:44, as provided below. In some embodiments, a Tn3 catalytic domain is encoded by a variant of SEQ ID NO:44. In some embodiments, a Tn3 catalytic domain is encoded by a polynucleotide (or a variant thereof) that encodes the polypeptide corresponding to SEQ ID NO:45. In some embodiments, the catalytic domain comprises a Hin recombinase that is encoded by a nucleotide sequence comprising, in part or in whole, SEQ ID NO:46, as provided below. In some embodiments, a Hin catalytic domain is encoded by a variant of SEQ ID NO:46. In some embodiments, a Hin catalytic domain is encoded by a polynucleotide (or a variant thereof) that encodes the polypeptide corresponding to SEQ ID NO:47. In some embodiments, the catalytic domain comprises a Gin recombinase (e.g., Gin beta recombinase) that is encoded by a nucleotide sequence comprising, in part or in whole, SEQ ID NO:48, as provided below. In some embodiments, a Gin catalytic domain is encoded by a variant of SEQ ID NO:48. In some embodiments, a Gin catalytic domain is encoded by a polynucleotide (or a variant thereof) that encodes the polypeptide corresponding to SEQ ID NO:49. Other exemplary compositions and methods of using dCas9-recombinase fusion proteins can be found in U.S. patent application U.S. Ser. No. 14/320,467; titled "Cas9 Variants and Uses Thereof," filed Jun. 30, 2014; the entire contents of which are incorporated herein by reference.

Stark Tn3 recombinase (nucleotide: SEQ ID NO: 44; amino acid: SEQ ID NO: 45):
(SEQ ID NO: 44)
ATGGCCCTGTTTGGCTACGCACGCGTGTCTACCAAGTCAACAGT
CACTCGATTTGCAAGTGAGGGCTCTTAAAGATGCCGAGTGAA
GGCAACAGAAATTTTACTGATAAGGCCAGCGGAAGCAGCACA
GACAGAGAGGGGCTGGATCTCCTGAGAATGAAGGTAAAGGAGG
GTGATGTGATCTTGGTCAAAAAATTGGATCGACTGGGGAGAGA
CACAGCTGATATGCTTCAGCTTATTAAGAGTTTGACGCTCAG
GGTGTGCCGTGAGGTTTATCGATGACGGCATCTCAACCGACT
CCTACATTTGGTCTTATGTTTGTGACAATTTTGTCCGCTGTGGC
TCAGGCTGAGCGGAGAAGGATTCTCGAAAGGACGAATGAGGGA

66

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CGGCAAGCAGCTAAGTTGAAAGGTATCAAATTTGGCAGACGAA
GG
(SEQ ID NO: 45)
MALFGYARVSTSQSLDLQVRALKDAGVKANRIFTDKASGSST
DREGLDLLRMKVKEGDVILVKKLDRGRDADMLQLIKEFDAQ
GVAVRFIDDGISTDSYIGLMFVTILSAVAQAERRRILERTNEG
RQAAKLKGIKFGRRR
Hin Recombinase (nucleotide: SEQ ID NO: 46;
amino acid: SEQ ID NO: 47):
(SEQ ID NO: 46)
ATGGCAACCATTGGCTACATAAGGGTGTCTACCATCGACCAAA
ATATCGACCTGCAGCGCAACGCTCTGACATCCGCCAATGCGA
TCGGATCTTCGAGGATAGGATCAGTGGCAAGATCGCCAACCGG
CCCGGTCTGAAGCGGCTCTGAAGTACGTGAATAAGGCGGATA
CTCTGGTTGTGTGAAGTTGGATCGCTTGGGTAGATCAGTGAA
GAATCTCGTAGCCCTGATAAGCGAGCTGCACGAGAGGGGTGCA
CATTTCCATTCTCTGACCGATTCCATCGATACGTCTAGCGCC
ATGGGCCGATTCTTCTTTTACGTATGTCCGCCCTCGTGAAA
TGGAGCGCGAAGTTATGTTGAACGGACTTTGGCTGGACTGGC
AGCGGCTAGAGCACAGGGCCGACTTGA
(SEQ ID NO: 47)
MATIGYIRVSTIDQNLQRLNLTANCDRIFEDRISGKIANR
PGLKRLKYVNKGDTLVVWKLDRGRSVKNLVALISELHERGA
HFHSLTDSIDTSSAMGRFFFFVMSALAEMERELIVERTLAGLA
AARAQRLG
Gin beta recombinase (nucleotide: SEQ ID
NO: 48; amino acid: SEQ ID NO: 49):
(SEQ ID NO: 48)
ATGCTCATTGGCTATGTAAGGGTCAGCACCAATGACCAAAACA
CAGACTTGCAACGCAATGCTTTGGTTTGCCTCGGATGTGAACA
GATATTTGAAGATAAAGTACGCGGCTCGGACAGACAGACCT
GGGCTTAAGAGAGCACTGAAAAGACTGCAGAAGGGGGACACC
TGGTCGTCTGAAACTGGATCGCTCGGACGAGCATGAAACA
TCTGATTAGCCTGGTTGGTGAGCTTAGGGAGAGAGGAATCAAC
TTCAGAAGCCTGACCGACTCCATCGACACAGTAGCCCATGG
GACGATTCTTCTTCTATGTGATGGGAGCACTTGCTGAGATGGA
AAGAGAGCTTATTATCGAAAGAACTATGGCTGGTATCGCTGCT
GCCCGGAACAAAGGCAGACGGTTCGGCAGACCGCCGAAGAGCG
GC
(SEQ ID NO: 49)
MLIGYVRVSTNDQNTDLQRLNLCAGCEQIFEDKLSGTRTDRP
GLKRLKRLQKGDTLVVWKLDRGRSMKHLISLVGELRERGIN
FRSLTDSIDTSSPMGRFFFFVVMGALAEMERELIERTMAGIAA
ARNKRRFRGRPPKSG

In some embodiments, the enzymatic domain comprises a deaminase or a catalytic domain thereof. For example, in

67

some embodiments, the general architecture of exemplary dCas9 fusion proteins with a deaminase enzyme or domain comprises the structure:

[NH₂]-[NLS]-[Cas9]-[deaminase]-[COOH],
[NH₂]-[NLS][deaminase]-[Cas9]-[COOH],
[NH₂]-[Cas9]-[deaminase]-[COOH], or
[NH₂]-[deaminase]-[Cas9]-[COOH];

wherein NLS is a nuclear localization signal, dCas9 comprises an intein as provided herein, NH₂ is the N-terminus of the fusion protein, and COOH is the C-terminus of the fusion protein. In some embodiments, a linker is inserted between the dCas9 and the deaminase domain. In some embodiments, a linker is inserted between the NLS and the deaminase and/or dCas9 domain. In some embodiments, the NLS is located C-terminal of the deaminase and/or the dCas9 domain. In some embodiments, the NLS is located between the deaminase domain and the dCas9 domain. Additional features, such as sequence tags, may also be present. One exemplary suitable type of nucleic acid-editing enzymes and domains are cytosine deaminases, for example, of the apolipoprotein B mRNA-editing complex (APOBEC)

68

family of cytosine deaminase enzymes, including activation-induced cytidine deaminase (AID) and apolipoprotein B editing complex 3 (APOBEC3) enzyme. Another exemplary suitable type of nucleic acid-editing enzyme and domain thereof suitable for use in the present invention include adenosine deaminases. For example, an ADAT family adenosine deaminase can be fused to a dCas9 domain comprising an intein. Some exemplary suitable nucleic-acid editing enzymes and domains, e.g., deaminases and deaminase domains, that can be fused to dCas9 domains comprising inteins according to aspects of this disclosure are provided below. It will be understood that, in some embodiments, the active domain of the respective sequence can be used, e.g., the domain without a localizing signal (nuclear localizing signal, without nuclear export signal, cytoplasmic localizing signal). Other exemplary compositions and methods of using dCas9-nuclease fusion proteins can be found in U.S. patent application U.S. Ser. No. 14/325,815; titled "Fusions of Cas9 Domains and Nucleic Acid-Editing Domains," filed Jul. 8, 2014; the entire contents of which are incorporated herein by reference.

Human AID:

(SEQ ID NO: 50)

MDSLLMNRKFLYQFKNVRWAKGRRETYLCYVVKRRDSATSFSLDFGYLRNKGCHVELLFLRYISDWDLDPGRC

YRVTWFTSWSPCYDCARHVADFLRGPNLSLRIFTARLYFCEDRKAEPGLRRLHRAGVQIAIMTFKDYFYCWNT

FVENHERTFKAWEGHLENSVRLSRQLRRILLPLYEVDDLRDAFRTLGL

(underline: nuclear localization signal; double underline: nuclear export signal)

Mouse AID:

(SEQ ID NO: 51)

MDSLLMKQKKFLYHFKNVRWAKGRHETLYCYVVKRRDSATSCSLDFGHLRNKSGCHVELLFLRYISDWDLDPGRC

YRVTWFTSWSPCYDCARHVAEFLRWPNLSLRIFTARLYFCEDRKAEPGLRRLHRAGVQIGIMTFKDYFYCWNT

FVENRERTFKAWEGHLENSVRLTRQLRRILLPLYEVDDLRDAFRMLGF

(underline: nuclear localization signal; double underline: nuclear export signal)

Dog AID:

(SEQ ID NO: 52)

MDSLLMKQKFLYHFKNVRWAKGRHETLYCYVVKRRDSATSFSLDFGHLRNKSGCHVELLFLRYISDWDLDPGRC

YRVTWFTSWSPCYDCARHVADFLRGYPNLSLRIFAARLYFCEDRKAEPGLRRLHRAGVQIAIMTFKDYFYCWNT

FVENREKTFKAWEGHLENSVRLSRQLRRILLPLYEVDDLRDAFRTLGL

(underline: nuclear localization signal; double underline: nuclear export signal)

Bovine AID:

(SEQ ID NO: 53)

MDSLLKKQKFLYQFKNVRWAKGRHETLYCYVVKRRDSPTSFSLDFGHLRNKAGCHVELLFLRYISDWDLDPGRC

YRVTWFTSWSPCYDCARHVADFLRGYPNLSLRIFTARLYFCEDRKAEPGLRRLHRAGVQIAIMTFKDYFYCWNT

TFVENHERTFKAWEGHLENSVRLSRQLRRILLPLYEVDDLRDAFRTLGL

(underline: nuclear localization signal; double underline: nuclear export signal)

Mouse APOBEC-3:

(SEQ ID NO: 54)

MGPFCLGCSHRKCYSPIRNLISQETFKFHKNLGYAKGRKDTFLCYEVTRKDCDSPVSLHHGVFKNKDNIHAEIC

FLYWPHDKVLKVLSPREEFKITWYMSWSPCFEAEQIVRFLATHHNLSDIFSSRLYNVQDPETQQNLCLRVQEG

AQVAAMDLYEFKKCKWKFDVNGGRRFRPWKRLLTNFRYQDSKLQEILRCPYIPVPSSSSSTLSNICLTGKLPETR

FCVEGRMDPLSEEFYSQFYNQVVKHLCYHYRMKPYLCYQLEQFNGQAPLKGCLLSEKQKQAEILFLDKIRSM

ELSQTITCYLTSWPCPNCAWQLAAFKRDRPDLILHIYTSRLYFHWKRPQKGLCSLWQSGILVDVMDLPQFTDC

WTNFVNPKRPFWPWKGLEIISRRTQRRRLRRIKESWGLQDLVNDFGNLQLGPPMS

(underline: nucleic acid editing domain)

- continued

Rat APOBEC-3:

(SEQ ID NO: 55)

MGPFCLGCSHRKCYSPIRNLISQETFKFHKNLRYAIDRKDTFLCYEVTRKDCDSPVSLHHGVFKNDNIHAEIC
FLYWFHDKVLKVLSPREEFKITWYMSWSPCFECAEQVLRFLATHHNLSDIFSSRLYNIRDPENQONLCRLVQEG
 AQVAAMDLYEFKKCKWKKFDVNGGRRFRPWKKLLTNFRYQDSKLQEILRPCYIPVSSSSSTLSNICLTGKLPETR
 FCVERRRVHLLSEEEFYQFYNQVRVKHLCYYHGVKPYLCYQLEQFNGQAPLKGCLLSEKKGQHAEILFLDKIRSM
ELSQVIITCYLTWSPCPCNAWQLAAPKRDRLILHIYTSRLYFHWKRPQKGLCSLWQSGILVDVMDLPQFTDC
 WTNFVNPKRPFWPWKGLEIISRRRTQRRRLHRIKESWGLQDLVNDFGNLQLGPPMS
 (underline: nucleic acid editing domain)

Rhesus macaque APOBEC-3G:

(SEQ ID NO: 56)

MVEPMDPRTFVSFNFNRPILSGLNTVWLCCVKTDPGSPPLDAKIFQGVYSKAKYHPEMRFLRWFKWRQLHH
DQEKVVTWYVSWSPCTRCANSVATFLAKDPKVTLTIFVARLYYFWKPDYQQALRILCQKRGGPHATMKIMNYNEF
 QDCWNKFVDGRGKPKPRNNLPKHITLLQATLGELLRLHMDPGTFTSNFNKPNWVSGQHETLYCYKVERLHNDTW
 VPLNQHRGFLRNQAPNIHGFPGKGRHAELCFLDLIPFWKLDGQQYRVTCFTSWSPCFSCAQEMAKFISNNEHVSLC
 IFAARIYDDQGRYQEGRLALHRDGAKIAMMNYSEFEYCWDTFVDRQGRPFQPDGLDEHSQALSGRLRAI
 (bold italic: nucleic acid editing domain; underline: cytoplasmic
 localization signal)

Chimpanzee APOBEC-3G:

(SEQ ID NO: 57)

MKPHFRNPVERMYQDTSFDFYRNPILSHRNTVWLCYEVKTGSPSRPPLDAKIFRGQVYSKLYHPEMRFFHWFS
KWRKLHRDQEYEVTTWYISWSPCTKCTRDVATFLAEDPKVTLTIFVARLYYFWDPDYQALRSLCQKRDGPRATMK
 IMNYDEFQHCWKSFKVYSQRELFEPPWNNLPKYYILLHIMLGELIRHSMDDPTFTSNFNELWVRGRHETLYCYEVE
 RLHNDTWVLLNQRRGFLCNQAPHKHGFLGRHAELCFLDVIPFWKLDLHQDYRVTCFTSWSPCFSCAQEMAKFIS
 NNKHVSLCIFAARIYDDQGRQCQEGRLTLAKAGAKISIMTYSEFKHCWDTFVDHQGCPFPQPDGLDEHSQALSGRL
 RAILQNQGN
 (underline: nucleic acid editing domain; double underline: cytoplasmic
 localization signal)

Green monkey APOBEC-3G:

(SEQ ID NO: 58)

MNPQIRNMVEQMEPDIFVYYFNRPILSGRNTVWLCYEVKTGDPGSPPLDANIFQGLYPEAKDHPMKFLHWFR
KWRQLHRDQEYEVTTWYVSWSPCTKCANSVATFLAEDPKVTLTIFVARLYYFWKPDYQQALRILCQERGGPHATMK
 IMNYNEFQHCWNEFVDGQKPKPRKNLPKHITLLHATLGELLRHVMDPGTFTSNFNKPNWVSGQRETYLYCYKVE
 RSHNDTWVLLNQHRGFLRNQAPDRHGFPKGRHAELCFLDLIPFWKLDGQQYRVTCFTSWSPCFSCAQKMAKFISN
 NKHVSLCIFAARIYDDQGRQCQEGRLTLHRDGAKIAVMNYSEFEYCWDTFVDRQGRPFQPDGLDEHSQALSGRLR
 AI
 (underline: nucleic acid editing domain; double underline: cytoplasmic
 localization signal)

Human APOBEC-3G:

(SEQ ID NO: 59)

MKPHFRNTVERMYRDTFSYNFYNRPILSRNTVWLCYEVKTGSPSRPPLDAKIFRGQVYSELKYHPEMRFFHWFS
KWRKLHRDQEYEVTTWYISWSPCTKCTRDMAFLAEDPKVTLTIFVARLYYFWDPDYQALRSLCQKRDGPRATMK
 IMNYDEFQHCWKSFKVYSQRELFEPPWNNLPKYYILLHIMLGELIRHSMDDPTFTFNFNNEPWRGRHETLYCYEVE
 RMHNDTWVLLNQRRGFLCNQAPHKHGFLGRHAELCFLDVIPFWKLDLQDYRVTCFTSWSPCFSCAQEMAKFIS
 NNKHVSLCIFTARIYDDQGRQCQEGRLTLAAGAKISIMTYSEFKHCWDTFVDHQGCPFPQPDGLDEHSQDLSGRL
 RAILQNQEN
 (underline: nucleic acid editing domain; double underline: cytoplasmic
 localization signal)

- continued

Human APOBEC-3F:

(SEQ ID NO: 60)

MKPHFRNTVERMYRDTFSYNFYNRPILSRRNTVWLCYEVKTKGPSRPRLDAKIFRGQVYSQPEHHAEMCFLSWFC
GNQLPAYKCFQITWVFSWTPCPDCCVAKLAEFLAEHPNVTLTISAARLYYWERYRRALCRLSQAGARVKIMDDE
 EFAYCWFENFVYSEGGQPFMPWYKFDDNYAFLHRTLKEILRNPMMEAMYPHIFYFHFKNLRKAYGRNESWLCFTMEVV
 KHHSPVSWKRGVFRNQVDPETHCHAERCFLSWFCDDILSPNTNYEVTWYTSWSPCECAGEVAEFLARHSNVNLT
 IFTARLYYFWDTDYQEGRLSLSQEGASVEIMGYKDFKYCWENFVYNDDEPFKPKWGLKYNFLFLDSKLQEILE
 (underline: nucleic acid editing domain)

Human APOBEC-3B:

(SEQ ID NO: 61)

MNPQIRNPMERMYRDTFYDNFENEPILYGRSYTWLCYEVKIKRGRSNLLWDTGVFRGQVYFKPQYHAEMCFLSWF
CGNQLPAYKCFQITWVFSWTPCPDCCVAKLAEFLSEHPNVTLTISAARLYYWERYRRALCRLSQAGARVTIMDY
 EEFAYCWFENFVYNEGGQFMPWYKFDDNYAFLHRTLKEILRYLMDPDTFTFNFNDPLVLRRTYLCYEVERLDN
 GTWVLMQHMGLFCNEAKNLLCGFYGRHAELRFLDLVPSLQLDPAQIYRVTFISWSPCFSWGCAGEVRAFLQEN
 THVRLRIFAARIYDYDPLYKEALQMLRDAGAQSIMTYDEFEYCWDTFVYRQGCPCFPQPDGEEHSQALSGLRLA
 ILQNQGN
 (underline: nucleic acid editing domain)

Human APOBEC-3C:

(SEQ ID NO: 62)

MNPQIRNPMKAMYPTFYFQFKNLWEANDRNETWLCFTVEGIKRRSVVSWKTGVFRNQVDSETHCHAERCFLSWF
CDDILSPNTKYQVTWYTSWSPCPDCCAGEVAEFLARHSNVNLTIFTARLYYFYPCYQEGRLSLSQEGVAEIMDY
 EDFKYCWENFVYNDNEPFKPKWGLKTNFRLLKRRRLRESLQ
 (underline: nucleic acid editing domain)

Human APOBEC-3A:

(SEQ ID NO: 63)

MEASPASGRHLMDPHIFTSNFNNGIGRHKTYLCYEVERLDNGTSVKMDQHRGFLHNQAKNLLCGFYGRHAELRF
LDLVPSLQLDPAQIYRVTFISWSPCFSWGCAGEVRAFLQENTHVRLRIFAARIYDYDPLYKEALQMLRDAGAQV
 SIMTYDEFKHCWDTFVDHQGCPFPQPDGDEHSQALSGLRLRAILQNQGN
 (underline: nucleic acid editing domain)

Human APOBEC-3H:

(SEQ ID NO: 64)

MALLTAETFRLQFNNKRRRLRRPYPRKALLCYQLTPQNGSTPTRGYFENKKKCHAEICFINEIKSMGLDETQCYQ
VTCYLTWSPCSSCAWELVDFIKAHDHLNLGIFASRLYYHWCKPQQKGLRLLCGSQVPVEVMGFPKFADCWENFVD
 HEKPLSFNPKMLEELDKNSRAIKRRLERIKIPGVRAQGRYMDILCDAEV
 (underline: nucleic acid editing domain)

Human APOBEC-3D:

(SEQ ID NO: 65)

MNPQIRNPMERMYRDTFYDNFENEPILYGRSYTWLCYEVKIKRGRSNLLWDTGVFRGPVLPKRQSNHRQEVYFRF
ENHAEMCFLSWFCGNRLPANRRFQITWVFSWNPCLPCVVKTKFLAEHPNVTLTISAARLYYRDRDWRWVLLRL
 HKAGARVKIMDYEDFAYCWFENFVYCNEGQPFMPWYKFDDNYASLHRTLKEILRNPMMEAMYPHIFYFHFKNLLKACG
 RNESWLCFTMEVTKHHSVFRKRGVFRNQVDPETHCHAERCFLSWFCDDILSPNTNYEVTWYTSWSPCECAGEV
 AEFLARHSNVNLTIFTARLYYFWDTDYQEGRLSLSQEGASVKIMGYKDFVSCWKNFVYSDDEPFKPKWGLQTNFR
 LLKRRRLREILQ
 (underline: nucleic acid editing domain)

Human APOBEC-1:

(SEQ ID NO: 66)

MTSEKGPSTGDPTRLRRRIEPWEDFVYDPRERLKEACLLYEIKWMSRKIWRSSGKNTTNHVEVNFIIKFTSERD
 FHPSMCSITWFLSWSPCEWCSQAIREFLSRHPGVTLVIYVARLPWHMDQQNRQGLRDLVNSGVTIQIMRASEYY
 HCWRNFVNYPPGDEAHWPQYPLWMLYALELHCIILSLPPCLKISRRWQNHLLTFFRLHLQNCYQTIIPPHILLA
 TGLIHPSVAWR

- continued

Mouse APOBEC-1:

(SEQ ID NO: 67)

MSSETGPVAVDPTLRRIEPHEFEVFFDPRELKRETCCLYEINWGGRRHSVWRHTSQNTSNHVEVNFLEKFTTERTY
 FRPNTRCSITWFLSWSPGCECSRAITEFLSRHPYVTLFIYIARLYHHTDQRNRQGLRDLISSGVTIQIMTEQEYC
 YCWRNFVNYPSPNEAYWPRYPHLWVKLYVLELYCIILGLPPCLKILRRKQPQLTFFTITLQTCHYQRIPPHLLWA
 TGLK

Rat APOBEC-1:

(SEQ ID NO: 68)

MSSETGPVAVDPTLRRIEPHEFEVFFDPRELKRETCCLYEINWGGRRHSIWRHTSQNTNKHVEVNFIEKFTTERTY
 FCPNTRCSITWFLSWSPGCECSRAITEFLSRYPHTVTLFIYIARLYHHADPRNRQGLRDLISSGVTIQIMTEQESG
 YCWRNFVNYPSPNEAHWPYPHLWVRLYVLELYCIILGLPPCLNILRRKQPQLTFFTIALQSCHYQRLPPHILWA
 TGLK

Human ADAT-2:

(SEQ ID NO: 69)

MEAKAAPKPAASGACSVSAEETEKWMEEMHMAKEALENTEVPVGLMVYNNNEVVGKGRNEVNQTKNATRHAEMV
 AIDQVLWDWCRQSGKSPSEVFTEHTVLYVTVEPCIMCAALRLMKIPLVVYGCQNERFGGCGSVLNIASADLPNTGR
 PFQCIPGYRAEEAVEMLKTFYKQENPNAPKSKVRKKECQKS

Mouse ADAT-2:

(SEQ ID NO: 70)

MEEKVESTTTPDGPCVSVSQETEKWMEEMHMAKEALENIEVPVGLMVYNNNEVVGKGRNEVNQTKNATRHAEMV
 AIDQVLWDWCHQHQSPSTVFEHTVLYVTVEPCIMCAALRLMKIPLVVYGCQNERFGGCGSVLNIASADLPNTGR
 PFQCIPGYRAEEAVEMLKTFYKQENPNAPKSKVRKKDCQKS

Mouse ADAT-1:

(SEQ ID NO: 71)

MWTADEIAQLCYAHYNVRLPKQKPEPNREWTLLAAVVKIQASANQACDIPEKEVQVTKEVVSMTGTGTCIGQSK
MRESGDILNDSHAIEIARRSFQRYLLHQLHLAAVLKEDSIFVPGTQRLWRLRPDLISFVFFSSHTPCGDASII PM
LEFEEQPCCPVIRSWANNSPVQETENLEDSDKRNCEDPASPAKKMRLGTPARSLSNCAVHGTQESGPVKPDV
SSSDLTKEEPDANGIASGSRFVVDVYRTGAKCVPGETGDLREPGAAYHQVGLLRVKPGRGDRTCMSKSDKMAR
WNLGCGQALLMHFLEKPIYLSAVVIGKCPYSQEAMRRALTGRCEETLVLPGRFGVQLEIQSGLLFEQSRCAV
HRKRGDSPGRLVPCGAASISWAVPQQPLDVTANGFPQGTTKKEIGSPRARSISKVELFRSFQKLLSSIADDEQP
DSIRVTKKLDITYQEKDAASAYQEAAGALRRIQPFASWIRNPPDYHQFK
 (underline: nucleic acid editing domain)

Human ADAT-1:

(SEQ ID NO: 72)

MWTADEIAQLCYEHYGIKPKKGPENHEWTLLAAVVKIQSPADKACDTPDKPVQVTKEVVSMTGTGTCIGQSK
MRKNGDILNDSHAIEIARRSFQRYLLHQLQLAATLKEDSIFVPGTQKGVWKLRRDLIFVFFSSHTPCGDASII PM
LEFEDQPCCPVFRNWAHNSSEASSNEAPGNERKCEDPDSVPTKKMRLEPGTAAREVTNGAAHHQSFGKQKSGP
ISPGIHSCLDITVEGLATVTRIAPGSAKVIDVYRTGAKCVPGEAGDSGKPGAAPHQVGLLRVKPGRGDRTRSMSCS
DKMARWNLGCGQALLMHLEPIYLSAVVIGKCPYSQEAMQALIGRCQNVSAIPKGFQVQELKILQSDLLFEQ
SRSAVQAKRADSPGRLVPCGAASISWAVPEQPLDVTANGFPQGTTKKTIGSLQARSQISKVELFRSFQKLLSRIA
RDKWPHSLRVQKLDITYQEKAAASSYQEAWSLTKQVFGSWIRNPPDYHQFK
 (underline: nucleic acid editing domain)

In some embodiments, the enzymatic domain comprises one or more of a transcriptional activator. For example, in some embodiments, the general architecture of exemplary dCas9 fusion proteins with a transcriptional activator domain comprises the structure:

$[\text{NH}_2]\text{-}[\text{NLS}]\text{-}[\text{Cas9}]\text{-}[(\text{transcriptional activator})_n]\text{-}[\text{COOH}]$,
 $[\text{NH}_2]\text{-}[\text{NLS}]\text{-}[(\text{transcriptional activator})_n]\text{-}[\text{Cas9}]\text{-}[\text{COOH}]$,

$[\text{NH}_2]\text{-}[\text{Cas9}]\text{-}[(\text{transcriptional activator})_n]\text{-}[\text{COOH}]$, or
 $[\text{NH}_2]\text{-}[(\text{transcriptional activator})_n]\text{-}[\text{Cas9}]\text{-}[\text{COOH}]$;
 wherein NLS is a nuclear localization signal, dCas9 comprises an intein as provided herein, NH_2 is the N-terminus of the fusion protein, and COOH is the C-terminus of the fusion protein. In some embodiments, the fusion proteins comprises one or more repeats of the transcriptional activator, for example wherein $n=1\text{-}10$ (e.g., n is 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10). In some embodiments, $n=1\text{-}20$. In some

75

embodiments, a linker is inserted between the dCas9 and the transcriptional activator domain. In some embodiments, a linker is inserted between the NLS and the transcriptional activator and/or dCas9 domain. In some embodiments, the NLS is located C-terminal of the transcriptional activator and/or the dCas9 domain. In some embodiments, the NLS is located between the transcriptional activator domain and the dCas9 domain. Additional features, such as sequence tags, may also be present. In some embodiments, the transcriptional activator is selected from the group consisting of VP64, (SEQ ID NO:73), VP16 (SEQ ID NO:74), and p65 (SEQ ID NO:75).

VP64
(SEQ ID NO: 73)
GSGRADALDDFDLMLGSDALDDFDLMLGSDALDDFDLDM
LGSDALDDFDLMLIN
VP16
(SEQ ID NO: 74)
APPTDVS LGDELHLDGEDVAMAHADALDDFDLMLGSDGSP
GPGFTPHDSAPYGALDMADFEFEQMFTDALGIDEYGGEFPG
IRR
p65:
(SEQ ID NO: 75)
PSGQISNQALALAPSSAPVLAQTMVPSSAMVPLAQPPAPAP
VLTPGPPQSLAPVPKSTQAGEGTLSEALLHLQFDEDELG
ALLGNSTDPGVFTDLASVDNSEFQQLLNQGVSMHSTAEPM
LMEYPEAITRLVTGSQRPPDPAPTPLGTSGLPNGLSGDEDF
SSIADMDFSALLSQISSSGQ

In some embodiments, the enzymatic domain comprises one or more of a transcriptional repressor. For example, in some embodiments, the general architecture of exemplary dCas9 fusion proteins with a transcriptional repressor domain comprises the structure:

$[NH_2]-[NLS]-[Cas9]-[(transcriptional\ repressor)_n]-[COOH]$,
 $[NH_2]-[NLS]-[(transcriptional\ repressor)_n]-[Cas9]-[COOH]$,
 $[NH_2]-[Cas9]-[(transcriptional\ repressor)_n]-[COOH]$, or
 $[NH_2]-[(transcriptional\ repressor)_n]-[Cas9]-[COOH]$;

wherein NLS is a nuclear localization signal, dCas9 comprises an intein as provided herein, NH_2 is the N-terminus of the fusion protein, and COOH is the C-terminus of the fusion protein. In some embodiments, the fusion proteins comprises one or more repeats of the transcriptional repressor, for example wherein $n=1-10$ (e.g., n is 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10). In some embodiments, $n=1-20$. In some embodiments, a linker is inserted between the dCas9 and the transcriptional repressor domain. In some embodiments, a linker is inserted between the NLS and the transcriptional repressor and/or dCas9 domain. In some embodiments, the NLS is located C-terminal of the transcriptional repressor and/or the dCas9 domain. In some embodiments, the NLS is located between the transcriptional repressor domain and the dCas9 domain. Additional features, such as sequence tags, may also be present. In some embodiments, the transcriptional repressor is selected from the group consisting of the KRAB (Krüppel associated box) domain of Kox1, SID (mSin3 interaction domain), the CS (Chromo Shadow) domain of HP1 α , or the WRPW domain of Hes1. These and other repressor domains are known in the art, and in some

76

embodiments correspond to those described in Urrutia, KRAB-containing zinc-finger repressor proteins. *Genome Biol.* 2003; 4(10):231; Gilbert et al. CRISPR-mediated modular RNA-guided regulation of transcription in eukaryotes. *Cell.* 2013; 154, 442-451; Konermann et al., Optical control of mammalian endogenous transcription and epigenetic states. *Nature.* 2013; 500, 472-476; and published U.S. patent application U.S. Ser. No. 14/105,017, published as U.S. 2014/0186958 A1, the entire contents of which are incorporated herein by reference. In some embodiments, the transcription repressor domain comprises one or more repeats (e.g., 2, 3, 4, 5, 6, 7, 8, 9, or 10 repeats) of a KRAB domain. In some embodiments, the KRAB domain comprises an amino acid sequence selected from the group consisting of SEQ ID NOs:76-79. In some embodiments, the transcriptional repressor domains comprises one or more repeats of a SID protein. In some embodiments, the SID protein comprises an amino acid sequence set forth as SEQ ID NO:80. In some embodiments, the repressor domain comprises 2, 3, 4, 5, 6, 7, 8, 9, or 10 repeats of a SID protein (e.g., SEQ ID NO:80). In some embodiments, the repressor domain comprises four repeats of SID (e.g., SID4x; SEQ ID NO:81).

KRAB (human; GenBank: AAD20972.1)
(SEQ ID NO: 76)
MNMFKAVTFKDVAVAFTEELGLLGAQRKLYRD
VMVENFRNLLSVGHPPFKQDVSPIERNEQLWIMTT
ATRRQGNLDTLPVKALLLYDLAQT
KRAB protein domain, partial
(human; GenBank: CAB52478.1):
(SEQ ID NO: 77)
EQVSFKDVCVDFTEQEEWYLLDPAQKILYRDVILEN
YSNLVSVGYCITKPEVIFKIEQGEEPWILEKGFPS
QCHP
KRAB A domain, partial
(human; GenBank: AAB03530.1):
(SEQ ID NO: 78)
EAVTFKDVAVVFTEELGLLDPQQRKLYRDVMLEN
FRNLLSV
KRAB (mouse; C2H2 type domain containing
protein; GenBank: CAM27971.1):
(SEQ ID NO: 79)
MDLVITYDDVHVNFTQDEWALLDPSQKSLYKGVMLE
TYKNLTAIGYIWEHTIEDHFQTSRSHGSKKTH
SID repressor domain:
(SEQ ID NO: 80)
GSGMNIQMLLEAADYLERREREAEHGYASMLP
SID4x repressor domain:
(SEQ ID NO: 81)
GSGMNIQMLLEAADYLERREREAEHGYASMLPGSG
MNIQMLLEAADYLERREREAEHGYASMLPGSGMNI
QMLLEAADYLERREREAEHGYASMLPGSGMNIQML
LEAADYLERREREAEHGYASMLPSR

In some embodiments, the enzymatic domain comprises an epigenetic modifier or a catalytic domain thereof. For example, in some embodiments, the general architecture of exemplary dCas9 fusion proteins with an epigenetic modifier or domain comprises the structure:

[NH₂]-[NLS]-[Cas9]-[epigenetic modifier]-[COOH],
 [NH₂]-[NLS]-[epigenetic modifier]-[Cas9]-[COOH],
 [NH₂]-[Cas9]-[epigenetic modifier]-[COOH], or
 [NH₂]-[epigenetic modifier]-[Cas9]-[COOH];

wherein NLS is a nuclear localization signal, dCas9 comprises an intein as provided herein, NH₂ is the N-terminus of the fusion protein, and COOH is the C-terminus of the fusion protein. In some embodiments, a linker is inserted between the dCas9 and the epigenetic modifier domain. In some embodiments, a linker is inserted between the NLS and the epigenetic modifier and/or dCas9 domain. In some embodiments, the NLS is located C-terminal of the epigenetic modifier and/or the dCas9 domain. In some embodiments, the NLS is located between the epigenetic modifier domain and the dCas9 domain. Additional features, such as sequence tags, may also be present. Epigenetic modifiers are well known in the art, and typically catalyze DNA methylation (and demethylation) or histone modifications (e.g., histone methylation/demethylation, acetylation/deacetylation, ubiquitylation, phosphorylation, sumoylation, etc.). The presence of one more epigenetic modifications can affect the transcriptional activity of one or more genes, for example turning genes from an "on" state to an "off" state, and vice versa. Epigenetic modifiers include, but are not limited to, histone demethylase, histone methyltransferase, hydroxylase, histone deacetylase, and histone acetyltransferase. Exemplary epigenetic modifying proteins can be found in Konermann et al., Optical control of mammalian endogenous transcription and epigenetic states. *Nature*.

2013; 500, 472-476; Mendenhall et al., Locus-specific editing of histone modifications at endogenous enhancers. *Nat. Biotechnol.* 2013; 31, 1133-1136; and Maeder et al., Targeted DNA demethylation and activation of endogenous genes using programmable TALE-TET1 fusion proteins. *Nat. Biotechnol.* 2013; 31, 1137-1142; the entire contents of each are incorporated herein by reference. In some embodiments, the epigenetic modifier domain is LSD1 (Lysine (K)-specific demethylase 1A) histone demethylase, which in some embodiments, comprises in whole or in part, an amino acid sequence set forth as SEQ ID NO:82 or SEQ ID NO:83. In some embodiments, the epigenetic modifier domain is TET1 hydroxylase catalytic domain, which in some embodiments, comprises an amino acid sequence set forth as SEQ ID NO:84. In some embodiments, the epigenetic modifier is a histone deacetylase (HDAC) effector domain. In some embodiments, the HDAC effector domain comprises in whole in in part, an amino acid sequence corresponding to any of the HDAC effector proteins provided in Supplementary Table 2 of Konermann et al., Optical control of mammalian endogenous transcription and epigenetic states. *Nature*. 2013; 500, 472-476; SEQ ID NOs:85-96. In some embodiments, the epigenetic modifier is a histone methyltransferase (HMT) effector domain. In some embodiments, the HMT effector domain comprises in whole in in part, an amino acid sequence corresponding to any of the HDAC effector proteins provided in Supplementary Table 3 of Konermann et al., Optical control of mammalian endogenous transcription and epigenetic states. *Nature*. 2013; 500, 472-476; SEQ ID NOs:97-106.

LSD1, isoform a (human):

(SEQ ID NO: 82)

MLSGKAAAAAAAAAATGTEAGPGTAGGSENGSEVAAQAGLSGPAEVGPGAVGERTPRKKEPPRA
 SPPGGLAEPPGSAGPQAGPTVPGSATPMETGIAETPEGRRTSRRKRAKVEYREMDSELANLSEDEYY
 SEEERNAKAEKEKLLPPPPQAPPEEENESEPEEPSGQAGGLQDDSSGGYGDGQASGVEGAAPQSRLP
 HDRMTSQEAACFPDIISGPQQTQKVFLLIRNRTLQWLWDNPKIQLTFEATLQQLEAPYNSDTVLVHRV
 HSYLERHGLINFGIYKRIKPLPTKKTGKVIIIGSGVSGLAARQLQSFQMDVTLLEARDVGGRVATF
 RKGNYVADLGAMVVTGLGGMNAVVSQVNMELAKIKQKCPLYEANGQADTVKVPKEKDEMVEQEFNR
 LLEATSYLSHQDLDFNVLNKPVSLGQALEVVVIQLQEKHVKDEQIEHWKKIVKTQEBLKELLNKMVNLK
 EKIKELHQYKEASEVKPPRDIITAEFLVKS KHRDLTALCKEYDELAETQGLKEKLQLEANPPSDVY
 LSSDRQILDWHFANLEFANATPLSTLSLKHWDQDDDFEFTGSHLTVRNGYSCVPVALAEGLDIKLNT
 AVRQVRYTASGCEIVANTRSTSQTFIYKCDAVLCTPLGLVKQPPAVQFVPPLEPWKTSAVQRMGF
 GNLNKVVLCPDRVFDWPSVNLFGHVGSTTASRGELFLFWNLYKAPILLALVAGEAAGIMENISDDVIV
 GRCLAILKGIFGSSAVPQPKETVVSRRWADPWARGSYSYVAAGSSGNDYDLMAQPIPTGPSIPGAPQP
 IPRLFFAGEHTIRNYPATVHGALLSGLREAGRIADQFLGAMYTLPRQATPGVPAQQSPSPM

LSD1, isoform b (human):

(SEQ ID NO: 83)

MLSGKAAAAAAAAAATGTEAGPGTAGGSENGSEVAAQAGLSGPAEVGPGAVGERTPRKKEPPRA
 SPPGGLAEPPGSAGPQAGPTVPGSATPMETGIAETPEGRRTSRRKRAKVEYREMDSELANLSEDEYY
 SEEERNAKAEKEKLLPPPPQAPPEEENESEPEEPSGVEGAAPQSRPLPHDRMTSQEAACFPDIISGPQ
 QTQKVFLLIRNRTLQWLWDNPKIQLTFEATLQQLEAPYNSDTVLVHRVHVSYLERHGLINFGIYKRIK
 LPTKKTGKVIIIGSGVSGLAARQLQSFQMDVTLLEARDVGGRVATFRKGNYVADLGAMVVTGLGGM
 PMAVVSQVNMELAKIKQKCPLYEANGQAVPKEKDEMVEQEFNRLLEATSYLSHQDLDFNVLNKPVSL
 GQALEVVVIQLQEKHVKDEQIEHWKKIVKTQEBLKELLNKMVNLKEKIKELHQYKEASEVKPPRDIITAE

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EFLVKS KHRDLTALCKEYDELAETQGKLEELQELEANPPSDVYLSSRRDRQILDWHFANLEFANATPL
 STL SLKHWDDDDFEFTGSHLTVRNGYSCVPVALAEGLDIKLNTAVRQVRYTASGCEVIAVNTRSTSQ
 TFIYKDAVLCTLPGLVLKQQPPAVQFVPPLEWKTSAVQRMGFGNLNKVVLCFDRVFWDP SVNLF GH
 VGSTTASRGELFLFWNLYKAPILLALVAGEAAGIMENI SDDVIVGRCLA ILKGIFGSSAVPQPKETVV
 SRWRADPWARGSYSYVAAGSSGNDYDLMAQPI TPGPSIPGAPQPI PRLFFAGEHTIRNYPATVHGALL
 SGLREAGRIADQFLGAMTYLPRQATPGVPAQQSPSM

TET1 catalytic domain:

(SEQ ID NO: 84)

SIVAQLSRPDPALAAALTNHDLVALACLGGRPALDAVKKGLPHAPALIKRTNRRIPERTSHRVADHAQV
 VRVLGFFQCHSHPAQAFDDAMTQFGMSGGSLPTCSCLDRVIQKDKGPYYTHLGAGPSVAAREIMEN
 RYQKGNAIRIEIVVYTGKEGKSSHGCPKIAKWVLRSSDEEKVLCVLRQRTGHHCTAVMVVLMVWD
 GIPLMADRLYTEL TENLSYNGHPTDRCTLNENRTCTCQGIDPETCGASFSGCSWSMYFNGCKFG
 RSPSPRRFRIDPSSPLHEKNLEDNLQSLATRLAPIYKQYAPVAYQNQVEYENVARECRLGSKGRPFS
 GVTACLD FCAHPHRDIHNNNGSTVVC TLTRDNRS LGVIPQDEQLHVLPLYKLSDTDEFGSKGMEA
 KIKSGAIEVLAPRRKRKTCFTQPVPRSGKKRAAMTEVLAHKIRAVEKKPIPRIKRKNNSTTTNSKP
 SSLPTLGSNTETVQPEVKS ETEPHFILKSSDNTKTYSLMPSAPHPVKEASPGFSWSPKTASATPAPLK
 NDATASCGFSERSSTPHCTMPSGRLSGANAAAADGPGISQLGEVAPLPTLSAPVMEPLINSEPTGVT
 EPLTPHQPNHQPSFLTSPQDLASSPMEEDEQHSEADEPPSDEPLSDDPLSPAEEKLPHIDEYWSDEH
 IFLDANIGGVAIPAHGSVLIECARRELHATTPVEHPNRNHPTRL SLVFYQHKNLKPQHGFELNKIK
 FEAKEAKNKKMKASEQKDQAANEGPEQSSEVNELNQIPSHKALTLTHDNVVTVPYALTHVAGPYNHW
 V

HDAC effector domains:

HDAC8 (*X. laevis*):

(SEQ ID NO: 85)

ASSPKKKRKVEASMSRVVKPKVASMEEMAAFH TDAYLQHLHKVSEEGDND DPETLEYGLGYDCPITEG
 IYDYAAAVGGATLTAAEQ LIEGKTRIAVNWPGGWHHAKKDEASGFCYLNDAVLGILKLRKFDRVLYV
 DMDLHHGDGVEDAFSFTSKVMTVSLHKFSPGFFPGTGDVSDIGLGKGRYYSINVPLQDGIQDDKYYQI
 CEGVLKEVFTTFNPEAVVLQLGADTIAGDPMCSFNMTPEGIGKCLKYVLQWQLPTLILGGGGYHLPNT
 ARCWTYLTALIVGRTLSEIPDHEFPTEYGPDYVLEITPSCRPRDNDTQKVQEILQSIKGNLKRVEF
 RPD3 (*S. cerevisiae*):

(SEQ ID NO: 86)

ASSPKKKRKVEASRRVAYFYDADVGNVYAGAGHPMKPHRIRMAHSLIMNYGLYKKMEIYRAKPAKQE
 MCQFHTDEYIDFLSRVTPDNLEMPKRESVKFNVGDDCPVFDGLYEYCSISGGGSMEGAARLNRGKCDV
 AVNYAGGLHHAKKSEASGFCYLN DIVLGIIELLYHPRVLYIDIDVHHGDGVEEAFYTTDRVMTCSFH
 KYGEFFPGTGELRDIGVGAGKNYAVNVPLRDGIDDATYRSVFEPV IKKIMEWYQPSAVVLQCGGDSLS
 GDRLCGFNL SMEGHANCVN YVKSFGIPMMVGGGGYTMRNVARTWCFETGLLNNVLDKDLPEYF

MesoLo4 (*M. loti*):

(SEQ ID NO: 87)

ASSPKKKRKVEASMLQIVHHPDYDAGFATNHRFPMSKYPLLMEALRAGLASPDALNTTEPAPASWL
 KLAHAADYVDQVISCVPKEIEREIGFPVGPVSLRAQLATGGTILAAARLALRHGIACNTAGGSHHAR
 RAQGAGFCTFNDVAVASLVLLDEGAAQNILVVDLDVHQGDGTADILSDEPGVFTFSMHGERNYPVRKI
 ASDLDIALPDGTGDAAYLRRLATILPELSARARWDIVFYNAGVDVHAEDRLGRLALSNGGLRARDEM
 IGHFRALGIPVCGVIGGGYSTDVPALASRHAILFEVASTYAEF

-continued

HDAC11 (human):

(SEQ ID NO: 88)

ASSPKKKRKVEASMLHTTQLYQHVPETRWPIVYSPRYNITFMGLEKLHPFDAGKWGKVINFLKEEKL
 SDSMLVEAREASEEDLLVVHTRRYLNELKWSFAVATITEIPPVIFLPNFLVQRKVLRLRTQTGGTIM
 AGKLAVERGWAINVGGGFHHCSSDRGGGFCAYADITLAIKFLFERVEGISRATIIDLDAHQGNNGHERD
 FMDDKRVIIMDVYNRHIYPGDRFAKQAIRKVELEWGTEDDEYLDKVERNIKKSLQEHLPDVVVYNAG
 TDILEGDRLGGLSISPAGIVKRDELVFRMVRGRRVPILMVTSGGYQKRTARIADSILNLFGLGLIGP
 ESPSVSAQNSDTPLLPPAVPEF

HDT1 (*A. thaliana*):

(SEQ ID NO: 89)

ASSPKKKRKVEASMEFWGIEVKSQKPVTVTPEEGILIHVSQASLGECKNKKGEFVPLHVKVGNQNLVL
 GTLSTENIPQLFCDLVFDKEFELSHTWGKGSVYFVGKTPNIEPQGYSEEEEEEEVPAAGNAKAVA
 KPKAKPAEVKPAVDDEDESDSDGMDDESDGEDSEEEPTPKKPASSKKRANETTPKAPVSAKKAKV
 AVTPQKTDEKKKGKKAQSEF

SIRT3 (human):

(SEQ ID NO: 90)

ASSPKKKRKVEASVMGAGISTPSGIPDFRSPGSGLYSNLQQYDLPEAIFELPFFHNPFPFFTLAK
 ELYPGNYKPNVTHYFLRLHDKGLLLRLYTQNIIDGLERVSGIPASKLVEAHGTASATCTVCQRPFP
 EDIRADVMADRVPRCPVCTGVVCPDIVFFGEPLPQRFLHVVDFPMADLLILGTSLEVEPFASLSEA
 VRSSVPRLLINRDLVGLAWHPRSRDVAQLGDVVHGVESLVELLGVTEMRDLVQRETGKLDGPDKEF

HST2 (*S. cerevisiae*):

(SEQ ID NO: 91)

ASSPKKKRKVEASTEMSVRKIAAHMKSNNPNAKVIIMVGAGISTSCGIPDFRSPGTGLYHNLARLKL
 PEAVFDVDFQSDPLPFYTLAKELYPGNFRPSKFHYLLKLFQDKDVLKRVYTQNIIDTLERQAGVKDDL
 IIEAHGSPAHCICGCKVYPQVFKSLAEHPIKDFVKCDVCGELVKPAIVFFGEDLPDSFSETWLN
 DSEWLREKITTSGKHPQQLVIVVGTSLAVYPPASLPEEIPRKVKRVLNLETVGDFKANKRPTDLIV
 HQYSDEFABQLVEELGWQEDFEKILTAQGGMGFEF

CobB (*E. coli* (K12)):

(SEQ ID NO: 92)

ASSPKKKRKVEASMEKPRVLVLTGAGISAESGIRTFRAADGLWEEHRVEDVATPEGFDRDPQLVQAFY
 NARRQLQQPEIQPNAAHLALAKLQDALGDRFLVLTQNIIDNLHERAGNTNVIHMHGELLKVRCSQSGQ
 VLDWTGDTVPTPKCHCCQFPAPLRPHVWVFGEMPLGMDIYMALSMADIFIAIGTSGHVYPAAGFVHE
 AKLHGAHTVELNLEPSQVGNFAEKYYGPASQVVPFVEKLLKGLKAGSIAEF

HST2 (*C. albicans*):

(SEQ ID NO: 93)

ASSPKKKRKVEASMPSLDDILKPVAEAVKNGKKVTFNGAGISTGAGIPDFRSPDTGLYANLAKNL
 FAEAVFDIDFFKEDPKPFYTLAEELYPGNFAPTKFHHFIKLLQDQGSLLKRVYTQNIIDTLERLAGVEDK
 YIVEAHGFSFASNHCDCHKEMTTETLKTMYMDKKIPSCQHCEGYVKPDIIVFFGEGLPVKFFDLWEDDC
 EDVEVAIVAGTSLTVFPFASLPGEVNKKCLRVLVNKEKVGTFKHEPRKSDIIALHDCDIVAERLCTLL
 GLDDKLNEVYEKEKIKYSAETKEIKMHEIEDKLKEAHLKEDKHTTKVDKKEKQNDANDKELEQLID
 KAKAEF

SIRT5 (human):

(SEQ ID NO: 94)

ASSPKKKRKVEASSSMADFRKFFAKAKHIVIISGAGVSAESGVPTFRGAGGYWRKWQAQDLATPLAF
 AHNPSRVWEFYHYRREVMGSKEPNAGHRAIAECETRLGKQGRVVVITQNIIDELHRKAGTKNLLLEIHG

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SLFKTRCTSCGVVAENYKSPICPALSGKGAPEPGTQDASIPVEKLPRCEEAGCGGLLRPHVVWFGENL
DPAILEEVDRLEAHCDLCLVVGTSVVVYPAAMFAPQVAARGVPVAEFNTETTPATNRRFRHFQGPCGT
TLPEALACHENETVSEF

Sir2A (*P. falciparum*):

(SEQ ID NO: 95)

ASSPKKKRKVEASMGNLMISFLKQDTSITLEELAKIIKKCKHVVALTGSGTSAESNIPSRGSSNSI
WSKYDPRIYGTIWFGWKYPEKIWEVIRDISSDYEIEINNGHVALSTLESGLYKLSVVTQNVDDLHEAS
GNTKVISLHGNVFEAVCCTCNKIVKLNKIMLQKTSHEFMHQLPPECPCGGIFKPNIIILFGEVVSDDLK
EABEEIAKCDLLLIGTSSSTVSTATNLCHFACKKKKIVEINISKTYITNKMSDYHVCASFSELTQVA
NILKGSSEKNKKIMEF

SIRT6 (human):

(SEQ ID NO: 96)

ASSPKKKRKVEASMSVNYAAGLSPYADKGKCGLPEIFDPPEELERKVVWELARLVQSSSVVFHTGAGI
STASGIPDFRPHGVWMEERGLAPKFDITTFESARPTQTHMALVQLERVGLLRFLVSQNVDDLHVRSG
FPRDKLAELHGNMFVEEACCKTQYVRDVTVMGLKATGRCTVAKARGLRACRGELRDTILDWEDS
LPDRDLALADEASRNADLSITLGTSLQIRPSGNLPLATKRRGGRLVIVNLQPTKHDRHADLRHGYVD
EVMTRLMKHLGLEIPAWDGPVRLERLPPLEF

HMT effector domains:

NUE (*C. trachomatis*):

(SEQ ID NO: 97)

ASSPKKKRKVEASMTNSTQDTLYLSLHGGIDSAIPYVRRVEQLLQFSFLPELQFQNAAVKQRIQRL
CYREEKRLAVSSLAKWLGLHQLRQRLAPKNPPVAICWINSYVGYGVFARESIPAWSYIGEYTGILRRR
QALWLDENDYCFRYPVPYRYSFRYFTIDSGMQGNVTRFINHSDNPNEAIGAFENGIFHIIIRAIKDIL
PGEELCYHYGPLYWKHRKKREEFVPQEEF

vSET (*P. bursaria chlorella virus*):

(SEQ ID NO: 98)

ASSPKKKRKVEASMFNDRVIVKKSPLGGYGVFARKSFEKGELVEECLCIVRHNDWGTAEEDYLFGRK
NMSAMALGFGAIFNHSKDPNARHELTAGLKRMRIFTIKPIAIGEEITISYGGDYWLSRPLTQNEF

SUV39H1 (human):

(SEQ ID NO: 99)

ASSPKKKRKVEASNLKCVRIKQFHKDLERELLRRHRSKTPRHLDPSLANYLVQKAKQRRALRRWEQ
ELNAKRSHLGRITVENEVDLDGPPRAFVYINEYRVGEGITLNQVAVGCECQDCLWAPTGGCCPGASLH
KFAYNDQGQVRLRAGLPIYECNSRCRCGYDCPNRVVQKGIRYDLCIFRTDDGRGWGVRTLEKIRKNSF
VMEYVGEIITSEEAERRGQIYDRQATYLFDLDYVEDVYTVDAAYYGNISHFVNHSCDNLQVYNVFI
DNLDERLPRIAFFATRTIRAGEELTFDYNMQVDPVDMESTRMDSNFGLAGLPGSPKKRVRIECKCGTE
SCRKYLFEF

DIM5 (*N. crassa*):

(SEQ ID NO: 100)

ASSPKKKRKVEASMEKAFRPHFFNHGKPDANPKEKKNCHWCQIRSFATHAQLPISIVNREDDAFLNPN
FRFIDHSIIGKNVPVADQSFVGCSCASDEECMYSTCQCLDEMAPDSDEEADPYTRKKRFAYYSQGAK
KGLLRDRVLQSQEPIYIECHQCACSKDCPNRVVERGRVPLQIFRTKDRGWGKCPVNIKRGQFVDRY
LGEIITSEADRRAESTIARKDVYLFALDKFSDPDSLPLLAGQPLEVDGEYMSGPTRFINHSCDP
NMAIFARVGDHADKHIHDLALFAIKDIPKGTETLFDYVNGLTGLESDAHDPKISEMTKCLCGTAKCR
GYLWEF

KYP (*A. thaliana*):

(SEQ ID NO: 101)

ASSPKKKRKVEASDISGGLEFGIPATNRVDDSPVSPSTSGFTYIKSLIIEPNVVIIPKSSTGCNCRGSC
TDSKKCACAKLNGGNFPYVDLNDGRLIESRDVVFECGPHCGCGPKCVNRTSQKRLRFNLEVFRSAKKG

-continued

WAVRSWEYIPAGSPVCEYIGVVRTADVDTISDNEYIFEIDCQQTMQGLGGRQRLRDVAVPMNNGVS
QSSDENAPEFCIDAGSTGNFARFINHSCEPNLFVQCVLSSHQDIRLARVVLFAADNISPMQELTYDY
GYALDSVHEF

SUVR4 (*A. thaliana*):

(SEQ ID NO: 102)

ASSPKKKRKVEASQSAYLHVSILARISDEDCANCKGNCLSDFPCTCARETSGEYAYTKEGLLKEKFL
DTCLMKKKEPDSFPKYCKDCPLERDHDGTYGKCDGHLIRKFIKECWKRCGCDMQCGNRVVQRGIRC
QLQVYFTQEGKGWGLRTLQDLPGKTFICEYIGEILTNTELYDRNVRSSSERHTYPTLTDADWGSEKDL
KDEEALCLDATICGNVARFINHRCEDANMIDIPIEIEIETPDRHYHIAFFTLLRDVKAMDELTDYDMIDF
NDKSHPVKAFRCCGSESCDRKIKGSQKSIERRKIVSAKKQGSKEVSKKRKEF

Set4 (*C. elegans*):

(SEQ ID NO: 103)

ASSPKKKRKVEASMQLEHQIANISVTFNDIPRSDHSMPTTELCYFDDFATTLVVDVSLNFTTHKMSKK
RRYLYQDEYRTARTVMKTFREQRDWTNAIYGLLTLRSVSHFLSKLPPNKLFEFRDHIVRFLNMFILDS
GYTIQECKRYSQEGHQGAKLVSTGVSRGDKIERLSGVVCLLSSSEDEDSILAQEGSDFSVMYSTRKRC
STLWLPGGAYINHDCRPTCEFVSHGSTAHIRVLRDMVPGDEITCFYGEFFGPNNDCECCTCEKNMN
GAFSYLRGNENAEPII SEKKTKYELRSRSEF

Set1 (*C. elegans*):

(SEQ ID NO: 104)

ASSPKKKRKVEASMKVAAKLATSRMRKDRAAASPSSDIENSENPSSLASHSSSSGRMTPSKNTRSR
KGVSVKDVSNHKITEFFQVRRSNRKTQKISDEAKHALRDTVLKGTNERLLEVYKDVVKGRGIRTKVN
FEKGFVVEYRGVMMMEYSEAKVIEEQYSNDEEIGSYMYFFEHNKKWCIDATKESPWKGRLINHSVLR
PNLKTQVVEIDGSHHLILVARRQIAQGEELLYDYGDRSAETIAKNPWLVTNEF

SETD8 (human)

(SEQ ID NO: 105)

ASSPKKKRKVEASSCDSTNAIAKQALKPKIKGQAPRKAQGGTQQNRKLTDFYPVRRSSRKSKAEL
QSEERKRIDELIESGKEGMDLIDGKGRGVIATKQFSRGDFVVEYHGDLEITDAKKREALYAQDP
STGCYMYFQYLSKTYCVDATRETNRLGRLINHSKCGNCQTKLHDIDGVPHLILIASRDIAGEELLY
DYGDRSKASIEAFPWLKHEF

TgSET8 (*T. gondii*):

(SEQ ID NO: 106)

ASSPKKKRKVEASASRRTGEFLRDAQAPSRWLKRSTGQDDGAFCLETWLAGAGDDAAGGERGRDREG
AADKAKQREERQKELEERFEEMKVEFEKAQORMIARRAALTGEIYSDGKGSKKPRVPSLPENDDAL
IEIIIDPEQGILKWLPSVMSIRQRTVIYQECLRRDLTACIHLTKVPGKGRAVFAADTILKDDFVVEYK
GELCSEREAREREREQRNRSKVPMSGFMYFKNGSRMMAIDATDEKQDFGPARLINHSRRNPNTPRAI
TLGDFNSEPRLIFVARRNIEKGEELLVDYGERDPDVIKEHPWLNSEF

In some embodiments, ligand-dependent Cas9-intein variants are provided herein that exhibit decreased off-target activity. For example, in some embodiment, Cas9-intein variants are provided herein that comprise a Cas9 nuclease domain, or a nuclease-deficient Cas9 domain and a heterologous nucleic acid-editing domain, such as, for example, a heterologous nuclease domain, a recombinase domain, or a deaminase domain. In some such embodiments, the ligand-dependent Cas9-inteins provided herein exhibit decreased, minimal, or no off-target activity in the presence of a ligand at a concentration effective to effect excision of the intein from the Cas9-intein variant, or at a concentration effective to induce a desired modification (e.g., cleavage, nicking, recombination, or deamination) of a target site. In some embodiments, the ligand-dependent Cas9-intein variants

provided herein exhibit an off-target activity in their active state (e.g., in the presence of or after being contacted with a suitable ligand) that is decreased as compared to the off-target activity of wild-type Cas9. For example, in some embodiments, the off-target activity of a Cas9-intein variant provided herein is decreased to less than 80%, less than 75%, less than 50%, less than 45%, less than 40%, less than 35%, less than 30%, less than 25%, less than 24%, less than 23%, less than 22%, less than 21%, less than 20%, less than 19%, less than 18%, less than 17%, less than 16%, less than 15%, less than 14%, less than 13%, less than 12%, less than 11%, less than 10%, less than 9%, less than 8%, less than 7%, less than 6%, less than 5%, less than 4%, less than 3%, less than 2%, or less than 1% of wild-type Cas9 under the same conditions.

Pharmaceutical Compositions

In some embodiments, any of the ligand-dependent site-specific enzymes described herein are provided as part of a pharmaceutical composition. For example, some embodiments provide pharmaceutical compositions comprising a Cas9 protein comprising an intein, or fusion proteins comprising a dCas9 protein with an intein fused to a nuclease, recombinase, deaminase, or a transcriptional activator as provided herein, or a nucleic acid encoding such a protein, and a pharmaceutically acceptable excipient. Pharmaceutical compositions may further comprise one or more gRNA(s).

In some embodiments, compositions provided herein are administered to a subject, for example, to a human subject, in order to effect a targeted genomic modification within the subject. In some embodiments, cells are obtained from the subject and are contacted with an inventive ligand-dependent site-specific enzyme *ex vivo*. In some embodiments, cells removed from a subject and contacted *ex vivo* with an inventive ligand-dependent site-specific enzyme are re-introduced into the subject, optionally after the desired genomic modification has been effected and/or detected in the cells. Although the descriptions of pharmaceutical compositions provided herein are principally directed to pharmaceutical compositions which are suitable for administration to humans, it will be understood by the skilled artisan that such compositions are generally suitable for administration to animals of all sorts. Modification of pharmaceutical compositions suitable for administration to humans in order to render the compositions suitable for administration to various animals is well understood, and the ordinarily skilled veterinary pharmacologist can design and/or perform such modification with merely ordinary, if any, experimentation. Subjects to which administration of the pharmaceutical compositions is contemplated include, but are not limited to, humans and/or other primates; mammals, domesticated animals, pets, and commercially relevant mammals such as cattle, pigs, horses, sheep, cats, dogs, mice, and/or rats; and/or birds, including commercially relevant birds such as chickens, ducks, geese, and/or turkeys.

Formulations of the pharmaceutical compositions described herein may be prepared by any method known or hereafter developed in the art of pharmacology. In general, such preparatory methods include the step of bringing the active ingredient into association with an excipient, and then, if necessary and/or desirable, shaping and/or packaging the product into a desired single- or multi-dose unit.

Pharmaceutical formulations may additionally comprise a pharmaceutically acceptable excipient, which, as used herein, includes any and all solvents, dispersion media, diluents, or other liquid vehicles, dispersion or suspension aids, surface active agents, isotonic agents, thickening or emulsifying agents, preservatives, solid binders, lubricants and the like, as suited to the particular dosage form desired. *Remington's The Science and Practice of Pharmacy*, 21st Edition, A. R. Gennaro (Lippincott, Williams & Wilkins, Baltimore, MD, 2006; incorporated in its entirety herein by reference) discloses various excipients used in formulating pharmaceutical compositions and known techniques for the preparation thereof. See also PCT application PCT/US2010/055131, incorporated in its entirety herein by reference, for additional suitable methods, reagents, excipients and solvents for producing pharmaceutical compositions comprising a nuclease. Except insofar as any conventional excipient medium is incompatible with a substance or its derivatives, such as by producing any undesirable biological effect or otherwise interacting in a deleterious manner with any other

component(s) of the pharmaceutical composition, its use is contemplated to be within the scope of this disclosure.

In some embodiments, compositions in accordance with the present invention may be used for treatment of any of a variety of diseases, disorders, and/or conditions, including, but not limited to, autoimmune disorders (e.g. diabetes, lupus, multiple sclerosis, psoriasis, rheumatoid arthritis); inflammatory disorders (e.g. arthritis, pelvic inflammatory disease); infectious diseases (e.g. viral infections (e.g., HIV, HCV, RSV), bacterial infections, fungal infections, sepsis); neurological disorders (e.g. Alzheimer's disease, Huntington's disease; autism; Duchenne muscular dystrophy); cardiovascular disorders (e.g. atherosclerosis, hypercholesterolemia, thrombosis, clotting disorders, angiogenic disorders such as macular degeneration); proliferative disorders (e.g. cancer, benign neoplasms); respiratory disorders (e.g. chronic obstructive pulmonary disease); gastrointestinal disorders (e.g. inflammatory bowel disease, ulcers); musculoskeletal disorders (e.g. fibromyalgia, arthritis); endocrine, metabolic, and nutritional disorders (e.g. diabetes, osteoporosis); genitourinary disorders (e.g. renal disease); psychological disorders (e.g. depression, schizophrenia); skin disorders (e.g. wounds, eczema); and blood and lymphatic disorders (e.g. anemia, hemophilia); etc.

Methods

In another aspect of this disclosure, methods for site-specific nucleic acid (e.g., DNA) modification are provided. In some embodiments, the methods comprise contacting a DNA with any of the ligand-dependent Cas9 proteins (complexed with a gRNA) described herein, either before or after contacting the protein with a ligand that induces self-excision of the ligand-dependent intein thereby activating the nuclease. For example, in some embodiments, the method comprises (a) contacting a RNA-guided nuclease (e.g., a Cas9 protein including Cas9 nickase) comprising a ligand-dependent intein with a ligand, wherein binding of the ligand to the intein induces self-excision of the intein; and (b); contacting a DNA with the RNA-guided nuclease, wherein the RNA-guided nuclease is associated with a gRNA; whereby self-excision of the intein from the RNA-guided nuclease in step (a) allows the RNA-guided nuclease to cleave the DNA, thereby producing cleaved DNA. In some embodiments, for examples those involving the use of an intein containing Cas9 nickase, the method produces a single strand break in the DNA. In some embodiments, the method produces a double strand break in the DNA. In some embodiments, the RNA-guided nuclease is able to bind a gRNA and optionally bind a target nucleic acid prior to being contacted with a ligand that induces self-excision of the intein, but the RNA-guided nuclease is unable to cleave the target nucleic acid until self-excision of the intein occurs. In some embodiments, the RNA-guided nuclease is unable to bind a gRNA and therefore is unable to bind a target nucleic acid until the RNA-guided nuclease is contacted with a ligand that induces self-excision of the intein. In some embodiments, the RNA-guided nuclease is any nuclease comprising Cas9 (or a variant or a fragment thereof) which comprises a ligand-dependent intein as provided herein.

In some embodiments, the method involves the use of fusion proteins comprising a nuclease-inactivated Cas9 (e.g., dCas9) fused to a nuclease domain (e.g., FokI), wherein the fusion protein comprises a ligand-dependent intein (e.g., in the dCas9 domain as provided herein), and the fusion protein lacks one or more activities (as described herein) prior to excision of the intein. In some embodiments, the fusion protein is any fusion protein described herein. In some embodiments, the method comprises contacting a

target nucleic acid (e.g., DNA) with two such fusion proteins, each comprising a distinct gRNA that targets the nucleic acid, and the method comprises contacting the target nucleic acid with two such fusion proteins. The method increases the specificity of cleavage, and therefore decreases off target effects, as two fusions are required to bind the target site to elicit any nuclease activity as the nuclease domains fused to the dCas9 domain typically must dimerize at the target site to induce cleavage. In some embodiments, the method comprises contacting the fusion proteins with a ligand that induces self-excision of the intein, either before or after the gRNAs bind the fusion proteins, and/or before or after the fusion proteins bind the target nucleic acid. Once the fusion proteins are activated following excision of the intein, the nuclease domains (e.g., the FokI domains) dimerize and cleave and the target nucleic acid. Compositions and methods of using dCas9-FokI fusions are known to those of skill in the art (see, e.g., U.S. patent application Ser. No. 14/320,498, titled "CAS9 VARIANTS AND USES THEREOF" which was filed on Jun. 30, 2014; the entire contents of which are incorporated herein by reference). Those of skill in the art are routinely able to design appropriate gRNAs that target two of the fusion proteins to a target nucleic acid, and understand that in some aspects the gRNAs are designed to hybridize to regions of the target nucleic acid that are no more than 10, no more than 15, no more than 20, no more than 25, no more than 30, no more than 40, no more than 50, no more than 60, no more than 70, no more than 80, no more than 90, or no more than 100 base pairs apart.

In some embodiments, any of the methods provided herein can be performed on DNA in a cell. For example, in some embodiments the DNA contacted by any of the inventive ligand-dependent site-specific Cas9 enzymes provided herein is in a eukaryotic cell. In some embodiments, the eukaryotic cell is in an individual. In some embodiments, the individual is a human. In some embodiments, any of the methods provided herein are performed *in vitro*. In some embodiments, any of the methods provided herein are performed *in vivo*.

In some embodiments of this disclosure, methods for site-specific nucleic acid (e.g., DNA) recombination are provided. In some embodiments, the methods are useful for inducing recombination of or between two or more regions of two or more nucleic acids (e.g., DNA). In some embodiments, the methods are useful for inducing recombination of or between two or more regions in a single nucleic acid molecule (e.g., DNA). Because the recombinase fusion proteins used in the methods are ligand-dependent, the timing of recombination can be controlled to minimize off-target effects. In some embodiments, the recombination of one or more target nucleic acid molecules requires the formation of a tetrameric complex at the target site. Typically, the tetramer comprises four (4) inventive RNA-guided recombinase fusion proteins (e.g., a complex of any four inventive recombinase fusion proteins provided herein). In some embodiments, each recombinase fusion protein of the tetramer targets a particular DNA sequence via a distinct gRNA bound to each recombinase fusion protein. In some embodiments, the fusion proteins are first contacted with a ligand that induces self-excision of the intein, thereby allowing the fusion proteins to (i) bind a gRNA, (ii) bind a target nucleic acid(s), and (iii) form a complex to induce recombination between the target nucleic acid(s). In some embodiments, the fusion proteins are able to bind a gRNA prior to excision of the intein and optionally are able to bind the target nucleic acid(s) but are unable to induce recombination until the intein is excised (e.g., through the addition of a

ligand that binds the ligand-dependent intein). Any of the ligand-dependent recombinase fusion proteins provided herein are useful for methods for site-specific recombination.

In some embodiments, the method for site-specific recombination between two DNA molecules comprises (a) contacting a first DNA with a first ligand-dependent RNA-guided recombinase fusion protein, wherein the nuclease-inactivated Cas9 domain binds a first gRNA that hybridizes to a region of the first DNA; (b) contacting the first DNA with a second ligand-dependent RNA-guided recombinase fusion protein, wherein the nuclease-inactivated Cas9 domain of the second fusion protein binds a second gRNA that hybridizes to a second region of the first DNA; (c) contacting a second DNA with a third ligand-dependent RNA-guided recombinase fusion protein, wherein the nuclease-inactivated Cas9 domain of the third fusion protein binds a third gRNA that hybridizes to a region of the second DNA; and (d) contacting the second DNA with a fourth ligand-dependent RNA-guided recombinase fusion protein, wherein the nuclease-inactivated Cas9 domain of the fourth fusion protein binds a fourth gRNA that hybridizes to a second region of the second DNA. In some embodiments, the fusion proteins are first contacted with a ligand that induces self-excision of the intein prior to forming a complex with a gRNA and/or prior to hybridizing with a target DNA. In some embodiments, the method comprises contacting the fusion proteins with the ligand after the fusion proteins form a complex and/or hybridizes to a target DNA. Typically, the binding of the fusion proteins in steps (a)-(d) results in the tetramerization of the recombinase catalytic domains of the fusion proteins, such that the DNAs are recombined (i.e., following excision of the intein). In some embodiments, the gRNAs of steps (a) and (b) hybridize to opposing strands of the first DNA, and the gRNAs of steps (c) and (d) hybridize to opposing strands of the second DNA. In some embodiments, the target sites of the gRNAs of steps (a)-(d) are spaced to allow for tetramerization of the recombinase catalytic domains. For example, in some embodiments, the target sites of the gRNAs of steps (a)-(d) are no more than 10, no more than 15, no more than 20, no more than 25, no more than 30, no more than 40, no more than 50, no more than 60, no more than 70, no more than 80, no more than 90, or no more than 100 base pairs apart. In some embodiments, the two regions of the two DNA molecules being recombined share homology, such that the regions being recombined are at least 80%, at least 90%, at least 95%, at least 98%, or are 100% homologous.

In some embodiments, methods for site-specific recombination between two regions of a single DNA molecule are provided. In some embodiments, the methods comprise (a) contacting a DNA with a first dCas9-recombinase fusion protein, wherein the dCas9 domain binds a first gRNA that hybridizes to a region of the DNA; (b) contacting the DNA with a second dCas9-recombinase fusion protein, wherein the dCas9 domain of the second fusion protein binds a second gRNA that hybridizes to a second region of the DNA; (c) contacting the DNA with a third dCas9-recombinase fusion protein, wherein the dCas9 domain of the third fusion protein binds a third gRNA that hybridizes to a third region of the DNA; and (d) contacting the DNA with a fourth dCas9-recombinase fusion protein, wherein the dCas9 domain of the fourth fusion protein binds a fourth gRNA that hybridizes to a fourth region of the DNA. In some embodiments, the fusion proteins are first contacted with a ligand that induces self-excision of the intein prior to forming a complex with a gRNA and/or prior to hybridizing

with a target DNA. In some embodiments, the method comprises contacting the fusion proteins with the ligand after the fusion proteins form a complex and/or hybridizes to a target DNA. Typically, the binding of the fusion proteins in steps (a)-(d) results in the tetramerization of the recombinase catalytic domains of the fusion proteins, such that the DNA is recombined (e.g. following the excision of the intein). In some embodiments, two of the gRNAs of steps (a)-(d) hybridize to the same strand of the DNA, and the other two gRNAs of steps (a)-(d) hybridize to the opposing strand of the DNA. In some embodiments, the gRNAs of steps (a) and (b) hybridize to regions of the DNA that are no more than 10, no more than 15, no more than 20, no more than 25, no more than 30, no more than 40, no more than 50, no more than 60, no more than 70, no more than 80, no more than 90, or no more than 100 base pairs apart, and the gRNAs of steps (c) and (d) hybridize to regions of the DNA that are no more than 10, no more than 15, no more than 20, no more than 25, no more than 30, no more than 40, no more than 50, no more than 60, no more than 70, no more than 80, no more than 90, or no more than 100 base pairs apart. In some embodiments, the two regions of the DNA molecule being recombined share homology, such that the regions being recombined are at least 80%, at least 90%, at least 95%, at least 98%, or are 100% homologous.

In some embodiments, any of the inventive methods for site-specific recombination are amenable for inducing recombination, such that the recombination results in excision (e.g., a segment of DNA is excised from a target DNA molecule), insertion (e.g., a segment of DNA is inserted into a target DNA molecule), inversion (e.g., a segment of DNA is inverted in a target DNA molecule), or translocation (e.g., the exchange of DNA segments between one or more target DNA molecule(s)). In some embodiments, the particular recombination event (e.g., excision, insertion, inversion, translocation, etc.) depends, inter alia, on the orientation (e.g., with respect to the target DNA molecule(s)) of the bound RNA-guided recombinase fusion protein(s). In some embodiments, the orientation, or direction, in which a RNA-guided recombinase fusion protein binds a target nucleic acid can be controlled, e.g., by the particular sequence of the gRNA bound to the RNA-guided recombinase fusion protein(s). Methods for controlling or directing a particular recombination event are known in the art, and include, for example, those described by Turan and Bode, "Site-specific recombinases: from tag-and-target- to tag-and-exchange-based genomic modifications." *FASEB J.* 2011; December; 25(12):4088-107, the entire contents of which are hereby incorporated by reference.

In some embodiments, any of the methods for site-specific recombination can be performed in vivo or in vitro. In some embodiments, any of the methods for site-specific recombination are performed in a cell (e.g., recombining genomic DNA in a cell). The cell can be prokaryotic or eukaryotic. The cell, such as a eukaryotic cell, can be in an individual, such as a subject, as described herein (e.g., a human subject). The methods described herein are useful for the genetic modification of cells in vitro and in vivo, for example, in the context of the generation of transgenic cells, cell lines, or animals, or in the alteration of genomic sequence, e.g., the correction of a genetic defect, in a cell in or obtained from a subject. In some embodiments, a cell obtained from a subject and modified according to the methods provided herein, is re-introduced into a subject (e.g., the same subject), e.g., to treat a disease, or for the production of genetically modified organisms in agricultural, medical, or biological research.

In applications in which it is desirable to recombine two or more nucleic acids so as to insert a nucleic acid sequence into a target nucleic acid, a nucleic acid comprising a donor sequence to be inserted is also provided, e.g., to a cell. By a "donor sequence" it is meant a nucleic acid sequence to be inserted at the target site induced by one or more RNA-guided recombinase fusion protein(s). In some embodiments, e.g., in the context of genomic modifications, the donor sequence will share homology to a genomic sequence at the target site, e.g., 1%, 2%, 3%, 4%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95%, or 100% homology with the nucleotide sequences flanking the target site, e.g., within about 100 bases or less of the target site, e.g. within about 90 bases, within about 80 bases, within about 70 bases, within about 60 bases, within about 50 bases, within about 40 bases, within about 30 bases, within about 15 bases, within about 10 bases, within about 5 bases, or immediately flanking the target site. In some embodiments, the donor sequence does not share any homology with the target nucleic acid, e.g., does not share homology to a genomic sequence at the target site. Donor sequences can be of any length, e.g., 10 nucleotides or more, 50 nucleotides or more, 100 nucleotides or more, 250 nucleotides or more, 500 nucleotides or more, 1000 nucleotides or more, 5000 nucleotides or more, 10000 nucleotides or more, 100000 nucleotides or more, etc.

Typically, the donor sequence is not identical to the target sequence that it replaces or is inserted into. In some embodiments, the donor sequence contains at least one or more single base changes, insertions, deletions, inversions, or rearrangements with respect to the target sequence (e.g., target genomic sequence). In some embodiments, donor sequences also comprise a vector backbone containing sequences that are not homologous to the DNA region of interest and that are not intended for insertion into the DNA region of interest.

The donor sequence may comprise certain sequence differences as compared to the target (e.g., genomic) sequence, for example, restriction sites, nucleotide polymorphisms, selectable markers (e.g., drug resistance genes, fluorescent proteins, enzymes etc.), which can be used to assess successful insertion of the donor sequence at the target site or in some cases may be used for other purposes (e.g., to signify expression at the targeted genomic locus). In some embodiments, if located in a coding region, such nucleotide sequence differences will not change the amino acid sequence, or will make silent amino acid changes (e.g., changes which do not affect the structure or function of the protein). In some embodiments, these sequence differences may include flanking recombination sequences such as FLPs, loxP sequences, or the like, that can be activated at a later time for removal of e.g., a marker sequence. The donor sequence may be provided to the cell as single-stranded DNA, single-stranded RNA, double-stranded DNA, or double-stranded RNA. It may be introduced into a cell in linear or circular form. If introduced in linear form, the ends of the donor sequence may be protected (e.g., from exonucleolytic degradation) by methods known to those of skill in the art. For example, one or more dideoxynucleotide residues are added to the 3' terminus of a linear molecule and/or self-complementary oligonucleotides are ligated to one or both ends. See, e.g., Chang et al., *Proc. Natl. Acad. Sci. USA.* 1987; 84:4959-4963; Nehls et al., *Science.* 1996; 272:886-889. In some embodiments, a donor sequence is introduced into a cell as part of a vector molecule having additional sequences such as, for example, replication origins, promoters, and genes encoding antibiotic resistance. In

some embodiments, donor sequences can be introduced as naked nucleic acid, as nucleic acid complexed with an agent such as a liposome or polymer (e.g., poloxamer), or can be delivered by viruses (e.g., adenovirus, AAV, etc.).

In some embodiments, any of the methods provided herein can be performed on DNA in a cell. For example, in some embodiments the DNA contacted by any RNA/gRNA-comprising complex provided herein is in a eukaryotic cell. In some embodiments, the eukaryotic cell is in an individual. In some embodiments, the individual is a human. In some embodiments, any of the methods provided herein are performed in vitro. In some embodiments, any of the methods provided herein are performed in vivo.

In some embodiments of this disclosure, methods for site-specific nucleic acid (e.g., DNA) editing are provided. In some embodiments, the fusion protein is used to introduce a point mutation into a nucleic acid by deaminating a target nucleobase, e.g., a cytidine (C) residue. In some embodiments, the method comprises contacting a DNA molecule with a ligand-dependent fusion protein comprising a nucleic acid inactivated RNA-guided nuclease (e.g., dCas9), which comprises a ligand dependent intein, fused to a deaminase, and (b) a gRNA targeting the fusion protein of step (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the deamination of a nucleotide base. Any of the fusion proteins comprising a gene editing domain as provided herein are amenable for use in the methods. In some embodiments, the method first comprises contacting the fusion protein with a ligand that induces self-excision of the intein prior to forming a complex with the gRNA. In some embodiments, the method comprises contacting the fusion protein with a ligand that induces self-excision of the intein after the fusion protein has formed a complex with the gRNA.

In some embodiments, the deamination of the target nucleobase results in the correction of a genetic defect, e.g., in the correction of a point mutation that leads to a loss of function in a gene product. In some embodiments, the genetic defect is associated with a disease or disorder, e.g., a lysosomal storage disorder or a metabolic disease, such as, for example, type I diabetes. In some embodiments, the methods provided herein are used to introduce a deactivating point mutation into a gene or allele that encodes a gene product that is associated with a disease or disorder. For example, in some embodiments, methods are provided herein that employ a DNA editing fusion protein to introduce a deactivating point mutation into an oncogene (e.g., in the treatment of a proliferative disease). A deactivating mutation may, in some embodiments, generate a premature stop codon in a coding sequence, which results in the expression of a truncated gene product, e.g., a truncated protein lacking the function of the full-length protein.

In some embodiments, the purpose of the methods provided herein is to restore the function of a dysfunctional gene via genome editing. Compositions and methods of using gene editing enzymes fused e.g., to Cas9 are known, and include those described in U.S. patent application Ser. No. 14/325,815 titled "FUSIONS OF CAS9 DOMAINS AND NUCLEIC ACID-EDITING DOMAINS," and filed on Jul. 8, 2014; the entire contents of which are incorporated herein by reference. The fusion proteins provided herein (comprising ligand-dependent inteins) can be validated for gene editing-based human therapeutics in vitro, e.g., by correcting a disease-associated mutation in human cell culture. It will be understood by the skilled artisan that the

fusion proteins provided herein, e.g., the fusion proteins comprising a dCas9 domain (e.g., comprising a ligand-dependent intein) and a nucleic acid deaminase domain can be used to correct any single point T→C or A→G mutation. In the first case, deamination of the mutant C back to U corrects the mutation, and in the latter case, deamination of the C that is base-paired with the mutant G, followed by a round of replication, corrects the mutation.

An exemplary disease-relevant mutation that can be corrected by the provided fusion proteins in vitro or in vivo is the H1047R (A3140G) polymorphism in the PI3KCA protein. The phosphoinositide-3-kinase, catalytic alpha subunit (PI3KCA) protein acts to phosphorylate the 3-OH group of the inositol ring of phosphatidylinositol. The PI3KCA gene has been found to be mutated in many different carcinomas, and thus it is considered to be a potent oncogene (Lee et al., PIK3CA gene is frequently mutated in breast carcinomas and hepatocellular carcinomas. *Oncogene*. 2005; 24(8): 1477-80). In fact, the A3140G mutation is present in several NCI-60 cancer cell lines, such as, for example, the HCT116, SKOV3, and T47D cell lines, which are readily available from the American Type Culture Collection (ATCC) (Ike-diobi et al., Mutation analysis of 24 known cancer genes in the NCI-60 cell line set. *Mol Cancer Ther*. 2006; 5(11): 2606-12.

In some embodiments, a cell carrying a mutation to be corrected, e.g., a cell carrying a point mutation, e.g., an A3140G point mutation in exon 20 of the PI3KCA gene, resulting in a H1047R substitution in the PI3KCA protein is contacted with an expression construct encoding a ligand-dependent Cas9 deaminase fusion protein and an appropriately designed gRNA targeting the fusion protein to the respective mutation site in the encoding PI3KCA gene. Control experiments can be performed where the gRNAs are designed to target the fusion enzymes to non-C residues that are within the PI3KCA gene. Genomic DNA of the treated cells can be extracted, and the relevant sequence of the PI3KCA genes PCR amplified and sequenced to assess the activities of the fusion proteins in human cell culture.

It will be understood that the example of correcting point mutations in PI3KCA is provided for illustration purposes and is not meant to limit the instant disclosure. The skilled artisan will understand that the instantly disclosed ligand-dependent DNA-editing fusion proteins can be used to correct other point mutations and mutations associated with other cancers and with diseases other than cancer including other proliferative diseases.

The successful correction of point mutations in disease-associated genes and alleles opens up new strategies for gene correction with applications in therapeutics and basic research. Site-specific single-base modification systems like the disclosed fusions of ligand-dependent Cas9 and deaminase enzymes or domains also have applications in "reverse" gene therapy, where certain gene functions are purposely suppressed or abolished. In these cases, site-specifically mutating Trp (TGG), Gln (CAA and CAG), or Arg (CGA) residues to premature stop codons (TAA, TAG, TGA) can be used to abolish protein function in vitro, ex vivo, or in vivo.

The instant disclosure provides methods for the treatment of a subject diagnosed with a disease associated with or caused by a point mutation that can be corrected by a ligand-dependent Cas9 DNA editing fusion protein provided herein. For example, in some embodiments, a method is provided that comprises administering to a subject having such a disease, e.g., a cancer associated with a PI3KCA point mutation as described above, an effective amount of a ligand-dependent Cas9 deaminase fusion protein that cor-

rects the point mutation or introduces a deactivating mutation into the disease-associated gene, e.g., following subsequent administration of the small molecule (e.g., ligand) that activates the fusion protein. In some embodiments, the disease is a proliferative disease. In some embodiments, the disease is a genetic disease. In some embodiments, the disease is a neoplastic disease. In some embodiments, the disease is a metabolic disease. In some embodiments, the disease is a lysosomal storage disease. Other diseases that can be treated by correcting a point mutation or introducing a deactivating mutation into a disease-associated gene will be known to those of skill in the art, and the disclosure is not limited in this respect.

The instant disclosure provides methods for the treatment of additional diseases or disorders, e.g., diseases or disorders that are associated or caused by a point mutation that can be corrected by deaminase-mediated gene editing. Some such diseases are described herein, and additional suitable diseases that can be treated with the strategies and fusion proteins provided herein will be apparent to those of skill in the art based on the instant disclosure. Exemplary suitable diseases and disorders include, without limitation, cystic fibrosis (see, e.g., Schwank et al., Functional repair of CFTR by CRISPR/Cas9 in intestinal stem cell organoids of cystic fibrosis patients. *Cell stem cell*. 2013; 13: 653-658; and Wu et. al., Correction of a genetic disease in mouse via use of CRISPR-Cas9. *Cell stem cell*. 2013; 13: 659-662, neither of which uses a deaminase fusion protein to correct the genetic defect); phenylketonuria—e.g., phenylalanine to serine mutation at position 835 in phenylalanine hydroxylase gene (T>C mutation)—see, e.g., McDonald et al., *Genomics*. 1997; 39:402-405; Bernard-Soulier syndrome (BSS)—e.g., phenylalanine to serine mutation at position 55 in the platelet membrane glycoprotein IX (T>C mutation)—see, e.g., Noris et al., *British Journal of Haematology*. 1997; 97: 312-320; epidermolytic hyperkeratosis (EHK)—e.g., leucine to proline mutation at position 160 in keratin 1 (T>C mutation)—see, e.g., Chipev et al., *Cell*. 1992; 70: 821-828; chronic obstructive pulmonary disease (COPD)—e.g., leucine to proline mutation at position 55 in α_1 -antitrypsin (T>C mutation)—see, e.g., Poller et al., *Genomics*. 1993; 17: 740-743; Charcot-Marie-Toot disease type 4J—e.g., leucine to proline mutation at position 197 in FIG. 4 (T>C mutation)—see, e.g., Kundu et al., *3 Biotech*. 2013; 3: 225-234; neuroblastoma (NB)—e.g., isoleucine to threonine mutation at position 41 in Caspase-9 (T>C mutation)—see, e.g., Lenk et al., *PLoS Genetics*. 2011; 7: e1002104; von Willebrand disease (vWD)—e.g., cysteine to arginine mutation at position 509 in von Willebrand factor (T>C mutation)—see, e.g., Lavergne et al., *Br. J. Haematol*. 1992; 82: 66-72; myotonia congenital—e.g., cysteine to arginine mutation at position 277 in the muscle chloride channel gene CLCN1 (T>C mutation)—see, e.g., Weinberger et al., *The J. of Physiology*. 2012; 590: 3449-3464; hereditary renal amyloidosis—e.g., stop codon to arginine mutation at position 78 in apolipoprotein All (T>C mutation)—see, e.g., Yazaki et al., *Kidney Int*. 2003; 64: 11-16; dilated cardiomyopathy (DCM)—e.g., tryptophan to Arginine mutation at position 148 in the FOXD4 gene (T>C mutation), see, e.g., Minorette et. al., *Int. J. of Mol. Med*. 2007; 19: 369-372; hereditary lymphedema—e.g., histidine to arginine mutation at position 1035 in tyrosine kinase (A>G mutation), see, e.g., Irrthum et al., *Am. J. Hum. Genet*. 2000; 67: 295-301; familial Alzheimer's disease—e.g., isoleucine to valine mutation at position 143 in presenilin1 (A>G mutation), see, e.g., Gallo et. al., *J. Alzheimer's disease*. 2011; 25: 425-431; Prion disease—e.g., methionine to valine mutation at posi-

tion 129 in prion protein (A>G mutation)—see, e.g., Lewis et. al., *J. of General Virology*. 2006; 87: 2443-2449; chronic infantile neurologic cutaneous articular syndrome (CINCA)—e.g., Tyrosine to Cysteine mutation at position 570 in cryopyrin (A>G mutation)—see, e.g., Fujisawa et. al. *Blood*. 2007; 109: 2903-2911; and desmin-related myopathy (DRM)—e.g., arginine to glycine mutation at position 120 in aB crystallin (A>G mutation)—see, e.g., Kumar et al., *J. Biol. Chem*. 1999; 274: 24137-24141. The entire contents of each of the foregoing references and database entries are incorporated herein by reference.

According to another aspect, methods for transcriptional activation of a gene are provided. In some embodiments, the methods comprise contacting a DNA molecule comprising a gene with (a) a ligand-dependent dCas9 fusion protein comprising a transcriptional activator (e.g., any of those provided herein) and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the transcriptional activation of the gene. In some embodiments, the method further comprises contacting the fusion protein with a ligand that induces self-excision of the intein. In some embodiments, the fusion protein is contacted with the ligand prior to forming a complex with a gRNA. In some embodiments, the fusion protein is contacted with the ligand after forming a complex with a gRNA. In some embodiments, the gRNA targets the promoter region of a gene. Methods for inducing gene activation using fusion proteins comprising a transcriptional activator are known in the art, and include those described by Perex-Pinera et al., "RNA-guided gene activation by CRISPR-Cas9-based transcription factors." *Nature Methods*. 2013; 10, 973-976; the entire contents of which are incorporated herein by reference.

According to another aspect, methods for transcriptional repression of a gene are provided. In some embodiments, the methods comprise contacting a DNA molecule comprising a gene with (a) a ligand-dependent dCas9 fusion protein comprising a transcriptional repressor (e.g., any of those provided herein) and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the transcriptional repression of the gene. In some embodiments, the method further comprises contacting the fusion protein with a ligand that induces self-excision of the intein. In some embodiments, the fusion protein is contacted with the ligand prior to forming a complex with a gRNA. In some embodiments, the fusion protein is contacted with the ligand after forming a complex with a gRNA. In some embodiments, the gRNA targets the promoter region of a gene. Methods for inducing gene repression using fusion proteins comprising a transcriptional repressor are known in the art, and include those described by Gilbert et al., CRISPR-mediated modular RNA-guided regulation of transcription in eukaryotes. *Cell*. 2013; 154, 442-451; the entire contents of which are incorporated herein by reference.

According to another aspect, methods for epigenetic modification of DNA are provided. In some embodiments, the methods comprise contacting a DNA molecule comprising with (a) a ligand-dependent dCas9 fusion protein comprising an epigenetic modifier (e.g., any of those provided herein) and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the

gRNA in an amount effective and under conditions suitable for the epigenetic modification of the DNA. In some embodiments, the DNA comprises one or more genes. In some embodiments, the method further comprises contacting the fusion protein with a ligand that induces self-excision of the intein. In some embodiments, the fusion protein is contacted with the ligand prior to forming a complex with a gRNA. In some embodiments, the fusion protein is contacted with the ligand after forming a complex with a gRNA. In some embodiments, the gRNA targets the promoter region of a gene. In some embodiments, the epigenetic modification that results is methylation of DNA. In some embodiments, the epigenetic modification that results is demethylation of DNA. In some embodiments, the epigenetic modification that results is methylation of histone protein. In some embodiments, the epigenetic modification that results is demethylation of histone protein. In some embodiments, the epigenetic modification that results is acetylation of histone protein. In some embodiments, the epigenetic modification that results is deacetylation of histone protein. Methods for inducing epigenetic modifications using fusion proteins comprising an epigenetic modifier are known in the art, and include those described by Konermann et al., Optical control of mammalian endogenous transcription and epigenetic states. *Nature*. 2013; 500, 472-476; Mendenhall et al., Locus-specific editing of histone modifications at endogenous enhancers. *Nat. Biotechnol.* 2013; 31, 1133-1136; and Maeder et al., Targeted DNA demethylation and activation of endogenous genes using programmable TALE-TET1 fusion proteins. *Nat. Biotechnol.* 2013; 31, 1137-1142; the entire contents of which are incorporated herein by reference.

In some embodiments, any of the methods provided herein can be performed on DNA in a cell. For example, in some embodiments the DNA contacted by any RNA/gRNA-comprising complex provided herein is in a eukaryotic cell. In some embodiments, the eukaryotic cell is in an individual. In some embodiments, the individual is a human. In some embodiments, any of the methods provided herein are performed in vitro. In some embodiments, any of the methods provided herein are performed in vivo.

In some embodiments of the methods provided herein, the ligand-dependent Cas9 protein, e.g., the Cas9-intein or the Cas9-intein fusion protein, is contacted with the ligand at a concentration effective to excise the intein from the Cas9-intein variant, or at a concentration effective to induce a desired modification (e.g., cleavage, nicking, recombination, or deamination) of a target site. In some embodiments, a ligand-dependent Cas9 protein provided herein is contacted with a suitable ligand at a concentration resulting in decreased off-target activity of the Cas9 protein as compared to the off-target activity of wild-type Cas9. For example, in some embodiments, a method provided herein comprises contacting a population of ligand-dependent Cas9 proteins in vitro or in vivo in the presence of a target nucleic acid to be modified with a suitable ligand at a concentration resulting in the desired modification of the target nucleic acid, and in either no off-target activity (i.e., no modification of any non-target nucleic acids) or in an off-target activity of less than 80%, less than 75%, less than 50%, less than 45%, less than 40%, less than 35%, less than 30%, less than 25%, less than 24%, less than 23%, less than 22%, less than 21%, less than 20%, less than 19%, less than 18%, less than 17%, less than 16%, less than 15%, less than 14%, less than 13%, less than 12%, less than 11%, less than 10%, less than 9%, less than 8%, less than 7%, less than 6%, less than 5%, less than 4%, less than 3%, less than 2%, or less than 1% of the

off-target activity observed or expected under the same conditions when using wild-type Cas9.

Polynucleotides, Vectors, Cells, Kits

In another aspect of this disclosure, polynucleotides encoding one or more of the inventive proteins and/or gRNAs are provided. For example, polynucleotides encoding any of the proteins described herein are provided, e.g., for recombinant expression and purification of isolated nucleases, recombinases, gene editing enzymes, and other nucleic acid modifying enzymes, e.g., comprising Cas9 variants (e.g., dCas9) comprising ligand-dependent inteins. In some embodiments, an isolated polynucleotide comprises one or more sequences encoding a ligand dependent RNA-guided nuclease (e.g., Cas9). In some embodiments, an isolated polynucleotide comprises one or more sequences encoding a Cas9 fusion protein, for example, any of the Cas9 fusion proteins described herein (e.g., those comprising a nuclease-inactivated Cas9 fused to a nuclease, recombinase, deaminase domain, or transcriptional activator). In some embodiments, an isolated polynucleotide comprises one or more sequences encoding a gRNA, alone or in combination with a sequence encoding any of the proteins described herein.

In some embodiments, vectors encoding any of the proteins described herein are provided, e.g., for recombinant expression and purification of Cas9 proteins, and/or fusions comprising Cas9 proteins (e.g., variants). In some embodiments, the vector comprises or is engineered to include an isolated polynucleotide, e.g., those described herein. In some embodiments, the vector comprises one or more sequences encoding a Cas9 protein (as described herein), a gRNA, or combinations thereof, as described herein. Typically, the vector comprises a sequence encoding an inventive protein operably linked to a promoter, such that the fusion protein is expressed in a host cell.

In some embodiments, cells are provided, e.g., for recombinant expression and purification of any of the Cas9 proteins provided herein. The cells include any cell suitable for recombinant protein expression, for example, cells comprising a genetic construct expressing or capable of expressing an inventive protein (e.g., cells that have been transformed with one or more vectors described herein, or cells having genomic modifications, for example, those that express a protein provided herein from an allele that has been incorporated into the cell's genome). Methods for transforming cells, genetically modifying cells, and expressing genes and proteins in such cells are well known in the art, and include those provided by, for example, Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)) and Friedman and Rossi, *Gene Transfer: Delivery and Expression of DNA and RNA, A Laboratory Manual* (1st ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2006)).

Some aspects of this disclosure provide kits comprising a ligand-dependent Cas9 variant (e.g., a ligand dependent Cas9 nuclease (or nickase), and/or a ligand-dependent dCas9 variant fused to a nuclease, recombinase, deaminase, or a transcriptional activator as provided herein. In some embodiments, the kit comprises a polynucleotide encoding an inventive Cas9 variant, nuclease, recombinase, and/or deaminase e.g., as provided herein. In some embodiments, the kit comprises a vector for recombinant protein expression, wherein the vector comprises a polynucleotide encoding any of the proteins provided herein. In some embodiments, the kit comprises a cell (e.g., any cell suitable for expressing Cas9 proteins or fusions comprising Cas9 pro-

teins, such as bacterial, yeast, or mammalian cells) that comprises a genetic construct for expressing any of the proteins provided herein. In some embodiments, any of the kits provided herein further comprise one or more gRNAs and/or vectors for expressing one or more gRNAs. In some embodiments, the kit comprises an excipient and instructions for contacting the Cas9 proteins or dCas9 fusions with the excipient to generate a composition suitable for contacting a nucleic acid with the inventive protein. In some embodiments, the composition is suitable for delivering an inventive protein to a cell, or for delivering a nucleic acid encoding the protein to a cell. In some embodiments, the composition is suitable for delivering an inventive protein to a subject. In some embodiments, the excipient is a pharmaceutically acceptable excipient.

EXAMPLES

Example 1: Small Molecule-Controlled Cas9

Cas9 variants that can be activated in the presence of a small molecule were engineered, allowing spatiotemporal control over DNA cleavage. These engineered Cas9 variants contain a small-molecule-regulated intein (Buskirk et al., *Proc. Natl. Acad. Sci. USA*. 2004; 101, 10505-10510), which has been optimized for mammalian cells (Peck et al., *Chem. Biol.* 2011; 18 (5), 619-630), that renders the protein inactive as a nuclease. Upon addition of the cell-permeable molecule, 4-hydroxytamoxifen (4-HT), the intein excises itself from the protein and ligates the flanking extein sequences, restoring Cas9 activity. Because these Cas9 variants can be active over a smaller time window than wild-type Cas9, the likelihood of having off-target cleavage is reduced.

The 37R3-2 intein was inserted at 15 different positions into human codon-optimized *Streptococcus pyogenes* Cas9 (e.g., SEQ ID NO:2). The intein was inserted in place of a single cysteine, alanine, serine, or threonine residue. Upon excision, the intein leaves a cysteine residue. Thus, the primary structure generated following protein splicing is either identical to the unmodified version of Cas9 when the intein is inserted in place of cysteine, or it is one amino acid different when the intein is inserted in place of alanine, serine, or threonine.

Plasmid constructs were generated in which the intein replaced amino acid residues: Cys80, Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622, Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, or Ser1274 (e.g., in the amino acid sequence set forth as SEQ ID NO:2). These plasmids express the Cas9 variant with a nuclear localization signal (NLS) and 3xFLAG tag from the CMV promoter.

HEK293-GFP stable cells were transfected with the Cas9 expression plasmid, a gRNA (targeting Emerald GFP; Guilinger et al., *Nature Biotechnology* (2014)), and iRFP670 (transfection control), using Lipofectamine 2000. Twelve hours after transfection, media, either containing 4-HT (1 μ M) or without 4-HT, was added.

Five days after transfection, cells were trypsinized and analyzed on a flow cytometer. Cells lacking GFP indicated genome modification. Cas9 variants that induced minimal genome modification in the absence of 4-HT but induce significant genome modification in the presence of 4-HT were deemed small-molecule-regulated variants in this Example. Of fifteen targeted insertions, five demonstrated minimal genome modification in the absence of 4-HT. These variants are highlighted in bold in the Table I below.

Additionally, a time course was performed in which incubation with 4-HT was limited to 2, 4, 8, 12 or 24 hours,

after which point the media was replaced. Presumably, the shorter time an "active" cas9 is present, the less off-target cleavage. As depicted in Table II below, treating with 4-HT for 2 hours is sufficient for on-target cleavage and longer treatment periods do not show significant increased cleavage in this assay.

To assess the ability of the ligand-dependent Cas9 proteins to affect genomic modifications in the presence of absence of ligand, HEK293-GFP stable cells (GenTarget) were transfected with Cas9 expression plasmids and sgRNAs targeting EMX, VEGF, or CLTA genomic sites using Lipofectamine 2000 as previously described (Guilinger et al., Fusion of catalytically inactive Cas9 to FokI nuclease improves the specificity of genome modification *Nature Biotechnology*. 2014; 32(6):577-82). 4-HT (1 μ M) was added during transfection for +4-HT samples. 12 hours after transfection the media was replaced. 60 hours after transfection, cells were trypsinized and genomic DNA was isolated using the DNeasy kit (Agencourt). 40-80 ng of genomic DNA was used as a template to PCR amplify the targeted genomic loci with flanking Survey primer pairs as previously described (Guilinger et al., Fusion of catalytically inactive Cas9 to FokI nuclease improves the specificity of genome modification *Nature Biotechnology*. 2014; 32(6): 577-82). PCR products were purified with a QIAquick PCR Purification Kit (Qiagen) and quantified with a Quant-iT PicoGreen dsDNA Kit (Life Technologies). 200 ng of purified PCR DNA was then combined with 2 μ L of NEBuffer 2 (NEB) in a total volume of 19 μ L and denatured then re-annealed with thermocycling at 95° C. for 5 min, 95-85° C. at 2° C./s, 85-20° C. at 0.2° C./s. The re-annealed DNA was incubated with 1 μ L of T7 Endonuclease I (10U/ μ L, NEB) at 37° C. for 15 min. 10 μ L of 50% glycerol was added to the T7 Endonuclease reaction and 15 μ L was analyzed on a 5% TBE 18-well Criterion PAGE gel (Bio-Rad) electrophoresed for 30 min at 200V and stained with EtBr for 15 min.

As shown in FIG. 2, the addition of 4-HT to ligand-dependent Cas9:Intein variants (Cas9:Intein with 37R3-2 replacing S219 (SEQ ID NO:30) and Cas9:Intein with 37R3-2 replacing C574 (SEQ ID NO:33)) resulted in genomic modification of the target sites, comparable to modification by wild-type Cas9. In the absence of 4-HT, the Cas9:Intein variants displayed minimal or no modification of the EMX and VEGF genomic target sites, while some background cleavage was observed for the CLTA genomic target site. Gene modification levels can be estimated by comparing the intensities of the cleaved (two smaller fragments) and uncleaved bands. These results demonstrate that Cas9 cleavage of genomic target sites can be controlled by the addition of ligand (here, 4-HT) which activates the proteins.

TABLE 1

Cas9 Variant	Cells without GFP (%)	
	-4-HT	+4-HT
None	4.65	3.42
wild-type cas9	48.49	40.49
intein(Cys80)-Cas9 (SEQ ID NO: 27)	7.08	4.96
intein(Ala127)-Cas9 (SEQ ID NO: 28)	7.97	19.73
intein(Thr146)-Cas9 (SEQ ID NO: 29)	8.77	21.60
intein(Ser219)-Cas9 (SEQ ID NO: 30)	6.53	23.98
intein(Thr333)-Cas9 (SEQ ID NO: 31)	4.96	9.17
intein(Thr519)-Cas9 (SEQ ID NO: 32)	9.49	25.96
intein(Cys574)-Cas9 (SEQ ID NO: 33)	5.74	21.44
intein(Thr622)-Cas9 (SEQ ID NO: 34)	5.67	3.96

TABLE 1-continued

Cas9 Variant	Cells without GFP (%)	
	-4-HT	+4-HT
intein(Ser701)-Cas9 (SEQ ID NO: 35)	6.54	9.56
intein(Ala728)-Cas9 (SEQ ID NO: 36)	20.82	41.89
intein(Thr995)-Cas9 (SEQ ID NO: 37)	14.95	21.39
intein(Ser1006)-Cas9 (SEQ ID NO: 38)	6.80	12.61
intein(Ser1154)-Cas9 (SEQ ID NO: 39)	21.14	41.94
intein(Ser1159)-Cas9 (SEQ ID NO: 40)	5.65	13.21
intein(Ser1274)-Cas9 (SEQ ID NO: 41)	3.08	5.00

TABLE 2

Cas9 Variant	Cells without GFP (%)					
	-4-HT	2 hrs	4 hrs	8 hrs	12 hrs	24 hrs
None	2.98	2.60	3.78	3.71	2.76	2.82
wild-type cas9	34.65	42.28	37.68	33.89	33.81	37.26
intein(Ala127)-Cas9 (SEQ ID NO: 28)	5.03	18.95	18.57	18.64	18.35	16.83
intein(Thr146)-Cas9 (SEQ ID NO: 29)	4.28	16.29	15.07	13.65	14.44	19.57
intein(Ser219)-Cas9 (SEQ ID NO: 30)	3.92	17.25	17.07	15.12	15.28	24.39
intein(Thr519)-Cas9 (SEQ ID NO: 32)	4.29	14.55	13.98	14.74	13.93	18.04
intein(Cys574)-Cas9 (SEQ ID NO: 33)	2.91	14.57	13.11	16.91	16.10	14.52

Example 2: Small Molecule-Controlled Cas9 Protein with Improved Genome-Editing Specificity

Cas9 nucleases that are activated by the presence of a cell-permeable small molecule were developed by inserting an evolved 4-hydroxytamoxifen (4-HT)-responsive intein at specific positions in Cas9. In human cells, conditionally active Cas9s modify target genomic sites with up to 25-fold higher specificity than wild-type Cas9.

The RNA-guided endonuclease Cas9 from the type II CRISPR-Cas system enables simple and efficient genome editing in a wide variety of organisms. Virtually any target DNA locus can be cleaved by programming Cas9 with a single guide RNA (sgRNA) that contains a stretch of ~20 nucleotides complementary to the target sequence¹⁻³. Due to its simplicity and robustness, the Cas9 system has been widely adopted for biological research and therapeutic development. The DNA cleavage specificity of Cas9 is imperfect⁴⁻⁸, however, raising concerns over off-target genome modification that may limit its usefulness in therapeutic or research applications. Cas9 off-target activity has been reduced through protein⁹⁻¹² and sgRNA¹³ engineering, and by direct delivery of Cas9:sgRNA protein:RNA complexes into cells¹⁴⁻¹⁶.

A complementary, underexplored strategy to improve Cas9 specificity is to reduce its activity once it has had sufficient opportunity to modify the target DNA locus. Indeed, higher concentrations of Cas9 in cells have been observed to degrade specificity⁴⁻⁶ (defined as the ratio of on-target:off-target DNA cleavage activity), presumably because any Cas9 protein present after the target locus has

been modified can only process off-target substrates. Unfortunately, wild-type Cas9 nucleases are not known to be regulated by other molecules and therefore are used in constitutively active form. While Cas9 can be regulated at the transcriptional level through the use of inducible promoters^{17,18}, transcriptional control cannot limit activity to the short temporal windows that may be necessary to maximize genome-editing specificity^{16,19}, in contrast with the high temporal resolution of post-translational strategies that directly control protein activity.

Engineered variants of Cas9 that can be controlled with a readily available, cell-permeable small molecule were developed. We previously evolved inteins that undergo protein splicing only in the presence of 4-hydroxytamoxifen (4-HT)²⁰. These inteins were developed by inserting the human estrogen receptor ligand-binding domain into the *M. tuberculosis* RecA intein and evolving the resulting inactive fusion protein into a conditionally active intein that requires the presence of 4-HT²⁰⁻²². Subsequent evolution at 37° ° C. yielded a second-generation intein, 37R3-2, with improved splicing properties in mammalian cells²². We envisioned that inserting the 37R3-2 intein into Cas9 at a location that disrupts Cas9 activity until protein splicing has taken place could result in conditionally active Cas9 nucleases that are active only in the presence of 4-HT (FIG. 3a).

We genetically inserted the 4-HT-dependent intein at each of fifteen positions in Cas9 (Cys80, Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622, Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, and Ser1274), chosen to distribute the location of the intein across the structural domains of Cas9²³ (FIG. 3b and Example 1). Because intein splicing leaves behind a single Cys residue, the intein was inserted in place of one Cas9 amino acid in each of the 15 candidate constructs. In addition to replacing natural Cys amino acids, we also favored replacing Ala, Ser, or Thr residues to minimize the likelihood that the resulting Cys point mutation resulting from protein splicing would disrupt Cas9 activity. The 15 intein-Cas9 candidates were expressed in HEK293-GFP cells together with a sgRNA that targets the genomic EGFP locus in these cells. Twelve hours post-transfection, cells were treated with or without 1 μ M 4-HT. Five days post-transfection, cells were analyzed on a flow cytometer for loss of GFP expression from Cas9-mediated EGFP cleavage and subsequent non-homologous end joining.

Eight of the candidates, corresponding to intein insertion at A127, T146, S219, T333, T519, C574, S1006, and S1159, demonstrated 4-HT-dependent loss of GFP expression consistent with 4-HT-triggered Cas9 activity (FIG. 3c). Interestingly, three intein-Cas9 proteins (insertion at A728, T995, and S1154) showed high DNA modification rates both in the presence and absence of 4-HT, suggesting that large protein insertions at these positions do not significantly inhibit nuclease activity, or that the intein lost its 4-HT dependence due to context-dependent conformational perturbations. We speculate that it may be possible to engineer split Cas9 variants by dividing the protein at these locations, given their tolerance of a 413-residue insertion. The lack of nuclease activity of the remaining four Cas9-inteins (insertion at C80, T622, S701, and S1274) in the presence or absence of 4-HT could result from the inability of the intein to splice in those contexts, the inability of Cas9 to refold properly following splicing, or intolerance of replacement of native Thr or Ser residues with Cys. We pursued two intein-Cas9 variants corresponding to insertion at S219 and

C574 (FIG. 3*b*). These two variants combined high activity in the presence of 4-HT and low activity in the absence of 4-HT.

To evaluate the genome modification specificity of conditionally active Cas9 variants, we expressed intein-Cas9 (S219), intein-Cas9(C574), and wild-type Cas9 in HEK293-GFP cells together with each of three previously described¹¹ sgRNAs that target the well-studied EMX, VEGF, and CLTA genomic loci. We assayed these Cas9:sgRNA combinations in human cells for their ability to modify the three on-target loci as well as 11 known off-target genomic sites (Table 3)^{4,5,10,13}. Cells were treated with or without 1 μ M 4-HT during transfection, and after 12 h the media was replaced with fresh media lacking 4-HT. We observed no cellular toxicity arising from 12 or 60 h of treatment with 1 μ M 4-HT in untransfected or transfected HEK293 cells (FIG. 5). Genomic DNA was isolated 60 h post-transfection and analyzed by high-throughput DNA sequencing

Overall on-target genome modification frequency of intein-Cas9(S219) and intein-Cas9 (C574) in the presence of 1 μ M 4-HT was similar to that of wild-type Cas9 (FIG. 4*a*, Tables 4 and 5). On-target modification frequency in the presence of 4-HT was 3.4- to 7.3-fold higher for intein-Cas9 (S219), and 3.6- to 9.6-fold higher for intein-Cas9(C574), than in the absence of 4-HT, whereas modification efficiency for wild-type Cas9 was 1.2- to 1.8-fold lower in the presence of 4-HT (FIG. 4*a*). Both intein-Cas9 variants exhibited a low level of background activity in the absence of 4-HT, consistent with previous reports²⁰⁻²². Western blot analysis of intein-Cas9(S219) from transfected HEK293 cells confirmed the presence of spliced product at the earliest assayed time point (4 h) following 4-HT treatment; no spliced product was detected in the absence of 4-HT (FIG. 6). Together, these results indicate that intein-Cas9(S219) and intein-Cas9(C574) are slightly less active than wild-type Cas9 in the presence of 4-HT, likely due to incomplete splicing (FIG. 6), but much less active in the absence of 4-HT.

High-throughput sequencing of 11 previously described off-target sites that are modified by wild-type Cas9:sgRNA complexes targeting the EMX, VEGF, and CLTA loci revealed that both intein-Cas9 variants when treated with 4-HT for 12 h exhibit substantially improved specificity compared to that of wild-type Cas9 (FIG. 7, Tables 4, 6, and 7). On-target:off-target indel modification ratios for both intein-Cas9 variants were on average 6-fold higher, and as much as 25-fold higher, than that of wild-type Cas9 (FIG. 4*b-d*). In the absence of 4-HT, the genome modification specificity of both intein-Cas9 variants was on average 14-fold higher than that of wild-type Cas9 in the absence of 4-HT (FIG. 8), presumably resulting from the much lower activity of the intein-Cas9 variants in the absence of 4-HT^{4,6}.

Since intein-Cas9s can result in slightly lower on-target modification rates compared to wild-type Cas9 (FIG. 4*a*), we sought to verify that the improvements in specificity among the intein-Cas9s were not simply a result of reduced activity. Both on- and off-target activity of Cas9 has been shown to be dependent on the amount of Cas9 expression plasmid transfected⁴⁻⁶. By transfecting lower amounts of the wild-type Cas9 expression plasmid, we compared intein-Cas9s with wild-type Cas9 under conditions that result in very similar levels of on-target modification. To minimize potential differences in transfection efficiency, we supplemented with a plasmid that does not express Cas9 so that the same total amount of plasmid DNA was transfected into each sample. High-throughput sequencing revealed that

wild-type Cas9 shows slightly improved specificity, as expected, as the on-target cleavage rate is reduced. The intein-Cas9 variants, however, remain substantially more specific than wild-type Cas9 at similar on-target DNA cleavage rates (FIGS. 9-11, Tables 6 and 8). For example, intein-Cas9(C574) and wild-type Cas9 (80 ng) have virtually identical on-target DNA cleavage rates (both 6.4%) at the EMX locus but all four off-target sites are modified at an average of 4-fold lower frequencies ($P < 1 \times 10^{-13}$) by intein-Cas9(C574) than by wild-type Cas9. These findings indicate that specificity improvements of intein-Cas9 variants do not simply arise from differences in overall genome editing activity.

Intein 37R3-2 can be activated by other estrogen receptor modulators. To enable intein-Cas9 applications in which endogenous β -estradiol is present, we inserted into the estrogen receptor ligand-binding domain a point mutation (G521R) that renders the domain more specific for 4-HT²⁴. This mutation slightly reduces affinity for 4-HT but almost abolishes affinity for β -estradiol. The addition of this mutation to intein-Cas9(S219) eliminates the ability of β -estradiol to trigger Cas9 activity (FIG. 12).

The intein-Cas9 variants developed here demonstrate small-molecule control of Cas9 function, thereby enhancing genome-modification specificity. The use of ligand-dependent Cas9 variants provides greater control over genomic modification efficiencies and specificities than is currently achievable with constitutively active or transcriptionally regulated genome editing. This approach can synergize with other specificity-augmenting strategies such as direct delivery of transient Cas9 protein into cells¹⁶, using truncated guide RNAs¹³, paired Cas9 nickases^{9,10}, or FokI-dCas9 fusions^{11,12}. This approach could also be applied to other genome engineering proteins to enable, for example, small-molecule control of TALE-based or Cas9-mediated transcriptional regulators.

TABLE 3

On-target and 11 known off-target substrates of Cas9: sgRNAs that target sites in EMX, VEGF, and CLTA. List of genomic on-target and off-target sites of the EMX, VEGF, and CLTA sites are shown with mutations from the on-target sequence shown in lower case. Protospacer-adjacent motifs (PAMs) are shown underlined.	
EMX On	GAGTCCGAGCAGAAGAAGAGGG (SEQ ID NO: 107)
EMX Off 1	GAGgCCGAGCAGAAGAAgACGG (SEQ ID NO: 108)
EMX Off 2	GAGTCctAGCAGgAGAAGAAGaG (SEQ ID NO: 109)
EMX Off 3	GAGTCtaAGCAGAAGAAGAAGaG (SEQ ID NO: 110)
EMX Off 4	GAGTtaGAGCAGAAGAAGAAAGG (SEQ ID NO: 111)
VEGF On	GGGTGGGGGAGTTTGCTCCTGG (SEQ ID NO: 112)
VEGF Off 1	GGaTGgaGGGAGTTTGCTCCTGG (SEQ ID NO: 113)
VEGF Off 2	GGGaGGGtGGAGTTTGCTCCTGG (SEQ ID NO: 114)
VEGF Off 3	cGGgGGaGGGAGTTTGCTCCTGG (SEQ ID NO: 115)
VEGF Off 4	GGGgaGGGGaAGTTTGCTCCTGG (SEQ ID NO: 116)
CLTA On	GCAGATGTAGTGTTCACAGGG (SEQ ID NO: 117)
CLTA Off 1	aCaTATGTAGTaTTTCACAGGG (SEQ ID NO: 118)

TABLE 3-continued

On-target and 11 known off-target substrates of Cas9: sgRNAs that target sites in EMX, VEGF, and CLTA. List of genomic on-target and off-targets sites of the EMX, VEGF, and CLTA sites are shown with mutations from the on-target sequence shown in lower case. Protospacer-adjacent motifs (PAMs) are shown underlined.		5
CLTA Off 2	cCAGATGTAGTATTcCCACAGGG (SEQ ID NO: 119)	10
CLTA Off 3	ctAGATGaAGTGcTTCCACATGG (SEQ ID NO: 120)	

TABLE 4

Raw sequence counts and modification frequencies for data plotted in FIGS. 4, 7, and 8. Total: total number of sequence counts. Modification frequency: number of indels divided by the total number of sequences listed as percentages.									
	-4-HT intein-Cas9 (S219)			-4-HT intein-Cas9 (C574)			-4-HT wt Cas9 (500 ng)		
	Indels	Total	Mod. Freq	Indels	Total	Mod. Freq	Indels	Total	Mod. Freq
EMX On	1123	59967	1.87%	561	56700	0.99%	15589	72127	21.61%
EMX Off 1	3	46360	0.01%	4	39544	0.01%	1143	55334	2.07%
EMX Off 2	8	52362	0.02%	3	36983	0.01%	540	89945	0.60%
EMX Off 3	32	66472	0.05%	10	49582	0.02%	5804	83231	6.97%
EMX Off 4	146	76633	0.19%	57	60976	0.09%	11817	86566	13.65%
VEGF On	359	34089	1.05%	379	44841	0.85%	3815	42732	8.93%
VEGF Off 1	214	49383	0.43%	117	40358	0.29%	14578	71764	20.31%
VEGF Off 2	29	34582	0.08%	10	21753	0.05%	2551	43775	5.83%
VEGF Off 3	18	47664	0.04%	4	43171	0.01%	1743	82128	2.12%
VEGF Off 4	58	56732	0.10%	33	44096	0.07%	14114	116598	12.10%
CLTA On	2087	48566	4.30%	930	51240	1.81%	16930	88447	19.14%
CLTA Off 1	8	79008	0.01%	8	72536	0.01%	3361	111154	3.02%
CLTA Off 2	3	69103	0.00%	0	76788	0.00%	75	78021	0.10%
CLTA Off 3	1	44342	0.00%	2	49937	0.00%	94	51070	0.18%
	+4-HT intein-Cas9 (S219)			+4-HT intein-Cas9 (C574)			+4-HT wt Cas9 (500 ng)		
	Indels	Total	Mod. Freq	Indels	Total	Mod. Freq	Indels	Total	Mod. Freq
EMX On	7434	54764	13.57%	5209	54997	9.47%	9820	55972	17.54%
EMX Off 1	185	43554	0.42%	116	42432	0.27%	1043	41387	2.52%
EMX Off 2	20	56997	0.04%	22	61504	0.04%	412	52780	0.78%
EMX Off 3	413	53819	0.77%	160	56140	0.29%	4149	67153	6.18%
EMX Off 4	2413	76405	3.16%	574	50867	1.13%	6561	62651	10.47%
VEGF On	1285	35095	3.66%	1179	38909	3.03%	1120	23157	4.84%
VEGF Off 1	2951	34729	8.50%	2272	48512	4.68%	6262	42489	14.74%
VEGF Off 2	288	19326	1.49%	273	35253	0.77%	1199	28117	4.26%
VEGF Off 3	167	45573	0.37%	107	56967	0.19%	679	42675	1.59%
VEGF Off 4	1465	37619	3.89%	1229	88062	1.40%	3159	33446	9.45%
CLTA On	5691	39290	14.48%	4348	56815	7.65%	7974	59031	13.51%
CLTA Off 1	286	79836	0.36%	32	72909	0.04%	1468	72166	2.03%
CLTA Off 2	0	25019	0.00%	11	64317	0.02%	18	36863	0.05%
CLTA Off 3	13	38264	0.03%	4	42814	0.01%	78	58340	0.13%

TABLE 5

P-values for comparisons between conditions in FIG. 2a. P-values were obtained using the Fisher exact test and adjusted for multiple comparisons using the Benjamini-Hochberg Method.				55
	intein-Cas9(S219) (+4-HT vs. -4-HT)	intein-Cas9(C574) (+4-HT vs. -4-HT)	wt Cas9 (+4-HT vs. -4-HT)	60
EMX On	<3.3 × 10 ⁻¹⁶	<3.3 × 10 ⁻¹⁶	1	65
VEGF On	<3.3 × 10 ⁻¹⁶	<3.3 × 10 ⁻¹⁶	1	
CLTA On	<3.3 × 10 ⁻¹⁶	<3.3 × 10 ⁻¹⁶	1	

TABLE 6

Raw sequence counts and modification frequencies for data plotted in FIG. 4b-d, and 9-11. Total: total number of sequence counts. Modification frequency: number of indels divided by the total number of sequences listed as percentages.											
	+4-HT intein- Cas9 (S219)			+4-HT intein- Cas9 (C574)			+4-HT wt Cas9 (500 ng)			+4-HT wt Cas9 (260 ng)	
	Indels	Total	Mod Freq	Indels	Total	Mod Freq	Indels	Total	Mod Freq	Indels	Total
EMX On	5446	66039	8.25%	4125	64260	6.42%	10453	63225	16.53%	6836	54232
EMX Off 1	134	65439	0.20%	115	65758	0.17%	466	37687	1.24%	488	44817
EMX Off 2	26	69924	0.04%	10	61188	0.02%	236	65936	0.36%	94	62522
EMX Off 3	438	81696	0.54%	173	68783	0.25%	4890	88690	5.51%	1760	74807
EMX Off 4	1907	87678	2.18%	708	82863	0.85%	6997	66384	10.54%	4131	80091
VEGF On	1633	51546	3.17%	1330	57690	2.31%	2072	37912	5.47%	1873	38181
VEGF Off 1	3132	67908	4.61%	1978	62133	3.18%	8471	76941	11.01%	5893	58006
VEGF Off 2	347	38567	0.90%	189	49925	0.38%	1008	33299	3.03%	668	31470
VEGF Off 3	84	52871	0.16%	44	58976	0.07%	1088	63365	1.72%	490	48793
VEGF Off 4	1067	52667	2.03%	845	92592	0.91%	3712	47327	7.84%	3838	74876
CLTA On	4230	46334	9.13%	3097	48752	6.35%	7586	59582	12.73%	5747	47919
CLTA Off 1	169	72881	0.23%	20	72486	0.03%	1247	88428	1.41%	576	79763
CLTA Off 2	2	40883	0.00%	2	56739	0.00%	27	63439	0.04%	27	64354
CLTA Off 3	5	45599	0.01%	3	39745	0.01%	72	49309	0.15%	16	47504
	+4-HT wt Cas9 (260 ng)			+4-HT wt Cas9 (140 ng)			+4-HT wt Cas9 (80 ng)			+4-HT wt Cas9 (50 ng)	
	Mod Freq	Indels	Total	Mod Freq	Indels	Total	Mod Freq	Indels	Total	Mod Freq	
EMX On	12.61%	6215	65222	9.53%	3674	57146	6.43%	3551	58687	6.05%	
EMX Off 1	1.09%	373	53270	0.70%	280	49924	0.56%	196	50343	0.39%	
EMX Off 2	0.15%	115	67406	0.17%	76	61509	0.12%	19	58791	0.03%	
EMX Off 3	2.35%	1400	77420	1.81%	507	69582	0.73%	312	63812	0.49%	
EMX Off 4	5.16%	3065	89659	3.42%	1372	83194	1.65%	933	78093	1.19%	
VEGF On	4.91%	1651	46948	3.52%	1209	44425	2.72%	994	45715	2.17%	
VEGF Off 1	10.16%	4073	60556	6.73%	2884	52518	5.49%	1792	53739	3.33%	
VEGF Off 2	2.12%	522	30231	1.73%	388	37944	1.02%	282	34935	0.81%	
VEGF Off 3	1.00%	293	48740	0.60%	183	46144	0.40%	143	50504	0.28%	
VEGF Off 4	5.13%	1892	66578	2.84%	1224	67397	1.82%	693	54510	1.27%	
CLTA On	11.99%	4593	56068	8.19%	3864	49534	7.80%	2243	46510	4.82%	
CLTA Off 1	0.72%	223	87021	0.26%	177	78827	0.22%	75	68762	0.11%	
CLTA Off 2	0.04%	11	74571	0.01%	7	38162	0.02%	6	46931	0.01%	
CLTA Off 3	0.03%	12	49085	0.02%	12	48776	0.02%	8	42490	0.02%	

TABLE 7

P-values for comparisons between conditions in FIG. 7. P-values were obtained using the Fisher exact test and adjusted for multiple comparisons using the Benjamini-Hochberg Method.				
	Independent Experiment 1		Independent Experiment 2	
	+4-HT intein- Cas9(S219) vs. +4-HT wt Cas9 (500 ng)	+4-HT intein- Cas9(C574) vs. +4-HT wt Cas9 (500 ng)	+4-HT intein- Cas9(S219) vs. +4-HT wt Cas9 (500 ng)	+4-HT intein- Cas9(C574) vs. +4-HT wt Cas9 (500 ng)
EMX On	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
EMX Off 1	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
EMX Off 2	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
EMX Off 3	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
EMX Off 4	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
VEGF On	2.8×10^{-12}	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
VEGF Off 1	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
VEGF Off 2	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
VEGF Off 3	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
VEGF Off 4	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
CLTA On	1	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
CLTA Off 1	$<2.4 \times 10^{-16}$	$<2.4 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
CLTA Off 2	9.1×10^{-5}	4.4×10^{-3}	1.4×10^{-4}	4.6×10^{-6}
CLTA Off 3	1.3×10^{-7}	1.5×10^{-14}	3.1×10^{-15}	3.5×10^{-15}

TABLE 8

P-values for comparisons between conditions in FIGS. 9 and 10. All conditions were treated with 4-HT. P-values were obtained using the Fisher exact test and adjusted for multiple comparisons using the Benjamini-Hochberg Method.					
	intein-Cas9(S219) vs. wt Cas9 (500 ng)	intein-Cas9(S219) vs. wt Cas9 (260 ng)	intein-Cas9(S219) vs. wt Cas9 (140 ng)	intein-Cas9(S219) vs. wt Cas9 (80 ng)	intein-Cas9(C574) vs. wt Cas9 (500 ng)
EMX On	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1	$<3.9 \times 10^{-16}$
EMX Off 1	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	6.7×10^{-9}
EMX Off 2	$<3.9 \times 10^{-16}$	4.3×10^{-12}	2.4×10^{-15}	1.7×10^{-8}	$<3.9 \times 10^{-16}$
EMX Off 3	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	2.0×10^{-6}	$<3.9 \times 10^{-16}$
EMX Off 4	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1	$<3.9 \times 10^{-16}$
VEGF On	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1.7×10^{-3}	1	$<3.9 \times 10^{-16}$
VEGF Off 1	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	3.8×10^{-12}	$<3.9 \times 10^{-16}$
VEGF Off 2	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	5.7×10^{-2}	$<3.9 \times 10^{-16}$
VEGF Off 3	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	5.9×10^{-13}	1.8×10^{-5}
VEGF Off 4	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1	$<3.9 \times 10^{-16}$
CLTA On	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1	1	$<3.9 \times 10^{-16}$
CLTA Off 1	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	0.21	0.74	$<3.9 \times 10^{-16}$
CLTA Off 2	1.4×10^{-4}	1.6×10^{-4}	0.13	9.4×10^{-2}	4.6×10^{-6}
CLTA Off 3	3.1×10^{-15}	2.3×10^{-2}	0.12	0.11	3.5×10^{-15}
		intein-Cas9(C574) vs. wt Cas9 (260 ng)	intein-Cas9(C574) vs. wt Cas9 (140 ng)	intein-Cas9(C574) vs. wt Cas9 (80 ng)	intein-Cas9(C574) vs. wt Cas9 (50 ng)
EMX On	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	0.56	1
EMX Off 1	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	3.7×10^{-12}
EMX Off 2	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1.0×10^{-13}	7.1×10^{-2}
EMX Off 3	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	7.5×10^{-13}
EMX Off 4	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1.0×10^{-11}
VEGF On	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1.9×10^{-5}	1
VEGF Off 1	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	9.6×10^{-2}
VEGF Off 2	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
VEGF Off 3	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$
VEGF Off 4	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	9.9×10^{-11}
CLTA On	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	1
CLTA Off 1	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	$<3.9 \times 10^{-16}$	2.0×10^{-9}
CLTA Off 2	5.4×10^{-6}	4.8×10^{-2}	3.5×10^{-2}	0.11	
CLTA Off 3	9.3×10^{-3}	5.7×10^{-2}	5.7×10^{-2}	0.16	

TABLE 9

Raw sequence counts and modification frequencies (for cells transfected with wild-type Cas9 (500 ng) but without a targeting sgRNA, in the presence of 4-HT).			
+4-HT wt Cas9 (500 ng) -sgRNA			
Indels	Total	Modification frequency	
EMX On	8	78943	0.01%
EMX Off 1	1	42232	0.00%
EMX Off 2	4	79008	0.01%
EMX Off 3	60	113629	0.05%
EMX Off 4	5	104159	0.00%
VEGF On	0	60667	0.00%
VEGF Off 1	2	111409	0.00%
VEGF Off 2	0	52048	0.00%
VEGF Off 3	4	88105	0.00%
VEGF Off 4	2	123559	0.00%

TABLE 9-continued

Raw sequence counts and modification frequencies (for cells transfected with wild-type Cas9 (500 ng) but without a targeting sgRNA, in the presence of 4-HT).			
+4-HT wt Cas9 (500 ng) -sgRNA			
Indels	Total	Modification frequency	
CLTA On	491	68600	0.72%
CLTA Off 1	10	116033	0.01%
CLTA Off 2	6	75723	0.01%
CLTA Off 3	4	53885	0.01%
Total: total number of sequence counts.			
Modification frequency: number of indels divided by the total number of sequences listed as percentages.			

Sequences

Intein 37R32:

TGCCTTGCCGAGGGTACCCGAATCTTCGATCCGGTCACTGGTACAACGCATCGCA
 TCGAGGATGTTGTCTGATGGGCGCAAGCCTATTTCATGTCGTGGCTGCTGCCAAGGA
 CGGAACGCTGCTCGCGCGGCCGTGGTGTCTGGTTCGACACGGGAACGCGGGA
 TGTGATCGGGTTGCGGATCGCCGGTGGCGCCATCGTGTGGGCGACACCCGATCAC
 AAGGTGCTGACAGAGTACGGCTGGCGTGCCGCGGGGAACCTCCGAAGGGAGAC
 AGGGTGGCCGACCCGGGTGGTCTGGTAACAGCCTGGCCTTGTCCCTGACGGCCG
 ACCAGATGGTCAGTGCCCTTGTGGATGTGAGCCCCCATCTATTTCGAGTA
 TGATCTACCACTCCCTTCAGTGAAGCTTCGATGATGGGCTTACTGACCAACCTG
 GCAGACAGGAGCTGGTTCACATGATCACTGGGCGAAGAGGGTGCCAGGCTTT
 GTGGATTTGACCTCCATGATCAGGCCACCTTCTAGAACGTGCTGGCTAGAGA
 TCCTGATGATTGGTCTCGTCTGGCGCTCCATGGAGCACCCAGGGAAGCTACTGTT
 TGCTCCTAACTTGCTCTTGGACAGGAACAGGGAATAATGTGTAGAGGGCATGGT
 GGAGATCTTCGACATGCTGCTGGCTACATCATCTCGGTTCCGCATGATGAATCTG
 CAGGGAGAGGAGTTTGTGTGCTCAAACTATTATTTTGTCTAATCTGGAGTGT
 ACACATTTCTGTCAGCACCTGAAGTCTCTGGAAGAGAAGGACCATATCCACCG
 AGCCCTGGACAAGATCACGGACACTTTGATCCACCTGATGGCCAAGGACAGGCT
 GACCTCGAGCAGCAGCACCGAGCGGCTGGCCAGCTCCTCCTCATCTCTCCAC
 ATCAGGCACATGAGTAAACAAGGAATGGAGCATCTGTACAGCATGAAGTACAAG
 AACGTGGTGCCCTCTATGACCTGCTGCTGGAGATGTGAGCGCCACCGCCTAC
 ATGCGGGTGGTTCTGGTGTAGCGCGTGAGGCGTTCGCGGATGCCCTGGATGA
 CAAATTCCTGACGACATGCTGGCGGAAGGACTCCGCTATTCCGTGATCCGAGAA
 GTGCTGCCAACCGCGCGGGCACGAACGTTCCGACCTCGAGGTCGAGGAAGTGCAC
 ACCCTCGTCGCCGAAGGGTTGTCTGTGCACACTGC (SEQ ID NO: 149)

Cas9-NLS-3xFLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKKFKVLGNTDRHSIKKNLIGALLFDSGE
 TAAATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLLESFLVEDKKHE
 RHPIPGNIVDEVAYHEKYPTIYHLRKKLVSDTKADLRILIYALAHMIKFRGHFLIEG
 DLNPDNSDVKLFILQVQTYNQLFEEENPINASGVDAKAILSARLSKSRRLLENLIAQLP
 GEKKNGLFNGLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYA
 DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVRQQLPE
 KYKEIFPDQSKNGYAGYIDGGASQEEFYKFKIPILKMDGTEELLVKLNREDLLRKQR
 TFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKILTFRIPIYVGLPARGNSRFA
 WMTRKSEETITPWNFEVVVKGASAQSFIERMTNFDKNLPNEKVLPHKSHLLYEYFTV
 YNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFD
 SVEISGVEDRFNASLGTYHDLKIIKDKDFLDNEENEDILEDIVLTLTLFPEDREMIERL
 KTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTIIDFLKSDGFANRN
 FMQLIHDDSLTFKEDIQKQVSGQGDLSLHEHIANLAGSPAIAKKGILQTVKVVDELVK
 VMGRHKPENIIVEMARENQTTQKGQKNSRERMKRIEIGIKELGSQILKEHPVENTQL
 QNEKLYLYLQNGRDMYVDQELDINRLSDYDVIDHIVPQSFLKDDSIDNKVLRSDK
 NRKSDNVPSSEVVKKMKNYWRQLLNAKLITQRKFDNLTKAERGLSELDKAGFIK
 RQLVETRQITKHVAQILD SRMNTKYDENDKLIREVVKVITLKSCLVSDFRKDFQFYKV
 REINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVVYDVRKMIKSEQEI
 GATKYPFYSNIMNPFKTEITLANGEIRKPLIETNGETGEIVWDKGRDFATVRKVLSP
 PQVNVVKKTEVQTTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVV
 AKVEKGSKKLKSVELLGITIMERSSEFEKNPIDFLEAKGYKEVKDILIKLPKYSLPE
 LENGRRMLASAGELQKGNELALPSKYVNFYLYLASHYEKLKGS PEDNBQKQLFVEQ
 PAAFKYFDTTIDRKRYTSTKEVLDTLIHQSI TGLYETRIDLSQLGGDGSPPKKRKRVS
 DYKDHGDYKDHIDYKDDDDKAAG (SEQ ID NO: 150)

Intein-Cas9 (C80) -NLS-3xFLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKKFKVLGNTDRHSIKKNLIGALLFDSGE
 TAAATRLKRTARRRYTRRKNRICLAEGTRI FPDVPTGTTTHRIEDVVDGRKPIHVVA
 KDGTLLARPVVSWFDQGTDRDVI GLRIAGGAIVWATPDHKVLTVEYGWRAAGELRKG
 DRVAGPGGSGNSLALSITADQMVSAALLDAEPPILYSEYDPTSPFSEASMMGLLTNLA
 DRELVHMINWAKRVPGFVDLTLDHQAHLLEAWLEILMIGLVWRSMEHPGKLLFAP
 NLLLDNRNQKCVGEMVEIFDMLLATSSRFRMMNLQGEFVCLKSIILLNSGVYTFLS
 STLKSLSEKDHIRALDKITDITLHMAKAGLTLOQQHQRLAQLLLILSHIRHMSNKG
 MEHLYSMKYKNVPLYDILLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHMDL
 AEGLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNCYLQEIFSNEMAKVDD
 SFHRLLESFLVEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVSDTKADLRILI
 YALAHMIKFRGHFLIEGDLNPDNSDVKLFILQVQTYNQLFEEENPINASGVDAKAIL
 SARLSKSRRLLENLIAQLPGEKKNGLFNGLIALSLGLTPNFKSNFDLAEDAKLQLSKDT
 YDDDLNLLAQIGDQYADLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEH
 HQDLTLKALVRQQLPEKYKEIFPDQSKNGYAGYIDGGASQEEFYKFKIPILKMDG
 TEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKIL
 TFRIPYVGLPARGNSRFAWMTRKSEETITPWNFEVVVKGASAQSFIERMTNFDKN
 LPNEKVLPHKSHLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRK
 VTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKIIKDKDFLDNEENEDILE
 DIVLTLTLFPEDREMIERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQ
 SGTITLDFLKSDFANRNFMQLIHDDSLTFKEDIQKQVSGQGDLSLHEHIANLAGSPA
 IAKKGILQTVKVVDELVKVMGRHKPENIIVEMARENQTTQKGQKNSRERMKRIEIGIK
 ELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDVIDHIVPQS
 FLKDDSIDNKVLRSDKNRGSNDNVPSSEVVKKMKNYWRQLLNAKLITQRKFDNLTK
 AERGLSELDKAGFIKRLVETRQITKHVAQILD SRMNTKYDENDKLIREVVKVITLKS
 KLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVVY
 DVRKMIKSEQEI GKATKYPFYSNIMNPFKTEITLANGEIRKPLIETNGETGEIVWD
 KGRDFATVRKVLSPQVNVVKKTEVQTTGGFSKESILPKRNSDKLIARKKDWDPKKY

Sequences

GGFDSPTVAYSVLVAVKVEKGSKKLKSVKELLGITIMERSSEFEKNPIDFLEAKGYKE
VKKDLIIKLPKYSLFELENGRKRMLASAGELQKGNELALPSKYVNFYLAHYEKLK
QAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLHQSIITGLYETRIDLSQ
LGGDGSPPKKRKVSDDYKDHGDYKDHIDYKDDDDKAAG (SEQ ID NO: 151)

Intein-Cas9 (A127) -NLS-3×FLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGE
TAEATRLKRTARRRYTRRNRI CYLQEIFSNEAKVDDSFHRLSEESFLVEEDKKHE
RHPIFGNIVDEVCLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVAADGTLLARPVV
SWFDQGRDVI GLRIAGGAIVWATPDHKVLT EYGWRAAGELRKGDRAVAGPGSGN
SLALSLTADQMVSALLDAEPPILYSEYDPTSPFSEASMMGLTLNLADRELVHMINWA
KRVPGFVDLT LHDQAHLLERAWLEILMIGLVWRSMHPGKLLFAPNLLDRNQKGC
VEGMVEIFDMLLATSSRFMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHI
HRAIDKITTDTLIHLMAKAGLTQQQHQRLAQLLLILSHIRHMSNKGMEHLYSMKYK
NVVPLYDLLLLLEMLDAHRLHAGSGASRVQAFADALDDKFLHDMLEGLRYSVIRE
VLPTRRARTFDLEVEELHTLV AEGVNVHNCYHEKYPTIYHLRKLKLVSDTKADLRLLI
YLALAHMIKFRGHFLIEGDLNPDNSDVKLFQVLQVTYNQLFEENPINASGVDAKAIL
SARLSKSRRLENLIAQLPGEKKNGLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKDT
YDDDLNLLAQGDQYADLF LAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEH
HQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDG
TEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKIL
TPRIPIYVGPLARGNSRFAMWTRKSEETITPWNFEVVDKGASQSFIERMTNFDKN
LNEKVLPHKSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRK
VTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKKI IKDKDFLDNEENEDILE
DIVLTTLTFEDREMIERLKT YAHLFDDKVMKQLKRRRYTGWRLSRKLINGIRDKQ
SGKTLDFLKSDGFANRNFMQLIHDDSLTFKEDIQKAQVSGQDLSLHEHIANLAGSPA
IKKGILQTVKVVDELVKVMGRHKPENIVIMARENQTTQKGQKNSRERMKRIEKGK
ELGSGILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDVDHIVPQSF
LKDDSIDNKVLT RSDKNRGKSDNVPSEEVKKMKNYWRQLNNAKLITQRKFDNLTK
AERGGSELDAKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKS
KLVSDFRQDFQYKVEINNYHHAHDAYLNAVVGTAIIKKYPKLESEFVYGDYKVY
DVRKMIKSEQIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWD
KGRDFATVRKVL SMPQVNI VKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKY
GGFDSPTVAYSVLVAVKVEKGSKKLKSVKELLGITIMERSSEFEKNPIDFLEAKGYKE
VKKDLIIKLPKYSLFELENGRKRMLASAGELQKGNELALPSKYVNFYLAHYEKLK
GSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVILADANLDKVL SAYNKHDKPIRE
QAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLHQSIITGLYETRIDLSQ
LGGDGSPPKKRKVSDDYKDHGDYKDHIDYKDDDDKAAG (SEQ ID NO: 152)

Intein-Cas9 (T146) -NLS-3×FLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGE
TAEATRLKRTARRRYTRRNRI CYLQEIFSNEAKVDDSFHRLSEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKLKLVSDCLAEGTRIFDPVTGTTTHRIEDVVDGRK
PIHVVAADGTLLARPVVSWFDQGRDVI GLRIAGGAIVWATPDHKVLT EYGWRA
AGELRKGDRAVAGPGSGNSLALSLTADQMVSALLDAEPPILYSEYDPTSPFSEASMM
GLTLNLADRELVHMINWAKRVPGFVDLT LHDQAHLLERAWLEILMIGLVWRSMH
PGKLLFAPNLLDRNQKGCVEGMVEIFDMLLATSSRFMMNLQGEFVCLKSIILLN
SGVYTFLSSTLKSLEEKDHIHRAIDKITTDTLIHLMAKAGLTQQQHQRLAQLLLILSHI
RHMSNKGMEHLYSMKYKNVVPLYDLLLLLEMLDAHRLHAGSGASRVQAFADALDD
KFLHDMLEGLRYSVIREVLPTRRARTFDLEVEELHTLV AEGVNVHNCADLRLLIY
LALAHMIKFRGHFLIEGDLNPDNSDVKLFQVLQVTYNQLFEENPINASGVDAKAIL
ARLSKSRRLENLIAQLPGEKKNGLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKDTY
DDDLNLLAQGDQYADLF LAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEH
QDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTE
ELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKIL
RIPYVGPLARGNSRFAMWTRKSEETITPWNFEVVDKGASQSFIERMTNFDKNLP
NEKVLPHKSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKV
TVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLKKI IKDKDFLDNEENEDILED
IVLTTLTFEDREMIERLKT YAHLFDDKVMKQLKRRRYTGWRLSRKLINGIRDKQS
GKTLDFLKSDGFANRNFMQLIHDDSLTFKEDIQKAQVSGQDLSLHEHIANLAGSPA
IKKGILQTVKVVDELVKVMGRHKPENIVIMARENQTTQKGQKNSRERMKRIEKGK
ELGSGILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDVDHIVPQSF
LKDDSIDNKVLT RSDKNRGKSDNVPSEEVKKMKNYWRQLNNAKLITQRKFDNLTK
AERGGSELDAKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKS
KLVSDFRQDFQYKVEINNYHHAHDAYLNAVVGTAIIKKYPKLESEFVYGDYKVY
DVRKMIKSEQIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWD
KGRDFATVRKVL SMPQVNI VKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKY
GGFDSPTVAYSVLVAVKVEKGSKKLKSVKELLGITIMERSSEFEKNPIDFLEAKGYKE
VKKDLIIKLPKYSLFELENGRKRMLASAGELQKGNELALPSKYVNFYLAHYEKLK
QAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLHQSIITGLYETRIDLSQ
LGGDGSPPKKRKVSDDYKDHGDYKDHIDYKDDDDKAAG (SEQ ID NO: 153)

Intein-Cas9 (S219) -NLS-3×FLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKFKVLGNTDRHSIKKNLIGALLFDSGE
TAEATRLKRTARRRYTRRNRI CYLQEIFSNEAKVDDSFHRLSEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKLKLVSDTKADLRLLIYLALAHMIKFRGHFLIEG
DLNPDNSDVKLFQVLQVTYNQLFEENPINASGVDAKAIL SARLSKCLAEGRIFDPV

Sequences

TGTTHRIEDVVDGRKPIHVVAADKGTLLARPVVSFWDQGTDRDVI GLRIAGGAIVW
 ATPDHKVLTEYGWRAAGELRKGDVAGPGGSGNSLALS LTADQMVSALLDAEPPIL
 YSEYDPTSPFSEASMMGLLTNLADRELVHMINWAKRVPGFVDLT LHDQAHLLERA
 WLEILMIGLVWRSMHPGKLLFAPNLLLDNRNQGKCV EGMVEIFDMLLATSSRFRMM
 NLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRA LDKITDTLIHLMAKAGLTL
 QQQHQRLAQQLLLILSHIRHMSNKGMEHLYSMKYKNV VPLYDLLLLLEMLDAHRLHAG
 GSGASRVQAFADALDDKFLHDMLEGLRYSVIREVLP TRRARTFDLEVEELHTLVAE
 GVVVHNCRRLENLIAQLPGEKKNGLFGNLIASLGLTP NFKSNFDLAEDAKLQLSKD
 TYDDDLNLLAQIGDQYADLFLAAKNLSDAILLSDILRV NTEITKAPLSASMIKRYDE
 HHQDLTLKALVRQQLPEKYKEIFPDQSKNGYAGYID GGASQEEFYKFIKPILEKMD
 GTEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAIL RRQEDFYFPLKDNREKIEKI
 LTFRIPIYYVGPLARGNSRFAWMTRKSEETITPWNFE EVVDKGASAQSFIERMTNFDK
 NLPNEKVLPHKSLLEYEFTVYNELTKVKYVTEGMRKPA FLSGEQKKAIVDLLFKTNR
 KVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGT YHDLKIKDKDFLDNEENEDI
 LEDIVLTLTLFEDREMIERLKYAHLFDDKVMKQLKR RRYTGWGRLSRKLINGIRD
 KQSGKTILDFLKSDFANRNFMLIHDDSLTFKEDIQKA QVSGQGDLSHEHIANLAG
 SPAIKKGIQLQTVKVVDELVKVMGRHKPENIVIEMARE NQTTQKGQKNSRERMKRIEE
 GIKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMY VVDQELDINRLSDYDVHDHIVP
 QSFLKDDSDINKVLTRSDKNRGKSDNVPSEEVVKMK NMYWRQLLNAKLITQKRFDN
 LTKAERGLSELDKAGFIKRQLVETRQITKHVAQILDS RMNTKYDENDKLIREVKITLKS
 KLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGTA LIKKYPKLESEFVYGDY
 KVDYVRKMIKSEQEI GKATAKYFFYSNIMNFFKTEIT LANGEIRKRPLIETNGETGEI
 VWDKGRDFATVRKVL SMPQVNIKKTEVQTTGGFSKES ILPKRNSDKLIARKDWDPKKY
 GGFDSPTVAYSVLVVAKEGKSKKLKSVKELLGITIM ERSSFEKNPIDFLEAK
 GYKEVKKDLIKLPHYSLFELENGRKRMLASAGELQKG NELALPSKYVNFLYLASHY
 PIREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYT STKEVLDTLIIHQSI TGLYETRI
 DLSQLGGDGSPPKKRKVSSDYKDHDGDYKDHDIDYK DDDDKAAG (SEQ ID NO: 154)

Intein-Cas9 (T333) -NLS-3×FLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVP SKKFKVLGNTDRHSIKKNLIGALLFDSGE
 TABATRLKRTARRRYTRRKNRICYLQEIFSNEMAKV DDFSFFHRLSEESFLVEEDKKHE
 RHPFGNIVDEVAYHEKYPTIYHLRKKLVSDTDKADLRLI YLALAHMIKFRGHFLIEG
 DLNPDNSDVKLFILQVQTYNQLFEENPINASGVDAKAIL SARLSKSRRLENLIAQLP
 GEKKNGLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKD TYDDDLNLLAQIGDQYA
 DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRY DEHHQDLCLAEGRIFDPVTG
 TTHRIEDVVDGRKPIHVVAADKGTLLARPVVSFWDQGT DRDVI GLRIAGGAIVWATP
 DHKVLTEYGWRAAGELRKGDVAGPGGSGNSLALS LTADQMVSALLDAEPPILYSE
 YDPTSPFSEASMMGLLTNLADRELVHMINWAKRVPGF VDLTLHDQAHLLERAWLEI
 LMIGLVWRSMHPGKLLFAPNLLLDNRNQGKCV EGMVEIFDMLLATSSRFRMMNLQ
 GEEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRA LDKITDTLIHLMAKAGLTLQQQ
 HQRLAQQLLLILSHIRHMSNKGMEHLYSMKYKNV VPLYDLLLLLEMLDAHRLHAGSG
 ASRVQAFADALDDKFLHDMLEGLRYSVIREVLP TRRARTFDLEVEELHTLVAEV
 VVHNCRRLENLIAQLPGEKKNGLFGNLIASLGLTP NFKSNFDLAEDAKLQLSKD
 TYDDDLNLLAQIGDQYADLFLAAKNLSDAILLSDILRV NTEITKAPLSASMIKRYDE
 HHQDLTLKALVRQQLPEKYKEIFPDQSKNGYAGYID GGASQEEFYKFIKPILEKMDG
 GTEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAIL RRQEDFYFPLKDNREKIEKI
 LTFRIPIYYVGPLARGNSRFAWMTRKSEETITPWNFE EVVDKGASAQSFIERMTNFDK
 NLPNEKVLPHKSLLEYEFTVYNELTKVKYVTEGMRKPA FLSGEQKKAIVDLLFKTNR
 KVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGT YHDLKIKDKDFLDNEENEDI
 LEDIVLTLTLFEDREMIERLKYAHLFDDKVMKQLKR RRYTGWGRLSRKLINGIRDKQ
 SGTILDFLKSDFANRNFMLIHDDSLTFKEDIQKA QVSGQGDLSHEHIANLAGSPA
 IKKGIQLQTVKVVDELVKVMGRHKPENIVIEMARE NQTTQKGQKNSRERMKRIE
 GIKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMY VVDQELDINRLSDYDVHDHIVP
 QSFLKDDSDINKVLTRSDKNRGKSDNVPSEEVVKMK NMYWRQLLNAKLITQKRFDN
 LTKAERGLSELDKAGFIKRQLVETRQITKHVAQILDS RMNTKYDENDKLIREVKITLKS
 KLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGTA LIKKYPKLESEFVYGDYK
 VYDVRKMIKSEQEI GKATAKYFFYSNIMNFFKTEIT LANGEIRKRPLIETNGETGEI
 VWDKGRDFATVRKVL SMPQVNIKKTEVQTTGGFSKES ILPKRNSDKLIARKDWDPKKY
 GGFDSPTVAYSVLVVAKEGKSKKLKSVKELLGITIM ERSSFEKNPIDFLEAKGYKE
 VKKDLIKLPHYSLFELENGRKRMLASAGELQKG NELALPSKYVNFLYLASHYKELK
 GSPEDNEQKQLFVEQHKHYLDEIEIQISEFSKRVI LADANLDKVL SAYNKHDKPIRE
 QAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYT STKEVLDTLIIHQSI TGLYETRI
 DLSQLGGDGSPPKKRKVSSDYKDHDGDYKDHDIDYK DDDDKAAG (SEQ ID NO: 155)

Intein-Cas9 (T519) -NLS-3×FLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVP SKKFKVLGNTDRHSIKKNLIGALLFDSGE
 TABATRLKRTARRRYTRRKNRICYLQEIFSNEMAKV DDFSFFHRLSEESFLVEEDKKHE
 RHPFGNIVDEVAYHEKYPTIYHLRKKLVSDTDKADLRLI YLALAHMIKFRGHFLIEG
 DLNPDNSDVKLFILQVQTYNQLFEENPINASGVDAKAIL SARLSKSRRLENLIAQLP
 GEKKNGLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKD TYDDDLNLLAQIGDQYA
 DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRY DEHHQDLTLKALVRQQLPE
 KYKEIFPDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQ
 RTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKI LTFRIPIYYVGPLARGNSRFA
 WMTRKSEETITPWNFEVVDKGASAQSFIERMTNFDKNLPNEKVLPHKSLLEYEYFCL
 AEGTRIFDPVTGTTHRIEDVVDGRKPIHVVAADKGTLLARPVVSFWDQGTDRDVI GL
 RIAGGAIVWATPDHKVLTEYGWRAAGELRKGDVAGPGGSGNSLALS LTADQMVS
 ALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRELVHMINWAKRVPGFVDLT LHD
 QAHLLERAWLEILMIGLVWRSMHPGKLLFAPNLLLDNRNQGKCV EGMVEIFDMLLA
 TSSRFRMMNLQGEFVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRA LDKITDTLIHL

Sequences

KSKLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYK
 VYDVRKMIKSEQEQIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIV
 WDKGRDPATVRKVLSPQVNIKKTEVQTTGGFSKESILPKRNSDKLIARKKDWDPK
 KYGGFDSPTVAYSVLVAKVEKGSKKLKSVKELLGITIMERSSEFKNPIDFLEAKG
 YKEVKKDLIIKLPKYSLEFLENGRKRMLASAGELQKGNELALPSKYVNFYLYASHYE
 KLKGSPEDEQKQLFVEQHKHYLDEIEQISEFSKRVILADANLDKVL SAYNKHDKP
 IREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQSI TGLYETRID
 LSQ LGGDGSPKKRKVSDDYKDHDGDYKDHDIDYKDDDDKAAG (SEQ ID NO: 158)

Intein-Cas9 (S701) -NLS-3×FLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKPKVLGNTDRHSIKKNLIGALLFDSGE
 TAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHE
 RHP I PGNIVDEVAYHEKYPTIYHLRKKLVSDTKADLRLIYLALAHMIKFRGHFLIEG
 DLNPDNSDVKLFIQLVQTYNQLFEENPINASGVDAKAIL SARLSKSRRLLENLIAQLP
 GEKKNGFLGNLIALSLGLTPNFKS NFDAEDAKLQLSKDTYDDDLNLLAQIGDQYA
 DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVRQQLPE
 KYKEIFPDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQR
 TFDNGSIPHQIHLGELHAILRRQEDFYPPFLKDNREKIEKILTFRIPIYVGPLARGNSRFA
 WMTRKSEETITPWNPEEVVDKGASAQSPIERMTNFDKNLPNEKVL PKHSLLEYFTV
 YNELTKVKVYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFD
 SVEISGVEDRFNLSGTYHDLKIIKD KDFLDNEENEDILEDIVLTTLTFEDREMI EERL
 KTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQSGKTI LDFLKS DGFANRN
 FMQLIHDDCLAEGRIFDPVTGTTHRIEDVVDGRKPIHVVAADGDTLLARPVVSF
 DQGTDRDVI GLRIAGGAI VWATPDHKVLTEYGWRAAGELRKGRVAGPGGSGNSLA
 LSLTADQMVSALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADRELVHMINWAKRV
 PGFVDLTLDHQAHLLERAWLEILMIGLVWRSMHPGKLLFAPNLLDRNQKCVEG
 MVEIFDMLLATSSRFMMNLQGEFEVCLKSIILLNSGVYTFLSSTLKSLEEKDHIHRA
 LDKITDTLIHLMAGKAGTLQOQHQRALQALLILSHIRHMSNKGMEHLYSMKYKNVV
 PLYDLLLEMLDAHRLHAGGSGASRVQAFADALDDKFLHDM LAEGLRYSVIREVLPT
 RRARTFDLEVEELHTLVAEGVVHNCITFKEDIQKAQVSGQGDSLHEHIANLAGSPA
 I KKGILQTVKVVDELVKVMGRHKPENI VIEMARENQTTQKGQKNSRERMKRIE EGIK
 ELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDVHDIVPQS F
 LKDDSIDNKVLT RSDKNRGKSDNVPSEEVVKKMKNYWRQLLNAKLITQRKFDNLTK
 AERGGSELDDKAGFIKRLVETRQITKHVAQILD SRMNTKYDENDKLIREVKVITLKS
 KLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYK
 VYDVRKMIKSEQEQIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVD
 KGRDPATVRKVLSPQVNIKKTEVQTTGGFSKESILPKRNSDKLIARKKDWDPK
 KYGGFDSPTVAYSVLVAKVEKGSKKLKSVKELLGITIMERSSEFKNPIDFLEAKG
 YKEVKKDLIIKLPKYSLEFLENGRKRMLASAGELQKGNELALPSKYVNFYLYASHYEKLK
 GSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVILADANLDKVL SAYNKHDKP
 IREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQSI TGLYETRIDLSQ
 LGGDGSPKKRKVSDDYKDHDGDYKDHDIDYKDDDDKAAG (SEQ ID NO: 159)

Intein-Cas9 (A728) -NLS-3×FLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKPKVLGNTDRHSIKKNLIGALLFDSGE
 TAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHE
 RHP I PGNIVDEVAYHEKYPTIYHLRKKLVSDTKADLRLIYLALAHMIKFRGHFLIEG
 DLNPDNSDVKLFIQLVQTYNQLFEENPINASGVDAKAIL SARLSKSRRLLENLIAQLP
 GEKKNGFLGNLIALSLGLTPNFKS NFDAEDAKLQLSKDTYDDDLNLLAQIGDQYA
 DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKALVRQQLPE
 KYKEIFPDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQR
 TFDNGSIPHQIHLGELHAILRRQEDFYPPFLKDNREKIEKILTFRIPIYVGPLARGNSRFA
 WMTRKSEETITPWNPEEVVDKGASAQSPIERMTNFDKNLPNEKVL PKHSLLEYFTV
 YNELTKVKVYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFD
 SVEISGVEDRFNLSGTYHDLKIIKD KDFLDNEENEDILEDIVLTTLTFEDREMI EERL
 KTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQSGKTI LDFLKS DGFANRN
 FMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANCLAEGRIFDPVTGTTHRIEDVV
 DGRKPIHVVAADGDTLLARPVVSF DQGTDRDVI GLRIAGGAI VWATPDHKVLTEY
 GWRAAGELRKGRVAGPGGSGNSLALSLTADQMVSALLDAEPPILYSEYDPTSPFSE
 ASMMGLLTNLADRELVHMINWAKRVPGFVDLTLDHQAHLLERAWLEILMIGLVWR
 SMEHPGKLLFAPNLLDRNQKCVEGMVEIFDMLLATSSRFMMNLQGEFEVCLKSI
 IILLNSGVYTFLSSTLKSLEEKDHIHRA LDKITDTLIHLMAGKAGTLQOQHQRALQALL
 ILSHIRHMSNKGMEHLYSMKYKNVVPLYDLLLEMLDAHRLHAGGSGASRVQAFAD
 ALDDKFLHDM LAEGLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVHNCISPAI
 KKGILQTVKVVDELVKVMGRHKPENI VIEMARENQTTQKGQKNSRERMKRIE EGIK
 ELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDVHDIVPQS F
 LKDDSIDNKVLT RSDKNRGKSDNVPSEEVVKKMKNYWRQLLNAKLITQRKFDNLTK
 AERGGSELDDKAGFIKRLVETRQITKHVAQILD SRMNTKYDENDKLIREVKVITLKS
 KLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYK
 VYDVRKMIKSEQEQIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVD
 KGRDPATVRKVLSPQVNIKKTEVQTTGGFSKESILPKRNSDKLIARKKDWDPK
 KYGGFDSPTVAYSVLVAKVEKGSKKLKSVKELLGITIMERSSEFKNPIDFLEAKG
 YKEVKKDLIIKLPKYSLEFLENGRKRMLASAGELQKGNELALPSKYVNFYLYASHYEKLK
 GSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVILADANLDKVL SAYNKHDKP
 IREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQSI TGLYETRIDLSQ
 LGGDGSPKKRKVSDDYKDHDGDYKDHDIDYKDDDDKAAG (SEQ ID NO: 160)

Sequences

Intein-Cas9 (T995) -NLS-3xFLAG:
MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKKFKVLGNTDRHSIKKNLIGALLFDSGE
TAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFPHRLEESFLVEEDKKHE
RHP1PGNIVDEVAYHEKYPTIYHLRKKLVDSITDKADLRILIYALAHMIKFRGHFLIEG
DLNPDNSDVKLFIQLVQTYNQLFEENPINASGVDKAILSARLSKSRRLLENLIAQLP
GEKKNGLFGNLIALSLGLTPNFKSINFDAEDAKLQLSKDTYDDDLNLLAQIGDQYA
DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPE
KYKEIFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQR
TFDNGSIPHQIHLGELHAILRRQEDFYPPFLKDNREKIEKILTFRIPIYVGLARGNSRFA
WMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPHKSHLLYEYFTV
YNELTKVKVVTTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFD
SVEISGVEDRFNLSLGTTHDLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMIERL
KTYAHLFDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTIIDFLKSDGFANRN
FMQLIHDDSLTFKEDIQKAQVSGQGSLEHEIANLAGSPAIIKKGILQTVKVVDELVK
VMGRHKPENIIVIMARENQTTQKGQKNSRERMKRIEIEGKELGSQLKEHPVENTQL
QNEKLYLYYLQNGRDMYVDQELDINRLSDYDVHDIVPQSFLKDDSIDNKVLTSDK
NRGKSDNVPSEEVVKMKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIK
RQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKV
REINNYHHAHDAYLNAVVGCLAEGTRIFDPVTGTTHRIEDVVDGRKPIHVAAAAD
GTLARPVVSWFDQGTDRDVIGLRIAGGAIVWATPDHKVLTTEYGWRAAGELRKGR
VAGPGGSGNSLALSITADQMVSAALLDAEPPILYSEYDPTSPFSEASMMGLLTNLADR
ELVHMINWAKRVPGFVDLTLDHQAHLLERAWLEILMIGLVWRSMHHPGKLLFAPNL
LLDRNQGKCEVGMVEIFDMLLATSSFRFRMNLQGEFVCLKSIILLNSGVYTFLSSTL
KSLEEKDHIHRALDKITDTLIHLMAKAGLTQQQHQRLAQLLLILSHIRHMSNKGME
HLYSMKYKNVPLDYDLLLEMLDAHLRHAGGSGASRVQAFADALDDKFLHDMALAE
GLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNCALIKKYPKLESEFVYGDY
KVYDVRKMIKSEQEI GKATKAYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEI
VWDKGRDFATVRKVLSPQVNIIVKTEVQTGGFSKESILPKRNSDKLIARKKDWDP
KYGGSFPTVAYSVLVAKVEKGSKKLSVKELLGITIMERSSEFKNPIDFLEAK
GYKEVKDLIIKLPKYSLEFELENGRKRMLASAGELQKGNELALPSKYVNFLYLASHY
PIREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQSIITGLYETRI
DLSQLGGDGSPPKKRKYSSDYKDHDGDYKDHDIDYKDDDDKAAG (SEQ ID NO: 161)

Intein-Cas9 (S1006) -NLS-3xFLAG:
MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKKFKVLGNTDRHSIKKNLIGALLFDSGE
TAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFPHRLEESFLVEEDKKHE
RHP1PGNIVDEVAYHEKYPTIYHLRKKLVDSITDKADLRILIYALAHMIKFRGHFLIEG
DLNPDNSDVKLFIQLVQTYNQLFEENPINASGVDKAILSARLSKSRRLLENLIAQLP
GEKKNGLFGNLIALSLGLTPNFKSINFDAEDAKLQLSKDTYDDDLNLLAQIGDQYA
DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPE
KYKEIFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQR
TFDNGSIPHQIHLGELHAILRRQEDFYPPFLKDNREKIEKILTFRIPIYVGLARGNSRFA
WMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPHKSHLLYEYFTV
YNELTKVKVVTTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFD
SVEISGVEDRFNLSLGTTHDLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMIERL
KTYAHLFDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTIIDFLKSDGFANRN
FMQLIHDDSLTFKEDIQKAQVSGQGSLEHEIANLAGSPAIIKKGILQTVKVVDELVK
VMGRHKPENIIVIMARENQTTQKGQKNSRERMKRIEIEGKELGSQLKEHPVENTQL
QNEKLYLYYLQNGRDMYVDQELDINRLSDYDVHDIVPQSFLKDDSIDNKVLTSDK
NRGKSDNVPSEEVVKMKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIK
RQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKV
REINNYHHAHDAYLNAVVGTAIIKKYPKLECLAEGTRIFDPVTGTTHRIEDVVDGRK
PIHVAAAADGTLARPVVSWFDQGTDRDVIGLRIAGGAIVWATPDHKVLTTEYGWRA
AGELRKGRDVAGPGGSGNSLALSITADQMVSAALLDAEPPILYSEYDPTSPFSEASMM
GLLTNLADRELHMINWAKRVPGFVDLTLDHQAHLLERAWLEILMIGLVWRSMH
PGKLLFAPNLLEDNRNQGKCEVGMVEIFDMLLATSSFRFRMNLQGEFVCLKSIILLN
SGVYTFLSSTLKSLEEKDHIHRALDKITDTLIHLMAKAGLTQQQHQRLAQLLLILSHI
RHMSNKGMEHLYSMKYKNVPLDYDLLLEMLDAHLRHAGGSGASRVQAFADALDD
KFLHDMALAEGLRYSVIREVLPTRRARTFDLEVEELHTLVAEGVVVHNCVYVYDYK
VYDVRKMIKSEQEI GKATKAYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIV
WDKGRDFATVRKVLSPQVNIIVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPK
KYGGSFPTVAYSVLVAKVEKGSKKLSVKELLGITIMERSSEFKNPIDFLEAKG
YKEVKDLIIKLPKYSLEFELENGRKRMLASAGELQKGNELALPSKYVNFLYLASHY
KLKGSPEDEQQLFVEQHKHYLDEIIIEQISEFSKRVIDADANLDKVLISAYNKHDKP
IREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQSIITGLYETRI
LSQLGGDGSPPKKRKYSSDYKDHDGDYKDHDIDYKDDDDKAAG (SEQ ID NO: 162)

Intein-Cas9 (S1154) -NLS-3xFLAG:
MDKKYSIGLDIGTNSVGWAVITDEYKVPSSKKFKVLGNTDRHSIKKNLIGALLFDSGE
TAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFPHRLEESFLVEEDKKHE
RHP1PGNIVDEVAYHEKYPTIYHLRKKLVDSITDKADLRILIYALAHMIKFRGHFLIEG
DLNPDNSDVKLFIQLVQTYNQLFEENPINASGVDKAILSARLSKSRRLLENLIAQLP
GEKKNGLFGNLIALSLGLTPNFKSINFDAEDAKLQLSKDTYDDDLNLLAQIGDQYA
DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPE
KYKEIFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQR
TFDNGSIPHQIHLGELHAILRRQEDFYPPFLKDNREKIEKILTFRIPIYVGLARGNSRFA
WMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPHKSHLLYEYFTV

Sequences

YNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFD
 SVEISGVEDRFNASLGTYHDLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMI EERL
 KTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQKSGKTI LDFLKSDGFANRN
 FMQLIHDDSLTFKEDIQKAQVSGQGDSLHEIHANLAGSPAIIKKGILQTVKVVDELVK
 VMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEBGIKELGSQILKEHPVENTQL
 QNEKLYLYYLQNGRDMYVDQELDINRLSDYDVHDHVPQSFLKDDSDNKNVLRSDK
 NRGKSDNVPSEEVVKMKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFTK
 RQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKV
 REINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVYDVRKMIAKSEQEIGK
 ATAKYFFYSNIMNPFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVL SM
 PQVNI VVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLV
 AKVEKGKCLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVA AAKDGTLLARPVVSWF
 DQGTDRDVI GLRIAGGAI VWPATPDHKVLT EYGWRAAGELRKGD RVAGPGGSGNSLA
 LSLTADQMVSA LLD AEPPI LYSEYDPTSPFSEASMMGLLTNLADREL VHMINWAKRV
 PGFVDLT LHDQAHL LERAWLEILMIGLVWRSM EHPGKLLFAPNLLDRNQK CVEG
 MVEIFDMLLATSSRFRMMNLQGEFVCLKSI ILLNSGVYTFLSSTLKSLEEKDHIHRA
 LDKITDTLTHLMAKAGLTQQQHQLAQLLILSHIRHMSNKGMEHLYSMKYKNV
 PLYDLLLEMLDAHRLHAGSGASRVQAFADALDDKFLHDM LAEGLRYSVIREVLP
 RRARPTFDLEVEELHTLV AEGVVVHNCCKLKS VKELLGITIMERS SFEKNPIDFLEAKG
 YKEVKKDLITIKLPKYSLFEL ENGRKRMLASAGELQKGNELALPSKYVNFPLYLASHYE
 IREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQSI TGLYETRID
 LSQLG DGSPKKRKVSSDYKDHDGDYKDHDIDYKDDDDKAAG (SEQ ID NO: 163)

Intein-Cas9 (S1159) -NLS-3xFLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVP SKFKVLGNTDRHSI KKNLIGALLFDSGE
 TAEATRLKRTARRRYTRRKNRICYLQE IFSNEMAKVDDSFHRL EESFLVEEDKKHE
 RHPIPGNI VDEVAYHEKYPTIYHLRKKLV DSTDKADLR LIYLALAHMIKFRGHFLIEG
 DLNPNDSVDVKLFIQLVQTYNQLFEENPINASGVDAKAIL SARLSKSRRL ENLIAQLP
 GEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYA
 DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPE
 KYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQR
 TFDNGSIHQIHLGELHAILRRQEDFY PFLKDNREKIEKILTFRI PYVVGPLARGNSRFA
 WMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVL PKHSLLEYEFTV
 YNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFD
 SVEISGVEDRFNASLGTYHDLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMI EERL
 KTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQKSGKTI LDFLKSDGFANRN
 FMQLIHDDSLTFKEDIQKAQVSGQGDSLHEIHANLAGSPAIIKKGILQTVKVVDELVK
 VMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEBGIKELGSQILKEHPVENTQL
 QNEKLYLYYLQNGRDMYVDQELDINRLSDYDVHDHVPQSFLKDDSDNKNVLRSDK
 NRGKSDNVPSEEVVKMKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFTK
 RQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKV
 REINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVYDVRKMIAKSEQEIGK
 ATAKYFFYSNIMNPFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVL SM
 PQVNI VVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLV
 AKVEKGKSKKLKCLAEGTRIFDPVTGTTTHRIEDVVDGRKPIHVVA AAKDGTLLARPV
 VSWFDQGTDRDVI GLRIAGGAI VWPATPDHKVLT EYGWRAAGELRKGD RVAGPGGSG
 NSLALS L TADQMVSA LLD AEPPI LYSEYDPTSPFSEASMMGLLTNLADREL VHMINW
 AKRVPGFVDLT LHDQAHL LERAWLEILMIGLVWRSM EHPGKLLFAPNLLDRNQK
 CVEGMVEIFDMLLATSSRFRMMNLQGEFVCLKSI ILLNSGVYTFLSSTLKSLEEKDH
 ITHRALDKITDTLTHLMAKAGLTQQQHQLAQLLILSHIRHMSNKGMEHLYSMKYK
 NVVPLYDLLLEMLDAHRLHAGSGASRVQAFADALDDKFLHDM LAEGLRYSVIRE
 VLPTRRARTFDLEVEELHTLV AEGVVVHNCVKEL LGITIMERS SFEKNPIDFLEAKGY
 KEVKKDLITIKLPKYSLFEL ENGRKRMLASAGELQKGNELALPSKYVNFPLYLASHYEK
 LKGPSPEDNEQQLFVEQHKHYLDEIEEQISEFSKRVILADANL DKVLSAYNKH RDKPI
 REQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVL DATLIHQSI TGLYETRID
 LSQLG DGSPKKRKVSSDYKDHDGDYKDHDIDYKDDDDKAAG (SEQ ID NO: 164)

Intein-Cas9 (S1274) -NLS-3xFLAG:

MDKKYSIGLDIGTNSVGWAVITDEYKVP SKFKVLGNTDRHSI KKNLIGALLFDSGE
 TAEATRLKRTARRRYTRRKNRICYLQE IFSNEMAKVDDSFHRL EESFLVEEDKKHE
 RHPIPGNI VDEVAYHEKYPTIYHLRKKLV DSTDKADLR LIYLALAHMIKFRGHFLIEG
 DLNPNDSVDVKLFIQLVQTYNQLFEENPINASGVDAKAIL SARLSKSRRL ENLIAQLP
 GEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYA
 DLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPE
 KYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQR
 TFDNGSIHQIHLGELHAILRRQEDFY PFLKDNREKIEKILTFRI PYVVGPLARGNSRFA
 WMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVL PKHSLLEYEFTV
 YNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFD
 SVEISGVEDRFNASLGTYHDLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMI EERL
 KTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQKSGKTI LDFLKSDGFANRN
 FMQLIHDDSLTFKEDIQKAQVSGQGDSLHEIHANLAGSPAIIKKGILQTVKVVDELVK
 VMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEBGIKELGSQILKEHPVENTQL
 QNEKLYLYYLQNGRDMYVDQELDINRLSDYDVHDHVPQSFLKDDSDNKNVLRSDK
 NRGKSDNVPSEEVVKMKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFTK
 RQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKV
 REINNYHHAHDAYLNAVVGITALIKKYPKLESEFVYGDYKVYDVRKMIAKSEQEIGK
 ATAKYFFYSNIMNPFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVL SM

-continued

Sequences

PQVNIYKKTVEVQTGGFSKESILPKRNSDKLIARKKDWDPKXYGGFDSPTVAYSVLVV
AKVEKSKSKKLKSVKELLGTTIMRSSSEKPNIDFLAEAKGYKEVKDQLIKLPKYSLF
LENGRKRMLASAGELQKGNELAPSKYVNFILYLAHSEKYGKSPEDNEBQKLFVBEQ
KHXYLDLIEIQICLAGETRIFPDVTGTHRIEDVVDRKGPJHVVAADKQDGLLPARPV
SWFDQGTRDVIGLRIAGGAIVWATPDHKVLTEYGWRAAGELRGDRVAGPGSGN
SLALSITADQMVSALLDAEPPILYSEYDPTSPFSBASMMGLLTNLADRELVHMNIWA
KRVPFGFVDTLHDDQAHLERAWLELIMLIGLVWSRMEHPGKLLFAPNLLDRNQK
VEGMVIEFDMLLTSSRPFRRMMLQGEFVCLKSIILNLSGVYTFPLSSLTSLGKECDHI
HRALDKITDTLILHLMAKAGLTQQQHQRLAQLLLLILSHIRHMSNKGMEHLYSMKYK
NVVPVLYDLLEMLDAHRLHLAGGSASRVQAFADALDKFLHDMLAELGRYSVIRE
VLPTRARNTFDLEVEELHTLVAEGSVVFNHCEFSKRVLTADANLKVLSAYNKHDRK
PIREQAANIILHFTLTNLGAPAAKFYVDTHIDRKRYTSKEVLDATLHQSIITGLYETRI
DLSQLGDDGSPKKKKRVSSDYKDHDGDYKDHDYKDDDDKAAG (SEQ ID NO: 165)

Intein-Cas9 (S219-**G521R**)) -NLS-3xFLAG:
MDKKYSIIGLDIGTNSVGWAVITDEYKVPSPKFKVLGNTRDHSIKNKLIGALLPDSGE
TAEATIRKTRARRYTRRKNRITCYLQEIFSNEMAKVDSFFHRLESFLVEEDKKHE
RHPIFGNIVDVEAYHEKYPTIYHLRKLRVSDTKADKLRLIYLALAHMIKFGPHLIEG
DLNPDNSDVKLPIQLVQTYNQLFEEENPINASGVDAKAIIKSARLSKCLAEGRTRIFDPV
TGTTHTRIEDVVDGRKPIHVVAAAKDGTLLARPVVSWPFGQGTDRVIGLRIAGGATVW
ATPDHKVLVTEYGWRAAGELRKGRDVAGPGGSSNLSLSTADQMVSALLDAEPIIL
YSEYDPTSPFSEASMMGLLTNLADRELGHVMINAWKRVPGFVDLTLDHQAHLLERA
WLEILMTLIGLVWRSMEHFGKLLFAPNLLLDNRNGQKCVGEMVEIFDMLLATSSRFMM
NLQGEEFVCLKSIILNNSGVYFTLSLSTLKSLEEKDHIHRALDKITDTLIHLMAKAGLTL

QQQHQRLAQLLLLSHIRHMSNRMEHLYSMKYKNVPLYLDLLEMLEDAHLRHAG
QQQVQVQAFADLDDKDLHDLMAELGRYSVIREVLPTRRATFDLVEEELHTLVAE
GVVVHNCRRLENLIAQLPGEKKNGLFGNLIAALSGLTPNFKSNFDLAEDAKLQLSKD
TYDDLLDNLALVQGDQYADGLFLAAKNLSDAILLSDILRVNTEIKAPLASMIKRYDE
HHQDLTLLKALVRQGLPEKYKEIFPDQSKNGYAGIYDGGASQEPFYKFIPILEKMD
GTBEELVKLNREDLLRKQRTFDNBSIPHQIHLGELHAI LRREQDFYPLKDNREKIKBI
LTPRIPIYVVGPLARGNSRFAMWTRKSEETIPTWNFEVEVDKGAASAQSFIERMTNFDK
NLNPEKVLPKHSLLIEYFTVYNELTKVKYVTEGMKRKPAFLSGEQKAIIVDLLFTKTNR
KVTVYKQLKDEDYFKKIECFDSVEISGVEDRFNASLGRYTHDLLKI IKDKDLFDNEENEDI
LEDIVLTLTLTFEDREMIERLTKYAHFLFDKVMYKQLKRRYTWGWRLSKLLINGIRD
KQSGKTLILDFLKSDFANRNPMLIHDHSLTFKEDIQKAQVSGQDLSLHEHIANLAG
SPAIKKGI LQTQVKVDVLYKVMGRHKPENI VIEMARENQTQKQGNSRERMKRIIE
GIKELGSSILKEHPVENTQLQNEKSLVLYLQNGRMDYVQDELINRLSDYQVDHIVP
QSPFLKDSIDNKVLTSDKNRKNLPSVEEVVKKMKNYWRQLNAKLITTKRFDNF
LTKAERGGSLDKAGFIKRQLVETRQITKHAQILDSRMNTKYDENDKLIREVKVIT
LKSKLVSDFRKDFQFYKVR EINNYYHAHDYALNAVGTGALIKKYPKLESFVYGDY
KVDVDRKMIAKSEBQELGAKATKYFFHNSDIMNVLITELANGEIRKRLPIETNDGETEI
VVDKGRDFATVRKVLSPMQVNI VKTTEVQTGFGFSESTLPRKNSDKLITAKNDWDP
KKYGGFDSPTVAYSVLVVAKEVGKSKLKSVELLGI TIMERSSFEKNPIDFLEAK
GYKEVKQDLIIILKPYSLEFELENGARKMLASAGELOKQNEALPSKYVNPLYLASHY
PIREQAENIIILPTLTNLNGAANAFKYFDITIDRKRYSSTKEVLDATLIIHQSIITGLYETRI
DLSQLGGDGSPPKKRKYSSDYKDHDGYKDHDIDYKDDDDKAAG (SEQ ID NO: 160)

Indel Calling Algorithm

[illegible]

Read 2 :

```
@M00265:68:000000000-AA85W:1:1101:14923:1642 2:N:0:1
CTCACTGGGCGAGAAAGGTAACTTATGTTTCAGTAGCCTCTTCTCATGTGCTT
CAACCCATCACGGCCTTTGCAAATAGAGCCCTTTATTATAGTAGACAAGAGTCT
AAGCAGAAGAGAGGCCACTACCCAACCATCTACTCTTCTTAATGGT (SEQ ID NO: 168)
+
3>>AAFCFBBBGGGGGGGGGGHNNNNNNNNNNNNNNNNNNNNNNNNNNNNHH
NNNNNNNNGGHNFDGFGGNNNNNNNNNNHGFFNNHGGGNNNNNNNNFNNNNN
NNNNNNNNNNNNNGGNNNGGNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN@
```

Step 1: Search for sequences (or reverse complements) flanking the on/off target sites in both Illumina reads from the following set:

	target site	5' flanking sequence	3' flanking sequence
EMX_On	GAGTCCGAGCAGAAGAAGAGGG (SEQ ID NO: 107)	AGCTGGAGGAGGAAGGCCT (SEQ ID NO: 174)	CTCCCATCACATCAACCGGT (SEQ ID NO: 188)
EMX_Off1	GAGGCCGAGCAGAAGAAGACGG (SEQ ID NO: 108)	CCCCTTCTTCTGCAAATGAG (SEQ ID NO: 175)	CGACAGATGTTGGGGGAGG (SEQ ID NO: 189)
EMX_Off2	GAGTCTAGCAGGAGAAGAAGAG (SEQ ID NO: 109)	GGCTGGGGCCAGCATGACCT (SEQ ID NO: 176)	GCAGCCTAGAGTCTTCTGTG (SEQ ID NO: 190)
EMX_Off3	GAGTCTAAGCAGAAGAAGAGAG (SEQ ID NO: 110)	CCTTTATTCTAGTAGACAA (SEQ ID NO: 177)	AGCCACTACCAACCATCTA (SEQ ID NO: 191)
EMX_Off4	GAGTTAGAGCAGAAGAAGAAAGG (SEQ ID NO: 111)	CATGGCAAGACAGATTGTCA (SEQ ID NO: 178)	CATGGAGTAAAGGCAATCTT (SEQ ID NO: 192)
VEGF_On	GGGTGGGGGAGTTTGTCTCTGG (SEQ ID NO: 112)	GGGAATGGGCTTTGGAAAGG (SEQ ID NO: 179)	ACCCCTATTCTGACCTCC (SEQ ID NO: 193)
VEGF_Off1	GGATGGAGGAGTTTGTCTCTGG (SEQ ID NO: 113)	CATCTAAGGACGATTGTG (SEQ ID NO: 180)	GGTGTGAGATGTCTGTCT (SEQ ID NO: 194)
VEGF_Off2	GGGAGGGTGGAGTTTGTCTCTGG (SEQ ID NO: 114)	CTGGTCAGCCATTATGATA (SEQ ID NO: 181)	GGATGGAAGGCGGCTCCG (SEQ ID NO: 195)
VEGF_Off3	CGGGGGAGGAGTTTGTCTCTGG (SEQ ID NO: 115)	CTGGAGAGAGGCTCCCATCA (SEQ ID NO: 182)	GGAACCTGTGATCCACAG (SEQ ID NO: 196)
VEGF_Off4	GGGGAGGGGAAGTTTGTCTCTGG (SEQ ID NO: 116)	CATTTTGTCTGTCAACTC (SEQ ID NO: 183)	CATTGAGTGGGTAGAGTCCA (SEQ ID NO: 197)
CLTA_On	GCAGATGTAGTGTTCACAGGG (SEQ ID NO: 117)	CTGAGTAGGATTAAGATATT (SEQ ID NO: 184)	TGGCTCTCAGTGCACCAGC (SEQ ID NO: 198)
CLTA_Off1	ACATATGTAGTATTTCCACAGGG (SEQ ID NO: 118)	GTTGGGAAGAGATGCATACA (SEQ ID NO: 185)	AATACAATGGACAATAACC (SEQ ID NO: 199)
CLTA_Off2	CCAGATGTAGTATTTCCACAGGG (SEQ ID NO: 119)	GCCTCCTTGATTGAGGTGTC (SEQ ID NO: 186)	GTCTGGCAGGCCCTCCTGT (SEQ ID NO: 200)
CLTA_Off3	CTAGATGAAGTCTTCCACATGG (SEQ ID NO: 120)	CTCATCTAGAGTTCTTTCCA (SEQ ID NO: 187)	CTTTCATTAGAGTTAGTCC (SEQ ID NO: 201)

Step 2: Extract the sequence between the target sites in both reads and ensure that it is identical (reverse complementary) in read 1 and read 2 and all positions within read 1 and read 2 have a quality score ≥ 30 (Phred score ≥ 30)

In above reads, CTCTTCTGCTTAGACTC (SEQ ID NO: 169) is reverse complement of GAGTCTAAGCAGAAGAG (SEQ ID NO: 170)

Step 3: Align extracted sequence to the reference sequence for the relevant on/off target sequence

```

reference sequence
GAGTCTAAGCAGAAGAAGAG (SEQ ID NO: 110)
sequence read
GAGTCTAAGC-----AGAAGAG (SEQ ID NO: 202)

```

Step 4: For deletions, count only if deletion occurred in close proximity to expected cleavage site (within 8 bp of 3' end of reference sequence)

Methods and Materials

Cas9, intein-Cas9, and sgRNA expression plasmids. A plasmid encoding the human codon-optimized *Streptococcus*

pyogenes Cas9 nuclease with an NLS and 3xFLAG tag (Addgene plasmid 43861)⁵ was used as the wild-type Cas9 expression plasmid. Intein 37R3-2 was subcloned at the

described positions into the wild-type Cas9 expression plasmid using USER (NEB M5505) cloning. sgRNA expression plasmids used in this study have been described previously¹¹. Plasmid constructs generated in this work will be deposited with Addgene.

Modification of genomic GFP. HEK293-GFP stable cells (GenTarget), which constitutively express Emerald GFP, served as the reporter cell line. Cells were maintained in "full serum media": Dulbecco's Modified Eagle's Media plus GlutaMax (Life Technologies) with 10% (vol/vol) FBS and penicillin/streptomycin (1x, Amresco). 5×10^4 cells were plated on 48-well collagen-coated Biocoat plates (Becton Dickinson). 16-18 h after plating, cells were transfected with Lipofectamine 2000 (Life Technologies) according to the manufacturer's protocol. Briefly, 1.5 μ L of Lipofectamine 2000 was used to transfect 650 ng of total plasmid: 500 ng Cas9 expression plasmid, 125 ng sgRNA expression plasmid, and 25 ng near-infrared iRFP670 expressing plasmid (Addgene plasmid 45457)²⁶. 12 h after transfection, the media was replaced with full serum media, with or without 4-HT (1 μ M, Sigma-Aldrich T176). The media was replaced again 3-4 days after transfection. Five days after transfection, cells were trypsinized and resuspended in full serum media and analyzed on a C6 flow cytometer (Accuri) with a 488-nm laser excitation and 520-nm filter with a 20-nm band pass. Transfections and flow cytometry measurements were performed in triplicate.

High-throughput DNA sequencing of genome modifications. HEK293-GFP stable cells were transfected with plasmids expressing Cas9 (500 ng) and sgRNA (125 ng) as described above. For treatments in which a reduced amount of wild-type Cas9 expression plasmid was transfected, pUC19 plasmid was used to bring the total amount of plasmid to 500 ng. 4-HT (1 μ M final), where appropriate, was added during transfection. 12 h after transfection, the media was replaced with full serum media without 4-HT. Genomic DNA was isolated and pooled from three biological replicates 60 h after transfection using a previously reported¹¹ protocol with a DNAAdvance Kit (Agencourt). 150 ng or 200 ng of genomic DNA was used as a template to amplify by PCR the on-target and off-target genomic sites with flanking HTS primer pairs described previously¹¹. PCR products were purified using RapidTips (Qiagen Genomics) and quantified using the PicoGreen dsDNA Assay Kit (Invitrogen). Purified DNA was PCR amplified with primers containing sequencing adaptors, purified with the MinElute PCR Purification Kit (Qiagen) and AMPure XP PCR Purification (Agencourt). Samples were sequenced on a MiSeq high-throughput DNA sequencer (Illumina), and sequencing data was analyzed as described previously⁴.

Western blot analysis of intein splicing. HEK293-GFP stable cells were transfected with 500 ng Cas9 expression plasmid and 125 ng sgRNA expression plasmid. 12 h after transfection, the media was replaced with full serum media, with or without 4-HT (1 μ M). Cells were lysed and pooled from three technical replicates 4, 8, 12, or 24 h after 4-HT treatment. Samples were run on a Bolt 4-12% Bis-Tris gel (Life Technologies). An anti-FLAG antibody (Sigma-Aldrich F1804) and an anti-mouse 800CW IRDye (LI-COR) were used to visualize the gel on an Odyssey IR imager.

Statistical analysis. Statistical tests were performed as described in the figure captions. All p-values were calculated with the R software package. p-values for the Fisher exact test were calculated using the `fisher.test` function, with a one-sided alternative hypothesis (`alternative="greater"` or `alternative="less"`, as appropriate). Upper bounds on p-values that are close to zero were determined manually. The Benjamini-Hochberg adjustment was performed using the R function `p.adjust (method="fdr")`.

Sensitivity limit of off-target cleavage assays. We used paired end sequencing to identify indels caused by genomic on- and off-target cleavage. Given that published studies (see the reference below) have shown that the Illumina platform has an indel rate that is several orders of magnitude lower than the ~0.1% substitution error rate, and our requirement that all called indels occur in both paired reads, the sensitivity of the high-throughput sequencing method for detecting genomic off-target cleavage in our study is limited by the amount genomic DNA (gDNA) input into the PCR amplification of each genomic target site. A 1 ng sample of human gDNA represents only ~330 unique genomes, and thus only ~330 unique copies of each genomic site are present. PCR amplification for each genomic target was performed on a total of 150 ng or 200 ng of input gDNA, which provides amplicons derived from at most 50,000 or 65,000 unique gDNA copies, respectively. Therefore, the high-throughput sequencing assay cannot detect rare genome modification events that occur at a frequency of less than approximately 1 in 50,000 (0.002%). When comparing between two conditions, such as wt Cas9 vs. intein-Cas9, this threshold becomes approximately 10 in 50,000 (0.02%) when using the Fisher exact test and a conservative multiple comparison correction (Bonferroni with 14 samples). See also Minoche, A. E., Dohm, J. C., & Himmelbauer, H.

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- 25 Zetsche, B., Volz, S. E. & Zhang, F. A split-Cas9 architecture for inducible genome editing and transcription modulation. *Nature biotechnology* 33, 139-142 (2015).
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All publications, patents, patent applications, publication, and database entries (e.g., sequence database entries) mentioned herein, e.g., in the Background, Summary, Detailed Description, Examples, and References sections, are incorporated by reference in their entirety as if each individual publication, patent, patent application, publication, and database entry was specifically and individually incorporated herein by reference. In case of conflict, the present application, including any definitions herein, will control.

EQUIVALENTS AND SCOPE

Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. The scope of the present invention is not intended to be limited to the above description, but rather is as set forth in the appended claims.

In the claims articles such as “a,” “an,” and “the” may mean one or more than one unless indicated to the contrary or otherwise evident from the context. Claims or descriptions that include “or” between one or more members of a group are considered satisfied if one, more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process unless indicated to the contrary or otherwise evident from the context. The invention includes embodiments in which exactly one member of the group is present in, employed in, or otherwise relevant to a given product or process. The invention also includes embodiments in which more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process.

Furthermore, it is to be understood that the invention encompasses all variations, combinations, and permutations in which one or more limitations, elements, clauses, descriptive terms, etc., from one or more of the claims or from

relevant portions of the description is introduced into another claim. For example, any claim that is dependent on another claim can be modified to include one or more limitations found in any other claim that is dependent on the same base claim. Furthermore, where the claims recite a composition, it is to be understood that methods of using the composition for any of the purposes disclosed herein are included, and methods of making the composition according to any of the methods of making disclosed herein or other methods known in the art are included, unless otherwise indicated or unless it would be evident to one of ordinary skill in the art that a contradiction or inconsistency would arise.

Where elements are presented as lists, e.g., in Markush group format, it is to be understood that each subgroup of the elements is also disclosed, and any element(s) can be removed from the group. It is also noted that the term “comprising” is intended to be open and permits the inclusion of additional elements or steps. It should be understood that, in general, where the invention, or aspects of the invention, is/are referred to as comprising particular elements, features, steps, etc., certain embodiments of the invention or aspects of the invention consist, or consist essentially of, such elements, features, steps, etc. For purposes of simplicity those embodiments have not been specifically set forth in haec verba herein. Thus for each embodiment of the invention that comprises one or more elements, features, steps, etc., the invention also provides embodiments that consist or consist essentially of those elements, features, steps, etc.

Where ranges are given, endpoints are included. Furthermore, it is to be understood that unless otherwise indicated or otherwise evident from the context and/or the understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value within the stated ranges in different embodiments of the invention, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise. It is also to be understood that unless otherwise indicated or otherwise evident from the context and/or the understanding of one of ordinary skill in the art, values expressed as ranges can assume any subrange within the given range, wherein the endpoints of the subrange are expressed to the same degree of accuracy as the tenth of the unit of the lower limit of the range.

In addition, it is to be understood that any particular embodiment of the present invention may be explicitly excluded from any one or more of the claims. Where ranges are given, any value within the range may explicitly be excluded from any one or more of the claims. Any embodiment, element, feature, application, or aspect of the compositions and/or methods of the invention, can be excluded from any one or more claims. For purposes of brevity, all of the embodiments in which one or more elements, features, purposes, or aspects is excluded are not set forth explicitly herein.

All publications, patents and sequence database entries mentioned herein, including those items listed above, are hereby incorporated by reference in their entirety as if each individual publication or patent was specifically and individually indicated to be incorporated by reference. In case of conflict, the present application, including any definitions herein, will control.

SEQUENCE LISTING

Sequence total quantity: 203

SEQ ID NO: 1 moltype = DNA length = 4104
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 mol_type = genomic DNA
 organism = Streptococcus pyogenes

SEQUENCE: 1

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RLSRKLINGI	RDQSGKTLT	DPLKSDGFAN	RNFMLIHDD	SLTFKEDIQK	AQVSGQGDLS		720
HEHIANLAGS	PAIKKGLQTL	VKVVDLVKV	MGRHKPENIV	IEMARENQTT	QKGQKNSRER		780
MKRIEEGIKE	LGSQILKEHP	VENTQLQNEK	LYLYYLQNGR	DMYVDQELDI	NRLSDYDVDH		840
IVPQSFLKDD	SIDNKVLTRS	DKNRGKSDNV	PSEEVVKMK	NYWRQLLNAK	LITQRKFDNL		900
TKAERGGLSE	LDKAGFLKRLQ	LVEITRQITKH	VAQILDSDRMN	TKYDENDKLI	REVKVITLKS		960
KLVSDFRKDF	QFYKVRINN	YHHAHDAYLN	AVVGTALIKK	YPKLESEFVY	GDYKVYDVRK		1020
MTAKSEQETG	KATKPYFFYS	NIMNFPKTEI	TLANGEIRKR	PLIETNGETG	EIVWDKGRDF		1080
ATVRKVLSP	QVNVKLVKTEV	QTGGFSKESI	LPKRNSDKLI	ARKKDWDPKK	YGGFDSPTVA		1140
YSVLVVAKE	KGKSKKLKSV	KELLGITIME	RSSFEPKNPID	FLEAKGYKEV	KKDLIIKLPK		1200
YSLFELENGR	KRMLASAGEL	QKGNELALPS	KYVNFLYLAS	HYEKLKGSPE	DNEQKQLFVE		1260
QHKHYLDEII	BQISEFSKRV	ILADANLDKV	LSAYNKHRRK	PIREQAENII	HLFTLTNLGA		1320
PAAPKYFDDT	IDRKRYTSTK	EVLDATLIHQ	SITGLYETRI	DLSQLGGD			1368

SEQ ID NO: 5 moltype = AA length = 1368
 FEATURE Location/Qualifiers
 REGION 1..1368
 note = Synthetic Polypeptide
 source 1..1368
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 5

MDKKYSIGLA	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM	AKVDDSPFHR	LEESFLVEED	KKHERHPIFG		120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD	LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD		180
VDKLFQVLQV	TYNQLFEEENP	INASGVDAKA	ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN		240
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT	YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI		300
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ	DLTLKALVR	QQLPEKYKEI	FFDQSKNGYA		360
GYIDGGASQE	EFYKFIKPIL	EKMDGTEELL	VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH		420
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY	YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE		480
VVDKGASQAS	FIERTNFDK	NLPNEKVLTPK	HSLLYEFYTV	YNELTKVKYV	TEGMRKPAPL		540
SGEQKKAIVD	LLFTKTNKSVT	VKQLKEDYFK	KIECFDSVEI	SGVEDRPNAS	LGTYHDLLEKI		600
IKDKDFLDNE	ENEDILEDIV	LTLTLFEDRE	MIEERLKTYA	HLFDDKVMKQ	LKRRRYTGWG		660

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RLSRKLINGI RDKQSGKITL DFLKSDGFAN RNFQMLIHDD SLTPKEDIQK AQVSGQGDSL 720
HEHIANLAGS PAIKKGILQT VKVDELVKV MGRHKPENIV IEMARENQTT QKGQKNSRER 780
MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMVVDQELDI NRLSDYDVDA 840
IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV PSEEVVKKMK NYWRQLLNAK LITQRKFDNL 900
TKAERGGGLSE LDKAGFIKRQ LVETRQITKH VAQILDSRMN TKYDENDKLI REVKVITLKS 960
KLVSDFRKDF QFYKVRINN YHHAHDAYLN AVVGITALIKK YPKLESEFVY GDYKVYDVRK 1020
MIAKSEQEIG KATAKYFFYS NIMNFPKTEI TLANGEIRKR PLIETNGETG EIVWDKGRDF 1080
ATVRKVLSPM QVNIVKKTEV QTGGFSKESI LPKRNSDKLI ARKKDWDPPK YGGFDSPTVA 1140
YSVLVVAKVE KGKSKKLKSV KELLGITIME RSSFEKNPID FLEAKGYKEV KKDIIKLPK 1200
YSLFELENGR KRMLASAGEL QKGNELALPS KYVNFYLAS HYEKLKGSPE DNEQKQLFVE 1260
QHKHYLDEII EQISEFSKRV ILADANLDKV LSAYNKHDK PIREQAENII HLFITLNLGA 1320
PAAFKYFDTT IDRKYRSTK EVLDATLIHQ SITGLYETRI DLSQLGGD 1368

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SEQ ID NO: 6          moltype = AA length = 198
FEATURE              Location/Qualifiers
REGION              1..198
                    note = Synthetic Polypeptide
source              1..198
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 6
GSQLVKSELE EKKSELRHKL KYVPHEYIEL IEIARNSTQD RILEMKVMEF FMKVYGYRGK 60
HLGGSRKPDG AIYTVGSPID YGVIVDTKAY SGGYNLPIGQ ADEMQRVVEE NQTRNKHINP 120
NEWWKVYPSS VTEPKFLFVS GHFKGNYKAQ LTRLNHITNC NGAVLVVEEL LIGGEMIKAG 180
TLTLEEVRRK FNNGEINF

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```

SEQ ID NO: 7          moltype = AA length = 413
FEATURE              Location/Qualifiers
REGION              1..413
                    note = Synthetic Polypeptide
source              1..413
                    mol_type = protein
                    organism = synthetic construct

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```

SEQUENCE: 7
CLAEGTRIFD PVTGTTHRIE DVVDGRKPIH VVAAAKDGTL LARPVVSWFD QGTRDVIGLR 60
IAGGAIVWAT PDHKVLTEYG WRAAGELRKG DRVAGPGGSG NSLALSLTAD QMVSALLDAE 120
PPILYSEYDP TSPFSEASMM GLLTNLADRE LVHMINWAKR VPGFVDLTLH DQAHLLCAW 180
LEILMIGLVW RSMHEPGKLL FAPNLLDRN QGKCEGMVE IFDMLLATSS RFRMMNLQGE 240
EFVCLKSIIL LNSGVYTFLS STLKSLEEKD HIHRALDKIT DTLIHLMKA GLTLQQQHQR 300
LAQLLLILSH IRHMSNKGME HLYSMKYKNV VPLYDLLLEM LDAHRLHAGG SGASRVQAF 360
DALDDKFLHD MLAEELRYSV IREVLPTTRA RTFDLEVEEL HTLVAEGVVV HNC 413

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SEQ ID NO: 8          moltype = AA length = 413
FEATURE              Location/Qualifiers
REGION              1..413
                    note = Synthetic Polypeptide
source              1..413
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 8
CLAEGTRIFD PVTGTTHRIE DVVDGRKPIH VVAAAKDGTL LARPVVSWFD QGTRDVIGLR 60
IAGGAIVWAT PDHKVLTEYG WRAAGELRKG DRVAGPGGSG NSLALSLTAD QMVSALLDAE 120
PPILYSEYDP TSPFSEASMM GLLTNLADRE LVHMINWAKR VPGFVDLTLH DQAHLLCAW 180
LEILMIGLVW RSMHEPGKLL FAPNLLDRN QGKCEGMVE IFDMLLATSS RFRMMNLQGE 240
EFVCLKSIIL LNSGVYTFLS STLKSLEEKD HIHRALDKIT DTLIHLMKA GLTLQQQHQR 300
LAQLLLILSH IRHMSNKGME HLYSMKYKNV VPLYDLLLEM LDAHRLHAGG SGASRVQAF 360
DALDDKFLHD MLAEELRYSV IREVLPTTRA RTFDLEVEEL HTLVAEGVVV HNC 413

```

```

SEQ ID NO: 9          moltype = AA length = 413
FEATURE              Location/Qualifiers
REGION              1..413
                    note = Synthetic Polypeptide
source              1..413
                    mol_type = protein
                    organism = synthetic construct

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```

SEQUENCE: 9
CLAEGTRIFD PVTGTTHRIE DVVDGRKPIH VVAAAKDGTL LARPVVSWFD QGTRDVIGLR 60
IAGGAIVWAT PDHKVLTEYG WRAAGELRKG DRVAGPGGSG NSLALSLTAD QMVSALLDAE 120
PPIPYSEYDP TSPFSEASMM GLLTNLADRE LVHMINWAKR VPGFVDLTLH DQAHLLCAW 180
LEILMIGLVW RSMHEPGKLL FAPNLLDRN QGKCEGMVE IFDMLLATSS RFRMMNLQGE 240
EFVCLKSIIL LNSGVYTFLS STLKSLEEKD HIHRALDKIT DTLIHLMKA GLTLQQQHQR 300
LAQLLLILSH IRHMSNKGME HLYSMKYKNV VPLYDLLLEM LDAHRLHAGG SGASRVQAF 360
DALDDKFLHD MLAEELRYSV IREVLPTTRA RTFDLEVEEL HTLVAEGVVV HNC 413

```

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SEQ ID NO: 10         moltype = AA length = 413
FEATURE              Location/Qualifiers
REGION              1..413
                    note = Synthetic Polypeptide

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source          1..413
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 10
CLAEGTRIFD PVTGTTHRIE DVVDGRKPIH VVAAAKDGTL LARPVVSFWD QGTRDVIGLR 60
IAGGATVWAT PDHKVLTEYG WRAAGELRKG DRVAGPGGSG NSLALSLTAD QMVSALLDAE 120
PPILYSEYDP TSPFSEASMM GLLTNLADRE LVHMINWAKR VPGFVDLTLH DQAHLLLECAW 180
LEILMIGLVW RSMHPGKLL FAPNLLDRN QGKCEGMVE IFDMLLATSS RFRMMNLQGE 240
EFVCLKSIIL LNSGVYTFLS STLKSLEEKD HIHRALDKIT DTLIHLMAKA GLTLQQQHQR 300
LAQLLLLILSH IRHMSNKGME HLYSMKYKNV VPLYDLLLEM LDAHRLHAGG SGASRVQAF 360
DALDDKFLHD MLAEELRYSV IREVLPTTRA RTFDLEVEEL HTLVAEGVVV HNC 413

SEQ ID NO: 11      moltype = AA length = 413
FEATURE           Location/Qualifiers
REGION            1..413
                  note = Synthetic Polypeptide

source            1..413
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 11
CLAEGTRIFD PVTGTTHRIE DVVDGRKPIH VVAAAKDGTL LARPVVSFWD QGTRDVIGLR 60
IAGGATVWAT PDHKVLTEYG WRAAGELRKG DRVAGPGGSG NSLALSLTAD QMVSALLDAE 120
PPIPYSEYDP TSPFSEASMM GLLTNLADRE LVHMINWAKR VPGFVDLTLH DQAHLLLECAW 180
LEILMIGLVW RSMHPGKLL FAPNLLDRN QGKCEGMVE IFDMLLATSS RFRMMNLQGE 240
EFVCLKSIIL LNSGVYTFLS STLKSLEEKD HIHRALDKIT DTLIHLMAKA GLTLQQQHQR 300
LAQLLLLILSH IRHMSNKGME HLYSMKYKNV VPLYDLLLEM LDAHRLHAGG SGASRVQAF 360
DALDDKFLHD MLAEELRYSV IREVLPTTRA RTFDLEVEEL HTLVAEGVVV HNC 413

SEQ ID NO: 12      moltype = AA length = 413
FEATURE           Location/Qualifiers
REGION            1..413
                  note = Synthetic Polypeptide

source            1..413
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 12
CLAEGTRIFD PVTGTTHRIE DVVDGRKPIH VVAAAKDGTL LARPVVSFWD QGTRDVIGLR 60
IAGGATVWAT PDHKVLTEYG WRAAGELRKG DRVAGPGGSG NSLALSLTAD QMVSALLDAE 120
PPILYSEYNP TSPFSEASMM GLLTNLADRE LVHMINWAKR VPGFVDLTLH DQAHLLERAW 180
LEILMIGLVW RSMHPGKLL FAPNLLDRN QGKCEGMVE IFDMLLATSS RFRMMNLQGE 240
EFVCLKSIIL LNSGVYTFLS STLKSLEEKD HIHRALDKIT DTLIHLMAKA GLTLQQQHQR 300
LAQLLLLILSH IRHMSNKGME HLYSMKYKNV VPLYDLLLEM LDAHRLHAGG SGASRVQAF 360
DALDDKFLHD MLAEGLRYSV IREVLPTTRA RTFDLEVEEL HTLVAEGVVV HNC 413

SEQ ID NO: 13      moltype = AA length = 413
FEATURE           Location/Qualifiers
REGION            1..413
                  note = Synthetic Polypeptide

source            1..413
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 13
CLAEGTRIFD PVTGTTHRIE DVVDGRKPIH VVAAAKDGTL LARPVVSFWD QGTRDVIGLR 60
IAGGAIVWAT PDHKVLTEYG WRAAGELRKG DRVAGPGGSG NSLALSLTAD QMVSALLDAE 120
PPILYSEYDP TSPFSEASMM GLLTNLADRE LVHMINWAKR VPGFVDLTLH DQAHLLERAW 180
LEILMIGLVW RSMHPGKLL FAPNLLDRN QGKCEGMVE IFDMLLATSS RFRMMNLQGE 240
EFVCLKSIIL LNSGVYTFLS STLKSLEEKD HIHRALDKIT DTLIHLMAKA GLTLQQQHQR 300
LAQLLLLILSH IRHMSNKGME HLYSMKYKNV VPLYDLLLEM LDAHRLHAGG SGASRVQAF 360
DALDDKFLHD MLAEGLRYSV IREVLPTTRA RTFDLEVEEL HTLVAEGVVV HNC 413

SEQ ID NO: 14      moltype = AA length = 413
FEATURE           Location/Qualifiers
REGION            1..413
                  note = Synthetic Polypeptide

source            1..413
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 14
CLAEGTRIFD PVTGTTHRIE DVVDGRKPIH VVAVAKDGTL LARPVVSFWD QGTRDVIGLR 60
IAGGATVWAT PDHKVLTEYG WRAAGELRKG DRVAGPGGSG NSLALSLTAD QMVSALLDAE 120
PPILYSEYDP TSPFSEASMM GLLTNLADRE LVHMINWAKR VPGFVDLTLH DQAHLLERAW 180
LEILMIGLVW RSMHPGKLL FAPNLLDRN QGKCEGMVE IFDMLLATSS RFRMMNLQGE 240
EFVCLKSIIL LNSGVYTFLS STLKSLEEKD HIHRALDKIT DTLIHLMAKA GLTLQQQHQR 300
LAQLLLLILSH IRHMSNKGME HLYSMKYKNV VPLYDLLLEM LDAHRLHAGG SGASRVQAF 360
DALDDKFLHD MLAEELRYSV IREVLPTTRA RTFDLEVEEL HTLVAEGVVV HNC 413

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SEQ ID NO: 15	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = Synthetic Polypeptide	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 15		
GGSGSGSGSG GSGSGGS		18
SEQ ID NO: 16	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Synthetic Polypeptide	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 16		
SGSETPGTSE SATPES		16
SEQ ID NO: 17	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = Synthetic Polypeptide	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 17		
SGSETPGTSE SA		12
SEQ ID NO: 18	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
REGION	1..21	
	note = Synthetic Polypeptide	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 18		
SGSETPGTSE SATPEGSGG S		21
SEQ ID NO: 19	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Synthetic Polypeptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 19		
VPFLLEPDNI NGKTC		15
SEQ ID NO: 20	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = Synthetic Polypeptide	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 20		
GSAGSAAGSG EF		12
SEQ ID NO: 21	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = Synthetic Polypeptide	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 21		
SIVAQLSRPD PA		12
SEQ ID NO: 22	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = Synthetic Polypeptide	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 22
MKIIEQLPSA 10

SEQ ID NO: 23 moltype = AA length = 10
FEATURE Location/Qualifiers
REGION 1..10
 note = Synthetic Polypeptide
source 1..10
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 23
VRHKLKRVGS 10

SEQ ID NO: 24 moltype = AA length = 12
FEATURE Location/Qualifiers
REGION 1..12
 note = Synthetic Polypeptide
source 1..12
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 24
GHGTGSTGSG SS 12

SEQ ID NO: 25 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
 note = Synthetic Polypeptide
source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 25
MSRPDPA 7

SEQ ID NO: 26 moltype = AA length = 4
FEATURE Location/Qualifiers
REGION 1..4
 note = Synthetic Polypeptide
source 1..4
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 26
GGSM 4

SEQ ID NO: 27 moltype = AA length = 1780
FEATURE Location/Qualifiers
REGION 1..1780
 note = Synthetic Polypeptide
source 1..1780
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 27
MDKKYSIGLD IGTNSVGWAV ITDEYKVP SK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC LAEGTRIFDP VTGTTTHRIED VVDGRKPIHV VAAAKDGTLL 120
ARPVVSWFDQ GTRDVIGLRI AGGAIVWATP DHKVLTEYGW RAAGELRKGD RVAGPGGSGN 180
SLALSLTADQ MVSALLDAEP PILYSEYDPT SPFSEASMMG LLTNLADREL VHMINWAKRV 240
PGFVDLTLHD QAHLLERAWL EILMIGLVWR SMEHPGKLLF APNLLLDNRQ GKCEVGMVEI 300
FDMLLATSSR FRMMNLQGEF FVCLKSIILL NSGVYTFLLS TLKSLEEKDH IHRALDKITD 360
TLIHLMAKAG LTLQQQHQLR AQLLLILSHI RHMSNKGMEH LYSMKYKNV VLYDLLLEML 420
DAHRLHAGGS GASRVQAFAD ALDDKFLHDM LAEGLRYSVI REVLPTRRAR TFDLEVEELH 480
TLVAEGVVVH NCYLQEIFSN EMAKVDDSPF HRLEESFLVE EDKKHERHPI FGNIVDEVAY 540
HEKYPTIYHL RKKLVDSYDK ADLRLIYLAL AHMIKFRGHF LIEQDLNPDN SDVDKLFQIL 600
VQTYNQLFEE NPINASGVDA KAILSARLSK SRRLENLIAQ LPGEKKNGLF GNLIASLGL 660
TPNFKSNFPL AEDAKLQLSK DTYDDDLNLL LAQIGDQYAD LFLAAKNLSD AILLSDILRV 720
NTEITKAPLS ASMIKRYDEH HQDLTLKAL VRQQLPEKYK EIFFDQSKNG YAGYIDGGAS 780
QEEFYKFIKP ILEKMDGTEE LLVKLNREDL LRKQRTFDNG SIPHQIHLGE LHAILRRQED 840
FYFPLKDNRE KIEKILTFRI PYYVGPLARG NSRFAMWTRK SEETITPWNF EEVVDKGASA 900
QSFIERMTNF DKNLPNEKVL PKHSLLYEYF TVYNELTKVK YVTEGMRKPA FLSGEQKKAI 960
VDLLFKTNRK VTVKQLKEDY FKKIECFDSV EISGVEDRFN ASLGTYHDLL KIIKDQDFLD 1020
NEENEDILED IVLTLTLFED REMIEERLKT YAHLPDDKVM KQLKRRRYTG WGRLSRKLIN 1080
GIRDKQSGKT ILDFLKSDGF ANRNFQQLIH DDSLTFKEDI QKAQVSGQGD SLHEHIANLA 1140
GSPAIIKGGIL QTVKVVDLV KVMGRHKPEN IVIEMARENQ TTQKQKQNSR ERMKRIEEGI 1200
KELGSQILKE HPVENTQLQN EKLYLYYLQN GRDMYVDQEL DINRLSDYDV DHIVPQSFLK 1260
DDSIDNKVLT RSDKNRGKSD NVPSEEVVKK MKNYWRQLLN AKLITQRKFD NLTKAERGGL 1320
SELDKAGFIK RQLVETRQIT KHVAQILDSR MNTKYDENDK LIREVKVITL KSKLVSDFRK 1380
DFQFYKVVREI NNYHHAHDAY LNAVVGTAII KKYPKLESEF VYGDYKYVDV RKMIAKSEQE 1440
IGKATAKYFF YSNIMNFFKT EITLANGEIR KRPLIETNGE TGEIVWDKGR DFATVRKVL 1500
MPQVNIKKKT EVQGGGFSKE SILPKRNSDK LTARKKDWDP KKYGGFDSPT VAYSVLVVA 1560
VEKGSKKLLK SVKELLGITI MERSSFENP IDFLKAGYK EVKDLIIKL PKYSLFELEN 1620

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GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD			1780

SEQ ID NO: 28 moltype = AA length = 1780
 FEATURE Location/Qualifiers
 REGION 1..1780
 note = Synthetic Polypeptide
 source 1..1780
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 28

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSPFHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVCLAE	GTRIFDPVTG	TTHRIEDVVD		GRKPIHVVA	AKDGTLLARP	VVSWFDQGT	180
DVIGLRIAGG	AIVWATPDHK	VLTEYGNRAA		GELRKGDVRA	GPGSGNSLA	LSLTADQMVS	240
ALLDAEPPIL	YSEYDPTSPF	SEASMMGLLT		NLADRELVHM	INWAKRVPGF	VDLTLDQAH	300
LLERAWLEIL	MIGLVWRSME	HPGKLLFAPN		LLLDNRNQKC	VEGMVEIFDM	LLATSSRFRM	360
MNLQGEFEVC	LKSIILLNSG	VYTFLSSTLK		SLEEKDHIHR	ALDKITDTLI	HLMAKAGLTL	420
QQQHQRQAQL	LLILSHIRHM	SNKGMEHLYS		MKYKNVVP	LY	DLLLEMLDAH	480
RVQAFADALD	DKFLHDM LAE	GLRYSVIREV		LPTRRARTFD	LEVEELHTLV	AEGVVVHNCY	540
HEKYPTIYHL	RKKLVDSDDK	ADLRLLIYLA		LAL	AHMIKFRGHF	LIEGDLNPDN	600
VQTYNQLFEE	NPINASGVDA	KAILSARLSK		SRRLENLIAQ	LPGEKKNGLF	GNLIALSLGL	660
TPNFKSNFDL	AEDAKLQLSK	DTYDDDLN		LAQIGDQYAD	LFLAAKNLSD	AILLSDILRV	720
NTEITKAPLS	ASMIKRYDEH	HQDLTLKAL		VRQQLPEKYK	EIFFDQSKNG	YAGYIDGGAS	780
QEEFYKFIKP	ILEKMDGT	TEE		LLVKLNREDL	LRKQRTFDNG	SIPHQIHLGE	840
FYPFLKDNRE	KIEKILTFRI	PYYVGPLARG		NSRFAWMTRK	SEETITPWNF	EEVVDKGASA	900
QSFIERMTNF	DKNLPNEKVL	PKHSLLYEYF		TVYNELTKVK	YVTEGMRKPA	FLSGEQKKAI	960
VDLLPKTNRK	VTVKQLKEDY	FKKIECFDSV		EISGVEDRPN	ASLGTYHDL	L	1020
NEENEDILED	IVLTLTFED	REMIERLKT		YAHLPDDKVM	KQLKRRRYTG	WGRLSRKLIN	1080
GIRDKQSGKT	ILDPLKSDGF	ANRNFMLQI		LIH	DDSLTFKEDI	QKAQVSGQGD	1140
GSPAIIKKGIL	QTVKVVDEL	KVMGRHKPEN		IVIEMARENQ	TTQKGQKNSR	ERMKRIEEGI	1200
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN		GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260
DDSIDNKVLT	RSDKNRGKSD	NVPSEEVVKK		MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320
SELDKAGFIK	RQLVETQIT	KHVAQILDSR		MNTKYDENDK	LIREVKVITL	KSKLVSDFRK	1380
DPQFYKVRRI	NNYHHADAY	LNAVVGTA		L	IKYPKLESEF	VYGDYKVYDV	1440
IGKATAKYFF	YSNIMNPFKT	EITFLANGEIR		KRPLIETNGE	TGEIVWDKGR	DFATVRKVL	1500
MPQVNIVKKT	EVQTTGGFSKE	SILPKRNSDK		LIARKKDWD	P	KKYGGFDSPT	1560
VEKGSKKKL	SVKELLGITI	MERSSFEKNP		IDFLEAKGYK	EVKKDLI	IKL	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL		ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR		DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET		RIDLSQLGGD			1780

SEQ ID NO: 29 moltype = AA length = 1780
 FEATURE Location/Qualifiers
 REGION 1..1780
 note = Synthetic Polypeptide
 source 1..1780
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 29

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSPFHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSCLAEG		TRIFDPVTGT	THRIEDVVDG	RKPIHVVA	180
KDGTLLARPV	VSWFDQGT	TRD		VIGLRIAGGA	IVWATPDHKV	LTEYGNRAAG	240
PGSGNSLAL	SLTADQMVA	LLDAEPPILY		SEYDPTSPFS	EASMMGLLTN	LADRELVHMI	300
NWAKRVPGFV	DLTLHDQAH	L		ERAWLEILM	IGLVWRSMEH	PGKLLFAPNL	360
EGMVEIFDML	LATSSRFRMM	NLQGEFEVCL		KSIILLNSGV	YTFLSSTLKS	LEEKDHIHRA	420
LDKITDTLH	LMKAGLTLQ	QQHQRQAQLL		LILSHIRHMS	NKGMEHLYSM	KYKNVVP	480
LLLEMLDAHR	LHAGSGGASR	VQAFADALDD		KFLHDM LAEG	L	RYSVIREVL	540
EVEELHTLVA	EGVVVHNC	DK		ADLRLLIYLA	L	AHMIKFRGHF	600
VQTYNQLFEE	NPINASGVDA	KAILSARLSK		SRRLENLIAQ	LPGEKKNGLF	GNLIALSLGL	660
TPNFKSNFDL	AEDAKLQLSK	DTYDDDLN		LAQIGDQYAD	LFLAAKNLSD	AILLSDILRV	720
NTEITKAPLS	ASMIKRYDEH	HQDLTLKAL		VRQQLPEKYK	EIFFDQSKNG	YAGYIDGGAS	780
QEEFYKFIKP	ILEKMDGT	TEE		LLVKLNREDL	LRKQRTFDNG	SIPHQIHLGE	840
FYPFLKDNRE	KIEKILTFRI	PYYVGPLARG		NSRFAWMTRK	SEETITPWNF	EEVVDKGASA	900
QSFIERMTNF	DKNLPNEKVL	PKHSLLYEYF		TVYNELTKVK	YVTEGMRKPA	FLSGEQKKAI	960
VDLLPKTNRK	VTVKQLKEDY	FKKIECFDSV		EISGVEDRPN	ASLGTYHDL	L	1020
NEENEDILED	IVLTLTFED	REMIERLKT		YAHLPDDKVM	KQLKRRRYTG	WGRLSRKLIN	1080
GIRDKQSGKT	ILDPLKSDGF	ANRNFMLQI		LIH	DDSLTFKEDI	QKAQVSGQGD	1140
GSPAIIKKGIL	QTVKVVDEL	KVMGRHKPEN		IVIEMARENQ	TTQKGQKNSR	ERMKRIEEGI	1200
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN		GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260
DDSIDNKVLT	RSDKNRGKSD	NVPSEEVVKK		MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320
SELDKAGFIK	RQLVETQIT	KHVAQILDSR		MNTKYDENDK	LIREVKVITL	KSKLVSDFRK	1380
DPQFYKVRRI	NNYHHADAY	LNAVVGTA		L	IKYPKLESEF	VYGDYKVYDV	1440
IGKATAKYFF	YSNIMNPFKT	EITFLANGEIR		KRPLIETNGE	TGEIVWDKGR	DFATVRKVL	1500
MPQVNIVKKT	EVQTTGGFSKE	SILPKRNSDK		LIARKKDWD	P	KKYGGFDSPT	1560

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VEKGKSKKLK	SVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIKL	PKYSLFELEN	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD			1780

SEQ ID NO: 30 moltype = AA length = 1780
 FEATURE Location/Qualifiers
 REGION 1..1780
 note = Synthetic Polypeptide
 source 1..1780
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 30

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVPK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM	AKVDDSFHHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD	LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFEEEN	INASGVDAKA	ILSARLSKCL	AEGTRIFDPV	TGTTTHRIEDV	240
VDGRKPIHVV	AAAKDGTLLA	RPVVSWFDDG	TRDVIGLRIA	GGAIVWATPD	HKVLTIEYGWR	300
AAGELRKGRD	VAGPGGSGNS	LALSLTADQM	VSALLDAEPP	ILYSEYDPTS	PFSEASMMGL	360
LTNLADRELV	HMINWAKRVP	GFVDLTLDHQ	AHLLERAWLE	ILMIGLVWRS	MEHPGKLLFA	420
PNLLLDNRNG	KCEVGMVEIF	DMLLATSSRF	RMMNLQGEFF	VCLKSIIILLN	SGVYTFLSST	480
LKSLLEEKDHI	HRALDKITDT	LIHLMAKAGL	TLQQQHQRLA	QLLLILSHIR	HMSNKGMEHL	540
YSMKYKVVVP	LYDLLLEMLD	AHRLHAGGSG	ASRVQAFADA	LDDKFLHDML	AEGLRYSVIR	600
EVLPTRRART	FDLEVEELHT	LVAEGVVVHN	CRRLENLIAQ	LPGEKKNGLF	GNLIALSLGL	660
TPNFKSNFDL	AEDAKLQLSK	DTYDDDLNLD	LAQIGDQYAD	LFLAAKNLSD	AILLSDILRV	720
NTEITKAPLS	ASMIKRYDEH	HQDLTLKAL	VRQQLPEKYK	EIFFDQSKNG	YAGYIDGGAS	780
QEEFYKFIKP	ILEKMDGTEE	LLVKLNREDL	LRKQRTFDNG	SIPHQIHLGE	LHAILRRQED	840
FYPFLKDNRE	KIEKILTFRI	PYYVGPLARG	NSRFAWMTRK	SEETITPNWF	EEVVDKGASA	900
QSFIERMTNF	DKNLPNEKVL	PKHSLLYEYF	TVYNELTKVK	YVTEGMRKPA	FLSGEQKKAI	960
VDLLFKTNRK	VTVQQLKEDY	FKKIECFDSV	EISGVEDRFN	ASLGTYHDL	KIIKDKDFLD	1020
NEENEDILED	IVLTLTLFED	REMIERLKT	YAHLPDDKVM	KQLKRRRYTG	WGRLSRKLIN	1080
GIRDKQSGKT	ILDPLKSDGF	ANRNFMLIHL	DDSLTFKEDI	QKAQVSGQGD	SLHEHIANLA	1140
GSPAIAKKGIL	QTVKVVDELV	KVMGRHKPEN	IVIEMARENQ	TTQKGQKNSR	ERMKRIEEGI	1200
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN	GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260
DDSIDNKVLT	RSDKNRGKSD	NVPSEEVVKK	MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320
SELDKAGFIK	RQLVETRQIT	KHVAQILDSR	MNTKYDENDK	LIREVKVITL	KSKLVSDFRK	1380
DFQFYKVRRI	NNYHHAHDAY	LNAVVGITALI	KKYPKLESEF	VYGDYKYVDV	RKMIKSEQE	1440
IGKATAKYFF	YSNIMNFFKT	EITLANGEIR	KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIKKTK	EVQTTGGFSKE	SILPKRNSDK	LTARKKDWD	KKYGGFDSPT	VAYSVLVVAK	1560
VEKGKSKKLK	SVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIKL	PKYSLFELEN	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD			1780

SEQ ID NO: 31 moltype = AA length = 1780
 FEATURE Location/Qualifiers
 REGION 1..1780
 note = Synthetic Polypeptide
 source 1..1780
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 31

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVPK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM	AKVDDSFHHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD	LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFEEEN	INASGVDAKA	ILSARLSKSR	RLENLIAQLP	GEKKNGLFNG	240
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT	YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ	DLCLAEGTRI	FDPVTGTTHR	IEDVVDGRKP	360
IHVVAACKDG	TLLARPVVSW	FDQGTDRDIG	LRIAGGAIVW	ATPDHKVLTE	YGWRAAGELR	420
KGDRVAGPGG	SGNSLALSLT	ADQMVSALLD	AEPPILYSEY	DPTSPFSEAS	MMGLLTNLAD	480
RELVHMINWA	KRVPGFVDLT	LHDQAHLLER	AWLEILMIGL	VWRSMHEPGK	LLFAPNLLLD	540
RNQGKCEVGM	VEIFDMLLAT	SSRFRMMNLQ	GEEFVCLKSI	ILLNSGVYTF	LSSTLKSLEE	600
KDHIHRALDK	ITDTLIHLMA	KAGLTLLQQH	QLRAQLLLIL	SHIRHMSNKG	MEHLYSMKYK	660
NVVPLYDLLL	EMLDAHRLHA	GGSGASRVQA	FADALDDKFL	HDMLAEGLRY	SVIREVLPT	720
RARTFDLEVE	ELHTLVAAEGV	VVHNCLLKAL	VRQQLPEKYK	EIFFDQSKNG	YAGYIDGGAS	780
QEEFYKFIKP	ILEKMDGTEE	LLVKLNREDL	LRKQRTFDNG	SIPHQIHLGE	LHAILRRQED	840
FYPFLKDNRE	KIEKILTFRI	PYYVGPLARG	NSRFAWMTRK	SEETITPNWF	EEVVDKGASA	900
QSFIERMTNF	DKNLPNEKVL	PKHSLLYEYF	TVYNELTKVK	YVTEGMRKPA	FLSGEQKKAI	960
VDLLFKTNRK	VTVQQLKEDY	FKKIECFDSV	EISGVEDRFN	ASLGTYHDL	KIIKDKDFLD	1020
NEENEDILED	IVLTLTLFED	REMIERLKT	YAHLPDDKVM	KQLKRRRYTG	WGRLSRKLIN	1080
GIRDKQSGKT	ILDPLKSDGF	ANRNFMLIHL	DDSLTFKEDI	QKAQVSGQGD	SLHEHIANLA	1140
GSPAIAKKGIL	QTVKVVDELV	KVMGRHKPEN	IVIEMARENQ	TTQKGQKNSR	ERMKRIEEGI	1200
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN	GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260
DDSIDNKVLT	RSDKNRGKSD	NVPSEEVVKK	MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320
SELDKAGFIK	RQLVETRQIT	KHVAQILDSR	MNTKYDENDK	LIREVKVITL	KSKLVSDFRK	1380
DFQFYKVRRI	NNYHHAHDAY	LNAVVGITALI	KKYPKLESEF	VYGDYKYVDV	RKMIKSEQE	1440
IGKATAKYFF	YSNIMNFFKT	EITLANGEIR	KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIKKTK	EVQTTGGFSKE	SILPKRNSDK	LTARKKDWD	KKYGGFDSPT	VAYSVLVVAK	1560
VEKGKSKKLK	SVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIKL	PKYSLFELEN	1620

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GRKRMASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD			1780

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SEQ ID NO: 32      moltype = AA  length = 1780
FEATURE           Location/Qualifiers
REGION            1..1780
                  note = Synthetic Polypeptide
source            1..1780
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 32
MDKKYSIGLD IGTSVGVAV ITDEYKVP SKFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSFPHR LEESFLVEED KKHERHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPDNDSD 180
VDKLFQQLVQ TYNQLFEEFN INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIALSLGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EYKFIKPIIL EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASQAS FIERMTNFDK NLPNEKVLPK HSLLYEYFCL AEGTRIFDPV TGTHRIEDV 540
VDGRKPIHVV AAADGDTLLA RPVVSFWDQG TRDVLGLRIA GGAIVWATPD HKVLTEYGWR 600
AAGELRKGDG VAGPGGSGNS LALSLTADQM VSALLDAEPP ILYSEYDPTS PFSEASMMGL 660
LTNLADREL V HMINWAKRVP GFVDLTLDHQ AHLLEAWLE ILMIGLVWRS MEHPGKLLFA 720
PNLLLDNRNG KCVEGMVEIF DMLLATSSRF RMMNLQGEFF VCLKSILLN SGVYTFLSST 780
LKSLEEKDHI HRALEDKITDT LIHLMAKAGL TLQQQHQR LA LLLILSHIR HMSNKGMEHL 840
YSMKYKNVVP LYDLLLEMLD AHR LHAGGSG ASRVQAFADA LDDKFLHDM L AEGLYSVIR 900
EVLPTRRART FDLEVEELHT LVAEGVVVHN CVYNELTKVK YVTEGMRKPA FLSGEQKKA I 960
VDLLPKTNRK VTVKQLKEDY FKKIECFDSV EISGVEDRPN ASLGTYHDL L KIIKDKDFLD 1020
NEENEDILED IVLTLTLFED REMIEERLKT YAHLPDDKVM KQLKRRRYTG WGRLSRKLIN 1080
GIRDQSGSGT ILDFLKSDFG ANRNFQMQLIH DDSLTFKEDI QKAQVSGQGD SLHEHIANLA 1140
GSPAIIKKGIL QTVKVVDDEL V KVMGRHKPEN IVIEMARENQ TTQKGQKNSR ERMKRIEEGI 1200
KELGSQLKE HPVENTQLQN EKLYLYYLQN GRDMYVDQEL DINRLSDYDV DHIVPQSFLK 1260
DDSIDNKVLT RSDKNRKGSD NVPSEEVVKK MKNYWRQLLN AKLITQRKFD NLTKAERGGL 1320
SELDKAGFIK RQLVETRQIT KHVAQILDSR MNTKYDENDK LIREVKVITL KSKLVSDFRK 1380
DFQFYKVR EI NNYHHAHDAY LNAVVGTA LI KKYPKLESEF VYGDYKYVDV RKMIAKSEQE 1440
IGKATAKYFF YSNIMNFFKT EITLANGEIR KRPLIETNGE TGEIVWDKGR DFATVRKVL S 1500
MPQVNIKKKT EVQTTGGFSKE SILPKRNSDK LIARKKDWD P KKYGGFDSPT VAYSVLVVA K 1560
VEKGSKKLK SVKELLGITI MERSSFEKNP IDFLEAKGYK EVKDLI I KL PKYSLFELEN 1620
GRKRMASAG ELQKGNELAL PSKYVNFLYL ASHYEKLKGS PEDNEQKQLF VEQHKHYLDE 1680
IEQISEFSK RVILADANLD KVLSAYNKHR DKPIREQAEN IIHLFTLTNL GAPAAFKYFD 1740
TTIDRKRYTS TKEVLDATLI HQSITGLYET RIDLSQLGGD 1780

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SEQ ID NO: 33      moltype = AA  length = 1780
FEATURE           Location/Qualifiers
REGION            1..1780
                  note = Synthetic Polypeptide
source            1..1780
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 33
MDKKYSIGLD IGTSVGVAV ITDEYKVP SKFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSFPHR LEESFLVEED KKHERHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPDNDSD 180
VDKLFQQLVQ TYNQLFEEFN INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIALSLGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EYKFIKPIIL EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASQAS FIERMTNFDK NLPNEKVLPK HSLLYEYFTV YNELTKVKYV TEGMRKPAFL 540
SGEQKKAIVD LLFKTNRKVT VKQLKEDYFK KIECLAEGTR IFDPVTGTTH RIEDVVDGRK 600
PIHVVAARD GTLLARPVS WFDQGTDRVI GLRIAGGAIV WATPDHKVLT EYGWRAAGEL 660
RKGD RVAGPG GSGNSLALS L TADQMVSA LL DAEPPILYSE YDTPSPFSEA SMMGLLTNLA 720
DREL VHM INW AKRVP GFVDL TLHDQAHLL E RAWLEILMIG LVWRSMEHPG KLLFAPNLLL 780
DRNQGKCV EG MVEIFDMLLA TSSRFRMMNL QGEEFVCLKS IILLNSGVYT FLSTLKSLE 840
EKDHIHRA LD KITDTLIHLM AKAGLTQQQ HQR LAQLLLI LSHIRHMSNK GMEHLYSMKY 900
KNVPLYDL L LEMLD AHR LH AGGSGASRVQ AFADALDDKF LHDMLAELR YSVIREVLP T 960
RRARTFDLEV BELHTLV AEG VVHNC FDSV EISGVEDRPN ASLGTYHDL L KIIKDKDFLD 1020
NEENEDILED IVLTLTLFED REMIEERLKT YAHLPDDKVM KQLKRRRYTG WGRLSRKLIN 1080
GIRDQSGSGT ILDFLKSDFG ANRNFQMQLIH DDSLTFKEDI QKAQVSGQGD SLHEHIANLA 1140
GSPAIIKKGIL QTVKVVDDEL V KVMGRHKPEN IVIEMARENQ TTQKGQKNSR ERMKRIEEGI 1200
KELGSQLKE HPVENTQLQN EKLYLYYLQN GRDMYVDQEL DINRLSDYDV DHIVPQSFLK 1260
DDSIDNKVLT RSDKNRKGSD NVPSEEVVKK MKNYWRQLLN AKLITQRKFD NLTKAERGGL 1320
SELDKAGFIK RQLVETRQIT KHVAQILDSR MNTKYDENDK LIREVKVITL KSKLVSDFRK 1380
DFQFYKVR EI NNYHHAHDAY LNAVVGTA LI KKYPKLESEF VYGDYKYVDV RKMIAKSEQE 1440
IGKATAKYFF YSNIMNFFKT EITLANGEIR KRPLIETNGE TGEIVWDKGR DFATVRKVL S 1500
MPQVNIKKKT EVQTTGGFSKE SILPKRNSDK LIARKKDWD P KKYGGFDSPT VAYSVLVVA K 1560
VEKGSKKLK SVKELLGITI MERSSFEKNP IDFLEAKGYK EVKDLI I KL PKYSLFELEN 1620

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GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD			1780

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SEQ ID NO: 34      moltype = AA  length = 1780
FEATURE           Location/Qualifiers
REGION            1..1780
                  note = Synthetic Polypeptide
source            1..1780
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 34
MDKKYSIGLD  IGTNSVGWAV  ITDEYKVP SK  KFKVLGNTDR  HSIKKNLIGA  LLFDSGETAE  60
ATRLKRTARR  RYTRRKNRIC  YLQEIFSNEM  AKVDDSPFHR  LEESFLVEED  KKHERHPIFG  120
NIVDEVAYHE  KYPTIYHLRK  KLVDSSTDKAD  LRLIYLALAH  MIKFRGHFLI  EGDLPDNDSD  180
VDKLPFQLVQ  TYNQLFEENP  INASGVDAKA  ILSARLSKSR  RLENLIAQLP  GEKKNGLFGN  240
LIALSLGLTP  NFKSNFDLAE  DAKLQLSKDT  YDDDLNLLA  QIGDQYADLF  LAAKNLSDAI  300
LLSDILRVNT  EITKAPLSAS  MIKRYDEHHQ  DLTLKALVR  QQLPEKYKEI  FFDQSKNGYA  360
GYIDGGASQE  EFYKFIKPIL  EKMDGTEELL  VKLNREDLLR  KQRTFDNGSI  PHQIHLGELH  420
AILRRQEDFY  PFLKDNREKI  EKILTFRIPY  YVGPLARGNS  RFAWMTRKSE  ETITPWNFEE  480
VVDKGASQAS  FIERMTNFDK  NLPNEKVLPK  HSLLYEYFTV  YNELTKVKYV  TEGMRKPAPL  540
SGEQKKAIVD  LLFKTNRKVT  VKQLKEDYFK  KIECPDSVEI  SGVEDRPNAS  LGTYHDL LKI  600
IKDKDFLDNE  ENEDILEDIV  LCLAEGTRIF  DPVTGTTHRI  EDVVDGRKPI  HVVAAAKDGT  660
LLARPVVSWF  DQGTDRVIGL  RIAGGAIVWA  TPDHKVLTVEY  GWRAAGELRK  GDRVAGPGGS  720
GNSLALSLTA  DQMVSAALDA  EPPILYSEYD  PTSPPSEASM  MGLLTNLADR  ELVHMINWAK  780
RVPGFVDLTL  HDQAHL LERA  WLEILMIGLV  WRSMEHPGKL  LFAPNLL LDR  NQKGCVEGMV  840
EIFDMLLAT S  SFRFRMNLQ G  EEFVCLKSII  LLNSGVYTF L  SSTLKSLEEK  DHIHRALDKI  900
TDTLIHLMAK  AGLTLQQHQ  RLAQLLLILS  HIRHMSNKG M  EHLYSMKYK N  VVPLYDL LLE  960
MLDAHRLHAG  GSGASRVQAF  ADALDDKFLH  DMLAEGLRYS  VIREVLPTRR  ARTFDLEVEE  1020
LHTLVAEGVV  VHNCLTFED  REMIEERLKT  YAHLPDDKVM  KQLKRRRYT G  WGRLSRKLIN  1080
GIRDKQSGKT  LDPLKSDGF  ANRNFMLQI H  DDSLTFKEDI  QKAQVSGQGD  SLHEHIANLA  1140
GSPAICKGIL  QTVKVVDL V  KVMGRHKPEN  IVIEMARENQ  TTQKGQKNS R  ERMKRIEEGI  1200
KELGSQILKE  HPVENTQLQ N  EKLYLYYLQ N  GRDMYVDQEL  DINRLSDYD V  DHIVPQSFLK  1260
DDSIDNKVLT  RSDKNRGKSD  NVPSEEVVKK  MKNYWRQLLN  AKLITQRKFD  NLTKAERGGL  1320
SELDKAGFIK  RQLVETRQIT  KHVAQILDSR  MNTKYDENDK  LIREVKVITL  KSKLVSDFRK  1380
DPQFYKVRRI  NNYHHAHDAY  LNAVVGTA LI  KKYPKLESEF  VYGDYKVYDV  RKMIAKSEQE  1440
IGKATAKYFF  YSNIMNPFKT  EITFLANGEIR  KRPLIETNGE  TGEIVWDKGR  DFATVRKVL S  1500
MPQVNI VKKT  EVQTGGFSKE  SILPKRNSDK  LIARKKDWD P  KKYGGFDSPT  VAYSVLVVA K  1560
VEKGSKKKL K  SVKELLGITI  MERSSFEKNP  IDFLEAKGYK  EVKKDLI IKL  PKYSLFELEN  1620
GRKRMLASAG  ELQKGNELAL  PSKYVNFLYL  ASHYEKLKGS  PEDNEQKQLF  VEQHKHYLDE  1680
IEQISEFSK  RVILADANLD  KVLSAYNKHR  DKPIREQAEN  IIHLFTLTNL  GAPAAFKYFD  1740
TTIDRKRYTS  TKEVLDATLI  HQSITGLYET  RIDLSQLGGD  1780

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SEQ ID NO: 35      moltype = AA  length = 1780
FEATURE           Location/Qualifiers
REGION            1..1780
                  note = Synthetic Polypeptide
source            1..1780
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 35
MDKKYSIGLD  IGTNSVGWAV  ITDEYKVP SK  KFKVLGNTDR  HSIKKNLIGA  LLFDSGETAE  60
ATRLKRTARR  RYTRRKNRIC  YLQEIFSNEM  AKVDDSPFHR  LEESFLVEED  KKHERHPIFG  120
NIVDEVAYHE  KYPTIYHLRK  KLVDSSTDKAD  LRLIYLALAH  MIKFRGHFLI  EGDLPDNDSD  180
VDKLPFQLVQ  TYNQLFEENP  INASGVDAKA  ILSARLSKSR  RLENLIAQLP  GEKKNGLFGN  240
LIALSLGLTP  NFKSNFDLAE  DAKLQLSKDT  YDDDLNLLA  QIGDQYADLF  LAAKNLSDAI  300
LLSDILRVNT  EITKAPLSAS  MIKRYDEHHQ  DLTLKALVR  QQLPEKYKEI  FFDQSKNGYA  360
GYIDGGASQE  EFYKFIKPIL  EKMDGTEELL  VKLNREDLLR  KQRTFDNGSI  PHQIHLGELH  420
AILRRQEDFY  PFLKDNREKI  EKILTFRIPY  YVGPLARGNS  RFAWMTRKSE  ETITPWNFEE  480
VVDKGASQAS  FIERMTNFDK  NLPNEKVLPK  HSLLYEYFTV  YNELTKVKYV  TEGMRKPAPL  540
SGEQKKAIVD  LLFKTNRKVT  VKQLKEDYFK  KIECPDSVEI  SGVEDRPNAS  LGTYHDL LKI  600
IKDKDFLDNE  ENEDILEDIV  LTLTFEDRE  MIEERLKT YA  HLPDDKVMKQ  LKRRRYTGWG  660
RLSRKLINGI  RDKQSGKTIL  DFLKSDGFAN  RNFMQLIHDD  CLAEGTRIFD  PVTGTTHRIE  720
DVVDGRKPIH  VVAAAKDGT L  LARPVVSWF D  QGTDRVIGLR  IAGGAIVWAT  PDHKVLTVEY  780
WRAAGELRKG  DRVAGPGGSG  NSLALSLTAD  QMVSAALDAE  PPILYSEYDP  TSPFSEASMM  840
GLLTNLADRE  LVHMINWAKR  VPGFVDLTLH  DQAHL LERAW  LEILMIGLVW  RSMHPGKLL  900
FAPNLLLDN  QGKCVEGMVE  IFDMLLATSS  RFRMNLQGE  EBFVCLKSII L  LNSGVYTFLS  960
STLKSLEEK  HIHRALDKIT  DTLIHLMAKA  GLTLQQQHQR  LAQLLLLILSH  IRHMSNKGME  1020
HLYSMKYKNV  QTYD LLEEM  LDAHRLHAGG  SGASRVQAF A  DALDDKFLHD  MLAEGLRYSV  1080
IREVLPTRRA  RTFDLEVEEL  HTLVAEGVVV  HNCLTFKEDI  QKAQVSGQGD  SLHEHIANLA  1140
GSPAICKGIL  QTVKVVDL V  KVMGRHKPEN  IVIEMARENQ  TTQKGQKNS R  ERMKRIEEGI  1200
KELGSQILKE  HPVENTQLQ N  EKLYLYYLQ N  GRDMYVDQEL  DINRLSDYD V  DHIVPQSFLK  1260
DDSIDNKVLT  RSDKNRGKSD  NVPSEEVVKK  MKNYWRQLLN  AKLITQRKFD  NLTKAERGGL  1320
SELDKAGFIK  RQLVETRQIT  KHVAQILDSR  MNTKYDENDK  LIREVKVITL  KSKLVSDFRK  1380
DPQFYKVRRI  NNYHHAHDAY  LNAVVGTA LI  KKYPKLESEF  VYGDYKVYDV  RKMIAKSEQE  1440
IGKATAKYFF  YSNIMNPFKT  EITFLANGEIR  KRPLIETNGE  TGEIVWDKGR  DFATVRKVL S  1500
MPQVNI VKKT  EVQTGGFSKE  SILPKRNSDK  LIARKKDWD P  KKYGGFDSPT  VAYSVLVVA K  1560

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VEKGKSKKLK	SVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIKL	PKYSLFELEN	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD			1780

SEQ ID NO: 36 moltype = AA length = 1780
 FEATURE Location/Qualifiers
 REGION 1..1780
 note = Synthetic Polypeptide
 source 1..1780
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 36

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVPK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM	AKVDDSFHHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD	LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFEEEN	INASGVDAKA	ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240
LTALSLGLTP	NFKSNFDLAE	DAKLQLSKDT	YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ	DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360
GYIDGGASQE	EFYKFIKPIL	EKMDGTEELL	VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY	YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480
VVDKGASQAS	PIERMNTNFDK	NLPNEKVLPK	HSLLYEYFTV	YNELTKVKYV	TEGMRKPAFL	540
SGEQKKAIVD	LLPKTNRKVT	VKQLKEDYFK	KIECFDSVEI	SGVEDRFNAS	LGTYHDLLKI	600
IKDKDFLDNE	ENEDILEDIV	LTTLTLFEDRE	MIEERLKTYA	HLFDDKVMKQ	LKRRRYTGWG	660
RLSRKLINGI	RDQSGKTIL	DFLKSDFGAN	RNFMLIHDD	SLTFKEDIQK	AQVSGQGDLS	720
HEHIANLCLA	EGTRIFDPVT	GTTHRIEDVV	DGRKPIHVVA	AAKDGTLAR	PVVSFWDQGT	780
RDVIGLRIAG	GAIVWATPDH	KVLTEYGWRA	AGELRKGRDV	AGPGGSGNSL	ALSLTADQMV	840
SALLDAEPII	LYSEYDPTSP	FSEASMMGLL	TNLADRELVA	MINWAKRVP	FVDLTLDHQA	900
HLLERAWLEI	LMIGLVWRSM	EHPGKLLFAP	NLLLDNRQKQ	CVEGMVEIFD	MLLATSSRRF	960
MMNLQGEFV	CLKSIIILNS	GVYTFLSSTL	KSLEEKDHIH	RALDKITDTL	IHLMAKAGLT	1020
LQQQHQLRAQ	LLLILSHIRH	MSNKGMEHLY	SMKYKNVPL	YDLLEMLDA	HLRHAGSGSA	1080
SRVQAFADAL	DDKFLHDMLA	EGLRYSVIRE	VLPTRRARTF	DLEVEELHTL	VAEGVVVHNC	1140
GSPAIIKKGIL	QTVKVVDELV	KVMGRHKPEN	IVIMARENQ	TTQKGQKNSR	ERMKRIEEGI	1200
KELGSQILKE	HPVENTQLQN	EKLYLYLQNG	GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260
DDSIDNKVLT	RSDKNRGKSD	NVPSEEVVKK	MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320
SELDKAGFIK	RQLVETRQIT	KHVAQILDSR	MNTKYDENDK	LIREVKVITL	KSKLVSDPRK	1380
DFQFYKVRRI	NNYHHAHDAY	LNAVVGITALI	KKYPKLESEF	VYGDYKVDV	RKMIKSEQE	1440
IGKATAKYFF	YSNIMNFFKT	EITLANGEIR	KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIKKTK	EVQTTGGFSKE	SILPKRNSDK	LTARKKDWD	KKYGGFDSPT	VAYSVLVVAK	1560
VEKGKSKKLK	SVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIKL	PKYSLFELEN	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD			1780

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ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM	AKVDDSFHHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD	LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFEEEN	INASGVDAKA	ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240
LTALSLGLTP	NFKSNFDLAE	DAKLQLSKDT	YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ	DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360
GYIDGGASQE	EFYKFIKPIL	EKMDGTEELL	VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY	YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480
VVDKGASQAS	PIERMNTNFDK	NLPNEKVLPK	HSLLYEYFTV	YNELTKVKYV	TEGMRKPAFL	540
SGEQKKAIVD	LLPKTNRKVT	VKQLKEDYFK	KIECFDSVEI	SGVEDRFNAS	LGTYHDLLKI	600
IKDKDFLDNE	ENEDILEDIV	LTTLTLFEDRE	MIEERLKTYA	HLFDDKVMKQ	LKRRRYTGWG	660
RLSRKLINGI	RDQSGKTIL	DFLKSDFGAN	RNFMLIHDD	SLTFKEDIQK	AQVSGQGDLS	720
HEHIANLAGS	PAIKKGILQT	VKVVDLVKV	MGRHKPENIV	IEMARENQTT	QKGQKNSRER	780
MKRIEEGIK	LGSQLKEHP	VENTQLQNEK	LYLYYLQNGR	DMYVDQELDI	NRLSDYDVH	840
IVPQSFLKDD	SIDNKVLT	DKNRGKSDNV	PSEEVVKKMK	NYWRQLLNK	LITQRKFDNL	900
TKAERGGLSE	LDKAGFIKRQ	LVETRQITKH	VQILDSTRMN	TKYDENDKLI	REVKVITLKS	960
KLVSDFRKDF	QFYKVRINN	YHHAHDAYLN	AVVGCLAEGT	RIFDPVTGTT	HRIEDVVDGR	1020
KPIHVVAANK	GGTLLARPVV	SWFDQGTDRV	IGLRIAGGAI	VWATPDHKVL	TEYGWRAAGE	1080
LRKGRVAGP	DGSGNSLALS	LTADQMVSA	LDAEPPILYS	EYDPTSPFSE	ASMMGLLTNL	1140
ADRELVHMIN	WAKRVPGFVD	LTLDQAHLL	ERAWLEILMI	GLVWRSMHP	GKLLFAPNLL	1200
LDNRQKCVVE	GMVEIFDMLL	ATSSRFMMN	LQGEFFVCLK	SIIILLNSGVY	TFLSSTLKL	1260
EKKDHHRAL	DKITDTLHLL	MAKAGLTQQ	QHQLAQALLL	ILSHIRHMSN	KGMEHLYSMK	1320
YKNVVPYLDL	LLEMLDAHRL	HAGGSGASRV	QAFADALDDK	FLHDMLEAGL	RYSVIREVLP	1380
TRRARTFDLE	VEELHTLVAE	GVVVHNCALI	KKYPKLESEF	VYGDYKVDV	RKMIKSEQE	1440
IGKATAKYFF	YSNIMNFFKT	EITLANGEIR	KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIKKTK	EVQTTGGFSKE	SILPKRNSDK	LTARKKDWD	KKYGGFDSPT	VAYSVLVVAK	1560
VEKGKSKKLK	SVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIKL	PKYSLFELEN	1620

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GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD			1780

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SEQ ID NO: 38      moltype = AA  length = 1780
FEATURE            Location/Qualifiers
REGION             1..1780
                   note = Synthetic Polypeptide
source             1..1780
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 38
MDKKYSIGLD IGTSVGVAV ITDEYKVP SKFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSPFHR LEESFLVEED KKHERHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPDNDSD 180
VDKLPQLVQ TYNQLFERNP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIALSLGLTP NFKNFNLDAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EYFKFIKPI EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASQAS FIERMTNFDK NLPNEKVLPK HSLLYEYFTV YNELTKVKYV TEGMRKPAPL 540
SGEQKKAIVD LLFKTNRKVT VKQLKEDYFK KIECPDSVEI SGVEDRFNAS LGTYHDLKKI 600
IKDKDFLDNE ENEDILEDIV LTLTLFEDRE MIEERLKTYA HLFDDKVMKQ LKRRRYTGWG 660
RLSRKLINGI RDKQSGKTIL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGDSL 720
HEHIANLAGS PAIKKGILQT VKVVDLVKV MGRHKPENIV IEMARENQTT QKGQKNSRER 780
MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMYVDQELDI NRLSDYDVDH 840
IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV PSEEVVKMK NYWRQLLNAK LITQRKFDNL 900
TKAERGLSE LDKAGFIKRP LVETRQITKH VAQILDSRMN TKYDENDKLI REVKVITLKS 960
KLVSDFRKDF QFYKVRINN YHHAHDAYLN AVVGTAIIK YPKLECLAE TRIFDPVTGT 1020
THRIEDVVDG RKP IHVAAA KDGTLARPV VSWFDQGTDR VIGLRIAGGA IWATPDHKV 1080
LTEYGWRAAG ELRGDVRNAG PGSGNSLAL SLTADQMVSA LLDAEPPILY SEYDPTSPFS 1140
EASMMGLLTN LADRELHMI NWAQRVPGFV DLTLDQAH LERAWLEILM IGLVWRSMEH 1200
PGKLLFAPNL LLDRNQKCV EGMVEIFDML LATSSRFMRM NLQGEFVCL KSIILLNSGV 1260
YTFLSSTLKS LEEKDHILRA LKIDTDLIH LMAKAGTLQ QHQRLAQLL LILSHIRHMS 1320
NKGMEHLYSM KYKNVPLYD LLEMLDAHR LHAGGSGASR VQAFADALDD KFLHDMLEAG 1380
LRYSVIREVL PTRRARTFDL EVEELHTLVA EGVVVHNC EFVGDYKVDV RKMIAKSEQE 1440
IGKATAKYFF YSNIMNFKTI EITLANGEIR KRPLIETNGE TGEIVWDKGR DFATVRKVL 1500
MPQVNIKKTK EVQGTGGSKE SILPKRNSDK LIARKKWDWP KKYGGFDSPT VAYSVLVVA 1560
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IEQISEFSK RVILADANLD KVLSAYNKHR DKPIREQAEN IIHLFTLTNL GAPAAFKYFD 1740
TTIDRRKRYTS TKEVLDATLI HQSITGLYET RIDLSQLGGD 1780

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FEATURE            Location/Qualifiers
REGION             1..1780
                   note = Synthetic Polypeptide
source             1..1780
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 39
MDKKYSIGLD IGTSVGVAV ITDEYKVP SKFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSPFHR LEESFLVEED KKHERHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPDNDSD 180
VDKLPQLVQ TYNQLFERNP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIALSLGLTP NFKNFNLDAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EYFKFIKPI EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASQAS FIERMTNFDK NLPNEKVLPK HSLLYEYFTV YNELTKVKYV TEGMRKPAPL 540
SGEQKKAIVD LLFKTNRKVT VKQLKEDYFK KIECPDSVEI SGVEDRFNAS LGTYHDLKKI 600
IKDKDFLDNE ENEDILEDIV LTLTLFEDRE MIEERLKTYA HLFDDKVMKQ LKRRRYTGWG 660
RLSRKLINGI RDKQSGKTIL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGDSL 720
HEHIANLAGS PAIKKGILQT VKVVDLVKV MGRHKPENIV IEMARENQTT QKGQKNSRER 780
MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMYVDQELDI NRLSDYDVDH 840
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TKAERGLSE LDKAGFIKRP LVETRQITKH VAQILDSRMN TKYDENDKLI REVKVITLKS 960
KLVSDFRKDF QFYKVRINN YHHAHDAYLN AVVGTAIIK YPKLESEFVY GDYKVDVVRK 1020
MTAKSEQEIG KATAKYFFYS NIMNFKTEI TLANGEIRK PLIETNGETG EIVWDKGRDF 1080
ATVRKVLSPM QKNIVKTEV QTGGFSKESI LPKRNSDKLI ARKKDWDPKK YGGFDSPTVA 1140
YSVLVVAKE VQCLAEGRIT IFDPVTGTH RIEDVVDGRK PIHVVAACKD GTLLARPVS 1200
WFDQGTDRVI GLRIAGGAIW WATPDHKVLT EYGWRAAGEL RKGDRVAGPG GSGNSLALS 1260
TADQMVALL DAEPPILYSE YDPTSPFSEA SMMGLLTNLA DRELHVMINW AKRVPGFVLD 1320
TLHDQAHLL EAEILMIG LVWRSMEHPG KLLFAPNLLL DRNQKCVG MVEIFDMLLA 1380
TSSRFMRMNL QGEFVCLKS IILLNSGVYT FLSTLKSLE EKDHILRALD KITDTLIHLM 1440
AKAGTLQQLQ HQRLAQLLLI LSHIRHMSNK GMEHLYSMKY KNVVPLYDLL LEMLDARHLH 1500
AGGSGASRVQ AFADALDDKF LHMMLAEGLR YSVIREVLPT RRARTFDLEV EELHTLVAEG 1560
VVHNCCKLKV SVKELLGITI MERSSFEKNP IDFLKAKGYK EVKDKLIKL PKYSLFELEN 1620
GRKRMLASAG ELQKGNELAL PSKYVNFLYL ASHYEKLKGS PEDNEQKQLF VEQHKHYLDE 1680

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IIEQISEFSK	RVILADANLD	KVLSAYNKH	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDTALI	HQSITGLYET	RIDLSQLGGD			1780

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SEQ ID NO: 40      moltype = AA length = 1780
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source            1..1780
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SEQUENCE: 40
MDKKYSIGLD IGTNSVGWAV ITDEYKVP SK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSFHHR LEESFLVEED KKHHRHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPNDNSD 180
VDKLFQILVQ TYNQLFEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIAALSGLTP NFKSNFLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EFYKFIKPI EKMMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASASQ FIERMTNFDK NLPNEKVLKP HSLLYEYFTV YNELTKVKYV TEGMRKPAFL 540
SGEQKKAIVD LLFKNTRKVT VKQLKEDYFK KIECPDSVEI SGVEDRPNAS LGTYHDLLEKI 600
IKDKDFLDNE ENEDILEDIV LTLTLFEDRE MIEERLKTYA HLPDDKVMKQ LKRRRYTGWG 660
RLSRKLINGI RDKQSGKITL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGDSL 720
HEHIANLAGS PAIKKGILQT VKVVDLVKV MGRHKPENIV IEMARENQTT QKGQKNSRER 780
MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMYVDQELDI NRLSDYDVVDH 840
IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV PSEEVVKKMK NYWRQLLNAK LITQRKFDNL 900
TKAERGGGLSE LDKAGFIKRO LVETRQITKH VAQILDSTRM TKYDENDKLI REVKVITLKS 960
KLVSDFRKDF QFYKVFREIN YHHAHDAYLN AVVGTAIIKK YPKLESEFVY GDYKVDVVRK 1020
MIAKSEQEIG KATAKYFFYS NIMNFPKTEI TLANGEIRKR PLIETNGETG EIVWDKGRDF 1080
ATVRKVLSPM QVNVVKKTEV QTGGFSKESI LPKRNSDKLI ARKKDWDPPK YGGFDSPTVA 1140
YSVLVVAKVE KGKSKKLKCL AEGTRIFDPV TGTTHRIEDV VDGRKPIHVV AAKDGTLA 1200
RPVVSFWDQG TRDVLGLRIA GGAIVWATPD HKVLTEYGWR AAGELRKGDR VAGPGGSGNS 1260
LALSITADQM VSALLDAEPP ILYSEYDPTS PFSEASMMGL LTNLADRELH HMINWAKRVP 1320
GFVDLTLHDQ AHLLEAWLE ILMIGLVWRS MEHPGKLLFA PNLLLDNRNQ KCVEGMVEIF 1380
DMLLATSSRF RMMLQGEFF VCLKSIIILN SGVYTFLSST LKSLLEEKDHI HRALDKITDT 1440
LIHLMKAGL TLQQQHQRLA QLLILSHIR HMSNKGMEHL YSMKYKNVVP LYDLLLEMLD 1500
AHLRHAGGSG ASRVQAFADA LDDKFLHMDL AEGLRYSVIR EVLPTRRART FDLEVEELHT 1560
LVAEGVVVHN CVKELLGTIT MERSSFEKNP IDFLEAKGYK EVKDLI IKL PKYSLFELEN 1620
GRKRMLASAG ELQKGNELAL PSKYVNFLYL ASHYEKLKGS PEDNEQKQLF VEQHKHYLDE 1680
IIEQISEFSK RVILADANLD KVLSAYNKH DKPIREQAEN IIHLFTLTNL GAPAAFKYFD 1740
TTIDRKRYTS TKEVLDTALI HQSITGLYET RIDLSQLGGD 1780

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                  note = Synthetic Polypeptide
source            1..1780
                  mol_type = protein
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SEQUENCE: 41
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ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSFHHR LEESFLVEED KKHHRHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPNDNSD 180
VDKLFQILVQ TYNQLFEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIAALSGLTP NFKSNFLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EFYKFIKPI EKMMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASASQ FIERMTNFDK NLPNEKVLKP HSLLYEYFTV YNELTKVKYV TEGMRKPAFL 540
SGEQKKAIVD LLFKNTRKVT VKQLKEDYFK KIECPDSVEI SGVEDRPNAS LGTYHDLLEKI 600
IKDKDFLDNE ENEDILEDIV LTLTLFEDRE MIEERLKTYA HLPDDKVMKQ LKRRRYTGWG 660
RLSRKLINGI RDKQSGKITL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGDSL 720
HEHIANLAGS PAIKKGILQT VKVVDLVKV MGRHKPENIV IEMARENQTT QKGQKNSRER 780
MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMYVDQELDI NRLSDYDVVDH 840
IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV PSEEVVKKMK NYWRQLLNAK LITQRKFDNL 900
TKAERGGGLSE LDKAGFIKRO LVETRQITKH VAQILDSTRM TKYDENDKLI REVKVITLKS 960
KLVSDFRKDF QFYKVFREIN YHHAHDAYLN AVVGTAIIKK YPKLESEFVY GDYKVDVVRK 1020
MIAKSEQEIG KATAKYFFYS NIMNFPKTEI TLANGEIRKR PLIETNGETG EIVWDKGRDF 1080
ATVRKVLSPM QVNVVKKTEV QTGGFSKESI LPKRNSDKLI ARKKDWDPPK YGGFDSPTVA 1140
YSVLVVAKVE KGKSKKLKSV KELLGITIME RSSFEKNPID PLEAKGYKEV KKDILIKLPK 1200
YSLFELENGR KRMLASAGEL QKGNELALPS KYVNFLYLAS HYEKLKGSPE DNEQKQLFVE 1260
QHKHYLDEII EQICLAEQTR IFDPVTGTH RIEDVVDGRK PIHVVAARD GTLLARPVVS 1320
WFDQGTDRVI GLRIAGGAIW WATPDHKVLT EYGWRAAGEL RKGDRVAGPG GSGNSLALSL 1380
TADQMVSAAL DAEPPILYSE YDPTSPFSEA SMMGLLTNLA DRELHVHMINW AKRVPGFVDL 1440
TLHDQAHLLQ QAWLEILMIG LVWRSMHPG KLLFAPNLLL DRNQGKCVGE MVEIFDMLLA 1500
TSSRFRRMNL QGEEFVCLKS IILLNSGVYT FLSSTLKSL EKDHIHRA LD KITDTLIHLM 1560
AKAGLTLQQQ HQRLAQLLLI LSHIRHMSNK GMEHLYSMKY KNVVPYLDLL LEMLDARHLH 1620

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AGSGASRVQ AFADALDDKF LHDMLAELR YSVIREVLPT RRARTFDLEV EELHTLVAEG 1680
VVVHNCEFSK RVILADANLD KVL SAYNKH R DKPIREQAEN IIHLFTLTNL GAPAAFKYFD 1740
TTIDRKRYTS TKEVL DATLI HQSITGLYET RIDLSQLGGD 1780

```

```

SEQ ID NO: 42      moltype = AA  length = 16
FEATURE           Location/Qualifiers
REGION           1..16
                  note = Synthetic Polypeptide
source           1..16
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 42
MAPKKRKKVG IHRGVP 16

```

```

SEQ ID NO: 43      moltype = AA  length = 23
FEATURE           Location/Qualifiers
REGION           1..23
                  note = Synthetic Polypeptide
source           1..23
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 43
MDYKDHDGDY KDHDIDYKDD DDK 23

```

```

SEQ ID NO: 44      moltype = DNA  length = 432
FEATURE           Location/Qualifiers
misc_feature      1..432
                  note = Synthetic Polynucleotide
source           1..432
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 44
atggccctgt ttggctacgc acgcgtgtct accagtcaac agtcactcga ttgcaagtg 60
agggctctta aagatgccgg agtgaaggca aacagaattt ttactgataa ggccagcgga 120
agcagcacag acagagaggg gctggatctc ctgagaatga aggtaaagga gggatgatgtg 180
atcttggtca aaaaattgga tcgactgggg agagacacag ctgatatgct tcagcttatt 240
aaagagtttg acgctcaggg tgttgccgtg aggtttatcg atgacggcat ctcaaccgac 300
tcctacattg gtcttatgtt tgtgacaatt ttgtccgtg ttggtcaggc tgagcggaga 360
aggattctcg aaaggacgaa tgagggacgg caagcagcta agttgaaagg tatcaaat 420
ggcagacgaa gg 432

```

```

SEQ ID NO: 45      moltype = AA  length = 144
FEATURE           Location/Qualifiers
REGION           1..144
                  note = Synthetic Polypeptide
source           1..144
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 45
MALFGYARVS TSQQSLDLQV RALKDAGVKA NRIFTDKASG SSTDREGLDL LRMKVKEGDV 60
ILVKKLDRLG RDTADMLQLI KEFDAQGVAV RFIDDGISTD SYIGLMFVTI LSAVAQAERR 120
RILERTNEGR QAAKLKGIKF GRRR 144

```

```

SEQ ID NO: 46      moltype = DNA  length = 414
FEATURE           Location/Qualifiers
misc_feature      1..414
                  note = Synthetic Polynucleotide
source           1..414
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 46
atggcaacca ttggctacat aagggtgtct accatcgacc aaaatatcga cctgcagcgc 60
aacgctctga catccgcaa ctgcgatcgg atcttcgagg ataggatcag tggcaagatc 120
gccaaccggc cgggtctgaa gcggtctctg aagtagctga ataaggcgga tactctggtt 180
gtgtggaagt tggatcgctt gggtagatca gtgaagaatc tcgtagccct gataagcgag 240
ctgcacgaga ggggtgcaca ttccattct ctgaccgatt ccatcgatac gtctagcgcc 300
atgggcccga tcttctttta cgtcatgtcc gccctcgctg aaatggagcg cgaacttatt 360
gttgaacgga ctttgctggt actggcagcg gctagagcac agggccgact tgga 414

```

```

SEQ ID NO: 47      moltype = AA  length = 138
FEATURE           Location/Qualifiers
REGION           1..138
                  note = Synthetic Polypeptide
source           1..138
                  mol_type = protein
                  organism = synthetic construct

```

-continued

SEQUENCE: 47
 MATIGYIRVS TIDQNIDLQR NALTSANCDR IFEDRISGKI ANRPGLKRAL KYVNGDITLV 60
 VKLDRLGRS VKNLVALISE LHERGAHFHS LTDSIDTSSA MGRFFFYVMS ALAEMERELI 120
 VERTLAGLAA ARAQGRIG 138

SEQ ID NO: 48 moltype = DNA length = 432
 FEATURE Location/Qualifiers
 misc_feature 1..432
 note = Synthetic Polynucleotide
 source 1..432
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 48
 atgtcattg gctatgtaag ggtcagcacc aatgacccaa acacagactt gcaacgcaat 60
 gctttggttt gcgcgggatg tgaacagata ttgaagata aactgagcgg cactcggaca 120
 gacagacctg ggcttaagag agcactgaaa agactgcaga aggggggacac cctggctcgtc 180
 tggaaactgg atcgccctgg acgcagcatg aaacatctga ttagcctggt tggtagactt 240
 agggagagag gaatcaactt cagaagcctg accgactcca tcgacaccag tagcccccag 300
 ggacgattct tcttctatgt gatgggagca cttgctgaga tggaaagaga gcttattatc 360
 gaaagaacta tggctggtat cgctgctgcc cggaacaaag gcagacgggt cggcagaccg 420
 cgaagagcgc gc 432

SEQ ID NO: 49 moltype = AA length = 144
 FEATURE Location/Qualifiers
 REGION 1..144
 note = Synthetic Polypeptide
 source 1..144
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 49
 MLIGVVRVST NDQNTDLQRN ALVCAGCEQI FEDKLSGTRT DRPGLKRALK RLQKGDITLVV 60
 WKLDRLGRSM KHLISLVGEL RERGINFRSL TDSIDTSSPM GRFFFYVMGA LAEMERELII 120
 ERTMAGIAAA RNKGRRFRGP PKSG 144

SEQ ID NO: 50 moltype = AA length = 198
 FEATURE Location/Qualifiers
 source 1..198
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 50
 MDSLLMNRK FLYQFKNVRW AKGRRETYLC YVVKRRDSAT SFSLDFGYLR NKGCHVELL 60
 FLRYISDWDL DPGRCYRVTW FTSWSPCYDC ARHVADFLRG NPNLSLRIFT ARLYFCEDRK 120
 AEPEGLRRLH RAGVQIAIMT FKDYFYCWNT FVENHERTEFK AWEGLHENS VRLSRQLRRL 180
 LPLYEVDDL R DAFRTLGL 198

SEQ ID NO: 51 moltype = AA length = 198
 FEATURE Location/Qualifiers
 source 1..198
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 51
 MDSLLMKQK FLYHFKNVRW AKGRHETLYC YVVKRRDSAT SCSLDFGHLR NKGCHVELL 60
 FLRYISDWDL DPGRCYRVTW FTSWSPCYDC ARHVAEFLRW NPNLSLRIFT ARLYFCEDRK 120
 AEPEGLRRLH RAGVQIGIMT FKDYFYCWNT FVENRETEFK AWEGLHENS VRLSRQLRRL 180
 LPLYEVDDL R DAFRTLGL 198

SEQ ID NO: 52 moltype = AA length = 198
 FEATURE Location/Qualifiers
 source 1..198
 mol_type = protein
 note = sub-species familiaris
 organism = Canis lupus

SEQUENCE: 52
 MDSLLMKQK FLYHFKNVRW AKGRHETLYC YVVKRRDSAT SFSLDFGHLR NKGCHVELL 60
 FLRYISDWDL DPGRCYRVTW FTSWSPCYDC ARHVADFLRG YPNLSLRIFA ARLYFCEDRK 120
 AEPEGLRRLH RAGVQIAIMT FKDYFYCWNT FVENRETEFK AWEGLHENS VRLSRQLRRL 180
 LPLYEVDDL R DAFRTLGL 198

SEQ ID NO: 53 moltype = AA length = 199
 FEATURE Location/Qualifiers
 source 1..199
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 53
 MDSLLKKQK FLYQFKNVRW AKGRHETLYC YVVKRRDSPT SFSLDFGHLR NKAGCHVELL 60
 FLRYISDWDL DPGRCYRVTW FTSWSPCYDC ARHVADFLRG YPNLSLRIFT ARLYFCEDRK 120
 KAEPEGLRRL HRAGVQIAIM TFKDYFYCWNT FVENHERTEFK AWEGLHENS VRLSRQLRRI 180
 LLPLYEVDDL R DAFRTLGL 199

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SEQ ID NO: 54 moltype = AA length = 429
 FEATURE Location/Qualifiers
 source 1..429
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 54

MGPFCLGCSH	RKCYSPINRL	ISQETFKPHF	KNLGYAKGRK	DTFLCYEVTR	KDCDSPVSLH	60
HGVFNKNDNI	HAEICFLYWF	HDKVLKVLSP	REEFKITWYM	SWSPCFECAE	QIVRFLATHH	120
NLSLDIFSSR	LYNVQDPETQ	QNLCLRVQEG	AQVAAMDLYE	FKKCWKKFVD	NGGRRFRPWK	180
RLLTNFRYQD	SKLQEILRPC	YIPVPSSSSS	TLSNICLTKG	LPETRFCEVEG	RRMDPLSEEE	240
FYSQFYNQRV	KHLCYYHRMK	PYLCYQLEQF	NGQAPLKGCL	LSEKKGQHAE	ILFLDKIRSM	300
ELSQVTITCY	LTWSPCPNCA	WQLAAPKDRD	PDLILHIYTS	RLYFHWKRPF	QKGLCSLWQS	360
GILVDVMDLP	QFTDCWNTFV	NPKRPFWPWK	GLEIISRRTQ	RRLRRIKESW	GLQDLVNDFG	420
NLQLGPPMS						429

SEQ ID NO: 55 moltype = AA length = 429
 FEATURE Location/Qualifiers
 source 1..429
 mol_type = protein
 organism = Rattus rattus

SEQUENCE: 55

MGPFCLGCSH	RKCYSPINRL	ISQETFKPHF	KNLRYAIDRK	DTFLCYEVTR	KDCDSPVSLH	60
HGVFNKNDNI	HAEICFLYWF	HDKVLKVLSP	REEFKITWYM	SWSPCFECAE	QVLRFLATHH	120
NLSLDIFSSR	LYNIRDPENQ	QNLCLRVQEG	AQVAAMDLYE	FKKCWKKFVD	NGGRRFRPWK	180
KLLTNFRYQD	SKLQEILRPC	YIPVPSSSSS	TLSNICLTKG	LPETRFCEVER	RRVHLLSEEE	240
FYSQFYNQRV	KHLCYYHGK	PYLCYQLEQF	NGQAPLKGCL	LSEKKGQHAE	ILFLDKIRSM	300
ELSQVIITCY	LTWSPCPNCA	WQLAAPKDRD	PDLILHIYTS	RLYFHWKRPF	QKGLCSLWQS	360
GILVDVMDLP	QFTDCWNTFV	NPKRPFWPWK	GLEIISRRTQ	RRLHRIKESW	GLQDLVNDFG	420
NLQLGPPMS						429

SEQ ID NO: 56 moltype = AA length = 370
 FEATURE Location/Qualifiers
 source 1..370
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 56

MVEPMDPRTF	VSNFNNRPIL	SGLNTVWLCC	EVKTKDPSGP	PLDAKIFQ GK	VYSKAKYHPE	60
MRFLRWFKHW	RQLHHDQEQK	VTWYVSWSPC	TRCANSVATF	LAKDPKVTLT	IFVARLYYFW	120
KPDYQQALRI	LCQKRGGPHA	TMKIMNYNEF	QDCWNKFVDG	RGKPFKPRNN	LPKHYYTLQA	180
TLGELLRLHM	DPGTFTSNFN	NKPWVSGQHE	TYLCYKVERL	HNDTWVPLNQ	HRGFLRNQAP	240
NIHGPFKGRH	AELCFLDLIP	FWKLDGQQYR	VTCFTSWSPC	FSCAQEMAKF	ISNNEHVSLC	300
IFAARIYDDQ	GRYQEGRLAL	HRDGAKIAMM	NYSEFEYCWD	TFVDRQGRPF	QPWDGLDEHS	360
QALSGRLRAI						370

SEQ ID NO: 57 moltype = AA length = 384
 FEATURE Location/Qualifiers
 source 1..384
 mol_type = protein
 organism = Pan troglodytes

SEQUENCE: 57

MKPHFRNPVE	RMVQDTFSDN	FYNRPILSHR	NTVWLCYEVK	TKGPSRPPLD	AKIFRGQVYS	60
KLKYHPEMRF	PHWFSKWRKL	HRDQYEVETW	YISWSPCTKC	TRDVATFLAE	DPKVTLTIFV	120
ARLYYFWDPD	YQEARLSLCQ	KRDGPRATMK	IMNYDEFQHC	WSKFVYSQRE	LFEPWNNLPK	180
YYILLHIMLG	EILRHSMDPP	TFTSNFNNEL	WVRGRHETYL	CYEVEERLHND	TWVLLNQRRG	240
FLCNQAPHKH	GFLEGRHAEL	CFLDVIPFWK	LDLHQDYRVT	CFTSWSPCFSC	CAQEMAKFIS	300
NNKHVSLCIF	AARIYDDQGR	CQEGRLTLAK	AGAKISIMTY	SEFKHCWDTF	VDHQGCPFPQ	360
WDGLEEHSQA	LSGRLRAILQ	NQGN				384

SEQ ID NO: 58 moltype = AA length = 377
 FEATURE Location/Qualifiers
 source 1..377
 mol_type = protein
 organism = Chlorocebus sabaeus

SEQUENCE: 58

MNPQIRNMVE	QMEPDIFVY	FNNRPILSGR	NTVWLCYEVK	TKDPSGPPLD	ANIFQGLKYP	60
EAKDHPMEMF	LHWFRKWRQL	HRDQYEVETW	YVSWSPCTRC	ANSVATFLAE	DPKVTLTIFV	120
ARLYYFWKPD	YQALRLILCQ	ERGPHATMK	IMNYNEFQHC	WNEFVDGQ GK	PFKPRKNLPK	180
HYTLLHATLG	ELLRHVMDPG	TFTSNFNNKP	WVSGQRETYL	CYKVERSHND	TWVLLNQHRG	240
FLRNQAPDRH	GFPKGRHAEL	CFLDLIPFWK	LDDQYRVTC	FTSWSPCFSC	AQKMAKFISN	300
NKHVSLCIFA	ARIYDDQGR	QEGRLTLHRD	GAKIAVMNYS	EFYECWDTFV	DRQGRPFQPW	360
DGLDEHSQAL	SGRLRAI					377

SEQ ID NO: 59 moltype = AA length = 384
 FEATURE Location/Qualifiers
 source 1..384
 mol_type = protein
 organism = Homo sapiens

-continued

SEQUENCE: 59
 MKPHFRNTVE RMYRDTFSYN FYNRPILSRR NTVWLCYEVK TKGPSRPPLD AKIFRGQVYS 60
 ELKYHPEMRF FHWFSKWRKL HRDQYEYEVW YISWSPCTKC TRDMATFLAE DPKVTLTIFV 120
 ARLYYFWDPD YQEALRSLCQ KRDGPRATMK IMNYDEFQHC WSKFVYSQRE LFEFWNNLPK 180
 YYILLHIMLG EILRHSMDPP TTFNFMNNEP WVRGRHETYL CYEVERMHND TWVLLNQRRG 240
 FLCNQAPHKH GFLEGRHAEL CFLDVIPFWK LDLDQDYRVT CFTSWSPCFS CAQEMAKFIS 300
 KNKHVSLCIF TARIYDDQGR CQEGRLTAE AGAKISIMTY SEFKHCWDTF VDHQGCFFQP 360
 WDGLDEHSQD LSGRLRAILQ NQEN 384

SEQ ID NO: 60 moltype = AA length = 373
 FEATURE Location/Qualifiers
 source 1..373
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 60
 MKPHFRNTVE RMYRDTFSYN FYNRPILSRR NTVWLCYEVK TKGPSRPPLD AKIFRGQVYS 60
 QPEHHAEMCF LSWFCGNQLP AYKCFQITWF VSWTPCPDCV AKLAEFLAEH PNVTLTISAA 120
 RLYYYWERDY RRALCRLSQA GARVKIMDDE EFAYCWENFV YSEGQPFMPW YKFDNYAFL 180
 HRTLKEILRN PMEAMYPHIF YPHFKNLRKA YGRNESWLCF TMEVVKHSP VSWKRGVFRN 240
 QVDPETHCHA ERCFLSWFCD DILSPNTNVE VTWYTSWSPC PECAGEVAEF LARHSNVNLT 300
 IFTARLYYFW DTDYQEGLSR LSQEGASVEI MGYKDFKYCW ENFVYNDDPE FKPWKGLKYN 360
 FLFLDSKLQE ILE 373

SEQ ID NO: 61 moltype = AA length = 382
 FEATURE Location/Qualifiers
 source 1..382
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 61
 MNPQIRNPME RMYRDTFYDN FENEPILYGR SYTWLCYEVK IKGRSNLLW DTGVFRGQVY 60
 FKPQYHAEMC FLSWFCGNQL PAYKCFQITW FVSWTPCPDC VAKLAELFSE HPNVTLTISA 120
 ARLYYYWERD YRRALCRLSQ AGARVTIMDY EEFAYCWENF VYNEGQQFMP WYKFDENYAF 180
 LHRTLKEILR YLMDPDPTFT NFNNPLVLR RRQTYLCYEV ERLDNGTWVL MDQHMGLFCN 240
 EAKNLLCGFY GRHAELRFLD LVPSLQLDPA QIYRVTFWIS WSPCFSWGCA GEVRAFLQEN 300
 THVRLRIFAA RIYDYDPLYK EALQMLRDAG AQVSIMTYDE FEYCWDTFVY RQGCPPQPWD 360
 GLEHSQALS GRLRAILQNQ GN 382

SEQ ID NO: 62 moltype = AA length = 190
 FEATURE Location/Qualifiers
 source 1..190
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 62
 MNPQIRNPMK AMYPGTFYFQ FKNLWEANDR NETWLCFTVE GIKRRSVVSW KTGVPFRNQVD 60
 SETHCHAERC FLSWFCDDIL SPNTKYQVTW YTSWSPCPDC AGEVAEFLAR HSNVNLTIFT 120
 ARLYYFQYPC YQEGRLSLSQ EGVAVEIMDY EDPKYCWENF VYNDNEPKP WKGLKTNFRL 180
 LKRLRLRESLQ 190

SEQ ID NO: 63 moltype = AA length = 199
 FEATURE Location/Qualifiers
 source 1..199
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 63
 MEASPASGPR HLMDPHIFTS NFNNIGIRHK TYLCYEVERL DNGTSVKMDQ HRGFLHNQAK 60
 NLLCGFYGRH AELRFLDLVP SLQLDPAQIY RVTWFISWSP CFSWGCAGEV RAFLQENTHV 120
 RLRIFAARIY DYDPLYKEAL QMLRDAGAQV SIMTYDEFKH CWDTFVDHQG CFPQPWDGLD 180
 EHSQALSGLR RAILQNQGN 199

SEQ ID NO: 64 moltype = AA length = 200
 FEATURE Location/Qualifiers
 source 1..200
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 64
 MALLTAETFR LQFNNKRRLR RPYYPKALL CYQLTPQNGS TPTRGYFENK KKCHAEICFI 60
 NEIKSMGLDE TQCYQVTCYL TWSPCSSCAW ELVDFIKAHD HNLGIFASR LYHHWCKPQQ 120
 KGLRLLCGSQ VPVEVMGFPK FADCWENFVD HEKPLSFNPM KMLEELDKNR RAIKRLERI 180
 KIPGVRAQGR YMDILCDAEV 200

SEQ ID NO: 65 moltype = AA length = 386
 FEATURE Location/Qualifiers
 source 1..386
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 65
 MNPQIRNPME RMYRDTFYDN FENEPILYGR SYTWLCYEVK IKGRSNLLW DTGVFRGPVL 60
 PKRQSNHRQE VYFRFENHAE MCFLSWFCGN RLPANRRFQI TWFVSWNPCL PCVVVKTKFL 120
 AEHPNVTLTI SAARLYYYRD RDWRWVLLRL HKAGARVKIM DYEDFAYCWE NFVCNEGQPF 180

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MPWYKFDNDY ASLHRTLKEI LRNPMEAMYP HIFYFHFKNL LKACGRNESW LCFTMEVTKH 240
HSAVFRKRGV FRNQVDPETH CHAERCFLSW FCDDILSPNT NYEVTWYTSW SPCPECAGEV 300
AEFLARHSNV NLTIPTARLC YFWDTDYQEG LCSLSQEGAS VKIMGYKDFV SCWKNFVYSD 360
DEPFKPWKGL QTNFRLLKRR LREILQ 386

```

```

SEQ ID NO: 66      moltype = AA  length = 236
FEATURE           Location/Qualifiers
source            1..236
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 66
MTSEKGPSTG DPTLRRRIEP WEFDVFPYDPR ELRKEACLLY EIKWGMRSRKI WRSSGKNTTN 60
HVEVNFIIKK TSERDFHPSM SCSITWFLSW SPCWECSQAI REFLSRHPGV TLVIYVARLF 120
WHMDQONRQG LRDLVNSGVT IQIMRASEYY HCWRNFVNYP PGDEAHWPQY PPLWMMLYAL 180
ELHCIILSLP PCLKISRRWQ NHLTFPRLHL QNCHYQTIPP HILLATGLIH PSVAWR 236

```

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SEQ ID NO: 67      moltype = AA  length = 229
FEATURE           Location/Qualifiers
source            1..229
                  mol_type = protein
                  organism = Mus musculus

```

```

SEQUENCE: 67
MSSETGPPAV DPTLRRRIEP HEFEVFFDPR ELRKETCLLY EINWGGRRHSV WRHTSQNTSN 60
HVEVNFLEKF TTERYFPRNT RCSITWFLSW SPCGECSRAI TEFLSRHPYV TLFYIARLY 120
HHTDQRNRQG LRDLISSGVT IQIMTEQEYC YCWRNFVNYP PSNEAYWPRY PHLWVKLYVL 180
ELYCIILGLP PCLKILRRKQ PQLTFFITL QTCHYQRIPP HLLWATGLK 229

```

```

SEQ ID NO: 68      moltype = AA  length = 229
FEATURE           Location/Qualifiers
source            1..229
                  mol_type = protein
                  organism = Rattus rattus

```

```

SEQUENCE: 68
MSSETGPPAV DPTLRRRIEP HEFEVFFDPR ELRKETCLLY EINWGGRRHSI WRHTSQNTNK 60
HVEVNFIEKF TTERYFPCNT RCSITWFLSW SPCGECSRAI TEFLSRYPHV TLFYIARLY 120
HHADPRNRQG LRDLISSGVT IQIMTEQESG YCWRNFVNYS PSNEAHWPY PHLWVRLYVL 180
ELYCIILGLP PCLNILRRKQ PQLTFFITL QSCHYQLRPP HILWATGLK 229

```

```

SEQ ID NO: 69      moltype = AA  length = 191
FEATURE           Location/Qualifiers
source            1..191
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 69
MEAKAAPKPA ASGACVSAA ETEKWMEEAM HMAKEALENT EVPVGCLMVY NNEVVGKGRN 60
EVNQTKNATR HAEMVAIDQV LDWCRQSGKS PSEVFHTVL YVTVEPCIMC AAALRLMKIP 120
LVVYGCQNER FGGCGSVLNI ASADLPNTGR PFQCIPGYRA EEAVEMLKTF YKQENPNAPK 180
SKVRKKECQK S 191

```

```

SEQ ID NO: 70      moltype = AA  length = 191
FEATURE           Location/Qualifiers
source            1..191
                  mol_type = protein
                  organism = Mus musculus

```

```

SEQUENCE: 70
MEEKVESTTT PDGPCVVSQV ETEKWMEEAM RMAKEALENI EVPVGCLMVY NNEVVGKGRN 60
EVNQTKNATR HAEMVAIDQV LDWCHQHQS PSTVFEHTVL YVTVEPCIMC AAALRLMKIP 120
LVVYGCQNER FGGCGSVLNI ASADLPNTGR PFQCIPGYRA EEAVELLKTF YKQENPNAPK 180
SKVRKKECQK S 191

```

```

SEQ ID NO: 71      moltype = AA  length = 499
FEATURE           Location/Qualifiers
source            1..499
                  mol_type = protein
                  organism = Mus musculus

```

```

SEQUENCE: 71
MWTADEIAQL CYAHYNVRLP KQKPEPNRE WTLAAVVKI QASANQACDI PEKEVQVTK 60
VVSMGTGTC IGQSKMRESG DILNDSHA EI IARRSFQRYL LHQLHLAAVL KEDSIFVPGT 120
QRGLWRLRPD LSFVFFSSHT PCGDASIIPM LEFEEQPCCP VIRSWANNSP VQETENLED 180
KDKRNCEDPA SPVAKMRLG TPARSLSNCV AHHGTEQESG VKPDVSSDL TKEEPDAANG 240
IASGSFRVVD VYRTGAKCVP GETGDLREPG AAYHQVGLLR VKPGRGDRTC SMSCSDKMAR 300
WNVLCGQCAL LMHFLEKFIY LSAVVIKCP YSQEAMRRAL TGRCEETLVL PRGFGVQELE 360
IQQSGLLFEQ SRCVHRKRG DSPGRLVPCG AAISWSAVPQ QPLDVTANGF PQGTTKKEIG 420
SPRARSRIK VELFRSFQKL LSSIADDEQP DSIRVTKKLD TYQEYKDAAS AYQEAWGALR 480
RIQPFASWIR NPPDYHQFK 499

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SEQ ID NO: 72 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 72
 MWTADEIAQL CYEHYGIRLP KKGKPEPNHE WTLAAVVKI QSPADKACDT PDKPVQVTKE 60
 VVSMGTGTHK IGQSKMRKNG DILNDSHAEV IARRSFQRYL LHQLQLAATL KEDSIFVPGT 120
 QKGVWKLRRD LIFVFFSSHT PCGDASIIPM LEFEDQPCCP VFRNWAHNS VEASSNLEAP 180
 GNERKCEDPD SPVTKKMRLE PGTAAREVTN GAAHHQSPGK QKSGPISPFI HSCDLTVEGL 240
 ATVTTRIAPGS AKVIDVYRTG AKCVPGEAGD SGKPGAAPHQ VGLLRVKPGR GDRTRSMSCS 300
 DKMARWNVLG CQGALLMHLL EEPYILSAVV IGKCPYSQEA MQRALIGRCQ NVSALPKGFG 360
 VQELKILQSD LLFEQSRSAV QAKRADSPGR LVPCGAAISW SAVPEQPLDV TANGFPQGT 420
 KKTIGSLQAR SQISKVELFR SFQKLLSRIA RDKWPHSLRV QKLDYQYK EAASSYQEA 480
 STLKQVFGS WIRNPPDIHQ FK 502

SEQ ID NO: 73 moltype = AA length = 57
 FEATURE Location/Qualifiers
 REGION 1..57
 note = Synthetic Polypeptide
 source 1..57
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 73
 GSGRADALDD FDLMLGSDA LDDFDLMLG SDALDDFDLD MLGSDALDDF DLDMLIN 57

SEQ ID NO: 74 moltype = AA length = 85
 FEATURE Location/Qualifiers
 REGION 1..85
 note = Synthetic Polypeptide
 source 1..85
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 74
 APPTDVSLGD ELHLDGEDVA MAHADALDDF DLDMLGDGDS PGPGFTPHDS APYGALDMAD 60
 FEFQMFTDA LGIDEYGGF PGIRR 85

SEQ ID NO: 75 moltype = AA length = 184
 FEATURE Location/Qualifiers
 REGION 1..184
 note = Synthetic Polypeptide
 source 1..184
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 75
 PSQGISNQAL ALAPSSAPVL AQTMPSSAM VPLAQPPAPA PVLTPGPPQS LSAPVPKSTQ 60
 AGEGLTSEAL LHLQFDADER LGALLGNSTD PGVFTDLASV DNSEFQQLN QGVSMHSTA 120
 EPMLMEYPEA ITRLVTGSQR PPDPAPTPLG TSGLPNGLSG DEDFSSIADM DFSALLSQIS 180
 SSGQ 184

SEQ ID NO: 76 moltype = AA length = 94
 FEATURE Location/Qualifiers
 source 1..94
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 76
 MNMFKEAVTF KDVAFAFTEE ELGLLGPAQR KLYRDVMVEN FRNLLSVGHP PFKQDVSPIE 60
 RNEQLWIMTT ATRRQGNLDT LPVKALLLYD LAQT 94

SEQ ID NO: 77 moltype = AA length = 74
 FEATURE Location/Qualifiers
 source 1..74
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 77
 EQVSPKDVCV DFTQEEWYLL DPAQKILYRD VILENYSNLV SVGYCITKPE VIFKIEQGEE 60
 PWILEKGFP S QCHP 74

SEQ ID NO: 78 moltype = AA length = 42
 FEATURE Location/Qualifiers
 source 1..42
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 78
 EAVTFKDVAV VFTEELGLL DPAQRKLYRD VMLNFRNLL SV 42

SEQ ID NO: 79 moltype = AA length = 69
 FEATURE Location/Qualifiers

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source                1..69
                      mol_type = protein
                      organism = Mus musculus

SEQUENCE: 79
MDLVTYDDVH VNFTQDEWAL LDPSQKSLYK GVMLETYKNL TAIGYIWEH TIEDHFQTSR 60
SHGSNKKTH                                           69

SEQ ID NO: 80        moltype = AA length = 32
FEATURE              Location/Qualifiers
REGION              1..32
                      note = Synthetic Polypeptide
source              1..32
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 80
GSGMNIQMLL EAADYLERRE REAEHGYASM LP                                     32

SEQ ID NO: 81        moltype = AA length = 130
FEATURE              Location/Qualifiers
REGION              1..130
                      note = Synthetic Polypeptide
source              1..130
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 81
GSGMNIQMLL EAADYLERRE REAEHGYASM LPGSGMNIQM LLEAADYLER REREAEHGYA 60
SMLPGSGMNI QMLLEAADYL ERREREAEHG YASMLPGSGM NIQMLLEAAD YLERREREAE 120
HGYASMLPSR                                           130

SEQ ID NO: 82        moltype = AA length = 876
FEATURE              Location/Qualifiers
source              1..876
                      mol_type = protein
                      organism = Homo sapiens

SEQUENCE: 82
MLSGKKAAAA AAAAAAATG TEAGPGTAGG SENGSEVAAQ PAGLSGPAEV GPGAVGERTP 60
RKKEPPRAS PGGLAEPPTS AGPQAGPTVV PGSATPMETG IAETPEGRRRT SRRKRAKVEY 120
REMDSELANL SEDEYYSEEE RNAKAEKEKK LPPPPPQAPP EEENESEPEE PSGQAGGLQD 180
DSSGGYGDGQ ASGVEGAAPF SRLPHDRMTS QEAACFPDII SGPQQTQKVF LFIRNRTLQL 240
WLDNPKIQLT FEATLQQLEA PYNSDTVLVH RVHSYLERHG LINFGIYKRI KPLPTKKTGK 300
VIIIGSGVSG LAAARQLQSF GMDVTLLER DRVGGRVATF RKGNYVADLG AMVVTGLGGN 360
PMAVSKQVN MELAKIKQKC PLYEANGQAD TVKVPKEKDE MVEQEFNRL EATSYLSQL 420
DPNVLNNKPV SLGQALEVVI QLQEKHVKDE QIEHWKKIVK TQBELKELLN KMVNLKEKIK 480
ELHQYQKEAS EVKPPRDITA EFLVSKSHRD LTALCKEYDE LAETQGKLEE KLQELEANPP 540
SDVYLSRRDR QILDWHFANL EFANATPLST LSLKHWDQDD DFETGSHLT VRNGYSCVPV 600
ALAEGLDIKL NTAVRQVRYT ASGCEVIAVN TRSTSQTFIY KCDAVLCTLP LGVLKQPPA 660
VQFVPLPEW KTSVQRMGF GNLNKVVLCP DRVFWDPVSN LFGHVGSTTA SRGELFLFWN 720
LYKAPILLAL VAGEAAGIME NISDDVIVGR CLAILKGIFG SSAPVQPKET VVSRWRADPW 780
ARGSYSYVAA GSSGNDYDLM AQPIPTGPSI PGAPQPIRL FFAGEHTIRN YPATVHGALL 840
SGLREAGRIA DQFLGAMITL PRQATPGVPA QQSPSM                               876

SEQ ID NO: 83        moltype = AA length = 852
FEATURE              Location/Qualifiers
source              1..852
                      mol_type = protein
                      organism = Homo sapiens

SEQUENCE: 83
MLSGKKAAAA AAAAAAATG TEAGPGTAGG SENGSEVAAQ PAGLSGPAEV GPGAVGERTP 60
RKKEPPRAS PGGLAEPPTS AGPQAGPTVV PGSATPMETG IAETPEGRRRT SRRKRAKVEY 120
REMDSELANL SEDEYYSEEE RNAKAEKEKK LPPPPPQAPP EEENESEPEE PSGVEGAAPF 180
SRLPHDRMTS QEAACFPDII SGPQQTQKVF LFIRNRTLQL WLDNPKIQLT FEATLQQLEA 240
PYNSDTVLVH RVHSYLERHG LINFGIYKRI KPLPTKKTGK VIIIGSGVSG LAAARQLQSF 300
GMDVTLLER DRVGGRVATF RKGNYVADLG AMVVTGLGGN PMAVSKQVN MELAKIKQKC 360
PLYEANGQAV PKEKDEMVEQ EFNRLLEATS YLSHQDLFNV LNNKPVSLGQ ALEVVIQLQE 420
KHVKDEQIEH WKIVKTQEE LKELLNKMVN LKEKIKELHQ QYKEASEVKK PRDITAEFLV 480
KSKHRDLTAL CKEYDELAET QGKLEELQE LEANPPSDVY LSSRDRQILD WHFANLEFAN 540
ATPLSTLSLK HWDQDDDEF TGSHTVRNG YSCVPVALAE GLDIKLNTAV RQVRYTASGC 600
EVIANTRST SQTFIYKCDV VLCTPLPLGV LKQPPAVQFV PPLPEWKTSA VQRMGFGNLM 660
KVVLCFDRVF WDPVSNLFGH VGSTTASRGE LFLFWNLKYA PILLALVAGE AAGIMENISD 720
DVIVGRCLAI LKGIFGSSAV PPKETVVS RWRADPWARG YSYVAAGSSG NDYDLMAQPI 780
TPGPSIPGAP QPIRLFFAG EHTIRNYPAT VHGALLSGLR EAGRIADQFL GAMYTLPQRA 840
TPGVPAQQSP SM                                           852

SEQ ID NO: 84        moltype = AA length = 817
FEATURE              Location/Qualifiers
REGION              1..817
                      note = Synthetic Polypeptide

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source          1..817
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 84
SIVAQLSRPD PALAALTNDH LVALACLGGF PALDAVKKGL PHAPALIKRT NRRIPERTSH 60
RVADHAQVVR VLGGFFQCHSH PAQAFDDAMT QFGMSGGGSL PTCSCLDRIV QKDKGPYYTH 120
LGAGPSVAAV REIMENRYGQ KGNARIEIV VYTGKEGKSS HGCPIAKWVL RRSDEEKVL 180
CLVRQRTGHH CPTAVMVVLI MVWDGIPLPM ADRLYTELTE NLKSYNGHPT DRRCTLNENR 240
TCTCQGIDPE TCGASFSGFC SWSMYFNGCK FGRSPSPRRF RIDPSSPLHE KNLEDNLQSL 300
ATRLAPIYKQ YAPVAYQNQV EYENVARECR LGSKEGRPFS GVTACLDPCA HPHRDIHNMN 360
NGSTVVCTLT REDNRSLGVI PQDEQLHVLP LYKLSDTDEF GSKEGMEAKI KSGAIEVLAP 420
RRKKRTCTFT PVPRSGKKRA AMMTEVLAHK IRAVEKKPIP RIKRKNNSTT TNNKPSLSP 480
TLGSNTETVQ PEVKSETEPH FILKSSDNTK TYSLMPSAPH PVKEASPGFS WSPKTASATP 540
APLKN DATAS CGFSERSSTP HCTMPSGRLS GANAAAADGP GISQLGEVAP LPTLSAPVME 600
PLINSEPTG VTEPLTPHQ NHQPSFLTSP QDLASSPMEE DEQHSEADEP PSDEPLSDDP 660
LSPAEKELPH IDEYWSDEH IFLDANIGGV AIAPAHGSVL IECARRELHA TTPVEHPNRM 720
HPTRLSLVPY QHKNLNKPKH GFELNKKIKFE AKEAKNKKMK ASEQKDQAAN EGPEQSSEVN 780
ELNQIPSHKA LTLTHDNVVT VSPYALTHVA GPYNHWV 817

SEQ ID NO: 85      moltype = AA length = 340
FEATURE           Location/Qualifiers
source            1..340
                  mol_type = protein
                  organism = Xenopus laevis

SEQUENCE: 85
ASSPKKKRKV EASMSRVVKP KVASMEEMAA FHTDAYLQHL HKVSEEGDND DPETLEYGLG 60
YDCPITEGIY DYAAAVGGAT LTAAEQLEIG KTRIAVNWPG GWHHAKKDEA SGFCYLNDAV 120
LGILKLREKF DRVLYVMDL HHGDGVEDAF SFTSKVMTVS LHKFSPGFPP GTGDVSDIGL 180
GKGRYYSINV PLQDGIQDDK YYQICEGVK EVFTTFNPEA VVLQLGADTI AGDPMCSFNM 240
TPEGIGKCLK YVLQWQLEPTL ILGGGGYHLP NTARCWTYLT ALIVGRTLSS EIPDHEFFTE 300
YGPDVVLEIT PSCRPRDRND QKVQEILQSI KGNLKRVEVF 340

SEQ ID NO: 86      moltype = AA length = 337
FEATURE           Location/Qualifiers
source            1..337
                  mol_type = protein
                  organism = Saccharomyces cerevisiae

SEQUENCE: 86
ASSPKKKRKV EASRRVAYFY DADVGNYAYG AGHPMKPHRI RMAHSLIMNY GLYKKMEIYR 60
AKPATKQEMC QFHTDEYIDF LSRVTPDNLE MFKRESVKFN VGDDCPVFDG LYEYCSISGG 120
GSMEGAARLN RGKCDVAVNY AGGLHHAKKS EASGFCYLNLD IVLGIIELLR YHPRVLYIDI 180
DVHHGDGVVEE AFYTTDRVMT CSFHKYGEFF PGTGELRDIG VGAGKNYAVN VPLRDGIDDA 240
TYRSVFEPVI KKIMEWYQPS AVVLQCGGDS LSGDRLGCFN LSMEGHANCV NYVKSFGIMP 300
MVVGGGGYTM RNVARTWCFE TGLLNNVVLD KDLPEYF 337

SEQ ID NO: 87      moltype = AA length = 315
FEATURE           Location/Qualifiers
source            1..315
                  mol_type = protein
                  organism = Mesorhizobium loti

SEQUENCE: 87
ASSPKKKRKV EASMLPQIVH HPDYDAGFAT NHRFPMSKYP LLMEALRARG LASPDALNTT 60
EPAPASWLKL AHAADYVDQV ISCSVPEKIE REIGFPVGPR VSLRAQLATG GTILAAARLAL 120
RHGIACNTAG GSHHARRAQG AGFCTFNDVA VASLVLLDEG AAQNILVVDL DVHQGDGTAD 180
ILSDEPGVFT FSMHGERNYP VRKIASDLDI ALPDGTGDA YLRLRLATIL EL SARARWDI 240
VFYNAGVDVH AEDRLGRLAL SNGGLRARDE MVIGHFRALG IPVCGVIGGG YSTDVPALAS 300
RHAILFEVAS TYAEF 315

SEQ ID NO: 88      moltype = AA length = 362
FEATURE           Location/Qualifiers
source            1..362
                  mol_type = protein
                  organism = Homo sapiens

SEQUENCE: 88
ASSPKKKRKV EASMLHTTQL YQHPVETRWP IVYSPRYNIT FMGLEKLHPF DAGKWGKVIN 60
FLKEEKLLSD SMLVEAREAS EEDLLVVHTR RYLNELKWSF AVATITEIPP VIFLPNFLVQ 120
RKVLRPLRTQ TGGTIMAGKL AVERGWAINV GGGFHHCCSD RGGGFCAAYD ITLAIKFLFE 180
RVEGISRATI IDLDAQHNG HERDFMDDKR VYIMDVYNRH IYPGDRFAKQ AIRRKVELEW 240
GTEDDEYLDK VERNIKKSLQ EHLDPVVVYN AGTDILEGDR LGGLSISPAG IVKRDELVFR 300
MVRGRRPVIL MVTSGGYQKR TARIIDASIL NLFGLGLIGP ESPSVSAQNS DTPLLPAPV 360
EF 362

SEQ ID NO: 89      moltype = AA length = 226
FEATURE           Location/Qualifiers
source            1..226
                  mol_type = protein
                  organism = Arabidopsis thaliana

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SEQUENCE: 89
ASSPKKKRKV EASMEFWGIE VKSGKPVTVT PEEGILIHVS QASLGECKNK KGEFVPLHVK 60
VGNQNLVLGT LSTENIPQLF CDLVFDKEFE LSHTWGKGSV YFVGKTPNI EPQGYSEEEE 120
EEEEVPAGN AAKAVAKPFA KPAEVKPAVD DEEDSDSDG MDEDDSDGED SEEEPTPKK 180
PASSKKRANE TTPKAPVSAK KAKVAVTPQK TDEKKKGGA ANQSEF 226

```

```

SEQ ID NO: 90      moltype = AA  length = 272
FEATURE           Location/Qualifiers
source            1..272
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 90
ASSPKKKRKV EASVMGAGIS TPGSIPDFRS PGSGLYSNLQ QYDLPYPEAI FELPFFFHNP 60
KPFFTLAKEL YPGNYKPNVT HYFLRLLHDK GLLRLRYTQN IDGLERVSGI PASKLVEAHG 120
TFASATCTVC QRPFPGEDIR ADVMADRVPR CPVCTGVVKP DIVFFGEPLP QRFLLVHVD 180
PMADLLILG TSLEVEPFAS LTEAVRSSVP RLLINRDLVG PLAWHPRS RD VAQLGDVVHG 240
VESLVELLGW TEEMRDLVQR ETGKLDGPDK EF 272

```

```

SEQ ID NO: 91      moltype = AA  length = 306
FEATURE           Location/Qualifiers
source            1..306
                  mol_type = protein
                  organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 91
ASSPKKKRKV EASTEMSVRK IAAHMKSNPN AKVIFMVAG ISTSCGIPDF RSPGTGLYHN 60
LARLKLYPE AVFDVDFQSD DPLPFYTLAK ELYPGNFRPS KFHLLKLFQ DKDVLKRVYT 120
QNIDTLERGA GVKDDLIEA HGSFAHCHCI GCGKYVPPQV FKSCLAHEPI KDFVKCDVCG 180
ELVKPAIVF GEDLPDSFSE TWLNDSEWLR EKITTSKGHP QQLVIVVGT SLAVYPFASL 240
PEEIPRKVKR VLCNLETVDG FKANKRPTDL IVHQYSDEFA EQLVEELGWQ EDFEKILTAQ 300
GGMGEF 306

```

```

SEQ ID NO: 92      moltype = AA  length = 257
FEATURE           Location/Qualifiers
source            1..257
                  mol_type = protein
                  organism = Escherichia coli

```

```

SEQUENCE: 92
ASSPKKKRKV EASMEKPRVL VLTGAGISAE SGIRTFRAD GLWEEHRVED VATPEGFDRD 60
PELVQAFYNA RRRQLQQPEI QPNAALHALA KLQDALGDRF LLVTQNIIDNL HERAGNTNVI 120
HMHGELLKVR CSQSGQVLDW TGDVTPEDKC HCCQFPAPLR PHVWFGGEMP LGMDEIYMAL 180
SMADIFIAIG TSGHVYPAAG FVHEAKLHGA HTVELNLEPS QVGNEFAEKY YGPASQVVPE 240
FVEKLLKGLK AGSIAEF 257

```

```

SEQ ID NO: 93      moltype = AA  length = 346
FEATURE           Location/Qualifiers
source            1..346
                  mol_type = protein
                  organism = Candida albicans

```

```

SEQUENCE: 93
ASSPKKKRKV EASMPSLDDI LKPVAEAVKN GKKVTFFNGA GISTGAGIPD FRSPDTGLYA 60
NLAKNLNPPA EAVFDIDFFK EDPKPFYTLA EELYPGNFAP TKFHHFIKLL QDQGS�KRVY 120
TONIDTLERL AGVEDKYIVE AHGSFASNHC VDCHKEMTTE TLKTYMKDKK IPSCQHCEGY 180
VKPDIVFFGE GLPVKFFDLW EDDCEDVEVA IVAGTSLTVF PFASLPGEVN KKCLRVLVNK 240
EKVGTFFKHEP RKSDIIALHD CDIVAERLCT LLGLDDKLINE VYEKEKIKYS KAETKEIKMH 300
ETEDKLKEEA HLKEDKHTTK VDKKEKQND A NDKELEQLID KAKAEF 346

```

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SEQ ID NO: 94      moltype = AA  length = 289
FEATURE           Location/Qualifiers
source            1..289
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 94
ASSPKKKRKV EASSSSMADF RKFFAKAKHI VIISGAGVSA ESGVPTFRGA GGYWRKWQAQ 60
DLATPLAFAH NPSRVWEFYH YRREVMSKE PNAGHRAIAE CETRLGKQGR RVVITQNI 120
ELHRKAGTKN LLEIHGSLFK TRCTSCGVVA ENYKSPICPA LSGKGAPEPG TQDASIPVEK 180
LPRCEEAGCG GLLRPHVVWF GENLDPAILE EVDRELAHCD LCLVVGTSV VYPAAMFAPQ 240
VAARGVPVAE PNTETTPATN RFRPHFQGPC GTTLPEALAC HENETVSEF 289

```

```

SEQ ID NO: 95      moltype = AA  length = 288
FEATURE           Location/Qualifiers
source            1..288
                  mol_type = protein
                  organism = Plasmodium falciparum

```

```

SEQUENCE: 95
ASSPKKKRKV EASMGNLMS FLKDDTQSIT LEELAKIIKK CKHVVALTGS GTSAESNIPS 60
FRGSSNSIWS KYDPRIYGTI WGFWKYPEKI WEVIRDISS YEIEINNGHV ALSTLESIGY 120
LKSVVTONVD GLHEASGNTK VISLHGNVFE AVCCCTCNKIV KLNKIMLQKT SHFMHQLPPE 180
CPCGGIFKFN IILFGEVVSS DLLKEAEEEE AKCDLLLVIG TSSTVSTATN LCHFACKKKK 240
KIVEINISKT YITNKMSDYH VCAKFSSELTK VANILKGSSE KNKKIMEF 288

```

SEQ ID NO: 96	moltype = AA length = 304	
FEATURE	Location/Qualifiers	
source	1..304	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 96		
ASSPKKKRKV	EASMSVNYAA	GLSPYADKKG CGLPEIFDPP EELERKVWEL ARLVWQSSSV 60
VFHTGAGIST	ASGIPDFRGP	HGVTWMEERG LAPKFDITFE SARPTQTHMA LVQLERVGLL 120
RFLVSQNVDG	LHVRSGFPRD	KLAELHGNMF VEECAKCKTK YVRDVTVVGTM GLKATGRLCT 180
VAKARGLRAC	RGELRDTILD	WEDSLPDRDL ALADBASRNA DLSITLGTSL QIRPSGNLPL 240
ATKRRGRGLV	IVNLQPTKHD	RHADLRIHGY VDEVMTRLMK HLGLEIPAWD GPRVLERALP 300
PLEF		
SEQ ID NO: 97	moltype = AA length = 234	
FEATURE	Location/Qualifiers	
source	1..234	
	mol_type = protein	
	organism = Chlamydia trachomatis	
SEQUENCE: 97		
ASSPKKKRKV	EASMTTNSTQ	DTLYLSLHGG IDSAIYPYVR RVEQLLQFSF LPQLQFQNA 60
VKQRIQLRCY	REEKRLAVSS	LAKWLQQLHK QRLRAPKNPP VAICWINSYV GYGVPFAESI 120
PAWSYIGEYT	GILRRRQALW	LDENDYCFRY PVPRYSFRYF TIDSGMQGNV TRFINHSDNP 180
NLEAIGAFEN	GIFHIIIRAI	KDILPGEELC YHYGPLYWKH RKKREEFVPQ EEEF 234
SEQ ID NO: 98	moltype = AA length = 134	
FEATURE	Location/Qualifiers	
source	1..134	
	mol_type = protein	
	organism = Paramecium bursaria	
SEQUENCE: 98		
ASSPKKKRKV	EASMFNDRVI	VKKSPLGGYG VFARKSFEKG ELVEECLCIV RHNDWDGTAL 60
EDYLFSRKNM	SAMALGFGAI	FNHSDKPNAR HELTAGLAKMR RIPTIKPIAI GEEITISYGD 120
DYWLSRPRLT	QNEF	
SEQ ID NO: 99	moltype = AA length = 349	
FEATURE	Location/Qualifiers	
source	1..349	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 99		
ASSPKKKRKV	EASNLKCVRI	LKQFHKDLER ELLRRHHRSK TPRHLDPSLA NYLVQKAKQR 60
RALRRWEQEL	NAKRSHLGRI	TVENEVDLDG PPRAPVYINE YRVGEGITLN QVAVGCECQD 120
CLWAPTGGCC	PGASLHKFAY	NDQGGQVRLRA GLPIIYECNSR CRCGYDCPNR VVQKGIRYDL 180
CIFRTDDGRG	WGVRTLKIRI	KNSFVMEYVG EIITSEEAER RGQIYDRQGA TYLFDLDYVE 240
DVYTVDAAYY	GNISHFVNHS	CDPNLQVYNV FIDNLDERLP RIAFFATRTI RAGEELTFDY 300
NMQVDPVME	STRMDSNFG	AGLPGSPKKR VRIECKGTE SCKRYLFEF 349
SEQ ID NO: 100	moltype = AA length = 346	
FEATURE	Location/Qualifiers	
source	1..346	
	mol_type = protein	
	organism = Neurospora crassa	
SEQUENCE: 100		
ASSPKKKRKV	EASMEKAFRP	FFFNHGKCPDA NPKEKKNCWH CQIRSFATHA QLPISIVNRE 60
DDAFLNPNFR	FIDHSIIIGN	VPVADQSFVR GCSCASDEEC MYSTCQCQDE MAPDSDEEAD 120
PYTRKKRFAY	YSQGAKKGLL	RDRVLQSQEP IYECHQGCAC SKDCPNRVVE RGRVTPLQIF 180
PTMQGLGRGV	CPVNLKRQGF	VDRLYGLGII SEEADRRRAE STIARKDQVY LFALDKFSDP 240
DSLDPLLAGQ	PLEVDGEYMS	GPTRFINHSC DPNMAIFARV GDHADKHIHD LALFAIKDIP 300
KGTELTFDYV	NGLTGLESDA	HDPSKISEMT KCLCGTAKCR GYLWEF 346
SEQ ID NO: 101	moltype = AA length = 282	
FEATURE	Location/Qualifiers	
source	1..282	
	mol_type = protein	
	organism = Arabidopsis thaliana	
SEQUENCE: 101		
ASSPKKKRKV	EASDISGGLE	FKGIPATNRV DDSPVSPSTG FTYIKSLIIE PNVIIPKST 60
GCNCRGSCD	SKKACACAKL	GGNFPYVDLN DGRLIESRDV VFECGPHCGC GPKCVNRTSQ 120
KRLRFNLLEV	RSAAKGWAVR	SWEYIPAGSF VCEYIGVVR TADVDITISDN EYIFEIDCQQ 180
TMQGLGGRQ	RLRDVAVPMN	NGVSQSEDE NAPEPCIDAG STGNFARFIN HSCEPNLFVQ 240
CVLSSHQDIR	LARVVLFAAD	NISPMQELTY DYGYALDSVH EF 282
SEQ ID NO: 102	moltype = AA length = 328	
FEATURE	Location/Qualifiers	
source	1..328	
	mol_type = protein	
	organism = Arabidopsis thaliana	

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SEQUENCE: 102
ASSPKKKRKV EASQSAYLHV SLARISDEDC CANCKGNCLS ADFPCTCARE TSGEYAYTKE 60
GLLKEKFLDT CLKMKKEPDS FPKVYCKDCP LERDHDKGTY GKCDGHLIRK FIKECWKRCG 120
CDMQCGNRVV QRGIRCQLQV YFTQEGKGWG LRTLQDLPKG TFICEYIGEI LTNTELYDRN 180
VRSSSERHTY PVTLDADWGS EKDLKDEEAL CLDATICGNV ARFINHRCED ANMIDIPIEI 240
ETPDRHYHYI AFFTLRVDKA MDELTWDYMI DFNDKSHPVK AFRCCCGSES CRDRKIKGSQ 300
GKSIERRKIV SAKKQQGSKE VSKKRKEF 328

SEQ ID NO: 103      moltype = AA  length = 303
FEATURE            Location/Qualifiers
source             1..303
                   mol_type = protein
                   organism = Caenorhabditis elegans

SEQUENCE: 103
ASSPKKKRKV EASQMLHEQI ANISVTFNDI PRSDHSMPTT ELCYFDDFAT TLVVDVLNLF 60
TTHKMSKKRR YLYQDEYRTA RTVMKTFREQ RDWTNAIYGL LTLRSVSHFL SKLPPNKLFE 120
FRDHIVRFLN MFILDSGYTI QECKRYSQEG HQGAKLVSTG VWSRGDKIER LSGVVCLLSS 180
EDEDGILAQE GSDFSVMYST RKRCSTLWLG PGAYINHDCR PTCEFVSHGS TAHIRVLRDM 240
VPGDEITCFY GSEFFGPNNI DCECCTCEKN MNGAFSYLRG NENAEPIISE KTKYELRSR 300
SEF 303

SEQ ID NO: 104      moltype = AA  length = 257
FEATURE            Location/Qualifiers
source             1..257
                   mol_type = protein
                   organism = Caenorhabditis elegans

SEQUENCE: 104
ASSPKKKRKV EASMKVAACK LATSRMRKDR AAAASPSSDI ENSENPSLA SHSSSSGRMT 60
PSKNTRSRKG VSVKDVSNHK ITEFFQVRRS NRKTSKQISD EAKHALRDTV LKGTNERLLE 120
VYKDVVKGKG IRTKVNFEKG DFVVEYRGVM MEYSEAKVIE EQYSNDEEIG SYMYFFEHNH 180
KKWCIDATKE SPWKGRLINH SVLRPNLTKT VVEIDGSHHL ILVARRQIAQ GEELLYDYGD 240
RSAETIAKNP WLVTNEF 257

SEQ ID NO: 105      moltype = AA  length = 224
FEATURE            Location/Qualifiers
source             1..224
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 105
ASSPKKKRKV EASSCDSTNA AIAKQALKKP IKGQAPRKK AQGKTQQNRK LTDFYPVRRS 60
SRKSKAELQS EERKRIDELI ESGKEEGMKI DLIDGKGRGV IATKQFSRGD FVVEYHGDLI 120
EITDAKKREA LYAQDPSTGC YMYFQYLSK TYCVDATRET NRLGRLINHS KCGNCQTKLH 180
DIDGVPHLIL IASRDIAAGE ELLYDYGDRS KASIEAFPWL KHEF 224

SEQ ID NO: 106      moltype = AA  length = 319
FEATURE            Location/Qualifiers
source             1..319
                   mol_type = protein
                   organism = Toxoplasma gondii

SEQUENCE: 106
ASSPKKKRKV EASASRRTGE FLRDAQAPSR WLKRSKTGQD DGAFCLETWL AGAGDDAAGG 60
ERGRDREGAA DKAQREERR QKELEERFEE MKVEFEEKAQ RMIARRALT GEIYSDGKGS 120
KKPRVPSLPE NDDDALIEII IDPEQGILKW PLSVMSIRQR TVIYQECLRR DLTACIHLTK 180
VPGKGRAVFA ADTILKDDFV VEYKGELCSE REAREREQRY NRSKVPMGSF MFYFKNGSRM 240
MAIDATDEKQ DFGPARLINH SRNPNTMPR AITLGDFNSE PRLIFVARRN IEKGEEELLVD 300
YGERDPDVIK EHPWLNSEF 319

SEQ ID NO: 107      moltype = DNA  length = 23
FEATURE            Location/Qualifiers
misc_feature       1..23
                   note = Synthetic Polynucleotide
source            1..23
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 107
gagtccgagc agaagaagaa ggg 23

SEQ ID NO: 108      moltype = DNA  length = 23
FEATURE            Location/Qualifiers
misc_feature       1..23
                   note = Synthetic Polynucleotide
source            1..23
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 108
gaggccgagc agaagaaaga cgg 23

SEQ ID NO: 109      moltype = DNA  length = 23
FEATURE            Location/Qualifiers

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misc_feature      1..23
                  note = Synthetic Polynucleotide
source            1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 109
gagtcctagc aggagaagaa gag                                23

SEQ ID NO: 110      moltype = DNA length = 23
FEATURE            Location/Qualifiers
misc_feature        1..23
                  note = Synthetic Polynucleotide
source              1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 110
gagtctaagc agaagaagaa gag                                23

SEQ ID NO: 111      moltype = DNA length = 23
FEATURE            Location/Qualifiers
misc_feature        1..23
                  note = Synthetic Polynucleotide
source              1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 111
gagtttagagc agaagaagaa agg                                23

SEQ ID NO: 112      moltype = DNA length = 23
FEATURE            Location/Qualifiers
misc_feature        1..23
                  note = Synthetic Polynucleotide
source              1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 112
gggtgggggg agtttgctcc tgg                                23

SEQ ID NO: 113      moltype = DNA length = 23
FEATURE            Location/Qualifiers
misc_feature        1..23
                  note = Synthetic Polynucleotide
source              1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 113
ggatggaggg agtttgctcc tgg                                23

SEQ ID NO: 114      moltype = DNA length = 23
FEATURE            Location/Qualifiers
misc_feature        1..23
                  note = Synthetic Polynucleotide
source              1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 114
gggaggggtgg agtttgctcc tgg                                23

SEQ ID NO: 115      moltype = DNA length = 23
FEATURE            Location/Qualifiers
misc_feature        1..23
                  note = Synthetic Polynucleotide
source              1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 115
cgggggaggg agtttgctcc tgg                                23

SEQ ID NO: 116      moltype = DNA length = 23
FEATURE            Location/Qualifiers
misc_feature        1..23
                  note = Synthetic Polynucleotide
source              1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 116
ggggagggga agtttgctcc tgg                                23

SEQ ID NO: 117      moltype = DNA length = 23

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FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = Synthetic Polynucleotide	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 117		
gcagatgtag tgtttcaca ggg		23
SEQ ID NO: 118	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = Synthetic Polynucleotide	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 118		
acatatgtag tatttcaca ggg		23
SEQ ID NO: 119	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = Synthetic Polynucleotide	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 119		
ccagatgtag tattccaca ggg		23
SEQ ID NO: 120	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = Synthetic Polynucleotide	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 120		
ctagatgaag tgcttcaca tgg		23
SEQ ID NO: 121	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 121		
agctggagga ggaagggct		20
SEQ ID NO: 122	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 122		
ccccctcttc tgcaaatgag		20
SEQ ID NO: 123	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 123		
ggctggggcc agcatgacct		20
SEQ ID NO: 124	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 124		
cctttattca tagtagacaa		20

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SEQ ID NO: 125	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 125		
catggcaaga cagattgtca		20
SEQ ID NO: 126	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 126		
gggaatgggc ttggaaagg		20
SEQ ID NO: 127	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 127		
catctaagga cggatttgtg		20
SEQ ID NO: 128	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 128		
ctggtcagcc cattatgata		20
SEQ ID NO: 129	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 129		
ctggagagag gctcccatca		20
SEQ ID NO: 130	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 130		
catttttgct gtcacaactc		20
SEQ ID NO: 131	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 131		
ctgagtagga ttaagatatt		20
SEQ ID NO: 132	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 132		
gttggaaga gatgcataca		20

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SEQ ID NO: 133	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 133		
gcctccttga ttgaggtgtc		20
SEQ ID NO: 134	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 134		
ctcatctaga gttctttcca		20
SEQ ID NO: 135	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 135		
ctcccatcac atcaaccggt		20
SEQ ID NO: 136	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 136		
cgacagatgt tggggggagg		20
SEQ ID NO: 137	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 137		
gcagcctaga gtcttctgtg		20
SEQ ID NO: 138	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 138		
agccactacc caaccatcta		20
SEQ ID NO: 139	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 139		
catggagtaa aggcaatctt		20
SEQ ID NO: 140	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	

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SEQUENCE: 140		
acccctatt tctgacctcc		20
SEQ ID NO: 141	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 141		
ggtgtcagaa tgcctgtct		20
SEQ ID NO: 142	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 142		
ggatggaagg gccggctccg		20
SEQ ID NO: 143	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 143		
ggaacctgtg atccccacag		20
SEQ ID NO: 144	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 144		
cattcagtgg gtagagtcca		20
SEQ ID NO: 145	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 145		
tggctcttca gtgcaccagg		20
SEQ ID NO: 146	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 146		
aatacaatgg acaaataacc		20
SEQ ID NO: 147	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 147		
gtctggcagg cccctcctgt		20
SEQ ID NO: 148	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	

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source          1..20
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 148
ctttcattag agtttagtcc                                     20

SEQ ID NO: 149      moltype = DNA length = 1239
FEATURE            Location/Qualifiers
misc_feature       1..1239
                    note = Synthetic Polynucleotide
source            1..1239
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 149
tgcccttgccg agggtagccg aatcttcgat ccggtcactg gtacaacgca tcgcatcgag 60
gatgttgctg atggggcgcaa gccattcat gtctgtggctg ctgccaagga cggaacgctg 120
ctcgcgccgc ccgtgggtgct ctgggttcgac cagggaaacgc gggatgtgat cgggttgccg 180
atcgccgggtg gcgccatcgt gtgggcgaca ccgatcaca aggtgtgac agagtacgac 240
tggcgtgccc ccggggaaact ccgcaaggga gacagggtgg ccgaccggg tggttctggt 300
aacagccctg ccttgctcct gacggccgac cagatgggta gtgccttgtt ggatgctgag 360
ccccccatcc tctattccga gtatgatcct accagtcctc tcagtgaagc ttcgatgatg 420
ggcttactga ccaacctggc agacagggag ctgggttcaca tgatcaactg ggcgaagagg 480
gtgccaggct ttgtggattt gacctccat gatcaggccc accttctaga acgtgcctgg 540
ctagagatcc tgatgattgg tctcgtctgg cgctccatgg agcaccagg gaagctactg 600
tttgctccta acttgctctt ggacaggaac cagggaatat gtgtagaggg catggtggag 660
atcttcgaca tgcgtctggc tatcatctct cgggtccgca tgatgaatct gcagggagag 720
gagtttgtgt gccctcaaatc tattattttg ctttaattctg gagtgtacac atttctgtcc 780
agcacccctg agtctctgga agagaaggac catatccacc gagccctgga caagatcacg 840
gacactttga tccacctgat ggccaaggca ggcctgaccc tgcagcagca gcaccagcgg 900
ctggcccagc tctctcctcat cctctccccc atcaggcaca tgagtaacaa aggaatggag 960
catctgtaca gcataagta caagaacgtg gtgcccctct atgacctgct gctggagatg 1020
ctggacgccc accgctctaca tgcgggtggt tctggtgcta gccgcgtgca ggcgttcgag 1080
gatgcccttg atgacaaatt cctgcacgac atgctggcgg aaggactccg ctattccgtg 1140
atccgagaag tgcgtccaac gcggcgggca cgaacgttcg acctcgaggt cgaggaactg 1200
cacaccctcg tgcgcgaagg ggtgtcgtg cacaactgc                                     1239

SEQ ID NO: 150      moltype = AA length = 1404
FEATURE            Location/Qualifiers
REGION            1..1404
                    note = Synthetic Polypeptide
source            1..1404
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 150
MDKKYSIGLD IGTNSVGWAV ITDEYKVP SKFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSFHHR LEESFLVEED KKHHRHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPNDNSD 180
VDKLFQILVQ TYNQLFEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIALSLGLTP NFKSNFLDAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EFYKFIKPIE EKMDOGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASASQ FIERMTNFDK NLPNEKVLPR HSLLYEYFTV YNELTKVKYV TEGMRKPAFL 540
SGEQKKAIVD LLFKNTRKVT VKQLKEDYFK KIECFDSVEI SGVEDRPNAS LGTYHDLKI 600
IKDKDFLDNE ENEDILEDIV LTLTLFEDRE MIEERLKYA HLFDKVMKQ LKRRRYTGWG 660
RLSRKLINGI RDKQSGKITL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGDSL 720
HEHIANLAGS PAIKKGILQT VKVVDLVKV MGRHKPENIV IEMARENQTT QKGQKNSRER 780
MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMYVDQELDI NRLSDYVDH 840
IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV PSEEVVKKMK NYWRQLLNAK LITQRKFDNL 900
TKAERGGLE LDKAGFIKRO LVETRQITKH VAQILDSDRM TKYDENDKLI REVKVITLKS 960
KLVSDFRKDF QFYKVRINN YHHAHDAYLN AVVGTAIIKK YPKLESEFVY GDYKVYDVRK 1020
MIAKSEQEIG KATAKYFFYS NIMNFPKTEI TLANGEIRKR PLIETNGETG EIVWDKGRDF 1080
ATVRKVLSP QVNIVKKTEV QTGGFSKESI LPKRNSDKLI ARKKDWDPPK YGGFDSPTVA 1140
YSVLVVAKEV KGSKKLKS V KELLGITIME RSSFEKNPID FLEAKGYKEV KKDILIKLPK 1200
YSLFELENGR KRMLASAGEL QKGNELALPS KYVNFYLLAS HYEKLKGSPE DNEQKQLFVE 1260
QHKHYLDEII EQISEFSKRV ILADANLDKV LSAYNKHDRK PIREQAENII HLFTLTNLGA 1320
PAAFKYFDTT IDRKYRSTK EVLADTLIHQ SITGLYETRI DLSQLGGDGS PKKKRKVSDD 1380
YKDHGDGYDK HDIDYKDDDD KAAG                                     1404

SEQ ID NO: 151      moltype = AA length = 1816
FEATURE            Location/Qualifiers
REGION            1..1816
                    note = Synthetic Polypeptide
source            1..1816
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 151
MDKKYSIGLD IGTNSVGWAV ITDEYKVP SKFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC LAEGTRIFDP VTGTHRIED VVDGRKPIHV VAAAKDGTLL 120

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ARPVSWFQDQ  GTRDVIGLRI  AGGAIVWATP  DHKVLTEYGW  RAAGELRKGD  RVAGPGGSGN  180
SLALSLTADQ  MVSALLDAEP  PILYSEYDPT  SPFSEASMMG  LLTNLADREL  VHMNNAKRV  240
PGFVDLTLHD  QAHLLERAWL  EILMIGLVWR  SMEHPGKLLF  APNLLDRNQ  GKCVGEMVEI  300
FDMLLATSSR  FRMMNLQGE  FVCLKSIILL  NSGVYTFLLS  TLKSLEEKDH  IHRALDKITD  360
TLIHLMAKAG  LTLQQQHRL  AQLLLILSHI  RHMSNKGMEH  LYSMKYKNV  PLYDLLLEML  420
DAHRLHAGGS  GASRVQAFAD  ALDDKFLHDM  LAEGLRYSVI  REVLPTRRAR  TFDLEVEELH  480
TLVAEGVVVH  NCYLQEIFSN  EMAKVDDSF  HRLEESFLVE  EDKKHERHPI  FGNIVDEVAY  540
HEKYPTIYHL  RKKLVDSSTK  ADLRLIYLAL  AHMIKFRGHF  LIEGDLNPDN  SDVDKLFQIL  600
VQTYNQLFEE  NPINASGVDA  KAILSARLSK  SRRLENLIAQ  LPGEKKNGLF  GNLIALLSLGL  660
TPNFKNFDL  AEDAKLQLSK  DTYDDDLN  LAQIGDQYAD  LFLAAKNLSD  AILLSDILRV  720
NTEITKAPLS  ASMIKRYDEH  HQDLTLLKAL  VRQQLPEKYK  EIFFDQSKNG  YAGYIDGGAS  780
QEEFYKFIKP  ILEKMDGTEE  LLVKLNREDL  LRKQRTFDNG  SIPHQIHLGE  LHAILRRQED  840
FYPFLKDNRE  KIEKILTFRI  PYYVGPLARG  NSRFAWMTRK  SEETITPWNF  EEVVDKGASA  900
QSFIERMTNF  DKNLPNEKVL  PKHSLLEYEF  TVYNELTKVK  YVTEGMRKPA  FLSGEQKKAI  960
VDLLPKTNRK  VTVKQLKEDY  FKKIECFDSV  EISGVEDRFN  ASLGTYHDL  KIIKDKDFLD  1020
NEENEDILED  IVLTLTLFED  REMIEERLKT  YAHLPDDKVM  KQLKRRRYTG  WGRLSRKLIN  1080
GIRDKQSGKT  ILDFLKSDF  ANRNFMLQIH  DDSLTFKEDI  QKAQVSGQGD  SLHEHIANLA  1140
GSPAIIKKGIL  QTVKVVDEL  KVMGRHKPEN  IVIEMARENQ  TTQKGQKNSR  ERMKRIEEGI  1200
KELGSQLKE  HPVENTQLQN  EKLYLYYLQN  GRDMYVDQEL  DINRLSDYDV  DHIVPQSFLK  1260
DDSIDNKVLT  RSDKNRGKSD  NVPSEEVVKK  MKNYWRQLLN  AKLITQRKFD  NLTKAERGGL  1320
SELDKAGFIK  RQLVETQIT  KHVAQILDSR  MNTKYDENDK  LIREVKVITL  KSKLVSDFRK  1380
DFQFYKVRRI  NNYHHAHDAY  LNAVVGITALI  KYPKLESEF  VYGDYKVYDV  RKMIAKSEQE  1440
IGKATAKYFF  YSNIMNFFKT  EITLANGEIR  KRPLIETNGE  TGEIVWDKGR  DFATVRKVLS  1500
MPQVNIVKKT  EVQTTGGFSKE  SILPKRNSDK  LIARKKDWD  KKYGGFSPT  VAYSVLVVAK  1560
VEKGKSKKIK  SVKELLGITI  MERSSFEKNP  IDFLEAKGYK  EVKDKLIIKL  PKYSLFELEN  1620
GRKRMLASAG  ELQKGNELAL  PSKYVNFLYL  ASHYEKLKGS  PEDNEQKQLF  VEQHKHYLDE  1680
IIEQISEFSK  RVILADANLD  KVL SAYNKH  DKPIREQAEN  IHLFTLTNL  GAPAAFKYFD  1740
TTIDRKRYTS  TKEVLDTALI  HQSITGLYET  RIDLSQLGGD  GSPKKKRKVS  SDYKDHGDY  1800
KDHDIDYKDD  DDKAAG  1816

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SEQ ID NO: 152      moltype = AA length = 1816
FEATURE             Location/Qualifiers
REGION              1..1816
                    note = Synthetic Polypeptide
source              1..1816
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 152
MDKKYSIGLD  IGTNSVGWAV  ITDEYKVPK  KFKVLGNTDR  HSIKKNLIGA  LLFDSGETAE  60
ATRLKRTARR  RYTRRKNRIC  YLQEIFSNEM  AKVDDSFHFR  LEESFLVEED  KKHERHPIFG  120
NIVDEVCLAE  GTRIFDPVTG  TTHRIEDVVD  GRKPIHVAA  AKDGTLLARP  VVSWFDQGTR  180
DVIGLRIAGG  AIVWATPDHK  VLTEYGWRAA  GELRKGDRA  GPGGSGNSLA  LSLTADQMVS  240
ALLDAEPPIL  YSEYDPTSPF  SEASMMGLLT  NLADRELVM  INWAKRVPGF  VDLTLHDQAH  300
LLERAWLEIL  MIGLVWRSME  HPGKLLFAPN  LLLDRNQKGC  VEGMVEIFDM  LLATSSRFRM  360
MNLQGEFVC  LKSIILNSG  VYTFLSSTLK  SLEEKDHIHR  ALDKITDTLI  HLMAGAGLTL  420
QQQHQLRAQL  LLLILSHIRM  SNKGMEHLYS  MKYKNVPLY  DLLLEMLDAH  RLHAGGSGAS  480
RVQAFADALD  DKFLHDMALAE  GLRYSVIREV  LPTRRARTFD  LEVEELHTLV  AEGVVVHNCY  540
HEKYPTIYHL  RKKLVDSSTK  ADLRLIYLAL  AHMIKFRGHF  LIEGDLNPDN  SDVDKLFQIL  600
VQTYNQLFEE  NPINASGVDA  KAILSARLSK  SRRLENLIAQ  LPGEKKNGLF  GNLIALLSLGL  660
TPNFKNFDL  AEDAKLQLSK  DTYDDDLN  LAQIGDQYAD  LFLAAKNLSD  AILLSDILRV  720
NTEITKAPLS  ASMIKRYDEH  HQDLTLLKAL  VRQQLPEKYK  EIFFDQSKNG  YAGYIDGGAS  780
QEEFYKFIKP  ILEKMDGTEE  LLVKLNREDL  LRKQRTFDNG  SIPHQIHLGE  LHAILRRQED  840
FYPFLKDNRE  KIEKILTFRI  PYYVGPLARG  NSRFAWMTRK  SEETITPWNF  EEVVDKGASA  900
QSFIERMTNF  DKNLPNEKVL  PKHSLLEYEF  TVYNELTKVK  YVTEGMRKPA  FLSGEQKKAI  960
VDLLPKTNRK  VTVKQLKEDY  FKKIECFDSV  EISGVEDRFN  ASLGTYHDL  KIIKDKDFLD  1020
NEENEDILED  IVLTLTLFED  REMIEERLKT  YAHLPDDKVM  KQLKRRRYTG  WGRLSRKLIN  1080
GIRDKQSGKT  ILDFLKSDF  ANRNFMLQIH  DDSLTFKEDI  QKAQVSGQGD  SLHEHIANLA  1140
GSPAIIKKGIL  QTVKVVDEL  KVMGRHKPEN  IVIEMARENQ  TTQKGQKNSR  ERMKRIEEGI  1200
KELGSQLKE  HPVENTQLQN  EKLYLYYLQN  GRDMYVDQEL  DINRLSDYDV  DHIVPQSFLK  1260
DDSIDNKVLT  RSDKNRGKSD  NVPSEEVVKK  MKNYWRQLLN  AKLITQRKFD  NLTKAERGGL  1320
SELDKAGFIK  RQLVETQIT  KHVAQILDSR  MNTKYDENDK  LIREVKVITL  KSKLVSDFRK  1380
DFQFYKVRRI  NNYHHAHDAY  LNAVVGITALI  KYPKLESEF  VYGDYKVYDV  RKMIAKSEQE  1440
IGKATAKYFF  YSNIMNFFKT  EITLANGEIR  KRPLIETNGE  TGEIVWDKGR  DFATVRKVLS  1500
MPQVNIVKKT  EVQTTGGFSKE  SILPKRNSDK  LIARKKDWD  KKYGGFSPT  VAYSVLVVAK  1560
VEKGKSKKIK  SVKELLGITI  MERSSFEKNP  IDFLEAKGYK  EVKDKLIIKL  PKYSLFELEN  1620
GRKRMLASAG  ELQKGNELAL  PSKYVNFLYL  ASHYEKLKGS  PEDNEQKQLF  VEQHKHYLDE  1680
IIEQISEFSK  RVILADANLD  KVL SAYNKH  DKPIREQAEN  IHLFTLTNL  GAPAAFKYFD  1740
TTIDRKRYTS  TKEVLDTALI  HQSITGLYET  RIDLSQLGGD  GSPKKKRKVS  SDYKDHGDY  1800
KDHDIDYKDD  DDKAAG  1816

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SEQ ID NO: 153      moltype = AA length = 1816
FEATURE             Location/Qualifiers
REGION              1..1816
                    note = Synthetic Polypeptide
source              1..1816
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 153

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFHHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSDCLAE	G	TRIFDPVTGT	THRIEDVVDG	RKPIHVVA	180
KDGTLLARPV	VSWFDQGTDR	VIGLRIAGGA		IVWATPDHKV	LTEYGWRAAG	ELRKGDVRVAG	240
PGSGNSLAL	SLTADQMVA	LLDAEPPILY		SEYDPTSPFS	EASMMGLLTN	LADRELVHMI	300
NWAKRVPGVF	DLTLHDQAH	LERAWLEILM		IGLVWRSMEH	PGKLLFAPNL	LLDRNQKCV	360
EGMVEIFDML	LATSSRFMRM	NLQGEFVCL		KSIIILLNSGV	YTFLSSTLKS	LEEKDHIHRA	420
LDKITDTLH	LMAKAGLTQ	QQHQRLAQLL		LILSHIRHMS	NKGMEHLYSM	KYKNVPLYD	480
LLLEMLDAHR	LHAGSGASR	VQAFADALDD		KPLHDMLEAG	LRYSVIREVL	PTRRARTFDL	540
EVEELHTLVA	EGVVVHNC	ADLRLLIYAL		AHMIKFRGHF	LIEGDLNPDN	SDVDKLFQI	600
VQTYNQLFEE	NPINASGVA	KAILSARLSK		SRRLLENLIAQ	LPGEKKNGLF	GNLIALSLGL	660
TPNFKSNFDL	AEDAKQLSK	DTYDDDLN		LAQIGDQYAD	LFLAAKNLSD	AILLSDILRV	720
NTEITKAPLS	ASMIKRYDEH	HQDLTLKAL		VRQQLPEKYK	EIFFDQSKNG	YAGYIDGGAS	780
QEEFYKFIKP	ILEKMDGTEE	LLVKLNREDL		LRKQRTFDNG	SIPHQIHLGE	LHAILRRQED	840
FYPFLKDNR	KIEKILTFRI	PYYVGPLARG		NSRFAWMTRK	SEETITPWNF	EEVVDKGASA	900
QSFIERMTNF	DKNLPNEKVL	PKHSLLYEYF		TVYNELTKVK	YVTEGMRKPA	FLSGEQKAI	960
VDLLPKTNRK	VTVQQLKEDY	FKKIECFDSV		EISGVEDRFN	ASLGTYHDL	KIICKDFLD	1020
NEENEDILED	IVLTLTLFED	REMIERLKT		YAHLPDDKVM	KQLKRRRYTG	WGRLSRKLIN	1080
GIRDKQSGKT	ILDPLKSDGF	ANRNFMLIH		DDSLTFKEDI	QKAQVSGQGD	SLHEHIANLA	1140
GSPAICKGIL	QTVKVVDEL	KVMGRHKPEN		IVIEMARENQ	TTQKGQKNSR	ERMKRIE	1200
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN		GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260
DDSIDNKVLT	RSDKNRGKSD	NVPSEEVVKK		MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320
SELDKAGFIK	RQLVETRQIT	KHVAQILDSR		MNTKYDENDK	LIREVKVITL	KSKLVSDFRK	1380
DPQFYKVR	NIYHHAHDAY	LNAVVGITALI		KKYPKLESEF	VYGDYKVYDV	RKMIKSE	1440
IGKATAKYFF	YSNIMNFFKT	EITLANGEIR		KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIVKKT	EVQTTGGFSKE	SILPKRNSDK		LIAARKDWD	KKYGGFSPT	VAYSVLVVAK	1560
VEKGSKKLK	SVKELLGITI	MERSSEKPN		IDFLEAKGYK	EVKKDLIKL	PKYSLFELEN	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL		ASHYEKLKGS	PEDNEQQLF	VEQHKHYLDE	1680
IIEQISEFSK	RVILADANLD	KVLSAYNKH		RDKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDA	TLI		HQSITGLYET	RIDLSQLGGD	GSPKKRKVS	1800
KDHDIDYKDD	DDKAAG						1816

SEQ ID NO: 154 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 154

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFHHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSDTKAD		LRLIYALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFQILVQ	TYNQLFENP	INASGVDAKA		ILSARLSKCL	AEGTRIFDPV	TGTHRIEDV	240
VDGRKPIHV	AAAKDGTLLA	RPVVSWFQDQ		TRDVIGLRIA	GGAIVWATPD	HKVLTEYGWR	300
AAGELRKGR	VAGPGSGNS	LALSLTADQM		VSALLDAEPP	ILYSEYDPTS	PFSEASMMGL	360
LTNLADREL	VMINWAKRVP	GFVDLTLHDQ		AHLLERAWLE	ILMIGLVWRS	MEHPGKLLFA	420
PNLLDRNQ	KCEGMVEIF	DMLLATSSRF		RMMNLQGEF	VCLKSIIILN	SGVYTFLSST	480
LKSLEEKDHI	HRAIDKITDT	LIHLMAKAGL		TLQQQHQLRA	QLLLILSHIR	HMSNKGMEHL	540
YSMKYKNVVP	LYDLLLEMLD	AHRLHAGSG		ASRVQAFADA	LDDKFLHDM	AEGLRYSVIR	600
EVLPTRRART	FDVEEELHT	LVAEGVVVHN		CRRLLENLIAQ	LPGEKKNGLF	GNLIALSLGL	660
TPNFKSNFDL	AEDAKQLSK	DTYDDDLN		LAQIGDQYAD	LFLAAKNLSD	AILLSDILRV	720
NTEITKAPLS	ASMIKRYDEH	HQDLTLKAL		VRQQLPEKYK	EIFFDQSKNG	YAGYIDGGAS	780
QEEFYKFIKP	ILEKMDGTEE	LLVKLNREDL		LRKQRTFDNG	SIPHQIHLGE	LHAILRRQED	840
FYPFLKDNR	KIEKILTFRI	PYYVGPLARG		NSRFAWMTRK	SEETITPWNF	EEVVDKGASA	900
QSFIERMTNF	DKNLPNEKVL	PKHSLLYEYF		TVYNELTKVK	YVTEGMRKPA	FLSGEQKAI	960
VDLLPKTNRK	VTVQQLKEDY	FKKIECFDSV		EISGVEDRFN	ASLGTYHDL	KIICKDFLD	1020
NEENEDILED	IVLTLTLFED	REMIERLKT		YAHLPDDKVM	KQLKRRRYTG	WGRLSRKLIN	1080
GIRDKQSGKT	ILDPLKSDGF	ANRNFMLIH		DDSLTFKEDI	QKAQVSGQGD	SLHEHIANLA	1140
GSPAICKGIL	QTVKVVDEL	KVMGRHKPEN		IVIEMARENQ	TTQKGQKNSR	ERMKRIE	1200
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN		GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260
DDSIDNKVLT	RSDKNRGKSD	NVPSEEVVKK		MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320
SELDKAGFIK	RQLVETRQIT	KHVAQILDSR		MNTKYDENDK	LIREVKVITL	KSKLVSDFRK	1380
DPQFYKVR	NIYHHAHDAY	LNAVVGITALI		KKYPKLESEF	VYGDYKVYDV	RKMIKSE	1440
IGKATAKYFF	YSNIMNFFKT	EITLANGEIR		KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIVKKT	EVQTTGGFSKE	SILPKRNSDK		LIAARKDWD	KKYGGFSPT	VAYSVLVVAK	1560
VEKGSKKLK	SVKELLGITI	MERSSEKPN		IDFLEAKGYK	EVKKDLIKL	PKYSLFELEN	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL		ASHYEKLKGS	PEDNEQQLF	VEQHKHYLDE	1680
IIEQISEFSK	RVILADANLD	KVLSAYNKH		RDKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDA	TLI		HQSITGLYET	RIDLSQLGGD	GSPKKRKVS	1800
KDHDIDYKDD	DDKAAG						1816

SEQ ID NO: 155 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide

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source          1..1816
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 155
MDKKYSIGLD IGTNSVGWAV ITDEYKVP SK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSPFHR LEESFLVEED KKHERHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPDNDSD 180
VDKLFIQLVQ TYNQLFEEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIAALSGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLCLAEGETRI FDPVTGTTHR IEDVVDGRKP 360
IHVVAAAKDG TLLARPVVSW FDQGRDVG LRIAGGAIVW ATPDHKVLTE YGWRAAGELR 420
KGD RVAGPGG SGNLSALSLT ADQMVSA LLD AEPPILYSEY DPTSPFSEAS MMGLLTNLAD 480
RELVHMINWA KRVPGFVDLT LHDQAHLLER AWLEILMIGL VWRSMHPGK LLPAPNLLLD 540
RNQ GKCV EGM VEI FDM LLAT SSRFRMMNLQ GEEFVCLKSI ILLNSGVYTF LSSTLKSLEE 600
KDHIHRALDK ITDTLIHLMA KAGLTLQQQH QRLAQLLLIL SHIRHMSNKG MEHLYSMKYK 660
NVVPLYD LLL EMLDAHRLHA GSGASRVQA FADALDDKFL HDMLAEGRLY SVIREVLPTR 720
RARTFDLEVE ELHTLV AEGV VVHNC LLKAL VRQQLPEKYK EIFFDQSKNG YAGYIDGGAS 780
QEEFYKFIKP ILEKMDGTEE LLVKLNREDL LRKQRTFDNG SIPHQIHLGE LHAILRRQED 840
FYPFLKDNRE KIEKILTFRI PYYVGPLARG NSRFAMWTRK SEETITPWNF EEVVDKGASA 900
QSFIERTMNF DKNLPNEKVL PKHSLLYEYF TVYNELTKVK YVTEGMRKPA FLSGEQKKAI 960
VDLLFKTNRK VTVKQLKEDY FKKIECFDSV EISGVEDRFN ASLGTYHDDL KIIKDKDFLD 1020
NEENEDILED IVLTLTLFED REMIEERLKT YAHLFDDKVM KQLKRRRYTG WGRLSRKLIN 1080
GIRDKQSGKT ILDFLKS DGF ANRNF MQLIH DDSLTFKEDI QKAQVSGQGD SLHEHIANLA 1140
GSPA I KKGIL QTVKVVD ELV KVMGRHKPEN IVIEMARENQ TTQKGQKNSR ERMKRIEEGI 1200
KELGSQILKE HPVENTQLQN EKLYLYYLQN GRDMYVDQEL DINRLSDYDV DHIVPQSFLK 1260
DDSIDNKVLT RSDKNRGKSD NVPSEEVVKK MKNYWRQLLN AKLITQRKFD NLTKAERGGL 1320
SELDKAGFIK RQLVETRQIT KHVAQILDSR MNTKYDENDK LIREVKVITL KSKLVSDFRK 1380
DFQFYKVR EI NNYHHAHDAY LNAVVG TALI KKYPKLESEF VYGDYKVYDV RKMIAKSEQE 1440
IGKATAKYFF YSNIMNFFKT EITLANGEIR KRPLIETNGE TGEIVWDKGR DFATVRKVL S 1500
MPQVNI VKKT EVQ TGGFSKE SILPKRNSDK LIARKKDWD P KKYGGFDSPT VAYSVLV VAK 1560
VEKGSKSKLK SVKELLGITI MERSSFEKNP IDFLEAKGYK EVK KDLI IKL PKYSLFELEN 1620
GKRRLMASAG ELQKGNELAL PSKYVNFLYL ASHYEKLKGS PEDNEQKQLF VEQHKHYLDE 1680
IIEQISEFSK RVILADANLD KVL SAYNKH R DKPIREQAEN I IHLFTLTNL GAPAAFKYFD 1740
TTIDRKRYTS TKEVLDATLI HQSITGLYET RIDLSQLGGD GSPKKKRKVS SDYKDHGDY 1800
KDHDIDYKDD DDKAAG 1816

SEQ ID NO: 156      moltype = AA length = 1816
FEATURE            Location/Qualifiers
REGION             1..1816
                   note = Synthetic Polypeptide

source             1..1816
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 156
MDKKYSIGLD IGTNSVGWAV ITDEYKVP SK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSPFHR LEESFLVEED KKHERHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPDNDSD 180
VDKLFIQLVQ TYNQLFEEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIAALSGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EBYKFIKPIE EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASQGS FIERMTNFDK NLPNEKVLPK HSLLYEYFCL AEGTRIFDPV TGTHRIEDV 540
VDGRKPIHVV AAADGTL LLA RPVVS WFDQG TRDVIGLRIA GGAIVWATPD HKVLTEYGWR 600
AAGELRKGRD VAGPGGSGNS LALS LTADQM VSALLDAEPP ILYSEYDPTS PFSEASM MGL 660
LTNLADREL V HMINWAKRVP GFVDLT LHDQ AHLLERAWLE ILMIGLVWRS MEHPGKLLFA 720
PNLLLD RNQG KCV EGM VEIF DMLLATSSRF RMMNLQGE EF VCLKSII LLN SGVYTFLSST 780
LKSLEEKDHI HRA LDKITDT LIHLMAGAGL TLQQQHQLRA QLLLI LSHIR HMSNKGMEHL 840
YSMKYKQVVP LYDLLLEMLD AHR LHAGGSG ASRVQAFADA LDDKFLH DML AGLRYSVIR 900
EVLPTRRART FDLEVEELHT LVAEGVVVHN CVYNELTKVK YVTEGMRKPA FLSGEQKKAI 960
VDLLFKTNRK VTVKQLKEDY FKKIECFDSV EISGVEDRFN ASLGTYHDDL KIIKDKDFLD 1020
NEENEDILED IVLTLTLFED REMIEERLKT YAHLFDDKVM KQLKRRRYTG WGRLSRKLIN 1080
GIRDKQSGKT ILDFLKS DGF ANRNF MQLIH DDSLTFKEDI QKAQVSGQGD SLHEHIANLA 1140
GSPA I KKGIL QTVKVVD ELV KVMGRHKPEN IVIEMARENQ TTQKGQKNSR ERMKRIEEGI 1200
KELGSQILKE HPVENTQLQN EKLYLYYLQN GRDMYVDQEL DINRLSDYDV DHIVPQSFLK 1260
DDSIDNKVLT RSDKNRGKSD NVPSEEVVKK MKNYWRQLLN AKLITQRKFD NLTKAERGGL 1320
SELDKAGFIK RQLVETRQIT KHVAQILDSR MNTKYDENDK LIREVKVITL KSKLVSDFRK 1380
DFQFYKVR EI NNYHHAHDAY LNAVVG TALI KKYPKLESEF VYGDYKVYDV RKMIAKSEQE 1440
IGKATAKYFF YSNIMNFFKT EITLANGEIR KRPLIETNGE TGEIVWDKGR DFATVRKVL S 1500
MPQVNI VKKT EVQ TGGFSKE SILPKRNSDK LIARKKDWD P KKYGGFDSPT VAYSVLV VAK 1560
VEKGSKSKLK SVKELLGITI MERSSFEKNP IDFLEAKGYK EVK KDLI IKL PKYSLFELEN 1620
GKRRLMASAG ELQKGNELAL PSKYVNFLYL ASHYEKLKGS PEDNEQKQLF VEQHKHYLDE 1680
IIEQISEFSK RVILADANLD KVL SAYNKH R DKPIREQAEN I IHLFTLTNL GAPAAFKYFD 1740
TTIDRKRYTS TKEVLDATLI HQSITGLYET RIDLSQLGGD GSPKKKRKVS SDYKDHGDY 1800
KDHDIDYKDD DDKAAG 1816

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SEQ ID NO: 157      moltype = AA  length = 1816
FEATURE            Location/Qualifiers
REGION             1..1816
                   note = Synthetic Polypeptide
source             1..1816
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 157
MDKKYSIGLD IGTSVGVAV ITDEYKVP SKFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSPFHR LEESFLVEED KKHERHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPNDNSD 180
VDKLFIQLVQ TYNQLFEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIALSLGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EFYKFIKPIL EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASQAS FIERMTNFDK NLPNEKVLPK HSLLYEYFTV YNELTKVKYV TEGMRKPAPL 540
SGEQKKAIVD LLFKNRKVT VKQLKEDYFK KIECLAEGTR IFDPVTGTH RIEDVVDGRK 600
PIHVVAANKD GTLLARPVVS WFDQGTDRVI GLRIAGGAIV WATPDHKVLT EYGWRAAGEL 660
RKGDRVAGPG GSGNSLALS LADQMVSAAL DAEPPILYSE YDPTSPFSEA SMMGLLTNLA 720
DRELVHMINW AKRVPGFVDL TLHDQAHLLR RAWLEILMIG LVWRSMHPG KLLFAPNLLL 780
DRNQGKCVGE MVEIFDMLLA TSSRFRMMNL QGEFEVCLKS IILLNSGVYT FLSSTLKSLE 840
EKDHIHRALD KITDTLIHLM AKAGLTQQQ HQRLAQLLLI LSHIRHMSNK GMEHLYSMKY 900
KNVVPLYDLL LEMLDARHLH AGSGASRVQ AFADALDDKF LHDMLAEGLR YSVIREVLPT 960
RRARTFDLEV EELHTLVAEG VVVHNCFDSV EISGVEDRPN ASLGTYHDL KIIKDKDFLD 1020
NEENEDILED IVLTLTFED REMIEERLKT YAHLPDDKVM KQLKRRRYTG WGRLSRKLIN 1080
GIRDQKSGKT LDFLKSDFG ANRNFMLIHL DDSLTFKEDI QKAQVSGQGD SLHEHIANLA 1140
GSPAIIKKGIL QTVKVVDELV KVMGRHKPEN IVIEMARENQ TTQKGQKNSR ERMKRIEEGI 1200
KELGSQILKE HPVENTQLQN EKLYLYYLQN GRDMYVDQEL DINRLSDYDV DHIVPQSFLK 1260
DDSIDNKVLT RSDKNRGKSD NVPSEEVVKK MKNYWRQLLN AKLITQRKFD NLTKAERGGL 1320
SELDKAGFIK RQLVETRQIT KHVAQILDSR MNTKYDENDK LIREVKVITL KSKLVSDFRK 1380
DFQFYKVR EI NNYHHAHDAY LNAVVGTA LI KYPKLESEF VYGDYKVYDV RKMIAKSEQE 1440
IGKATAKYFF YSNIMNFFKT EITLANGEIR KRPLIETNGE TGEIVWDKGR DFATVRKVL S 1500
MPQVNIKKIT EVQTTGGFSKE SILPKRNSDK LIARKKDWD P KKYGGFDSPT VAYSVLVVA 1560
VEKGSKKLKV SVKELLGITI MERSSFENP IDFLKAGYK EVKDLIKL PKYSLFELEN 1620
GRKRLASAG ELQKGNELAL PSKYVNFYLY ASHYEKLKGS PEDNEQKQLF VEQHKHYLDE 1680
IIEQISEFSK RVILADANLH KVLASAYNKH DKPIREQAEN IHLFTLTNL GAPAAFKYFD 1740
TTIDRKRYTS TKEVLDAITLI HQSITGLYET RIDLSQLGGD GSPKKKRVKS SDYKDHGDY 1800
KDHDIDYKDD DDKAAG 1816

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SEQ ID NO: 158      moltype = AA  length = 1816
FEATURE            Location/Qualifiers
REGION             1..1816
                   note = Synthetic Polypeptide
source             1..1816
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 158
MDKKYSIGLD IGTSVGVAV ITDEYKVP SKFKVLGNTDR HSIKKNLIGA LLFDSGETAE 60
ATRLKRTARR RYTRRKNRIC YLQEIFSNEM AKVDDSPFHR LEESFLVEED KKHERHPIFG 120
NIVDEVAYHE KYPTIYHLRK KLVDSSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPNDNSD 180
VDKLFIQLVQ TYNQLFEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN 240
LIALSLGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI 300
LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLKALVR QQLPEKYKEI FFDQSKNGYA 360
GYIDGGASQE EFYKFIKPIL EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 420
AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE 480
VVDKGASQAS FIERMTNFDK NLPNEKVLPK HSLLYEYFTV YNELTKVKYV TEGMRKPAPL 540
SGEQKKAIVD LLFKNRKVT VKQLKEDYFK KIECPDSVEI SGVEDRPNAS LGTYHDLKLI 600
IKDKDFLDNE ENEDILEDIV LCLAEGTRIF DPVTGTHRI EDVVDGRKPI HVVAAKDGT 660
LLARPVVSF DQGTDRVIGL RIAGGAIVWA TPDHKVLT EY GWRAAGELRK GDRVAGPGGS 720
GNSLALS LTA DQMVSAALLDA EPPILYSEYD PTPSPFSEASM MGLLTNLADR ELVHMINWAK 780
RVPGFVDLTL HDQAHLLERA WLEILMIGLV WRSMEHPGKL LFAPNLLLD R NQKCVGEMV 840
EIFDMLLAT SRFMMNLQGE EEFVCLKSII LLNSGVYTF L SSTLKSLEEK DHIHRALDKI 900
TDTLIHLMAG AGLTLQQQHQ RLAQLLLILS HIRHMSNKG M EHLYSMKYKN VVPLYDLLE 960
MLDAHRLHAG GSGASRVQAF ADALDDKFLH DMLAEGRLYS VIREVLPTRR ARTFDLEVEE 1020
LHTLVAEGVV VHNCLTFED REMIEERLKT YAHLPDDKVM KQLKRRRYTG WGRLSRKLIN 1080
GIRDQKSGKT LDFLKSDFG ANRNFMLIHL DDSLTFKEDI QKAQVSGQGD SLHEHIANLA 1140
GSPAIIKKGIL QTVKVVDELV KVMGRHKPEN IVIEMARENQ TTQKGQKNSR ERMKRIEEGI 1200
KELGSQILKE HPVENTQLQN EKLYLYYLQN GRDMYVDQEL DINRLSDYDV DHIVPQSFLK 1260
DDSIDNKVLT RSDKNRGKSD NVPSEEVVKK MKNYWRQLLN AKLITQRKFD NLTKAERGGL 1320
SELDKAGFIK RQLVETRQIT KHVAQILDSR MNTKYDENDK LIREVKVITL KSKLVSDFRK 1380
DFQFYKVR EI NNYHHAHDAY LNAVVGTA LI KYPKLESEF VYGDYKVYDV RKMIAKSEQE 1440
IGKATAKYFF YSNIMNFFKT EITLANGEIR KRPLIETNGE TGEIVWDKGR DFATVRKVL S 1500
MPQVNIKKIT EVQTTGGFSKE SILPKRNSDK LIARKKDWD P KKYGGFDSPT VAYSVLVVA 1560
VEKGSKKLKV SVKELLGITI MERSSFENP IDFLKAGYK EVKDLIKL PKYSLFELEN 1620
GRKRLASAG ELQKGNELAL PSKYVNFYLY ASHYEKLKGS PEDNEQKQLF VEQHKHYLDE 1680

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IIEQISEFSK	RVILADANLD	KVLSAYNKH	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDTALI	HQSITGLYET	RIDLSQLGGD	GSPKKKRKVS	SDYKDHDGDY	1800
KDHDIDYKDD	DDKAAG					1816

SEQ ID NO: 159 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 159

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60		
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFPHR	LEESFLVEED	KKHERHPIFG	120		
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD		LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180		
VDKLFIQLVQ	TYNQLFEEFN	INASGVDAKA		ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240		
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT		YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300		
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ		DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360		
GYIDGGASQE	EFYKFIKPIL	EKMDGTEELL		VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420		
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY		YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480		
VVDKGASQ	FIERTMTNFDK	NLPNEKVL	PK	HSLLYEYFTV	YNELTKVKYV	TEGMRKPAFL	540		
SGEQKKAIVD	LLFKTNRKVT	VKQLKEDYFK		KIECFDSVEI	SGVEDRFNAS	LGTYHDL	600		
IKDKDFLDNE	ENEDILEDIV	LTTLTFEDRE		MIEERLKTYA	HLFDDKVMKQ	LKRRRYTGWG	660		
RLSRKLINGI	RDQSGKTIL	DFLKSDGFAN		RNFMQLIHDD	CLAEGTRIFD	PVTGTTHRIE	720		
DVVDGRKPIH	VVAAAKDGT	LARPVVS	WFD	QGTRDVI	GLR	IAGGAIVWAT	PDHKVLT	780	
WRAAGELRKG	DRVAGPGGSG	NSLALS	LTAD	QMV	SALLDAE	PPILYSEYDP	TSPFSEASMM	840	
GLLTNLADRE	LVHMINWAKR	VPGFVDL	TLH	DQAHLLERAW	LEILMIGLVW	RSMEHPGKLL	900		
FAPNLLLDNR	GQKCV	EGMVE		IFDMLLATSS	RFRMMNLQGE	EFVCLKSIL	LNSGVYTF	960	
STLKSLEEKD	HIHRALDKIT	DTLIHLMAKA		GLTLQQQHQR	LAQLLLLISH	IRHMSNKGME	1020		
HLYSMKYKNV	VPLYDLLLEM	LDARHLHAGG		SGASRVQ	AFA	DALDDKFLHD	MLAEGRLYSV	1080	
IREVLPTRRR	RTFPLEVEEL	HTLVAEGVVV		HNCLTFKEDI	QKAQVSGQGD	SLHEHIANLA	1140		
GSPAIKKGIL	QTVKVVD	DEL		KVMGRHKPEN	IVIEMARENQ	TTQKGQKNSR	ERMKRIEEGI	1200	
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN		GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260		
DDSIDNKVIT	RSDKNRKGSD	NVPSEEVVKK		MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320		
SELDKAGFIK	RQLVETRQIT	KHVAQILDSR		MNTKYDENDK	LIREVKVITL	KSKLVSDFRK	1380		
DFQFYKVR	EI	NNYHHAH	DAY	LNAVVG	TALI	KKYPKLESEF	VYGDYKYVDV	RKMIAKSEQE	1440
IGKATAKYFF	YSNIMNFFKT	EITLANGEIR		KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500		
MPQVNI	VKK	EVQ	TGGFSKE	SILPKRNSDK	LIARKKDWD	P	KYGGFDSPT	VAYSVLV	1560
VEKGSK	KLK	SVKELLG	ITI	MERSSFEKNP	IDFLEAKGYK	EVVKDLI	IKL	PKYSLFELEN	1620
GRKRLMASAG	BLQKNSAL	PSKYVNF	LYL	ASHYEKLKGS	PEDNEQKQLF	VEQKH	HYLDE	1680	
IIEQISEFSK	RVILADANLD	KVLSAYNKH		DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740		
TTIDRKRYTS	TKEVLDTALI	HQSITGLYET		RIDLSQLGGD	GSPKKKRKVS	SDYKDHDGDY	1800		
KDHDIDYKDD	DDKAAG						1816		

SEQ ID NO: 160 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 160

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60		
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFPHR	LEESFLVEED	KKHERHPIFG	120		
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD		LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180		
VDKLFIQLVQ	TYNQLFEEFN	INASGVDAKA		ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240		
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT		YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300		
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ		DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360		
GYIDGGASQE	EFYKFIKPIL	EKMDGTEELL		VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420		
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY		YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480		
VVDKGASQ	FIERTMTNFDK	NLPNEKVL	PK	HSLLYEYFTV	YNELTKVKYV	TEGMRKPAFL	540		
SGEQKKAIVD	LLFKTNRKVT	VKQLKEDYFK		KIECFDSVEI	SGVEDRFNAS	LGTYHDL	600		
IKDKDFLDNE	ENEDILEDIV	LTTLTFEDRE		MIEERLKTYA	HLFDDKVMKQ	LKRRRYTGWG	660		
RLSRKLINGI	RDQSGKTIL	DFLKSDGFAN		RNFMQLIHDD	SLTFKEDIQK	AQVSGQGD	720		
HEHIANLCLA	EGTRIFDPVT	GTTHRIEDVV		DGRKPIHVVA	AAKDG	TTLLAR	PVVS	780	
RDVIGLRIAG	GAIWVATPDH	KVLTEYGWRA		AGELRKGDRV	AGPGGSGNSL	ALSLTADQMV	840		
SALLDAEPP	I	LYSEYDPTSP		FSEASMMGLL	TNLADREL	VH	MINWAKR	900	
HLLERAWLEI	LMIGLVWRSM	EHPGKLLFAP		NLLLDNRNQK	CVEGMVEIFD	MLLATSSRFR	960		
MMNLQGEFEV	CLKSII	LLNS		GVYTFLSSTL	KSLEEKDHIH	RALDKITD	TL	IHLMAKAGLT	1020
LQQQHQR	LQ	LAQ		LLLILSHIRH	MSNKGMEHLY	SMKYKNV	VPL	YD	1080
SRVQAFADAL	DDKFLHMDLA	EGRLYSVIRE		VLPTRRARTF	DLEVEELHTL	VAEGVVVHNC	1140		
GSPAIKKGIL	QTVKVVD	DEL		KVMGRHKPEN	IVIEMARENQ	TTQKGQKNSR	ERMKRIEEGI	1200	
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN		GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260		
DDSIDNKVIT	RSDKNRKGSD	NVPSEEVVKK		MKNYWRQLLN	AKLITQRKFD	NLTKAERGGL	1320		
SELDKAGFIK	RQLVETRQIT	KHVAQILDSR		MNTKYDENDK	LIREVKVITL	KSKLVSDFRK	1380		
DFQFYKVR	EI	NNYHHAH	DAY	LNAVVG	TALI	KKYPKLESEF	VYGDYKYVDV	RKMIAKSEQE	1440
IGKATAKYFF	YSNIMNFFKT	EITLANGEIR		KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500		
MPQVNI	VKK	EVQ	TGGFSKE	SILPKRNSDK	LIARKKDWD	P	KYGGFDSPT	VAYSVLV	1560
VEKGSK	KLK	SVKELLG	ITI	MERSSFEKNP	IDFLEAKGYK	EVVKDLI	IKL	PKYSLFELEN	1620

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GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD	GSPKKKRKVS	SDYKDHDGDY	1800
KDHDIDYKDD	DDKAAG					1816

SEQ ID NO: 161 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 161

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFHHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD		LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFEENP	INASGVDAKA		ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT		YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ		DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360
GYIDGGASQE	EFYKFIKPI	EKMDGTEELL		VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY		YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480
VVDKGASQ	FIERTMTNFDV	NLPNEKVL	PK	HSLLYEYFTV	YNELTKVKYV	TEGMRKPAFL	540
SGEQKKAIVD	LLPKTNRKVT	VKQLKEDYFK		KIECFDSVEI	SGVEDRPNAS	LGTYHDL	600
IKDKDFLDNE	ENEDILEDIV	LTLTLFEDRE		MIEERLKTYA	HLFDDKVMKQ	LKRRRYTGWG	660
RLSRKLINGI	RDQSGKTIL	DFLKSDGFAN		RNFMQLIHDD	SLTFKEDIQK	AQVSGQGD	720
HEHIANLAGS	PAIKKGILQT	VKVVDLVKV		MGRHKPENIV	IEMARENQTT	QKGQKNSRER	780
MKRIEEGIKE	LGSQILKEHP	VENTQLQNEK		LYLYYLQNGR	DMYVDQELDI	NRLSDYVDH	840
IVPQSFLKDD	SIDNKVLTRS	DKNRGKSDNV		PSEEVVKKMK	NYWRQLLNAK	LITQKFDNL	900
TKAERGGLSE	LDKAGFIKQ	LVETRQITKH		VAQILDSRMN	TKYDENDKLI	REVKVITLKS	960
KLVSDFRKDF	QFYKVRINN	YHHAHDAYLN		AVVGCLAEGT	RIFDPVTGTT	HRIEDVVDGR	1020
KPIHVVA	AAK	DKGTLLARPV		SWFDDQGTRDV	IGLRIAGGAI	VWATPDHKVL	1080
LRKGRDVAGP	GGSGNSLALS	LTADQMVSAL		LDAEPPILYS	EYDPTSPFSE	ASMMGLLTNL	1140
ADRELVHMIN	WAKRVP	GFV		DLTHDQAHLL	ERAWLEILMI	GLVWRSMEHP	1200
LDRNQKGCVE	GMEVIFDML	ATSSRFRMMN		LQGEFVCLK	SIILLNSGVY	TFLSSTLKS	1260
EEKDHIHRAL	DKITDTLIHL	MAKAGLTLQ		QHQLAQLLL	ILSHIRHMSN	KGMEHLYSMK	1320
YKNVVP	LYDL	LLEMLDAHRL		HAGGSGASRV	QAFADALDDK	FLHDMLEAGL	1380
TRRARTFDLE	VEELHTLVAE	GVVVHNCALI		KKYPKLESEF	VYGDYKVDV	RKMIKSEQE	1440
IGKATAKYFF	YSNIMNPFKT	EITLANGEIR		KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIVKKT	EVQTTGGFSKE	SILPKRNSDK		LJARKKDWD	KKYGGFDSPT	VAYSVLVAK	1560
VEKGSKKLK	LVQTKGCV	IT		MERSSFEKNP	IDFLEAKGYK	EVKDKLIKL	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL		ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEEQISEFSK	RVILADANLD	KVLSAYNKHR		DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET		RIDLSQLGGD	GSPKKKRKVS	SDYKDHDGDY	1800
KDHDIDYKDD	DDKAAG						1816

SEQ ID NO: 162 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 162

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFHHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD		LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFEENP	INASGVDAKA		ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT		YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ		DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360
GYIDGGASQE	EFYKFIKPI	EKMDGTEELL		VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY		YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480
VVDKGASQ	FIERTMTNFDK	NLPNEKVL	PK	HSLLYEYFTV	YNELTKVKYV	TEGMRKPAFL	540
SGEQKKAIVD	LLPKTNRKVT	VKQLKEDYFK		KIECFDSVEI	SGVEDRPNAS	LGTYHDL	600
IKDKDFLDNE	ENEDILEDIV	LTLTLFEDRE		MIEERLKTYA	HLFDDKVMKQ	LKRRRYTGWG	660
RLSRKLINGI	RDQSGKTIL	DFLKSDGFAN		RNFMQLIHDD	SLTFKEDIQK	AQVSGQGD	720
HEHIANLAGS	PAIKKGILQT	VKVVDLVKV		MGRHKPENIV	IEMARENQTT	QKGQKNSRER	780
MKRIEEGIKE	LGSQILKEHP	VENTQLQNEK		LYLYYLQNGR	DMYVDQELDI	NRLSDYVDH	840
IVPQSFLKDD	SIDNKVLTRS	DKNRGKSDNV		PSEEVVKKMK	NYWRQLLNAK	LITQKFDNL	900
TKAERGGLSE	LDKAGFIKQ	LVETRQITKH		VAQILDSRMN	TKYDENDKLI	REVKVITLKS	960
KLVSDFRKDF	QFYKVRINN	YHHAHDAYLN		AVVGTA	LIKK	YKLECLAEG	1020
THRIEDVDG	RKPIHVVA	AAK		DKGTLLARPV	VSWFDDQGTRD	VIGLRIAGGA	1080
LTEYGWRAAG	ELRKGDRVAG	PGGSGNSLAL		SLTADQM	VSA	LLDAEPPILY	1140
EASMMGLLTN	LADRELVHMI	NWAKRVP	GFV	DLTHDQAH	L	ERAWLEILM	1200
PGKLLFAPNL	LDRNQKGCV	EGMVEIFDML		LATSSRFRMM	NLQGEFVCL	KSIILLNSGV	1260
YTFLSSTLKS	LEEKDHIHRA	LDKITDTLIH		LMAKAGLTLQ	QQHQLAQLL	LILSHIRHMS	1320
NKGMEHLYSM	KYKNVVP	LYD		LLEMLDAHR	LHAGGSGASR	VQAFADALDD	1380
LRYSVIREVL	TRRARTFDL	EVEELHTLVA		EGVVHNC	EF	VYGDYKVDV	1440
IGKATAKYFF	YSNIMNPFKT	EITLANGEIR		KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIVKKT	EVQTTGGFSKE	SILPKRNSDK		LJARKKDWD	KKYGGFDSPT	VAYSVLVAK	1560

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VEKGKSKKLK	SVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIDL	PKYSLFELEN	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IEQISEFSK	RVILADANLD	KVLSAYNKH	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD	GSPKKKRKVS	SDYKDHGDY	1800
KDHDIDYKDD	DDKAAG					1816

SEQ ID NO: 163 moltype = AA length = 1816
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 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 163

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFPHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD		LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFENP	INASGVDAKA		ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT		YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ		DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360
GYIDGGASQE	EFYKFIKPI	EKMDGTEELL		VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY		YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480
VVDKGASAQ	FIERTMNFDP	NLPNEKVL	PK	HSLLYEYFTV	YNELTKVKYV	TEGMRKPAFL	540
SGEQKKAIVD	LLFKTNRKVT	VKQLKEDYFK		KIECFDSVEI	SGVEDRPNAS	LGTYHDLKI	600
IKDKDFLDNE	ENEDILEDIV	LTTLTFEDRE		MIEERLKYA	HLFDDKVMKQ	LKRRRYTGWG	660
RLSRKLINGI	RDQSGKTI	L	DLKSDGFAN	RNFQMQLI	HDD	SLTFKEDIQK	720
HEHIANLAGS	PAIKKGILQ	T	VKVVDLVKV	MGRHKPENIV	IEMARENQTT	QKGQKNSRER	780
MKRIEEGIKE	LGSQILKEHP	VENTQLQNEK		LYLYYLQNGR	DMYVDQELDI	NRLSDYVDVH	840
IVPQSFLKDD	SIDNKVLTRS	DKNRGKSDNV		PSEEVVKKMK	NYWRQLLNAK	LITQRKFDNL	900
TKAERGGLSE	LDKAGFIK	RQ	LVETRQITKH	VAQILD	SRMN	TKYDENDKLI	960
KLVSDFRKDF	QFYKVR	INN	YHHAHDAYLN	AVVGTALIKK	YPKLESEFVY	GDYKVDVVRK	1020
MIKSEQEI	G	KATAKYFFYS	NIMNFFKTEI	TLANGEIRKR	PLIETNGETG	EIVWDKGRDF	1080
ATVRKVL	SMP	QVNIVKKEV	QTGGFSKESI	LPKRNSDKLI	ARKKDWDPPK	YGGFDSPTVA	1140
YSVLVVAKVE	KF	IFDPVTGTH	RIEDVVDGRK	PIHVVA	AKD	GTLLARPVVS	1200
WFDQGT	RDVI	GLRIAGGAI	WATPDHKVLT	EYGWRAAGEL	RKGD	RVAGPG	1260
TADQMV	SALL	DAEPPILYSE	YDPTSPFSEA	SMMGLLTNLA	DREL	VHMINW	1320
TLHDQAHLLE	RAWLEIMIG	LVWRSM	EHPPG	KLLFAPNLLL	DRNQK	CEVEG	1380
TSSRFRMMNL	QGEEFVCLKS	IILLNSGVYT	FLSSTLKSLE	EKDHIHRALD	KITDTLIHLM		1440
AKAGLT	LQQQ	AFDALAQLLI	LSHIRHMSNK	GMEHLYSMKY	KNVVP	LYDLL	1500
AGSGASRVQ	Q	HLDM	LAEGLR	YSVIREVLT	RRARTF	DELEV	1560
VVVHNCKKLK	SVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIDL	PKYSLFELEN		1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE		1680
IEQISEFSK	RVILADANLD	KVLSAYNKH	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD		1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD	GSPKKKRKVS	SDYKDHGDY		1800
KDHDIDYKDD	DDKAAG						1816

SEQ ID NO: 164 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 164

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFPHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD		LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFENP	INASGVDAKA		ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT		YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300
LLSDILRVNT	EITKAPLSAS	MIKRYDEHHQ		DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360
GYIDGGASQE	EFYKFIKPI	EKMDGTEELL		VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY		YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480
VVDKGASAQ	FIERTMNFDP	NLPNEKVL	PK	HSLLYEYFTV	YNELTKVKYV	TEGMRKPAFL	540
SGEQKKAIVD	LLFKTNRKVT	VKQLKEDYFK		KIECFDSVEI	SGVEDRPNAS	LGTYHDLKI	600
IKDKDFLDNE	ENEDILEDIV	LTTLTFEDRE		MIEERLKYA	HLFDDKVMKQ	LKRRRYTGWG	660
RLSRKLINGI	RDQSGKTI	L	DLKSDGFAN	RNFQMQLI	HDD	SLTFKEDIQK	720
HEHIANLAGS	PAIKKGILQ	T	VKVVDLVKV	MGRHKPENIV	IEMARENQTT	QKGQKNSRER	780
MKRIEEGIKE	LGSQILKEHP	VENTQLQNEK		LYLYYLQNGR	DMYVDQELDI	NRLSDYVDVH	840
IVPQSFLKDD	SIDNKVLTRS	DKNRGKSDNV		PSEEVVKKMK	NYWRQLLNAK	LITQRKFDNL	900
TKAERGGLSE	LDKAGFIK	RQ	LVETRQITKH	VAQILD	SRMN	TKYDENDKLI	960
KLVSDFRKDF	QFYKVR	INN	YHHAHDAYLN	AVVGTALIKK	YPKLESEFVY	GDYKVDVVRK	1020
MIKSEQEI	G	KATAKYFFYS	NIMNFFKTEI	TLANGEIRKR	PLIETNGETG	EIVWDKGRDF	1080
ATVRKVL	SMP	QVNIVKKEV	QTGGFSKESI	LPKRNSDKLI	ARKKDWDPPK	YGGFDSPTVA	1140
YSVLVVAKVE	KF	AEGRIFDPV	TGTHRIEDV	VDGRKPIHVV	AAAKD	GTLLA	1200
RPVVS	WFDQ	G	TRDVIGLRIA	GGAIVWATPD	HKVLTEYGWR	AAGELRK	1260
LALSLTADQM	VSALLDAEPP	ILYSEYDPTS	PFSEASMMGL	LTNLADREL	V	HMINWAKRVP	1320
GFVDLT	LHDQ	AHLLERAWLE	ILMIGLVWRS	MEHPGKLLFA	PNLLLD	RNQG	1380
DMLLATSSRF	RMMNLQGEF	VCLKSIIILLN	SGVYTF	LSST	LKSLEE	KDHI	1440
LIHLM	AKAGL	T	LQQQHQR	LA	QLLLILSHIR	HMSNKGMEHL	1500

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AHRLHAGGSG	ASRVQAFADA	LDDKFLHDM	AEGLRYSVIR	EVLPTRRART	FDLEVEELHT	1560
LVAEGVVVHN	CVKELLGITI	MERSSFEKNP	IDFLEAKGYK	EVKKDLIIKL	PKYSLFELEN	1620
GRKRMLASAG	ELQKGNELAL	PSKYVNFLYL	ASHYEKLKGS	PEDNEQKQLF	VEQHKHYLDE	1680
IIIEQISEFSK	RVILADANLD	KVLSAYNKHR	DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET	RIDLSQLGGD	GSPKKKRKVS	SDYKDHDGDY	1800
KDHDIDYKDD	DDKAAG					1816

SEQ ID NO: 165 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 165

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFPHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD		LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFEEENP	INASGVDAKA		ILSARLSKSR	RLENLIAQLP	GEKKNGLFGN	240
LIALSLGLTP	NFKSNFDLAE	DAKLQLSKDT		YDDDLNLLA	QIGDQYADLF	LAAKNLSDAI	300
LLSDILRVNT	BITKAPLSAS	MIKRYDEHHQ		DLTLLKALVR	QQLPEKYKEI	FFDQSKNGYA	360
GYIDGGASQE	EFYKFIKPI	EKMDGTEELL		VKLNREDLLR	KQRTFDNGSI	PHQIHLGELH	420
AILRRQEDFY	PFLKDNREKI	EKILTFRIPY		YVGPLARGNS	RFAWMTRKSE	ETITPWNFEE	480
VVDKGASAQ	FIERTMTNFDK	NLPNEKVLPK		HSLLYEYFTV	YNELTKVKYV	TEGMRKPAPL	540
SGEQKKAIVD	LLFKTNRKVT	VKQLKEDYFK		KIECPDSVEI	SGVEDRPNAS	LGTYHDLKI	600
IKDKDFLDNE	ENEDILEDIV	LTLTLPEDRE		MIEERLKTYA	HLFDDKVMKQ	LKRRRYTGWG	660
RLSRKLINGI	RDQSGKTIL	DPLKSDGFAN		RNFMQLIHDD	SLTFKEDIQK	AQVSGQGDSL	720
HEHIANLAGS	PAIKKGILQT	VKVVDLVKV		MGRHKPENIV	IEMARENQTT	QKGQKNSRER	780
MKRIEEGIKE	LGSQILKEHP	VENTQLQNEK		LYLYYLQNGR	DMYVDQELDI	NRLSDYDVDH	840
IVPQSFLKDD	SIDNKVLTRS	DKNRGKSDNV		PSEEVVKKMK	NYWRQLLNAK	LITQRKFDNL	900
TKAERGGLE	LDKAGFIKRG	LVETRQITKH		VAQILDSRMN	TKYDENDKLI	REVKVITLKS	960
KLVSDFRKDF	QFYKVVREINN	YHHAHDAYLN		AVVGTALIKK	YPKLESEFVY	GDYKVYDVRK	1020
MTAKSEQEI	KATAKYFFYS	NIMNPFKTEI		TLANGEIRKR	PLIETNGETG	EIVWDKGRDF	1080
ATVRKVLSP	QNVIVKKEFV	QTGGFSKESI		LPKRNSDKLI	ARKKDWDPKK	YGGFDSPTVA	1140
YSVLVAVKE	KGKSKKLKSV	KELLGITIME		RSSFPEKNPID	FLEAKGYKEV	KKDLIIKLPK	1200
YSLFELENGR	KRMLASAGEL	QKGNELALPS		KYVNFLYLAS	HYEKLKGSPE	DNEQKQLFVE	1260
QHKHYLDEII	EQICLAEGTR	IFDPVTGTTT		RIEDVVDGRK	PIHVVAACKD	GTLLARPVVS	1320
WFDQGTDRVI	GLRIAGGAI	WATPDHKVLT		EYGWRAAGEL	RKGDRVAGPG	GSGNSLALSL	1380
TADQMVALL	DAEPPILYSE	YDPTSPFSEA		SMMGLLTNLA	DRELVMHINW	AKRVPGFVDL	1440
TLHDQAHLE	LWVRSMEHPG	KLLFAPNLLL		DRNQKCEVEG	MVEIFDMLLA		1500
TSSRFRMMNL	QGEEFVCLKS	IILLNSGVYT		FLSSTLKSLE	EKDHIHRALD	KITDTLIHLM	1560
AKAGLTQQQ	HQRLAQLLLI	LSHIRHMSNK		GMEHLYSMKY	KNVVPLYDLL	LEMLDAHRLH	1620
AGSGASRVQ	AFADALDDKF	LHDMLAEGLR		YSVIREVLPT	RRARTFDLEV	EELHTLVAEG	1680
VVVHNCFSK	RVILADANLD	KVLSAYNKHR		DKPIREQAEN	IIHLFTLTNL	GAPAAFKYFD	1740
TTIDRKRYTS	TKEVLDATLI	HQSITGLYET		RIDLSQLGGD	GSPKKKRKVS	SDYKDHDGDY	1800
KDHDIDYKDD	DDKAAG						1816

SEQ ID NO: 166 moltype = AA length = 1816
 FEATURE Location/Qualifiers
 REGION 1..1816
 note = Synthetic Polypeptide
 source 1..1816
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 166

MDKKYSIGLD	IGTNSVGWAV	ITDEYKVP	SK	KFKVLGNTDR	HSIKKNLIGA	LLFDSGETAE	60
ATRLKRTARR	RYTRRKNRIC	YLQEIFSNEM		AKVDDSFPHR	LEESFLVEED	KKHERHPIFG	120
NIVDEVAYHE	KYPTIYHLRK	KLVDSTDKAD		LRLIYLALAH	MIKFRGHFLI	EGDLNPDNSD	180
VDKLFIQLVQ	TYNQLFEEENP	INASGVDAKA		ILSARLSKCL	AEGTRIFDPV	TGTTTHRIEDV	240
VDGRKPIHV	AAAKDGTLLA	RPVVSWFDDQ		TRDVIGLRIA	GGAIVWATPD	HKVLTIEYGR	300
AAGELRKGRD	VAGPGSGSNS	LALSLTADQM		VSALLDAEPP	ILYSEYDPTS	PFSEASMMGL	360
LTNLADREL	HMNIWAKRVP	GFVDLTLDH		Q	AHLLEAWLE	ILMIGLVWRS	420
PNLLLDNRQG	KCEVGMVEIF	DMLLATSSRF		RMMNLQGEFF	VCLKSIIILLN	SGVYTFLSST	480
LKSLKEEKDHI	HRALDKITDT	LIHLMAKAGL		TLQQQHQR	LA	QLLLILSHIR	540
YSMKYKVVVP	LYDLLLEMLD	AHRLHAGGSG		ASRVQAFADA	LDDKFLHDM	AEGLRYSVIR	600
EVLPTRRART	FDLEVEELHT	LVAEGVVVHN		CRRLLENLIA	LPGEKKNGLF	GNLIALSLGL	660
TPNFKSNFDL	AEDAKQLSK	DTYDDDLN		LAQIGDQYAD	LFLAAKNLSD	AILLSDILRV	720
NTEITKAPLS	ASMIKRYDEH	HQDLTLLKAL		VRQQQLPEKYK	EIFFDQSKNG	YAGYIDGGAS	780
QEEFYKFIKP	ILEKMDGTTEE	LLVKNREDL		LRKQRTFDNG	SIPHQIHLGE	LHAILRRQED	840
FYPFLKDNRE	KIEKILTFRI	PYYVGPLARG		NSRFAWMTRK	SEETITPWNF	EEVVDKGASA	900
QSFIERMTNF	DKNLPNEKVL	PKHSLLYEYF		TVYNELTKVK	YVTEGMRKPA	FLSGEQKKAI	960
VDLLFKTNRK	VTVKQLKEDY	FKKIECPDSV		EISGVEDRFN	ASLGTYHDL	KIKDKDFLD	1020
NEENEDILED	IVLTLTFFED	REMIERLKT		YAHLPDDKVM	KQLKRRRYTG	WGRLSRKLIN	1080
GIRDKQSGKT	ILDPLKSDGF	ANRNFQMLIH		DDSLTFKEDI	QKAQVSGQGD	SLHEHIANLA	1140
GSPAIIKKGIL	QTVKVVDDEL	KVMGRHKPEN		IVIEMARENQ	TTQKGQKNSR	ERMKRIEEGI	1200
KELGSQILKE	HPVENTQLQN	EKLYLYYLQN		GRDMYVDQEL	DINRLSDYDV	DHIVPQSFLK	1260
DDSIDNKVLT	RSDKNRGKSD	NVPSEEVVKK		MKNYWRQLLN	AKLITQRKFD	NLTKAERGG	1320
SELDKAGFIK	RQLVETQIT	KHVAQILDSR		MNTKYDENDK	LIREVKVITL	SKSLVSDFRK	1380
DPQFYKVR	REI	NNYHHAHDAY		LNAVVGITALI	KKYPKLESEF	VYGDYKVYDV	1440

IGKATAKYFF	YSNIMNFFKT	EITLANGEIR	KRPLIETNGE	TGEIVWDKGR	DFATVRKVLS	1500
MPQVNIYKKT	SVQTKGGSKE	SILPKRNSDI	LIARKKDWDP	KYKGGDFSPT	VAYSVLVEAN	1560
VEKGKSVKKL	EYQELLGITI	MERSSEFPKN	IDFLAEAKGY	EVYKGLDIKL	PYKSLFELBN	1620
KGRKMLASAG	ELQKGNELAL	PSKYVNFYLI	ASHYKELKGS	PEDNQKQOLF	VEQHKHYLDE	1680
IEEQISEFSK	RVLQNDNLAD	KVLSAYNKRH	DKTIREQAEN	IILHFLTNTLN	GAAAPFYDLN	1740
TTIDRKRYTS	TEKVELDATLI	HQSITGLYET	RIDLSQLGGD	GSPKKKRKVS	SDYAKHDGDY	1800
KDHDIDRYKDD	DDKAAG					1816

```

SEQ ID NO: 167          moltype = DNA   length = 153
FEATURE                 Location/Qualifiers
misc_feature            1..153
                        note = Synthetic Polynucleotide
source                  1..153
                        mol_type = other DNA
                        organism = synthetic construct

```

SEQUENCE: 167

tgcagctctca	tgaacttggcc	ttttagtagaa	aacaccattta	gaagagtaga	tgggttgggta	60
gttggtctctt	tctgctttaca	ctcttggtcta	ctatgaataa	agggctcat	ttaaaaggc	120
cgtgctatqqt	tgaagcatag	ctcgaagaact	qct			153

```

SEQ ID NO: 168          moltype = DNA   length = 155
FEATURE                 Location/Qualifiers
misc_feature            1..155
                        note = Synthetic Polynucleotide
source                  1..155
                        mol_type = other DNA
                        organism = synthetic construct

```

SEQUENCE: 168

ctcaccctggg	cgagaaaggt	aacttatgtt	tcagtagcct	ctttctcaat	gtgcttcaac	60
ccatcacggc	ctttgcaaat	agagcccttt	attcatagta	gacacagcta	taagcagaag	120
aaagcccta	ccccaacat	tactctctta	atttgt			155

```

SEQ ID NO: 169          moltype = DNA   length = 17
FEATURE                 Location/Qualifiers
misc_feature            1..17
                        note = Synthetic Polynucleotide
source                 1..17
                        mol_type = other DNA
                        organism = synthetic construct

```

SEQUENCE: 169
ctcttctgct taagactc 17

```
SEQ ID NO: 170      moltype = DNA  length = 17
FEATURE             Location/Qualifiers
misc_feature        1..17
                    note = Synthetic Polynucleotide
source              1..17
                    mol_type = other DNA
                    organism = synthetic construct
```

SEQUENCE: 170
gaqtctaaqc aqaagag 17

```

SEQ ID NO: 171          moltype = DNA   length = 17
FEATURE                 Location/Qualifiers
misc_feature            1..17
                        note = Synthetic Polynucleotide
source                  1..17
                        mol_type = other DNA
                        organism = synthetic construct

```

SEQUENCE: 171
gaqtctaaqc aqaagag 17

```
SEQ ID NO: 172      moltype = AA  length = 248
FEATURE             Location/Qualifiers
source              1..248
                   mol_type = protein
                   organism = Homo sapiens
```

SEQUENCE: 172			
NSLALSVDLT	QWVSALLDAE	PPILYSEYDP	TSPPFSEASML
VPGFVSTDLTH	DQAHLECAW	LEILMIGLVW	RSMEHPGKLL
IFDMLLATSS	RFRMMNLQGE	EFVCLKSIIL	LNSGVYTFLS
DTLHLHMAKA	GLTLQQQHQR	LAQLLLLILSH	IRHMSNKGME
LDAHRLHA			

```
SEQ ID NO: 173      moltype = AA  length = 248
FEATURE             Location/Qualifiers
REGION              1..248
                    note = Synthetic Polypeptide
```

-continued

VARIANT 24
note = Xaa can be any amino acid

VARIANT 29
note = Xaa can be any amino acid

VARIANT 78
note = Xaa can be any amino acid

VARIANT 228
note = Xaa can be any amino acid

source 1..248
mol_type = protein
organism = synthetic construct

SEQUENCE: 173
NSLALSITAD QMVSALLDAE PPIXYSEYXP TSPFSEASMM GLLTNLADRE LVHMINWAKR 60
VPGFVDLTLH DQAHLLXAW LEILMIGLVW RSMHPGKLL FAPNLLDRN QGKCVGMVE 120
IFDMLLATSS RFRMMNLQGE EFVCLKSIIL LNSGVYTPLS STLKSLEEKD HIHRALDKIT 180
DTLIHLMAKA GLTLQQQHQR LAQLLLILSH IRHMSNKGME HLYSMKYXNV VPLYDLLLEM 240
LDAHRLHA 248

SEQ ID NO: 174 moltype = DNA length = 20
FEATURE Location/Qualifiers
misc_feature 1..20
note = Synthetic Polynucleotide
source 1..20
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 174
agctggagga ggaagggcct 20

SEQ ID NO: 175 moltype = DNA length = 20
FEATURE Location/Qualifiers
misc_feature 1..20
note = Synthetic Polynucleotide
source 1..20
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 175
cccttcttc tgcaaatgag 20

SEQ ID NO: 176 moltype = DNA length = 20
FEATURE Location/Qualifiers
misc_feature 1..20
note = Synthetic Polynucleotide
source 1..20
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 176
ggctggggcc agcatgacct 20

SEQ ID NO: 177 moltype = DNA length = 20
FEATURE Location/Qualifiers
misc_feature 1..20
note = Synthetic Polynucleotide
source 1..20
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 177
cctttattca tagtagacaa 20

SEQ ID NO: 178 moltype = DNA length = 20
FEATURE Location/Qualifiers
misc_feature 1..20
note = Synthetic Polynucleotide
source 1..20
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 178
catggcaaga cagattgtca 20

SEQ ID NO: 179 moltype = DNA length = 20
FEATURE Location/Qualifiers
misc_feature 1..20
note = Synthetic Polynucleotide
source 1..20
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 179
gggaatgggc tttggaaagg 20

SEQ ID NO: 180 moltype = DNA length = 20

-continued

FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 180		
catctaagga cggatttg		20
SEQ ID NO: 181	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 181		
ctggtcagcc cattatgata		20
SEQ ID NO: 182	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 182		
ctggagagag gtcccatca		20
SEQ ID NO: 183	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 183		
catttttgct gtcacaactc		20
SEQ ID NO: 184	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 184		
ctgagtagga ttaagatatt		20
SEQ ID NO: 185	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 185		
gttggaaga gatgcataca		20
SEQ ID NO: 186	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 186		
gcctcctga ttgaggtgc		20
SEQ ID NO: 187	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 187		
ctcatctaga gttctttcca		20

-continued

SEQ ID NO: 188	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 188		
ctcccatcac atcaaccggt		20
SEQ ID NO: 189	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 189		
cgacagatgt tggggggagg		20
SEQ ID NO: 190	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 190		
gcagcctaga gtcttctgtg		20
SEQ ID NO: 191	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 191		
agccactacc caaccatcta		20
SEQ ID NO: 192	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 192		
catggagtaa aggcaatctt		20
SEQ ID NO: 193	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 193		
acccctatt tctgacctcc		20
SEQ ID NO: 194	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 194		
ggtgtcagaa tgtcctgtct		20
SEQ ID NO: 195	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 195		
ggatggaagg gccggctccg		20

-continued

SEQ ID NO: 196	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 196		
ggaacctgtg atccccacag		20
SEQ ID NO: 197	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 197		
cattcagtgg gtagagtcca		20
SEQ ID NO: 198	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 198		
tggtcttca gtgcaccagc		20
SEQ ID NO: 199	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 199		
aatacaatgg acaaataacc		20
SEQ ID NO: 200	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 200		
gtctggcagg cccctcctgt		20
SEQ ID NO: 201	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Synthetic Polynucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 201		
ctttcattag agtttagtcc		20
SEQ ID NO: 202	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Synthetic Polynucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 202		
gagtctaagc agaagag		17
SEQ ID NO: 203	moltype = length =	
SEQUENCE: 203		
000		

What is claimed is:

1. A polynucleotide encoding a hybrid protein comprising an intein flanked by two extein sequences of a Cas9 protein, wherein the two extein sequences ligate to form the Cas9 protein upon excision of the intein.

2. The polynucleotide of claim 1, wherein an activity of the Cas9 protein is disrupted by the intein, and wherein the disrupted activity is restored upon excision of the intein from the Cas9 protein.

3. The polynucleotide of claim 1, wherein the Cas9 protein is capable of binding a guide RNA (gRNA) prior to excision of the intein.

4. The polynucleotide of claim 1, wherein the Cas9 protein has no or minimal gRNA binding activity prior to excision of the intein.

5. The polynucleotide of claim 1, wherein the intein is a ligand-dependent intein.

6. The polynucleotide of claim 1, wherein the intein comprises a ligand-binding domain.

7. The polynucleotide of claim 6, wherein the ligand-binding domain comprises an amino acid sequence having at least 90% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 7-14.

8. The polynucleotide of claim 1, wherein the Cas9 protein comprises a nuclease-inactivated Cas9 (dCas9) domain.

9. The polynucleotide of claim 1, wherein the Cas9 protein comprises a Cas9 nickase.

10. The polynucleotide of claim 1, wherein the hybrid protein further comprises a nucleic acid-editing domain.

11. The polynucleotide of claim 10, wherein the nucleic acid-editing domain comprises a heterologous nuclease domain, a recombinase domain, or a deaminase domain.

12. The polynucleotide of claim 11, wherein the deaminase domain comprises a cytidine deaminase domain.

13. The polynucleotide of claim 11, wherein the deaminase domain comprises an adenosine deaminase domain.

14. The polynucleotide of claim 11, wherein the intein is a ligand-dependent intein.

15. The polynucleotide of claim 11, wherein the Cas9 protein comprises a nuclease-inactivated Cas9 (dCas9) domain.

16. The polynucleotide of claim 11, wherein the Cas9 protein comprises a Cas9 nickase.

17. The polynucleotide of claim 11, wherein the Cas9 protein and the deaminase domain are separated by a linker.

18. The polynucleotide of claim 17, wherein the linker comprises a peptide linker.

19. The polynucleotide of claim 17, wherein the linker comprises a non-peptide linker.

20. The polynucleotide of claim 1, wherein (a) the intein is a ligand-dependent intein, (b) the Cas9 protein comprises a dCas9 domain or is a Cas9 nickase, and (c) the Cas9 protein further comprises a deaminase domain.

21. The polynucleotide of claim 20, wherein the deaminase domain comprises a cytidine deaminase domain.

22. The polynucleotide of claim 20, wherein the deaminase domain comprises an adenosine deaminase domain.

23. The polynucleotide of claim 1, wherein the Cas9 protein comprises a Cas9 domain having the amino acid

sequence selected from the group consisting of SEQ ID NO: 2 (Cas9), SEQ ID NO: 4 (Cas9 nickase) and SEQ ID NO: 5 (dCas9).

24. The polynucleotide of claim 1, wherein the intein is inserted at or replaces residue Cys80, Ala127, Thr146, Ser219, Thr333, Thr519, Cys574, Thr622, Ser701, Ala728, Thr995, Ser1006, Ser1154, Ser1159, or Ser1274 of a Cas9 protein having the amino acid sequence set forth as SEQ ID NO: 2, SEQ ID NO: 4, or SEQ ID NO: 5.

25. The polynucleotide of claim 1, wherein the hybrid protein comprises an amino acid sequence having at least 90% identity to any one of SEQ ID NOs: 27-41.

26. The polynucleotide of claim 17, wherein linker is a peptide linker.

27. The polynucleotide of claim 17, wherein linker is a non-peptide linker.

28. A method for site-specific DNA cleavage, comprising:

- introducing the polynucleotide of claim 1 into a cell under conditions where the hybrid protein is expressed;
- contacting the hybrid protein with a ligand, wherein binding of the ligand to the intein induces self-excision of the intein; and
- contacting a DNA with the Cas9 protein, wherein the Cas9 protein is associated with a gRNA, wherein self-excision of the intein from the hybrid protein in step (b) allows the Cas9 protein to cleave the DNA, thereby producing cleaved DNA.

29. A method of DNA editing, the method comprising:

- introducing the polynucleotide of claim 10 into a cell under conditions where the hybrid protein is expressed, wherein the intein is a ligand-dependent intein, and wherein an activity of the Cas9 protein selected from the group consisting of nuclease activity, target nucleic acid binding activity, and gRNA binding activity is disrupted by the intein;
- contacting the hybrid protein with a gRNA capable of targeting the Cas9 protein to a target nucleotide sequence in a DNA molecule;
- contacting the hybrid protein with a ligand that binds to the ligand-dependent intein and induces self-excision of the ligand-dependent intein, thereby restoring the activity of the Cas9 protein; and
- contacting the target nucleotide sequence with the restored Cas9 protein, thereby editing the DNA molecule.

30. The method of DNA editing of claim 29, wherein the nucleic acid-editing domain comprises a cytidine deaminase domain.

31. The method of DNA editing of claim 29, wherein the nucleic acid-editing domain comprises an adenosine deaminase domain.

32. The polynucleotide of claim 6, wherein the ligand-binding domain comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 7-14.

33. The polynucleotide of claim 1, wherein the Cas9 protein comprises an amino acid sequence having at least about 90% identical to a sequence selected from the group consisting of SEQ ID NO: 2, SEQ ID NO: 4 and SEQ ID NO: 5.

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